TRACK W (NONLINEAR WAVE): BLUEPRINT

SCOTT ARMSTRONG AND ZOHAR KOMARGODSKI

ABSTRACT. A proof plan for track W: two-frequency quasi-periodic solutions of the cubic Dirichlet wave equation $u_{tt} = u_{xx} - mu - u^3$ on $[0, \pi]$ for every fixed $m > 0$, excited in the modes $J = \{1, 2\}$ (owner decision D30). Part 0 fixes every normalization once and proves milestone W0: the cubic attachment, the quartic coefficients, the partial Birkhoff normal-form action coefficients, nondegeneracy for every $m > 0$, and the Duffing consistency check. The proposed headline W-NLW2 closes the document. Nothing here is frozen or approved. The milestone plan is in `PLAN.md`.

CONTENTS

1. Conventions and normalizations 3
1.1. Sources and locators 3
1.2. The equation 3
1.3. Sequence spaces 4
1.4. Coordinates, Hamiltonian and equations of motion 4
1.5. Complex coordinates, actions, angles 5
1.6. Comparison with the printed conventions of the sources 6
2. Milestone W0: cubic attachment and quartic normalization 6
2.1. Reconstruction 6
2.2. The quartic coefficients 7
2.3. The cubic gradient 8
2.4. PDE $\leftrightarrow$ coefficient system 9
2.5. Quartic nonresonance 10
2.6. Partial quartic Birkhoff normal form: the action coefficients 11
2.7. Nondegeneracy for every $m > 0$ 12
2.8. Duffing consistency check 13
2.9. Ledger: proved, cited, source gaps 13
3. Milestone W1: weighted sequence calculus 13
3.1. Realization in Lean and transport maps 14
3.2. External facts 14
3.3. Weighted spaces, embeddings, multipliers 16
3.4. Two-sided spaces, Young and the algebra 18
3.5. The cubic field 20
3.6. The sine system and analytic reconstruction 23
3.7. Analytic maps between weighted spaces 25
3.8. Support for the KAM step 28
3.9. Ledger 32
4. Milestone W2: PDE reconstruction 33
4.1. External facts 33
4.2. Reconstruction functionals 34
4.3. The coefficient system along an orbit 35
4.4. Classical solutions from analytic coefficient solutions 36
4.5. The W8 form: conclusion (5) from conclusions (3)–(4) 37

4.6. Lean totality, sources and summary 38
5. Milestone W3: the analytic partial Birkhoff normal form 39
5.1. External facts 39
5.2. Coordinates, symplectic form, Hamiltonian fields 41
5.3. The quartic Hamiltonian in two-sided coordinates 43
5.4. Quantitative small divisors 44
5.5. The generating Hamiltonian 45
5.6. The Lie series of the flow 46
5.7. The normal form 49
5.8. Interface to milestones W4–W6 53
5.9. Constants and their dependence on m 54
5.10. Summary of status 54
6. Milestone W4: the normal homological equation 54
6.1. External facts 55
6.2. Setting and hypotheses 55
6.3. Fourier series with Banach-space coefficients 59
6.4. Divisors and the multiplier solve 60
6.5. The Lie derivative along the normal form 62
6.6. Reading the components off the field norm 63
6.7. The homological equation 64
6.8. Comparison with P96-K and consequences for W5–W6 67
6.9. Interfaces 67
6.10. Ledger 68
7. Milestone W5: one KAM step 68
7.1. External facts 69
7.2. Phase space, symplectic form, smoothing Hamiltonians 69
7.3. The homological equation in the generality needed 70
7.4. Taylor approximation 72
7.5. The step data and the flow 74
7.6. The new Hamiltonian 76
7.7. The KAM step 78
7.8. The action–angle substitution and the initial Hamiltonian 79
7.9. Constants and ledger 81
8. Milestone W6: iteration, convergence, invariance 81
8.1. Sequences and the iterative lemma 82
8.2. Convergence 83
8.3. Invariance 84
8.4. The preparatory step and the interface to W7 85
8.5. Output to W8 87
8.6. Ledger 89
9. Milestone W7: measure of the resonance set 89
9.1. External facts 89
9.2. Indices, weights and the reindexing 90
9.3. Unperturbed frequencies and frequency coordinates 91
9.4. Constants and the order in which they are chosen 92
9.5. Interface: the frequency families delivered by W6 93
9.6. The slicing lemma and the elementary zone estimates 94
9.7. The preparatory zones 96
9.8. The iteration zones 97
9.9. The measure theorem 98

9.10. Action shells and density one 98
9.11. Diophantine frequencies and the frequency limit 100
9.12. Source check, deviations and suspected errata 101
9.13. Summary of the interface 101
10. Milestone W8: shells, patching, family injectivity, and the proof of W-NLW2 102
10.1. External facts 102
10.2. Shell data 103
10.3. The KAM iteration on one shell 105
10.4. The tori of one shell 106
10.5. Patching 109
10.6. Family injectivity 110
10.7. The proof of W-NLW2 111
10.8. Source check, deviations and ledger 113
11. The frozen headline W-NLW2 113
11.1. Definitions 114
11.2. Statement 114

1. Conventions and normalizations

This part fixes every normalization of track W *once*. Later parts, the proposed headline (§11) and all future Lean statements use exactly these conventions; any departure must be declared as a deviation. Section 1.6 compares them with the printed conventions of the wave source, which are not internally consistent (§1.6, and `docs/tracks/W/errata-check-W0.md`).

1.1. Sources and locators.

- **P96-W**: J. Pöschel, *Quasi-periodic solutions for a nonlinear wave equation*, Comment. Math. Helv. **71** (1996), 269–296 (ETH E-Periodica scan). Printed page = PDF page +267.
- **P96-K**: J. Pöschel, *A KAM-theorem for some nonlinear partial differential equations*, Ann. Scuola Norm. Sup. Pisa (4) **23** (1996), 119–148 (Numdam scan). Printed page = PDF page +117.
- **KP96**: S. Kuksin, J. Pöschel, *Invariant Cantor manifolds of quasi-periodic oscillations for a nonlinear Schrödinger equation*, Ann. of Math. **143** (1996), 149–179; *author version 2.1*, cited by its own page numbers (not the journal's).

Versions and hashes are those of `docs/explorations/pde-kam-and-leapfrogging.md`, §1. Every formula of a scanned source relied on below was read on a page image, not only on the OCR layer.

1.2. The equation.

Standing assumption 1.1 (Mass)**.** $m > 0$ is fixed. All constants may depend on m unless stated otherwise; no uniformity as $m \to 0$ or $m \to \infty$ is claimed anywhere.

Definition 1.2 (Cubic Dirichlet wave equation)**.** A *classical solution* on an open interval $I \subset \mathbb{R}$ is a function $u : I \times [0,\pi] \to \mathbb{R}$ such that $u, u_t, u_x, u_{tt}, u_{xx}$ exist and are continuous on $I \times [0,\pi]$ (one-sided in x at the endpoints) and

$$u_{tt} = u_{xx} - mu - u^3 \quad \text{on } I \times [0,\pi], \qquad u(t,0) = u(t,\pi) = 0 \quad (t \in I). \tag{1}$$

Source: P96-W (1) and p. 272 with $f(u) = u^3$; the source writes $u_{tt} = u_{xx} - mu - f(u)$ with f odd, analytic, $f(u) = au^3 + O(u^5)$, $a \neq 0$; we take $a = 1$ and no higher terms (declared specialization).

Definition 1.3 (Dirichlet basis and linear frequencies)**.** For $j \geq 1$ let $\phi_j(x) = \sqrt{2/\pi}\,\sin(jx)$ and $\lambda_j = \sqrt{j^2 + m} > 0$. Then $\int_0^\pi \phi_i \phi_j \, dx = \delta_{ij}$, $-\phi_j'' + m\phi_j = \lambda_j^2 \phi_j$ and $\phi_j(0) = \phi_j(\pi) = 0$.

Source: P96-W p. 277, identical.

We record the elementary facts about λ used repeatedly: $j < \lambda_j < j + \sqrt{m}$, $j/\lambda_j < 1$, the map $j \mapsto \lambda_j$ is strictly increasing, the function $f(s) = \sqrt{s^2 + m}$ is strictly convex on $\mathbb{R}$ with $0 < f'(s) = s/f(s) < 1$ for $s > 0$, f' is strictly increasing, and $\lambda_{j+1} - \lambda_j = (2j+1)/(\lambda_{j+1} + \lambda_j) < 1$.

1.3. **Sequence spaces.**

Definition 1.4 (Weighted sequence spaces)**.** For $a \geq 0$ and $p \in \mathbb{R}$, $\ell^{a,p}$ is the space of real sequences $w = (w_j)_{j \geq 1}$ with

$$\|w\|_{a,p}^2 = \sum_{j \geq 1} |w_j|^2 \, j^{2p} e^{2aj} < \infty,$$

and $\ell_{\mathbb{C}}^{a,p}$ its complex analogue. The phase space is $E_{a,p} = \ell^{a,p} \times \ell^{a,p}$ with $\|(q,p)\|_E = \max(\|q\|_{a,p}, \|p\|_{a,p})$. The symbol p in $\ell^{a,p}$ is the regularity exponent; the second component of $E_{a,p}$ is the momentum, also written p; context separates them.

Source: P96-W p. 277 (the source's s is our p).

Convention 1.5 (Headline weights)**.** The headline and milestone W0 use $a = 1$ and $p = 1$: phase space $E_{1,1}$, nonlinear field in $\ell^{1,2}$. The source allows any $a > 0$, $s > 0$; this specialization removes two headline parameters. Lower exponents $\ell^{1,p}$, $p \leq 1$, appear only as target spaces of the unbounded linear part.

Definition 1.6 (Two-sided spaces)**.** For $a \geq 0$, $s \in \mathbb{R}$, $\ell_b^{a,s}$ is the space of complex sequences $(c_l)_{l \in \mathbb{Z}}$ with

$$\|c\|_{a,s}^2 = \sum_{l \in \mathbb{Z}} \langle l \rangle^{2s} e^{2a|l|} |c_l|^2 < \infty, \qquad \langle l \rangle = \max(1, |l|).$$

At $l = 0$ the weight is 1, so this is exactly the norm $|c_0|^2 + \sum_{l \neq 0} |c_l|^2 |l|^{2s} e^{2|l|a}$ of P96-W p. 279. For $a \geq 0$, $s \geq 0$ and $c, d \in \ell_b^{a,s}$ the convolution is $(c * d)_l = \sum_{i \in \mathbb{Z}} c_i d_{l-i}$; the series converges absolutely for every l by Cauchy–Schwarz, since the weight is ≥ 1 and so $\|c\|_{\ell^2} \leq \|c\|_{a,s}$. (For $s < 0$ it can diverge, e.g. $a = 0$, $s = -1$, $c = d \equiv 1$; every use has $s \geq 0$, e.g. $s \geq 1$ in Lemma 2.4; W1's Lean version is guarded by summability, Definition 3.9.)

1.4. **Coordinates, Hamiltonian and equations of motion.**

Definition 1.7 (Reconstruction)**.** For $(q,p) \in E_{a,p}$ with $a > 0$ put

$$U_m(q)(x) = \sum_{j \geq 1} \frac{q_j}{\sqrt{\lambda_j}} \phi_j(x), \qquad V_m(p)(x) = \sum_{j \geq 1} \sqrt{\lambda_j} \, p_j \phi_j(x).$$

The intended dictionary is $u = U_m(q)$, $u_t = V_m(p)$. Equivalently, if $u = \sum x_j \phi_j$ and $u_t = \sum y_j \phi_j$, then $q_j = \sqrt{\lambda_j} \, x_j$ and $p_j = y_j / \sqrt{\lambda_j}$; this is a symplectic rescaling, $dq_j \wedge dp_j = dx_j \wedge dy_j$.

Source: P96-W p. 277 ($u = \mathcal{S}q$, $v = \mathcal{S}'p$), identical.

Lemma 2.1 shows that for $a > 0$ these series converge absolutely and uniformly on $[0, \pi]$ together with all their x-derivatives, for every $p \in \mathbb{R}$.

Definition 1.8 (Hamiltonian)**.** For $(q,p) \in E_{1,1}$,

$$H_m(q,p) = \Lambda(q,p) + G(q), \qquad \Lambda(q,p) = \frac{1}{2} \sum_{j \geq 1} \lambda_j \big(q_j^2 + p_j^2\big), \qquad G(q) = \frac{1}{4} \int_0^\pi U_m(q)(x)^4 \, dx.$$

Λ is finite on $E_{a,p}$ for $p \geq \frac{1}{2}$; in particular on $E_{1,1}$. The same formula defines $G(q)$ for $q \in \ell^{a,p}$ with $a > 0$, $p \in \mathbb{R}$ ($U_m(q)$ is continuous by Lemma 2.1); this is the domain on which Lemma 2.3 and Proposition 3.17 use G.

Source: P96-W (8) p. 280 and (9) p. 280 with $g(u) = u^4/4$; the energy $\frac{1}{2}\langle v, v\rangle + \frac{1}{2}\langle Au, u\rangle + \int g(u)$ of p. 277 in the coordinates of Definition 1.7.

Definition 1.9 (Cubic gradient)**.** For $a > 0$, $p \in \mathbb{R}$ and $q \in \ell^{a,p}$ let $G_q(q)$ be the sequence

$$G_q(q)_l = \frac{1}{\sqrt{\lambda_l}} \int_0^\pi U_m(q)(x)^3 \phi_l(x)\, dx \qquad (l \geq 1);$$

the integrand is continuous by Lemma 2.1 ($k = 0$), and the value depends only on the sequence q, not on the space $\ell^{a,p}$ containing it. The headline (and the frozen `cubicGrad`) uses $q \in \ell^{1,1}$. For $p \geq 0$, $G_q(q)_l = \partial G/\partial q_l$ by Lemma 2.3; for $p \geq \frac{1}{2}$, $G_q(q) \in \ell^{a,p+1}$ by Theorem 2.5, in particular $G_q(q) \in \ell^{1,2}$ for $q \in \ell^{1,1}$.

Definition 1.10 (Symplectic form and equations of motion)**.** The symplectic form is $\sum_j dq_j \wedge dp_j$ and the Hamiltonian equations are $\dot q_j = \partial H/\partial p_j$, $\dot p_j = -\partial H/\partial q_j$, i.e.

$$\dot q_j = \lambda_j p_j, \qquad \dot p_j = -\lambda_j q_j - G_q(q)_j \qquad (j \geq 1). \tag{2}$$

The linear part $q \mapsto (\lambda_j q_j)$ maps $\ell^{a,p+1}$ boundedly to $\ell^{a,p}$ (norm $\leq \sqrt{1+m}$) but not $\ell^{a,p}$ to itself; the system is read in $\ell^{1,0}$ for $(q,p) \in E_{1,1}$, or coordinatewise. The nonlinear part G_q is smoothing of order one (Theorem 2.5).

Source: P96-W (6) p. 277 and p. 280 ($X_G = \sum(\partial_{p_j} G\, \partial_{q_j} - \partial_{q_j} G\, \partial_{p_j})$), identical.

1.5. **Complex coordinates, actions, angles.**

Definition 1.11 (Complex coordinates)**.** $z_j = (q_j + ip_j)/\sqrt{2}$, $\bar z_j = (q_j - ip_j)/\sqrt{2}$, so $q_j = \sqrt{2}\,\mathrm{Re}\, z_j$, $p_j = \sqrt{2}\,\mathrm{Im}\, z_j$, $|z_j|^2 = (q_j^2 + p_j^2)/2$, and

$$\sum_j dq_j \wedge dp_j = i \sum_j dz_j \wedge d\bar z_j, \qquad \Lambda = \sum_j \lambda_j |z_j|^2.$$

Under the linear flow $\dot q_j = \lambda_j p_j$, $\dot p_j = -\lambda_j q_j$ one has $\dot z_j = -i\lambda_j z_j$.

Source: P96-W p. 281 (definition of z, form $i\sum dz \wedge d\bar z$, $\Lambda = \sum \lambda_j |z_j|^2$), identical; the recovery formula and the rotation sign printed later in P96-W differ, see §1.6.

Definition 1.12 (Actions and radian angles)**.** The (oscillator) action of mode j is $I_j = (q_j^2 + p_j^2)/2 = |z_j|^2$. On $\{I_j > 0\}$ the radian angle φ_j is defined by

$$q_j = \sqrt{2I_j}\cos\varphi_j, \qquad p_j = -\sqrt{2I_j}\sin\varphi_j, \qquad z_j = \sqrt{I_j}\, e^{-i\varphi_j}.$$

Then $dq_j \wedge dp_j = d\varphi_j \wedge dI_j$ (Jacobian determinant 1), $\Lambda = \sum \lambda_j I_j$, and the linear flow is $\dot\varphi_j = \lambda_j$, $\dot I_j = 0$. For tangential modes $j \in J = \{1, 2\}$ we write $I = (I_1, I_2)$; for normal modes $j \geq 3$ we write $Z_j = I_j$.

Convention 1.13 (Period-one torus angles and frequencies)**.** Torus parameters $\theta \in \mathbb{R}^2$ have period one: $\varphi_j = 2\pi\theta_j$. A torus with radian frequency vector ω (period 2π) has campaign frequency $\nu = \omega/(2\pi)$. The pair (θ, Y) with $Y = 2\pi I$ is canonical, $d\varphi \wedge dI = d\theta \wedge dY$. Small divisors in campaign units are $2\pi\, k \cdot \nu + l \cdot \Omega$. The linear torus of action $I \in (0,\infty)^2$ is

$$K_I^0(\theta) = (Q^0, P^0), \qquad Q_j^0 = \sqrt{2I_j}\cos(2\pi\theta_j), \quad P_j^0 = -\sqrt{2I_j}\sin(2\pi\theta_j) \quad (j = 1, 2),$$

and $Q_j^0 = P_j^0 = 0$ for $j \geq 3$, which is invariant under the linear flow with $\nu = (\lambda_1, \lambda_2)/(2\pi)$.

1.6. **Comparison with the printed conventions of the sources.** The following table maps each normalization to P96-W (and, where relevant, KP96). "Same" means checked identical on the page image. Details and derivations are in `docs/tracks/W/errata-check-W0.md`.

Object	This blueprint	P96-W (printed)
basis, λ_j	$\phi_j = \sqrt{2/\pi}\sin jx$, $\lambda_j = \sqrt{j^2+m}$	same (p. 277)
$\ell^{a,p}$	$\sum \lvert w_j\rvert^2 j^{2p} e^{2aj}$	same, exponent called s (p. 277)
two-sided $\ell^{a,s}_b$	weight $\langle l\rangle^{2s} e^{2a\lvert l\rvert}$	same (p. 279)
u, u_t	$U_m(q)$, $V_m(p)$	same, $\mathcal{S}q$, $\mathcal{S}'p$ (p. 277)
H	$\frac12\sum\lambda_j(q_j^2+p_j^2)+\frac14\int U^4$	same, (8)–(9) p. 280
quartic coefficient	$g_{ijkl}=\int_0^\pi \phi_i\phi_j\phi_k\phi_l$ (unweighted)	$G_{ijkl} = g_{ijkl}/\sqrt{\lambda_i\lambda_j\lambda_k\lambda_l}$, (10) p. 281; (11) correct
symplectic form	$\sum dq\wedge dp = i\sum dz\wedge d\bar z$	same (pp. 278, 281)
z	$(q+ip)/\sqrt2$	same (p. 281)
recovery of q,p	$q=\sqrt2\,\mathrm{Re}\,z$, $p=\sqrt2\,\mathrm{Im}\,z$	$q=\mathrm{Re}\,z$, $p=\mathrm{Im}\,z$ (p. 290): **disagrees**
action	$I_j=\lvert z_j\rvert^2=(q_j^2+p_j^2)/2$	$I_j=q_j^2+p_j^2$ (p. 276, informal); $I=\frac12\lvert z\rvert^2$ (p. 287); $\Lambda=\langle\alpha,I\rangle$ with $\alpha=\lambda$ (p. 290) requires $I=\lvert z\rvert^2$: **inconsistent**
quartic in z	$\frac{1}{16}\sum g_{ijkl}\prod(z+\bar z)/\sqrt{\lambda\lambda\lambda\lambda}$	$\frac14\sum G_{ijkl}\prod(z+\bar z)$, then $\sum' G_{ijkl}w^4$ (p. 282): **two factor-4 slips**
normal form	$\frac12\langle AI,I\rangle+\langle BI,Z\rangle$, $A_{ii}=\frac{9}{8\pi\lambda_i^2}$, $A_{ij}=B_{ji}=\frac{3}{2\pi\lambda_i\lambda_j}$	$\bar G_{ij}=\frac{6}{\pi}\frac{4-\delta_{ij}}{\lambda_i\lambda_j}$ (pp. 282, 284), $A=B=\bar G$ (p. 290): 16× ours
rotation sign	$\dot z_j=-i\lambda_j z_j$, φ increases	$\dot z=i\omega z$, flow $e^{i\omega t}$ (pp. 287, 289–290): time-reversed
torus angles	period one, $\nu=\omega/2\pi$	period 2π

Remark 1.14 (Why the disagreements are harmless for existence but fatal for constants). Every nondegeneracy condition used by the source (P96-W (A_1)–(A_3), p. 288) is invariant under multiplying (A,B) by a common positive constant and under rescaling I, and H is even in p, so the time reversal $(q,p)(t)\mapsto(q,-p)(-t)$ maps solutions to solutions. Hence the qualitative theorem is unaffected. The printed constants, frequency shifts, and the amplitude–action dictionary are not, and must not be copied.

2. Milestone W0: cubic attachment and quartic normalization

Throughout, Standing assumption 1.1 ($m>0$) and the conventions of §1 are in force. The results of this part are: (a) the coefficient system (2) is equivalent to the PDE (Proposition 2.6); (b) $G_q:\ell^{1,1}\to\ell^{1,2}$ is a bounded cubic polynomial with an explicit constant (Theorem 2.5); (c) closed form of the quartic coefficients (Lemma 2.2); (d) the action coefficients A,B of the partial quartic Birkhoff normal form for $J=\{1,2\}$ (Theorem 2.10); (e) the nondegeneracy conditions of the source for every $m>0$ (Theorem 2.13); (f) the Duffing consistency check (Proposition 2.15). All proofs are complete except where a step is explicitly cited to a pinned source and labelled as such; those citations belong to later milestones (W1, W3) and are listed at the end (§2.9).

2.1. Reconstruction.

Lemma 2.1 (Reconstruction is bounded into C^k)**.** *Let $a>0$, $p\in\mathbb{R}$, $k\in\mathbb{N}$. For $c\in\ell^{a,p}_{\mathbb{C}}$,*

$$\sum_{j\ge1}\lvert c_j\rvert\, j^k \le M_{a,p,k}\|c\|_{a,p},\qquad M_{a,p,k}^2=\sum_{j\ge1} j^{2k-2p}e^{-2aj}<\infty.$$

Consequently the series $\sum_j c_j\phi_j$ and all its termwise x-derivatives of order $\le k$ converge absolutely and uniformly on $[0,\pi]$; $c\mapsto\sum_j c_j\phi_j$ is a bounded linear map $\ell^{a,p}\to C^k([0,\pi])$ of norm

$\le \sqrt{2/\pi}\, M_{a,p,k}$, *its value vanishes at* $x = 0, \pi$, *and* $\int_0^\pi (\sum_j c_j\phi_j)\phi_l\,dx = c_l$. *In particular* U_m, V_m *and* $q \mapsto U_m(\lambda q)$ *are bounded linear maps* $\ell^{a,p} \to C^k([0,\pi])$ *for every* $p \in \mathbb{R}$, *because* $\lambda_j^{\pm 1/2} \le (1+m)^{1/4} j^{1/2}$ *and multiplication by* (λ_j^r) *maps* $\ell^{a,p}$ *into* $\ell^{a,p-|r|}$ *for* $|r| \le 2$, $r \ge 0$, *with norm* $\le (1+m)^{r/2}$.

Proof. Cauchy–Schwarz: $\sum |c_j| j^k = \sum (|c_j| j^p e^{aj})(j^{k-p} e^{-aj}) \le \|c\|_{a,p} M_{a,p,k}$, and $M_{a,p,k} < \infty$ because $a > 0$. Since $|\partial_x^r \phi_j| \le \sqrt{2/\pi}\, j^r$, the Weierstrass M-test gives uniform convergence of the derivative series of order $r \le k$, hence termwise differentiation. Boundary values and the coefficient formula follow termwise from $\phi_j(0) = \phi_j(\pi) = 0$ and orthonormality. The bound on λ_j^r is $j^2 + m \le (1+m) j^2$. □

Source: P96-W Lemma 1, p. 278, first two sentences (asserted there without the estimate).

2.2. **The quartic coefficients.**

Lemma 2.2 (Closed form, (c)). *For* $i, j, k, l \ge 1$ *let* $g_{ijkl} = \int_0^\pi \phi_i\phi_j\phi_k\phi_l\,dx$. *Then*

$$g_{ijkl} = \frac{1}{2\pi}\Big([i+l=j+k] + [i+k=j+l] + [i+j=k+l] \\ - [i=j+k+l] - [j=i+k+l] - [k=i+j+l] - [l=i+j+k] \Big),$$

where $[\cdot]$ *is the Iverson bracket. In particular* g *is symmetric,* $g_{ijkl} = 0$ *unless* $\pm i \pm j \pm k \pm l = 0$ *for some choice of signs,* $g_{iiii} = \frac{3}{2\pi}$, *and* $g_{iijj} = \frac{1}{\pi}$ *for* $i \ne j$. *Equivalently, for* $l \ge 1$ *and* $i, j, k \ge 1$,

$$g_{ijkl} = \frac{1}{2\pi} \sum_{\varepsilon \in \{\pm 1\}^3} s(\varepsilon)\, [\, l = \varepsilon_1 i + \varepsilon_2 j + \varepsilon_3 k\,], \qquad s(\varepsilon) = (-1)^{1+\#\{t:\varepsilon_t = -1\}}, \tag{3}$$

so that $|g_{ijkl}| \le \frac{1}{2\pi} \sum_\varepsilon [\, l = \varepsilon_1 i + \varepsilon_2 j + \varepsilon_3 k\,]$.

Proof. **Step S2.1: Product to sum.** $\sin\alpha \sin\beta = \frac{1}{2}(\cos(\alpha-\beta) - \cos(\alpha+\beta))$ gives

$$\phi_i\phi_j\phi_k\phi_l = \frac{1}{\pi^2}\big(\cos(i-j)x - \cos(i+j)x\big)\big(\cos(k-l)x - \cos(k+l)x\big).$$

Step S2.2: Cosine integrals. For integers A, B: $\int_0^\pi \cos Ax \cos Bx\,dx = \frac{\pi}{2}\big([A=B] + [A=-B]\big)$ (this includes $A = B = 0$, value π). Hence

$$g_{ijkl} = \frac{1}{2\pi}\Big([i-j=k-l] + [i-j=l-k] - [i-j=k+l] - [i-j=-k-l] \\ - [i+j=k-l] - [i+j=l-k] + [i+j=k+l] + [i+j=-k-l] \Big).$$

The last bracket vanishes for positive indices; rewriting the others gives the stated form. The special values are read off: for (i,i,i,i) the three positive brackets hold and no negative one; for (i,i,j,j), $i \ne j$, exactly $[i+l=j+k]$ and $[i+k=j+l]$ hold (a negative bracket would force one index to be the sum of the other three, e.g. $i = i + 2j$, impossible).

Step S2.3: Signed form. The seven relations, solved for l, are $l = -i+j+k$, $l = i-j+k$, $l = i+j-k$ (sign $+$, one minus), $l = i-j-k$, $l = -i+j-k$, $l = -i-j+k$ (sign $-$, two minuses), $l = i+j+k$ (sign $-$, no minus); the eighth pattern $l = -i-j-k$ is impossible for $l \ge 1$ and may be added with either sign. This is (3). □

Source: P96-W (10)–(11), p. 281 (only the selection rule and G_{iijj} are stated, "by an elementary calculation – see [6]"); KP96 p. 13 (computation of G_{ijij}). The weighted coefficient of P96-W is $G_{ijkl} = g_{ijkl}/\sqrt{\lambda_i\lambda_j\lambda_k\lambda_l}$; its (11) agrees with the lemma.

2.3. **The cubic gradient.**

Lemma 2.3 (G and its gradient as absolutely convergent sums)**.** *Let $a > 0$, $p \geq 0$, $q \in \ell^{a,p}$ and write $\hat{q}_j = q_j/\sqrt{\lambda_j}$; G and G_q are taken on $\ell^{a,p}$ (Definitions 1.8 and 1.9). Then*

$$G(q) = \frac{1}{4} \sum_{i,j,k,l \geq 1} g_{ijkl} \hat{q}_i \hat{q}_j \hat{q}_k \hat{q}_l, \qquad G_q(q)_l = \frac{1}{\sqrt{\lambda_l}} \sum_{i,j,k \geq 1} g_{ijkl} \hat{q}_i \hat{q}_j \hat{q}_k,$$

both sums converging absolutely; G is Gâteaux differentiable along every coordinate with $\partial G/\partial q_l = G_q(q)_l$; and $\int_0^\pi U_m(q)^3 \phi_l \, dx = \sqrt{\lambda_l}\, G_q(q)_l$.

Proof. By Lemma 2.1 with $k = 0$, $S = \sum_j |\hat{q}_j| < \infty$ and $U_m(q) = \sum \hat{q}_j \phi_j$ uniformly. Since $|\phi_j| \leq \sqrt{2/\pi}$, the quadruple series $\sum \hat{q}_i \hat{q}_j \hat{q}_k \hat{q}_l \phi_i \phi_j \phi_k \phi_l$ is dominated by $(2/\pi)^2 S^4$ uniformly in x; integrate termwise (dominated convergence for series) to obtain both formulas and their absolute convergence. Differentiating $\frac{1}{4} \int U_m(q + t e_l)^4$ at $t = 0$ under the integral (the integrand is a polynomial in t with coefficients continuous in x) gives $\int U_m(q)^3 \phi_l / \sqrt{\lambda_l}$, which is $G_q(q)_l$ by Definition 1.9 and the triple-series formula. □

Lemma 2.4 (Weighted convolution algebra)**.** *Let $a \geq 0$ and $s \geq 1$. For $c, d \in \ell_b^{a,s}$,*

$$\|c * d\|_{a,s} \leq K_s \|c\|_{a,s} \|d\|_{a,s}, \qquad K_s = 2^s \Big(1 + 2\zeta(2s)\Big)^{1/2}.$$

In particular $K_{3/2}^2 = 8(1 + 2\zeta(3))$. The convolution sum converges absolutely for each l.

Proof. Let $w_l = \langle l \rangle^s e^{a|l|}$, $C_l = w_l |c_l|$, $D_l = w_l |d_l|$.

Step S2.4: Weight inequality. For $i, l \in \mathbb{Z}$: $e^{a|l|} \leq e^{a|i|} e^{a|l-i|}$ and $\langle l \rangle \leq \langle i \rangle + \langle l - i \rangle$ (if $|l| \leq 1$ the left side is 1; otherwise $|l| \leq |i| + |l - i|$). By convexity of $t \mapsto t^s$, $(\alpha + \beta)^s \leq 2^{s-1}(\alpha^s + \beta^s)$, so $w_l \leq 2^{s-1}(w_i e^{a|l-i|} + e^{a|i|} w_{l-i}) \leq 2^{s-1}(w_i w_{l-i}/\langle l - i \rangle^s + w_i w_{l-i}/\langle i \rangle^s)$.

Step S2.5: Young and Cauchy–Schwarz. Hence $w_l |(c * d)_l| \leq 2^{s-1} \sum_i \big(C_i \, |d_{l-i}| e^{a|l-i|} + |c_i| e^{a|i|} D_{l-i} \big) = 2^{s-1} \big((C * D')_l + (C' * D)_l \big)$ with $C_i' = C_i/\langle i \rangle^s$, $D_i' = D_i/\langle i \rangle^s$. Young's inequality $\|X * Y\|_{\ell^2} \leq \|X\|_{\ell^2} \|Y\|_{\ell^1}$ and $\|D'\|_{\ell^1} \leq \|D\|_{\ell^2} (\sum_i \langle i \rangle^{-2s})^{1/2} = \|D\|_{\ell^2} (1 + 2\zeta(2s))^{1/2}$ give the claim; the same bounds give absolute convergence of each convolution sum. □

Source: P96-W Lemma 2, p. 279 and Appendix A, pp. 294–295; KP96 Lemma 2 p. 12 and Appendix A, pp. 31–32 (constant "depending only on s", not made explicit there). This lemma is the W1 node `bpW:w1:algebra`; it is proved here because (b) needs it.

Theorem 2.5 (Cubic gradient, (b))**.** *Let $a \geq 0$, $p \geq \frac{1}{2}$, and define the symmetric trilinear map T on $(\ell_{\mathbb{C}}^{a,p})^3$ by*

$$T(q, q', q'')_l = \frac{1}{\sqrt{\lambda_l}} \sum_{i,j,k \geq 1} g_{ijkl} \, \hat{q}_i \hat{q}_j' \hat{q}_k'' \qquad (\hat{q}_j = q_j/\sqrt{\lambda_j}).$$

The sums converge absolutely, T is a bounded trilinear map into $\ell_{\mathbb{C}}^{a,p+1}$, real on real arguments, and

$$\|T(q, q', q'')\|_{a,p+1} \leq C_* \|q\|_{a,p} \|q'\|_{a,p} \|q''\|_{a,p}, \qquad C_* = \frac{K_{p+1/2}^2}{\pi}.$$

For $a > 0$ and real $q \in \ell^{a,p}$, $G_q(q) = T(q, q, q)$ (Definition 1.9 on $\ell^{a,p}$, Lemma 2.3). Consequently, for $a = p = 1$, $G_q : \ell^{1,1} \to \ell^{1,2}$ is real analytic (indeed a continuous cubic polynomial), and

$$\|G_q(q)\|_{1,2} \leq C_m \|q\|_{1,1}^3, \qquad C_m = \frac{8(1 + 2\zeta(3))}{\pi} \approx 8.67.$$

The constant is independent of $m > 0$ (we only use $\lambda_j \geq j$).

Proof. **Step S2.6: Even majorants.** Given $q \in \ell^{a,p}_{\mathbb{C}}$ define the even two-sided sequence $x_i = |q_{|i|}|/\sqrt{\lambda_{|i|}}$ for $i \neq 0$, $x_0 = 0$; similarly x', x''. Using $\lambda_i \geq i$,

$$\|x\|^2_{a,p+1/2} = 2\sum_{i\geq 1} \frac{i}{\lambda_i}\, i^{2p} e^{2ai} |q_i|^2 \leq 2\|q\|^2_{a,p}.$$

Step S2.7: Selection rule bound. By (3), for $l \geq 1$,

$$\sum_{i,j,k\geq 1} |g_{ijkl}|\, |\hat{q}_i \hat{q}'_j \hat{q}''_k| \leq \frac{1}{2\pi} \sum_{i,j,k\geq 1} \sum_{\varepsilon} [\, l = \varepsilon_1 i + \varepsilon_2 j + \varepsilon_3 k\,]\, x_i x'_j x''_k \leq \frac{1}{2\pi}(x * x' * x'')_l,$$

since $(i,j,k,\varepsilon) \mapsto (\varepsilon_1 i, \varepsilon_2 j, \varepsilon_3 k)$ is injective into $(\mathbb{Z}\setminus 0)^3$ and x is even. Thus the sum defining T converges absolutely.

Step S2.8: Smoothing. With $X = x * x' * x''$ (even, by evenness of the factors) and $l/\lambda_l \leq 1$,

$$\|T\|^2_{a,p+1} \leq \sum_{l\geq 1} \frac{l^{2p+2} e^{2al}}{\lambda_l} \frac{X_l^2}{4\pi^2} \leq \frac{1}{4\pi^2} \sum_{l\geq 1} l^{2p+1} e^{2al} X_l^2 \leq \frac{1}{8\pi^2} \|X\|^2_{a,p+1/2}.$$

By Lemma 2.4 with $s = p + \frac{1}{2} \geq 1$, $\|X\|_{a,p+1/2} \leq K_s^2 \|x\|\|x'\|\|x''\| \leq K_s^2 2^{3/2} \|q\|\|q'\|\|q''\|$. Hence $\|T\|_{a,p+1} \leq (2\sqrt{2}\pi)^{-1} 2^{3/2} K_s^2 \prod \|\cdot\| = (K_s^2/\pi) \prod \|\cdot\|$.

Step S2.9: Analyticity. Trilinearity and symmetry are clear from the formula and the symmetry of g. A bounded symmetric trilinear map T between complex Banach spaces defines the continuous homogeneous cubic polynomial $q \mapsto T(q,q,q)$, which is analytic on the whole space, with the finite power series $T(q_0+h)^3$ expanded by multilinearity (in Lean: a `ContinuousMultilinearMap` is analytic). It maps real sequences to real sequences. For $a > 0$ the identification with G_q is Lemma 2.3. □

Source: P96-W Lemma 3, pp. 279–280 (with $O(\cdot)$ instead of a constant) and (13)–(14), p. 284; KP96 p. 16. The source deduces analyticity from coordinatewise analyticity and local boundedness [P96-W ref. 9, Appendix A]; the polynomial structure makes that principle unnecessary here.
SOURCE_GAP: P96-W p. 280 asserts $\|\mathcal{S}q\|_{a,s+1/2} \leq \|q\|_{a,s}$ in the two-sided norm. The correct bound is $\leq \sqrt{2}\|q\|_{a,s}$ (Step S2.6); the operator norm is exactly $\sqrt{2}$ for every fixed $m > 0$ (take $q = e_j$, $j \to \infty$), so the printed bound fails for every $m > 0$ (e.g. $q = e_1$, $a = s = 0$, $m = 1$). Constants only; recorded as new item N-1 in the errata check.

2.4. PDE ↔ coefficient system.

Proposition 2.6 (Equivalence, (a)). *Let $I \subset \mathbb{R}$ be an open interval and $q, p : I \to \ell^{1,1}$.*

- *(i) (Coefficients ⇒ PDE.) Suppose q, p are continuous into $\ell^{1,1}$, continuously differentiable into $\ell^{1,0}$, and satisfy (2) as an identity in $\ell^{1,0}$. Then $u(t,x) = U_m(q(t))(x)$ is a classical solution of (1) on I (Definition 1.2) with $u_t(t,\cdot) = V_m(p(t))$.*
- *(ii) (PDE ⇒ coefficients.) Suppose $u(t,x) = U_m(q(t))(x)$ is a classical solution of (1) on I and $u_t(t,\cdot) = V_m(p(t))$ for every t. Then every q_j, p_j is C^1 on I and (2) holds coordinatewise.*

In both directions the cubic term is realized exactly: for $q \in \ell^{1,1}$,

$$U_m(q)^3 = \sum_{l\geq 1} \sqrt{\lambda_l}\, G_q(q)_l\, \phi_l \quad \text{uniformly on } [0,\pi], \qquad -\partial_x^2 U_m(q) + m U_m(q) = U_m(\lambda^2 q), \tag{4}$$

where $(\lambda^2 q)_j = \lambda_j^2 q_j$ and the right sides converge in C^0 (first) and C^0 (second, as $\lambda^2 q \in \ell^{1,-1}$).

Proof. **Step S2.10: Force identity.** Let $f = U_m(q)^3 \in C([0,\pi])$. By Lemma 2.3, $\int f\phi_l = \sqrt{\lambda_l} G_q(q)_l$; by Theorem 2.5 and Lemma 2.1 (the sequence $(\sqrt{\lambda_l} G_q(q)_l)$ lies in $\ell^{1,1}$), $S = \sum_l \sqrt{\lambda_l} G_q(q)_l \phi_l$ converges uniformly to a continuous S with the same sine coefficients as

f. Since $(\phi_l)_{l\geq 1}$ is a complete orthonormal system of $L^2(0,\pi)$, $f = S$ almost everywhere, hence everywhere by continuity. The second identity is termwise differentiation (Lemma 2.1 with $k = 2$) and $-\phi_j'' + m\phi_j = \lambda_j^2\phi_j$.

Step S2.11: Time regularity. In (i), $\dot q = \lambda p$ is continuous into $\ell^{1,0}$, and since p is C^1 into $\ell^{1,0}$ and multiplication by λ is bounded $\ell^{1,0} \to \ell^{1,-1}$, $\dot q$ is C^1 into $\ell^{1,-1}$ with $\ddot q = \lambda\dot p = -\lambda^2 q - \lambda G_q(q)$. As $U_m : \ell^{1,-1} \to C^2([0,\pi])$ is bounded linear, $t \mapsto u(t,\cdot)$ is C^2 into $C^2([0,\pi])$; therefore $u, u_t, u_{tt}, u_x, u_{xx}$ exist and are jointly continuous, $u_t = U_m(\lambda p) = V_m(p)$ and $u_{tt} = U_m(\ddot q)$.

Step S2.12: Identification. By (4), $U_m(\lambda^2 q) = -u_{xx} + mu$ and $U_m(\lambda G_q(q)) = \sum_l \sqrt{\lambda_l} G_q(q)_l \phi_l = u^3$; hence $u_{tt} = u_{xx} - mu - u^3$. The boundary conditions hold by Lemma 2.1.

Step S2.13: Converse. In (ii), $\int_0^\pi u(t,x)\phi_l(x)\,dx = q_l(t)/\sqrt{\lambda_l}$ and $\int_0^\pi u_t\phi_l = \sqrt{\lambda_l}p_l$. Differentiation under the integral sign (u_t, u_{tt} continuous on compact sets) gives $\dot q_l = \sqrt{\lambda_l}\int u_t\phi_l = \lambda_l p_l$ and $\sqrt{\lambda_l}\dot p_l = \int u_{tt}\phi_l$. Two integrations by parts, using $u(t,0) = u(t,\pi) = 0 = \phi_l(0) = \phi_l(\pi)$, give $\int u_{xx}\phi_l = -l^2\int u\phi_l$, so $\int u_{tt}\phi_l = -\lambda_l^2 q_l/\sqrt{\lambda_l} - \sqrt{\lambda_l}G_q(q)_l$ by Lemma 2.3. Divide by $\sqrt{\lambda_l}$. □

Source: P96-W Lemma 1, p. 278, and (7), p. 278.
Expansion/strengthening of the source proof: P96-W Lemma 1 (p. 278) argues via convergence in complex neighbourhoods, termwise time differentiation and completeness, for real-analytic curves. The proof above expands these steps and proves a stronger C^1-curve version (i); this is not a source gap. (Separately, the second line of the source's u_{tt} display drops a minus sign on $\lambda_j^2 q_j$: errata item N-4; the proof above uses the correct sign.)
completeness of (ϕ_l) in $L^2(0,\pi)$ is a standard external fact (Fourier basis of L^2 on the circle applied to odd extensions). The all-time real-analytic version used by the headline is W2.

2.5. **Quartic nonresonance.**

Lemma 2.7 (Qualitative nonresonance of all quartic terms)**.** *Let $m > 0$, let $a, b, c, d \geq 1$ with $g_{abcd} \neq 0$, and let $\sigma \in \{\pm 1\}^4$. If $\sigma_1\lambda_a + \sigma_2\lambda_b + \sigma_3\lambda_c + \sigma_4\lambda_d = 0$, then the indices with $\sigma = +1$ and those with $\sigma = -1$ form equal multisets (in particular there are two of each).*

Proof. By Lemma 2.2, $g_{abcd} \neq 0$ implies a signed relation $\pm a \pm b \pm c \pm d = 0$; in particular each index is at most the sum of the other three.

Step S2.14: Four or three equal signs. All signs equal is impossible ($\lambda > 0$). If three signs agree, say $\lambda_a + \lambda_b + \lambda_c = \lambda_d$, then $d \leq a + b + c$, so $\lambda_d \leq \lambda_{a+b+c}$, while $\lambda_x + \lambda_y > \lambda_{x+y}$ for $x, y \geq 1$ (square both sides: $2\sqrt{(x^2+m)(y^2+m)} > 2xy - m$); hence $\lambda_a + \lambda_b + \lambda_c > \lambda_{a+b+c} \geq \lambda_d$.

Step S2.15: Two against two. Let $\lambda_a + \lambda_b = \lambda_c + \lambda_d$ with $\{a, b\} \neq \{c, d\}$ as multisets, and let $f(s) = \sqrt{s^2 + m}$. Up to renaming within and between the two sides, the signed relation is one of: (1) $a+b = c+d$: for fixed sum, $f(x)+f(y)$ is strictly increasing in $|x-y|$ by strict convexity of f, so equality forces $|a-b| = |c-d|$ and $\{a, b\} = \{c, d\}$; (2) $a-b = d-c =: t$ (that is, $a+c = b+d$): then $f(b+t)+f(b) = f(c)+f(c+t)$ and $y \mapsto f(y)+f(y+t)$ is strictly increasing on $\{y \geq 1,\ y+t \geq 1\}$, so $b = c$, $a = d$; (3) one index is the sum of the other three, and after renaming it is $a = b+c+d$ (the side containing the largest index is called the left side). Then $\lambda_a - \lambda_c = \int_c^{b+c+d} f'$ and $\lambda_d - \lambda_b = \int_b^d f'$ (nonpositive if $d \leq b$). Since $f' > 0$ is increasing, $\int_c^{c+b+d} f' > \int_{c+2b}^{c+b+d} f' \geq \int_b^d f'$ when $d > b$ (an interval of the same length $d - b$ shifted right by $c + b > 0$). So $\lambda_a + \lambda_b > \lambda_c + \lambda_d$. Every signed relation among four positive integers with two signs on each side is of type (1) or (2) after renaming, and every relation with one sign different is of type (3); this exhausts the cases. □

Remark 2.8. The lemma is qualitative and covers *all* indices. It does not give a uniform lower bound: divisors such as $\lambda_{j+1} - \lambda_j - \lambda_{k+1} + \lambda_k$ tend to 0, which is why P96-W (p. 276) says no complete quartic normal form is available. The quantitative bound for quartic terms with at least

one index in $\{1,\dots,n\}$, $|\lambda'_i+\lambda'_j+\lambda'_k+\lambda'_l| \geq cm/\sqrt{n^2+m}^3$, is P96-W Lemma 4, p. 283 (proof pp. 285–286); it belongs to W3 and is cited, not reproved, here.

2.6. **Partial quartic Birkhoff normal form: the action coefficients.** Write $\hat\lambda=(\lambda_j)$ and, in the complex coordinates of Definition 1.11, $q_j=(z_j+\bar z_j)/\sqrt{2}$. Then

$$G=\frac14\sum_{i,j,k,l}\frac{g_{ijkl}}{\sqrt{\lambda_i\lambda_j\lambda_k\lambda_l}}q_iq_jq_kq_l=\frac{1}{16}\sum_{i,j,k,l}\frac{g_{ijkl}}{\sqrt{\lambda_i\lambda_j\lambda_k\lambda_l}}(z_i+\bar z_i)(z_j+\bar z_j)(z_k+\bar z_k)(z_l+\bar z_l). \tag{5}$$

Expanding gives a sum of monomials $z^\alpha\bar z^\beta$ with $|\alpha|+|\beta|=4$. Along the linear flow ($\dot z_j=-i\lambda_jz_j$) such a monomial evolves by $e^{-i\langle\alpha-\beta,\hat\lambda\rangle t}$; it is *resonant* if $\langle\alpha-\beta,\hat\lambda\rangle=0$. Let $J=\{1,2\}$, $N=\{j\geq3\}$, and split $G=G^{\tan}+G^{\nor}$, where $G^{\nor}$ collects the monomials all of whose indices lie in N.

Definition 2.9 (Partial quartic normal form)**.** $\bar G$ is the sum of the resonant monomials of $G^{\tan}$ (with their coefficients), and the partial quartic normal form is $\Lambda+\bar G+G^{\nor}$.

Theorem 2.10 (Action coefficients, (d))**.** *For $J=\{1,2\}$ and every $m>0$, with $I=(I_1,I_2)$, $Z=(I_j)_{j\geq3}$ and $I_j=|z_j|^2$,*

$$\bar G=\frac12\langle AI,I\rangle+\langle BI,Z\rangle,\qquad A_{ii}=\frac{9}{8\pi\lambda_i^2},\quad A_{12}=A_{21}=\frac{3}{2\pi\lambda_1\lambda_2},\quad B_{ji}=\frac{3}{2\pi\lambda_j\lambda_i}\ (j\geq3,\ i\in J).$$

Equivalently $A=\frac{3}{8\pi}D(4\mathbf{1}\mathbf{1}^{\mathsf T}-\mathrm{Id})D$ with $D=\mathrm{diag}(\lambda_1^{-1},\lambda_2^{-1})$, and $(BI)_j=\frac{3}{2\pi\lambda_j}\big(\frac{I_1}{\lambda_1}+\frac{I_2}{\lambda_2}\big)$. More generally, the full angle average of G is

$$\langle G\rangle=\frac{3}{16\pi}\Big(4\Big(\sum_{j\geq1}\frac{I_j}{\lambda_j}\Big)^2-\sum_{j\geq1}\frac{I_j^2}{\lambda_j^2}\Big). \tag{6}$$

Proof. **Step S2.16: Resonant equals averaged.** By Lemma 2.7, a monomial $z^\alpha\bar z^\beta$ of (5) with nonzero coefficient is resonant iff $\alpha=\beta$, i.e. it is $|z_a|^2|z_b|^2=I_aI_b$. With $z_j=\sqrt{I_j}e^{-i\varphi_j}$ the average over independent angles of $z^\alpha\bar z^\beta$ is $\delta_{\alpha\beta}I^\alpha$. Hence $\bar G$ is the part of $\langle G\rangle$ (average over all angles, computed on finitely supported q and extended by the absolutely convergent formula) consisting of the terms that contain at least one tangential action.

Step S2.17: Averages of monomials. With $q_j=\sqrt{2I_j}\cos\varphi_j$: $\langle q_iq_jq_kq_l\rangle=0$ unless the indices pair up; $\langle q_i^4\rangle=4I_i^2\cdot\frac38=\frac32I_i^2$ and $\langle q_i^2q_j^2\rangle=I_iI_j$ for $i\neq j$. Among ordered quadruples, (i,i,i,i) occurs once and the multiset $\{i,i,j,j\}$ $(i<j)$ six times.

Step S2.18: Summation. Therefore, with $y_j=I_j/\lambda_j$ and Lemma 2.2,

$$\langle G\rangle=\frac14\Big(\sum_i\frac{3}{2\pi}\cdot\frac32y_i^2+\sum_{i<j}6\cdot\frac1\pi\,y_iy_j\Big)=\frac{9}{16\pi}\sum_iy_i^2+\frac{3}{2\pi}\sum_{i<j}y_iy_j,$$

which is (6). Matching with $\frac12\sum_{i,j}A_{ij}I_iI_j=\frac12\sum_iA_{ii}I_i^2+\sum_{i<j}A_{ij}I_iI_j$ on J and with $\sum_{j\geq3,i\in J}B_{ji}I_iZ_j$ on the mixed terms gives the coefficients. □

Proposition 2.11 (Formal partial Birkhoff transformation)**.** *Let F be the quartic polynomial whose coefficient on each nonresonant monomial $z^\alpha\bar z^\beta$ of $G^{\tan}$ with coefficient $c_{\alpha\beta}$ is $c_{\alpha\beta}/(-i\langle\alpha-\beta,\hat\lambda\rangle)$, and zero otherwise. Then F is real ($\overline{F}=F$ under $z\leftrightarrow\bar z$), $X_\Lambda F=G^{\tan}-\bar G$ where $X_\Lambda F=\frac{d}{dt}F$ along the linear flow, and formally*

$$H\circ\Phi_F^1=\Lambda+\bar G+G^{\nor}+(\textit{terms of degree }\geq6),$$

where Φ_F^1 is the time-one map of X_F. The quartic part of the normal form is independent of the choice of F among solutions of $X_\Lambda F=G^{\tan}-\bar G$.

Proof. The divisors are nonzero by Lemma 2.7; $X_\Lambda(z^\alpha \bar z^\beta) = -i\langle \alpha-\beta, \hat\lambda\rangle z^\alpha \bar z^\beta$ gives the homological identity. Reality: the coefficient of $\bar z^\alpha z^\beta$ is $\bar c_{\alpha\beta}$ (since G is real) and the divisor changes sign and is conjugated. The Lie series $H\circ\Phi_F^1 = H+\{H,F\}+\frac12\{\{H,F\},F\}+\dots$ with $\{H,F\} = X_F H = -X_H F$ has quartic part $G - X_\Lambda F = G^{\mathrm{nor}} + \bar G$. Any two solutions F differ by a resonant quartic F_0, and $X_\Lambda F_0 = 0$. □

Source: P96-W Main Proposition, pp. 281–282; proof pp. 282–284, (12) and the Taylor formula on p. 283. The analytic realization of Φ_F^1 on a neighbourhood of 0 in $E_{a,p}$, the smoothing of X_F, and the bound $O(\|z\|^6)$ on the remainder are W3 (they use P96-W Lemma 4 and (13)–(14)). The *coefficients* printed there, $\bar G_{ij} = \frac{6}{\pi}\frac{4-\delta_{ij}}{\lambda_i\lambda_j}$, are 16 times those of Theorem 2.10; see Remark 2.12.

Remark 2.12 (Comparison with the printed coefficients). P96-W p. 282 expands G as $\frac14\sum G_{ijkl}(z_i+\bar z_i)\cdots(z_l+\bar z_l)$; by (5) the prefactor is $\frac{1}{16}$. The next line rewrites this as $\sum' G_{ijkl}w_iw_jw_kw_l$ *without* any prefactor, and the multiplicity count of p. 284 ($\bar G_{ij} = 24G_{iijj}$, $\bar G_{ii} = 12G_{iiii}$) is correct for that line. Two factors of 4 are therefore lost; with the prefactor $\frac{1}{16}$ the same count gives exactly A and B above. The factor is common to A and B, so the nondegeneracy arguments of P96-W Lemmas 5–6 are unaffected (Remark 1.14).

2.7. **Nondegeneracy for every $m > 0$.** The Cantor Manifold Theorem used by P96-W (p. 288, assumption A) requires, with $\omega(I) = \lambda_J + AI$ and $\Omega(I) = \lambda_N + BI$, ($\mathrm{A}_1$) $\det A \neq 0$; (A_2) $\langle l, \lambda_N\rangle \neq 0$; ($\mathrm{A}_3$) $\langle k, \omega(I)\rangle + \langle l, \Omega(I)\rangle \not\equiv 0$ as an affine function of I, for all $(k,l) \in \mathbb{Z}^2\times\mathbb{Z}^N$ with $1 \le |l| \le 2$ ($|l| = \sum|l_j|$).

Theorem 2.13 (Nondegeneracy, (e))**.** *For $J = \{1,2\}$ and every $m > 0$:*

(i) $\det A = -\dfrac{63}{64\pi^2\lambda_1^2\lambda_2^2} = -\dfrac{63}{64\pi^2(1+m)(4+m)} \neq 0$*;*
(ii) $\langle l, \lambda_N\rangle \neq 0$ *for* $1 \le |l| \le 2$*;*
(iii) (A_3) *holds.*

Proof. **Step S2.19: Determinant.** $\det A = (\frac{3}{8\pi})^2\lambda_1^{-2}\lambda_2^{-2}\det(\begin{smallmatrix}3&4\\4&3\end{smallmatrix}) = -\frac{63}{64\pi^2}\lambda_1^{-2}\lambda_2^{-2}$. (For general n: $\det(4\mathbf{1}\mathbf{1}^{\mathsf T} - \mathrm{Id}) = (-1)^{n-1}(4n-1)$.)

Step S2.20: Pure normal. $\langle l, \lambda_N\rangle$ is $\pm\lambda_a$, $\pm2\lambda_a$, $\pm(\lambda_a+\lambda_b)$ or $\pm(\lambda_a-\lambda_b)$ with $a \neq b$ in N; all are nonzero since $\lambda > 0$ is strictly increasing.

Step S2.21: Reduction of ($\mathbf{A}_3$). Let $b = (\lambda_1^{-1}, \lambda_2^{-1})$ and $s = \sum_{j\ge3} l_j/\lambda_j$. Then $A = \frac{3}{8\pi}(4bb^{\mathsf T} - D^2)$ and $B^{\mathsf T} l = \frac{3}{2\pi} s\, b$, so the linear part of the affine map is $\frac{3}{8\pi}\big(4(\langle b,k\rangle + s)b - D^2k\big)$. It vanishes iff $k_i/\lambda_i = c$ for $i = 1,2$ with $c = 4(\langle b,k\rangle + s)$; then $\langle b,k\rangle = 2c$ and $s = -\frac74 c$. The constant part is $c(\lambda_1^2+\lambda_2^2) + \langle l, \lambda_N\rangle$.

Step S2.22: The case $|l| = 1$. $l = \sigma e_\nu$, $\nu \ge 3$: $s = \sigma/\lambda_\nu$, $c = -4\sigma/(7\lambda_\nu)$, and the constant part vanishes iff $7\lambda_\nu^2 = 4(\lambda_1^2+\lambda_2^2)$. Then $k_i^2 = c^2\lambda_i^2 = \frac{4\lambda_i^2}{7(\lambda_1^2+\lambda_2^2)} \in (0, \frac47)$, not the square of an integer. So (A_3) holds.

Step S2.23: The case $|l| = 2$. $|s| \le 2/\lambda_3 < \frac23$, so $|c| < \frac{8}{21}$, and $k_2 - k_1 = c(\lambda_2-\lambda_1)$ with $0 < \lambda_2 - \lambda_1 < 1$; integrality forces $k_1 = k_2$, hence $c = 0$, hence $s = 0$. But $s \neq 0$ for $|l| = 2$: $s = 0$ would need $l = e_a - e_b$ with $\lambda_a = \lambda_b$, $a \neq b$. So the linear part does not vanish. □

Source: P96-W assumption A p. 288; Lemma 5 pp. 291–292 ((A_1), with the printed matrix; the printed $(1-4n)\prod\lambda_j^{-2}$ agrees after scaling); Lemma 6 pp. 292–293 ((A_3) for admissible J). Our proof of (iii) is the specialization of Lemma 6 to $J = \{1,2\}$; the second branch of the source's $|l| = 2$ argument ($|\langle w,l\rangle| \ge (4n-1)/(4n-4)$) cannot occur for $J = \{1,2\}$ since $|\langle w,l\rangle| \le 2/\lambda_3 < 1$.

Remark 2.14 (Spectral asymptotics with the corrected B). $\Omega_j(I) = \lambda_j + \frac{3}{2\pi\lambda_j}\langle b, I\rangle = j + \frac{m_I}{j} + O(j^{-3})$ with $m_I = \frac{m}{2} + \frac{3}{2\pi}\langle b, I\rangle$, so $(\Omega_i - \Omega_j)/(i-j) = 1 + O(j^{-2})$ uniformly for bounded I: the exponent $\kappa = 2$ of P96-W p. 291 is unchanged by the correction (only $v = \frac{24}{\pi}b$ becomes $\frac{3}{2\pi}b$).

2.8. **Duffing consistency check.**

Proposition 2.15 (One-mode frequency shift, (f))**.** *Restrict (1) to $u = x(t)\phi_j(x)$ with only mode j excited and drop the forcing of other modes (which is $O(x^3)$ and does not enter the first-order frequency shift). The coefficient equation is the Duffing equation $\ddot{x} + \lambda_j^2 x + \varepsilon x^3 = 0$ with $\varepsilon = g_{jjjj} = \frac{3}{2\pi}$, whose Lindstedt–Poincaré frequency at amplitude a is $\lambda_j + \frac{3\varepsilon a^2}{8\lambda_j} + O(a^4)$. With $q_j = \sqrt{\lambda_j}x$ and $I_j = \frac{1}{2}\lambda_j a^2$ this is*

$$\lambda_j + \frac{9}{16\pi\lambda_j}a^2 = \lambda_j + \frac{9}{8\pi\lambda_j^2}I_j = \lambda_j + A_{jj}I_j,$$

the frequency $\partial_{I_j}(\lambda_j I_j + \frac{1}{2}A_{jj}I_j^2)$ of the normal form. The printed P96-W value $\bar{G}_{jj} = \frac{18}{\pi\lambda_j^2}$ would predict a shift 16 *times larger.*

Proof. $\int(x\phi_j)^3\phi_j = x^3 g_{jjjj}$ gives the Duffing coefficient. The Lindstedt shift $3\varepsilon a^2/(8\omega_0)$ is classical (e.g. by averaging εx^3 against cos over a period: $\frac{2}{T}\int \varepsilon a^3\cos^4 = \frac{3}{4}\varepsilon a^3$). Along a solution $x = a\cos\varphi$, $q_j = \sqrt{\lambda_j}a\cos\varphi$ and $p_j = \dot{q}_j/\lambda_j$, so to leading order $I_j = \frac{1}{2}(q_j^2 + p_j^2) = \frac{1}{2}\lambda_j a^2$. Substitute. □

Remark 2.16. The relation between I_j and the amplitude holds to leading order only; the agreement checked is of the coefficient of the first-order shift, which is what the normal form determines. `scripts/wave/w0_checks.py` also integrates the Duffing equation numerically for $m \in \{0.1, 1, 10\}$ and confirms $|\omega_{\text{num}} - (\lambda_1 + A_{11}I_1)| = O(I_1^2)$.

2.9. **Ledger: proved, cited, source gaps.**

- *Proved here*: Lemmas 2.1, 2.2, 2.3, 2.4, 2.7; Theorems 2.5, 2.10, 2.13; Propositions 2.6, 2.11 (formal level), 2.15.
- *External standard facts*: completeness of (ϕ_l) in $L^2(0,\pi)$; analyticity of continuous multilinear maps; Young's inequality; $\zeta(3) < \infty$; the classical Lindstedt shift.
- *Cited, for later milestones*: quantitative divisor bound (P96-W Lemma 4, W3); analytic realization of the Birkhoff map and its remainder (P96-W Main Proposition, W3).
- *Expansions of source proofs*: P96-W Lemma 1 (expanded and strengthened to C^1 curves).
- *Source gaps / normalization defects*: P96-W p. 280 constant (N-1); P96-W p. 278 sign typo (N-4); KP96 p. 13 factor 2 (N-5); P96-W p. 282/284 factor 16 (E-PDE-W-1); P96-W pp. 287, 290 action/recovery/sign conventions (E-PDE-W-2).

3. Milestone W1: weighted sequence calculus

Throughout, Standing assumption 1.1 ($m > 0$) and the conventions of §1 are in force, and $\mathbb{K}$ denotes $\mathbb{R}$ or $\mathbb{C}$. This part supplies the sequence-space analysis used by every later milestone: the weighted spaces $\ell^{a,p}$ as transports of ℓ^2 (the realization of the frozen `KAM.Wave.WaveSpace`), embeddings and diagonal multipliers, the two-sided spaces and the convolution algebra, the cubic gradient as a continuous trilinear map with its analyticity, smoothing and gradient structure, the Hamiltonian field, completeness of the sine system, analyticity of the reconstruction, and the Banach-space analytic toolkit (complexification, Cauchy estimate, continuity of derivatives, uniform limits, holomorphic flows, Hilbert- and Banach-valued Fourier coefficient bounds, weighted lattice sums, the Hilbert-matrix operator bound) that the Birkhoff normal form (W3), the KAM step and iteration of P96-K in the setting $d = 1$, $\delta = -1$, $\bar{p} - p = 1$ (W4–W6) and the assembly (W2, W8) consume. Every result is proved here or is an external fact from the list §3.2. W1 does *not* supply the operator-valued Fourier square-sum bound printed on P96-K p. 127: that bound is false

(Remark 3.38); W1 supplies the ℓ^1-type replacement of Lemmas 3.36 and 3.37, and W4 must use it.

Route. Every weighted space is the image of one Hilbert space $\ell^2(\mathbb{N}_+;\mathbb{K})$ under an explicit diagonal isometry, so that each map between weighted spaces becomes a map on ℓ^2, and a diagonal map becomes a bounded multiplier. The cubic gradient is a trilinear form in the coefficients whose coefficients obey the selection rule of Lemma 2.2; majorizing by an even two-sided sequence turns it into a triple convolution, which the algebra property controls with a gain of one power of j. Multilinearity then gives analyticity for free, over $\mathbb{R}$ and over $\mathbb{C}$, with no appeal to the principle "coordinatewise analytic and locally bounded implies analytic" used by the source. The analytic toolkit of §3.7 works with holomorphic maps on open subsets of complex Banach spaces, defined as complex Fréchet differentiable maps; it never needs the (absent in Mathlib) theorem that such maps are analytic on infinite-dimensional domains, because all analyticity conclusions of the headline have finite-dimensional domains.

3.1. **Realization in Lean and transport maps.**

Definition 3.1 (Weights, transport and coefficient maps)**.** For $a, p \in \mathbb{R}$ and $j \in \mathbb{N}_+$ let $\omega_{a,p}(j) = j^p e^{aj} > 0$ (frozen `KAM.Wave.ellWeight a p j`). Write $\ell^2_{\mathbb{K}} = \ell^2(\mathbb{N}_+;\mathbb{K})$ (Lean `lp (fun _ : PNat => K) 2`) and, for $a \geq 0$, let $\ell^{a,p}_{\mathbb{K}}$ be the space of Definition 1.4 ($\ell^{a,p} = \ell^{a,p}_{\mathbb{R}}$). Define

$$\tau_{a,p} : \ell^{a,p}_{\mathbb{K}} \to \ell^2_{\mathbb{K}}, \quad (\tau_{a,p} w)_j = \omega_{a,p}(j)\, w_j, \qquad \kappa_{a,p} : \ell^2_{\mathbb{K}} \to \ell^{a,p}_{\mathbb{K}}, \quad (\kappa_{a,p} y)_j = \frac{y_j}{\omega_{a,p}(j)}.$$

A vector $y \in \ell^2_{\mathbb{K}}$ *represents* $w = \kappa_{a,p} y \in \ell^{a,p}_{\mathbb{K}}$; we call y the *weighted coordinates* of w at weight (a,p). A map A from $\ell^{a,p}_{\mathbb{K}}$ to $\ell^{a',p'}_{\mathbb{K}}$ is represented by $\hat{A} = \tau_{a',p'} \circ A \circ \kappa_{a,p}$ on $\ell^2_{\mathbb{K}}$.

Source: ours (the frozen encoding, `docs/tracks/W/W-STATEMENTS.md`, design summary). Lean guard: the division in $\kappa_{a,p}$ is by $\omega_{a,p}(j) > 0$; no case split.

Realization table. Every space of this part is realized as an ℓ^2 transport; no new normed type is introduced. The frozen objects are the case $\mathbb{K} = \mathbb{R}$, $a = p = 1$.

Space	Lean carrier	Transport / coefficient map
$\ell^{a,p}_{\mathbb{K}}$	`lp (fun _ : PNat => K) 2`	$\tau_{a,p}$, $\kappa_{a,p}$; for $\mathbb{K} = \mathbb{R}$, $a = p = 1$: $\kappa_{1,1}$ = frozen `coeff`
$E_{a,p}$ (Def. 1.4)	`lp ... 2` × `lp ... 2` (max norm)	$\tau_{a,p} \times \tau_{a,p}$; $E_{1,1}$ = frozen `WaveSpace`
$E^{\mathbb{C}}_{a,p}$ (complexification)	`lp (fun _ : PNat => Complex) 2` × `lp ... 2`	same weights; ι, re of Def. 3.22
$\ell^{a,s}_b$ (Def. 1.6)	`lp (fun _ : Int => Complex) 2`	weight $\langle l\rangle^s e^{a\lvert l\rvert}$ (Def. 3.9)
ℓ^p_∞ (Def. 3.7)	`lp (fun _ : PNat => Real) ∞`	weight j^p
$\mathcal{P}^{a,p}$ (Def. 3.31)	`(Fin 2 -> Complex)` × `(Fin 2 -> Complex)` × `lp ... 2` × `lp ... 2`	weights $\omega_{a,p}$ on the last two factors

In Lean every weighted statement below is stated on the carrier, with κ applied where the mathematics speaks of coefficients. The non-frozen helper `KAM.Wave.coeffW a p y j := y j / ellWeight a p j` (over $\mathbb{K}$) satisfies `coeffW 1 1 = coeff` on real sequences by unfolding; it is a distinct name, not a substitute for a frozen declaration. Statements uniform in $\mathbb{K}$ are written once over `[RCLike K]`.

3.2. **External facts.**

The proofs of this part use the following external (Mathlib-level) facts, besides those of track K's Part 1 (`bp:p1:ext-*`, cited by label in the text).

(W1-E1) *ℓ^2 spaces.* $\ell^2(\iota;\mathbb{K})$ is a Hilbert space for every index type ι (`lp.instInnerProductSpace`, completeness `lp.completeSpace`), with $\langle f, g\rangle = \sum_i \overline{f_i} g_i$ (`lp.inner_eq_tsum`); $|f_i| \leq \|f\|$ (`lp.norm_apply_le_norm`); membership and norm bounds follow from bounds on finite

partial sums (`mem$\ell$p_gen_iff`, `lp.norm_le_of_forall_sum_le`); $f = \sum_i f_i\, e_i$ with $e_i =$ `lp.single 2 i 1` (`lp.hasSum_single`).

(W1-E2) *Series.* $\sum_{k\geq 1} k^{-2} = \pi^2/6$ (`hasSum_zeta_two`); $\sum_{k\geq 1} k^{-\sigma} < \infty$ for $\sigma > 1$ (`Real.summable_one_div_nat_rpow`); Tonelli for nonnegative double series and Fubini for absolutely summable ones (`tsum_prod'`, `Summable.tsum_comm`).

(W1-E3) *Analytic maps.* A continuous multilinear map is analytic (indeed a continuous polynomial) everywhere (`ContinuousMultilinearMap.analyticAt`, `.cpolynomialAt`); a continuous linear or bilinear map is analytic (`ContinuousLinearMap.analyticAt`, `.analyticAt_bilinear`); compositions and pairs of analytic maps are analytic (`AnalyticAt.comp`, `AnalyticAt.prod`); a continuous bilinear map applied to two analytic maps is analytic (`ContinuousLinearMap.analyticAt_bilinear` at the pair, then `AnalyticAt.comp` with `AnalyticAt.prod`); $\mathbb{C}$-analytic maps are $\mathbb{R}$-analytic (`AnalyticAt.restrictScalars`); analytic maps are Fréchet differentiable (`AnalyticAt.differentiableAt`).

(W1-E4) *One complex variable, Banach values.* For a complete complex normed space F: the Cauchy estimate: if $R > 0$, φ is continuous on the closed disc $|\zeta - c| \leq R$, holomorphic inside, and $\|\varphi(\zeta)\| \leq B$ on the circle, then $\|\varphi'(c)\| \leq B/R$ (`Complex.norm_deriv_le_of_forall_mem_sphere_norm_le`); the Cauchy integral formula (`bp:p1:ext-cauchy`, valid for complete F), in particular $\varphi'(c) = \frac{1}{2\pi i} \oint_{|\zeta-c|=R} \varphi(\zeta)(\zeta - c)^{-2} d\zeta$ (`DiffContOnCl.deriv_eq_smul_circleIntegral`); continuous linear maps commute with Bochner integrals (`ContinuousLinearMap.integral_comp_comm`) and the norm is attained by norm-one functionals (`NormedSpace.norm_le_dual_bound`); holomorphic maps on open subsets of $\mathbb{C}$ are analytic (`DifferentiableOn.analyticOnNhd`); a locally uniform limit of holomorphic maps on an open subset of $\mathbb{C}$ is holomorphic (`TendstoLocallyUniformlyOn.differentiableOn`).

(W1-E5) *Uniform limits and derivatives.* If $f_k \to f$ pointwise on an open set U of a normed space over $\mathbb{K}$, each f_k is differentiable on U and $Df_k \to g$ uniformly on U, then f is differentiable on U with $Df = g$ (`hasFDerivAt_of_tendstoUniformlyOn`). Mean value inequality on convex sets (`bp:p1:ext-mvt`).

(W1-E6) *Integrals.* Bochner integrals of continuous maps of a compact interval or box into a complete space, the estimate $\|\int F\| \leq \int \|F\|$, differentiation under the integral sign with a dominating bound on the derivative, a.e.-strongly measurable integrand and derivative (`hasFDerivAt_integral_of_dominated_of_fderiv_le`; continuity in the integration variable supplies the measurability), dominated convergence (`bp:p1:ext-integral`).

(W1-E7) *ODE.* Picard–Lindelöf existence and uniqueness for Lipschitz fields in a Banach space (`IsPicardLindelof`, `ODE_solution_unique`); Gronwall's inequality (`norm_le_gronwallBound_of_norm_deriv_right_le`).

(W1-E8) *Fourier analysis.* The characters $e_n(x) = e^{inx}$, $n \in \mathbb{Z}$, form a Hilbert basis of L^2 of the circle $\mathbb{R}/2\pi\mathbb{Z}$ with normalized Haar measure (`fourierBasis` on `AddCircle (2 * Real.pi)`), and the characters $e^{2\pi i k\cdot x}$, $k \in \mathbb{Z}^n$, one of $L^2(\mathbb{R}^n/\mathbb{Z}^n)$ (`UnitAddTorus.mFourierBasis`); Parseval (`tsum_sq_fourierCoeff`); continuous periodic functions with absolutely summable coefficients are the sums of their Fourier series (`bp:p1:ext-fourier`).

(W1-E9) *Cauchy's theorem for rectangles, Banach values.* For a complete complex normed space F, $f : \mathbb{C} \to F$ and $z, w \in \mathbb{C}$ with f complex differentiable on the closed rectangle $[\operatorname{Re} z, \operatorname{Re} w] \times [\operatorname{Im} z, \operatorname{Im} w]$, the oriented boundary integral vanishes: $\int_{\operatorname{Re} z}^{\operatorname{Re} w} f(x + i \operatorname{Im} z)\, dx - \int_{\operatorname{Re} z}^{\operatorname{Re} w} f(x + i \operatorname{Im} w)\, dx + i \int_{\operatorname{Im} z}^{\operatorname{Im} w} f(\operatorname{Re} w + iy)\, dy - i \int_{\operatorname{Im} z}^{\operatorname{Im} w} f(\operatorname{Re} z + iy)\, dy = 0$ (`Complex.integral_boundary_rect_eq_zero_of_differentiableOn`; interval integrals

are oriented, so either order of the endpoints is allowed; `bp:p1:ext-rectangle`). Fubini for continuous integrands on a box (`MeasureTheory.integral_prod`).

3.3. **Weighted spaces, embeddings, multipliers.**

Lemma 3.2 (The weighted spaces)**.** *Let $a \geq 0$, $p \in \mathbb{R}$.*

(i) $\tau_{a,p}$ and $\kappa_{a,p}$ are mutually inverse $\mathbb{K}$-linear bijections and $\|\tau_{a,p} w\|_{\ell^2} = \|w\|_{a,p}$. Hence $\ell^{a,p}_{\mathbb{K}}$, with the inner product $\langle w, w' \rangle_{a,p} = \sum_j \omega_{a,p}(j)^2 \, \overline{w_j} w'_j$, is a Hilbert space isometric to $\ell^2_{\mathbb{K}}$; it is complete and separable.

(ii) Coordinate evaluation is continuous: $|(\kappa_{a,p} y)_j| \leq \|y\| / \omega_{a,p}(j)$, and $y \mapsto (\kappa_{a,p} y)_j$ is the continuous linear functional $\omega_{a,p}(j)^{-1} \, \mathrm{ev}_j$.

(iii) Finite truncations converge: with $P_N y = \sum_{j \leq N} y_j e_j$, $\|y - P_N y\| \to 0$ as $N \to \infty$.

(iv) For $\mathbb{K} = \mathbb{R}$ and $a = p = 1$, $\kappa_{1,1}$ is the frozen `coeff`, and the frozen `WaveSpace` $= \ell^2_{\mathbb{R}} \times \ell^2_{\mathbb{R}}$ with its product (max) norm is isometric to $E_{1,1}$ with $\|(q,p)\|_E = \max(\|q\|_{1,1}, \|p\|_{1,1})$ via $\kappa_{1,1} \times \kappa_{1,1}$.

Proof. **Step S3.1: Isometry.** $\|\tau_{a,p} w\|^2 = \sum_j \omega_{a,p}(j)^2 |w_j|^2 = \|w\|^2_{a,p}$ by Definition 1.4; so $\tau_{a,p}$ maps $\ell^{a,p}_{\mathbb{K}}$ into $\ell^2_{\mathbb{K}}$, $\kappa_{a,p}$ maps back, and both compositions are the identity coordinatewise. Transport the inner product and completeness of (W1-E1).

Step S3.2: Separability. Finitely supported sequences with entries in $\mathbb{Q}$ (resp. $\mathbb{Q} + i\mathbb{Q}$) form a countable set; it is dense by (iii) and density of $\mathbb{Q}$ in each of finitely many coordinates. (Separability is not consumed by later milestones.)

Step S3.3: Coordinates and truncation. (ii) is (W1-E1) ($|y_j| \leq \|y\|$) divided by $\omega_{a,p}(j)$. (iii): $\|y - P_N y\|^2 = \sum_{j>N} |y_j|^2$ is the tail of a convergent series.

Step S3.4: Frozen case. `coeff y j` is $y_j / \omega_{1,1}(j)$ by its frozen definition, and Mathlib's norm on a product is the max of the norms. □

Source: P96-W p. 277 and P96-K p. 120 ("$\ell^{a,p}$ is the Hilbert space ..."), asserted without proof; the transport is ours.

Lemma 3.3 (Derivatives of coordinates)**.** *Let $c : \mathbb{R} \to \ell^2_{\mathbb{K}}$ be differentiable at t, and let (a,p), (a',p') be weights with $a \geq a'$, $p \geq p'$ and ι the inclusion of Lemma 3.5, represented by $\hat{\iota}$. For $v \in \ell^2_{\mathbb{K}}$ the following are equivalent: (a) for every j, $t \mapsto (\kappa_{a,p} c(t))_j$ has derivative $(\kappa_{a',p'} v)_j$ at t; (b) $\hat{\iota}\, c'(t) = v$. In particular, if $K : \mathbb{R}^2 \to$ `WaveSpace` is differentiable and X^w is the field of Proposition 3.18(v), then the frozen invariance conclusion (4) of `KAM.Wave.nlw2` at (I,θ) is equivalent to $\hat{\iota}\big(DK(\theta)\nu\big) = X^w(K(\theta))$ with $\hat{\iota}$ the representative of $E_{1,1} \subset E_{1,0}$.*

Proof. **Step S3.5: Coordinates of the derivative.** By Lemma 3.2(ii), $c \mapsto (\kappa_{a,p} c)_j$ is a continuous linear functional, so the coordinate curve has derivative $(\kappa_{a,p} c'(t))_j = (\kappa_{a',p'} \hat{\iota} c'(t))_j$ (the inclusion does not change coefficients, Lemma 3.5).

Step S3.6: Comparison. Hence (a) says $(\kappa_{a',p'} \hat{\iota} c'(t))_j = (\kappa_{a',p'} v)_j$ for all j, which is (b) because $\kappa_{a',p'}$ is injective. For the frozen statement apply this to $c(t) = K(\theta + t\nu)$ componentwise at $t = 0$, $c'(0) = DK(\theta)\nu$, with $(a,p) = (1,1)$, $(a',p') = (1,0)$, and compare with Proposition 3.18(v). □

Source: ours.

Lemma 3.4 (Diagonal multipliers)**.** *Let $\mu : \mathbb{N}_+ \to \mathbb{K}$ and $M \geq 0$ with $|\mu_j| \leq M$ for all j. Then $D_\mu y = (\mu_j y_j)_j$ is a bounded linear operator on $\ell^2_{\mathbb{K}}$ with $\|D_\mu\| \leq M$, and $\|D_\mu\| \geq |\mu_j|$ for every j. Consequently a diagonal map $w \mapsto (m_j w_j)$ from $\ell^{a,p}_{\mathbb{K}}$ to $\ell^{a',p'}_{\mathbb{K}}$ has representative D_μ with $\mu_j = m_j \, \omega_{a',p'}(j) / \omega_{a,p}(j)$; it is bounded with norm $\leq M$ whenever $|\mu_j| \leq M$ for all j, and unbounded if (μ_j) is unbounded.*

Proof. $\sum_{j\le N}|\mu_j y_j|^2 \le M^2\|y\|^2$ for every N, so $D_\mu y \in \ell^2$ with $\|D_\mu y\| \le M\|y\|$ ((W1-E1)); $\|D_\mu e_j\| = |\mu_j|$ gives the lower bound. The representative is computed coordinatewise: $\tau_{a',p'}(m\,\kappa_{a,p}y)_j = \omega_{a',p'}(j)m_j y_j/\omega_{a,p}(j)$. □

Source: standard.

Lemma 3.5 (Embeddings)**.** *Let $a \ge a' \ge 0$ and $p, p' \in \mathbb{R}$.*

- *(i) If $p \ge p'$, then $\ell^{a,p}_{\mathbb{K}} \subset \ell^{a',p'}_{\mathbb{K}}$ with $\|w\|_{a',p'} \le \|w\|_{a,p}$; the representative of the inclusion is D_μ with $\mu_j = j^{p'-p}e^{(a'-a)j} \in (0,1]$.*
- *(ii) If $a > a'$ and $\sigma = p' - p > 0$, then $\|w\|_{a',p'} \le \big(\sigma/(e(a-a'))\big)^\sigma \|w\|_{a,p}$.*
- *(iii) The inclusions do not change coefficients: $\kappa_{a',p'}D_\mu = \kappa_{a,p}$. They are injective and compose: the inclusion $(a,p) \to (a'',p'')$ is the composite through (a',p').*

Proof. (i) and (ii) are Lemma 3.4 with $m_j = 1$: $\mu_j = \omega_{a',p'}(j)/\omega_{a,p}(j)$. In (i) both factors of μ_j are at most 1. In (ii), $t^\sigma e^{-(a-a')t}$ is maximal on $(0,\infty)$ at $t = \sigma/(a-a')$, with value $(\sigma/(e(a-a')))^\sigma$. (iii) is the identity $\mu_j/\omega_{a',p'}(j) = 1/\omega_{a,p}(j)$; injectivity because $\mu_j \ne 0$; composition because the μ's multiply. □

Source: standard (implicit in P96-W pp. 277–280 and P96-K p. 121).

Lemma 3.6 (Frequency multipliers and the linear flow)**.** *Let $a \ge 0$, $p \in \mathbb{R}$, $r \in \mathbb{R}$, and let λ^r denote the multiplier $w \mapsto (\lambda_j^r w_j)$.*

- *(i) λ^r is bounded $\ell^{a,p+r}_{\mathbb{K}} \to \ell^{a,p}_{\mathbb{K}}$; its norm is $(1+m)^{r/2}$ for $r \ge 0$ (attained at $j = 1$) and at most 1 for $r < 0$; its representative is D_μ with $\mu_j = (\lambda_j/j)^r$. It is an isomorphism $\ell^{a,p+r} \to \ell^{a,p}$ with inverse λ^{-r}.*
- *(ii) For $r > 0$, λ^r is not bounded from $\ell^{a,p}$ to itself: $\|\lambda^r e_j\|_{a,p}/\|e_j\|_{a,p} = \lambda_j^r \to \infty$.*
- *(iii) (Complex rotations.) For $t \in \mathbb{R}$, $R_t : z \mapsto (e^{-i\lambda_j t}z_j)$ is a linear isometry of $\ell^{a,p}_{\mathbb{C}}$, $R_{t+s} = R_tR_s$, and $t \mapsto R_t z$ is continuous.*
- *(iv) (Real linear flow.) For $(q,p) \in E_{a,p}$ put $L_t(q,p) = \big(\cos(\lambda t)q + \sin(\lambda t)p,\ -\sin(\lambda t)q + \cos(\lambda t)p\big)$ (coordinatewise). Then L_t preserves $\|q\|^2_{a,p} + \|p\|^2_{a,p}$, $\|L_t(q,p)\|_E \le \sqrt{2}\,\|(q,p)\|_E$, $L_{t+s} = L_tL_s$, and every coordinate solves $\dot q_j = \lambda_j p_j$, $\dot p_j = -\lambda_j q_j$. In the complex coordinates of Definition 1.11, L_t is R_t.*

Proof. **Step S3.7: Ratio.** By Lemma 3.4, $\mu_j = \lambda_j^r\omega_{a,p}(j)/\omega_{a,p+r}(j) = (\lambda_j/j)^r$. Since $j^2 < \lambda_j^2 = j^2 + m \le (1+m)j^2$, $1 < \lambda_j/j \le \sqrt{1+m}$ with equality at $j = 1$; so $|\mu_j| \le (1+m)^{r/2} = \mu_1$ for $r \ge 0$ and $|\mu_j| \le 1$ for $r < 0$, and Lemma 3.4 applies. The inverse statement is the same computation with $-r$.

Step S3.8: Unboundedness and rotations. (ii) is immediate. (iii): $|e^{-i\lambda_j t}| = 1$, so R_t is D_μ with $|\mu_j| = 1$ (an isometry), and the group law is coordinatewise. Continuity: $\|R_tz - R_sz\|^2 = \sum_j |e^{-i\lambda_j t} - e^{-i\lambda_j s}|^2\omega_{a,p}(j)^2|z_j|^2 \to 0$ as $t \to s$ by dominated convergence for series (terms $\le 4\omega^2|z_j|^2$).

Step S3.9: Real flow. Coordinatewise $(q_j(t), p_j(t))$ is a rotation of (q_j, p_j), so $q_j(t)^2 + p_j(t)^2 = q_j^2 + p_j^2$; multiplying by $\omega_{a,p}(j)^2$ and summing gives the first claim, and then $\max(\|q(t)\|, \|p(t)\|)^2 \le \|q\|^2 + \|p\|^2 \le 2\|(q,p)\|_E^2$. The derivative and the group law are coordinatewise trigonometry; with $z_j = (q_j + ip_j)/\sqrt{2}$ one gets $z_j(t) = e^{-i\lambda_j t}z_j$. □

Source: P96-W p. 277 and Definition 1.10 (the multiplier bound is stated there without the norm); the factor $\sqrt{2}$ in (iv) is the price of the max norm of the frozen phase space and is not in the source, which uses $\ell^{a,s} \times \ell^{a,s}$ without specifying the product norm.

Definition 3.7 (Sup-weighted sequences)**.** For $p \in \mathbb{R}$, ℓ^p_∞ is the space of real sequences $(w_j)_{j\geq 1}$ with $|w|_p = \sup_j |w_j| j^p < \infty$, realized as `lp (fun _ : PNat => Real) ∞` by $w \mapsto (j^p w_j)$. (The norm is that of `lp` with exponent ∞, a supremum over a bounded set.) A map $\tilde\Omega : \Pi \to \ell^p_\infty$ on $\Pi \subset \mathbb{R}^2$ is *L-Lipschitz* ($L \geq 0$) if $|\tilde\Omega(\xi) - \tilde\Omega(\zeta)|_p \leq L|\xi - \zeta|$ for all $\xi, \zeta \in \Pi$ (max norm on $\mathbb{R}^2$); P96-K's semi-norm $|\tilde\Omega|^{\mathcal{L}}_{p,\Pi}$ is the least such L and is used only through such hypotheses.

Source: P96-K p. 120 (Assumption B) and p. 121 (Lipschitz semi-norms).

Lemma 3.8 (Spectral asymptotics of the wave normal frequencies)**.** *Let $b = (\lambda_1^{-1}, \lambda_2^{-1})$ and, for $\xi \in \mathbb{R}^2$ and $j \geq 3$, $\Omega_j(\xi) = \lambda_j + \frac{3}{2\pi\lambda_j}\langle b, \xi\rangle$ (the normal frequencies of Theorem 2.10, $\Omega = \lambda_N + B\xi$). Put $\bar\Omega_j = j$ and $\tilde\Omega_j = \Omega_j - \bar\Omega_j$ ($j \geq 3$). Then*

$$|\tilde\Omega(\xi)|_1 \leq \frac{m}{2} + \frac{3}{2\pi}|\langle b, \xi\rangle|, \qquad |\tilde\Omega(\xi) - \tilde\Omega(\zeta)|_1 \leq \frac{3}{2\pi}\Big(\frac{1}{\lambda_1} + \frac{1}{\lambda_2}\Big)|\xi - \zeta|.$$

So Assumption B of P96-K holds with $d = 1$, $\delta = -1$ (which satisfies $\delta < d - 1$), and, with $\bar p = p + 1$ for the perturbation field, the convention $p - \bar p \leq \delta$ of P96-K p. 121 holds with equality. Moreover $|\Omega_i(\xi) - \Omega_j(\xi) - (i - j)| \leq \big(m + \frac{3}{\pi}|\langle b, \xi\rangle|\big)/\min(i, j)$ for $i, j \geq 3$.

Proof. $\lambda_j - j = m/(\lambda_j + j) \in (0, m/(2j)]$, and $j \cdot \frac{3}{2\pi\lambda_j} \leq \frac{3}{2\pi}$ since $\lambda_j > j$. Multiply by j and take the supremum; the Lipschitz bound is the second term with $|\langle b, \xi - \zeta\rangle| \leq (\lambda_1^{-1} + \lambda_2^{-1})|\xi - \zeta|$. The last bound is $|\tilde\Omega_i| + |\tilde\Omega_j| \leq 2|\tilde\Omega|_1/\min(i, j)$. □

Source: ours; checks P96-K Assumption B (p. 120) for the wave normal form of P96-W p. 290 with the corrected B of Theorem 2.10. Normal modes are indexed by $j \geq 3$ here; P96-K indexes them from 1, and the shift $j \mapsto j - 2$ changes $\bar\Omega_j$ only by the constant 2, a lower-order term allowed by Assumption B. In Lean, normal-mode sequences are `PNat`-indexed with the entries $j = 1, 2$ set to zero.

3.4. **Two-sided spaces, Young and the algebra.**

Definition 3.9 (Two-sided transport)**.** For $a \geq 0$, $s \in \mathbb{R}$ let $\omega^b_{a,s}(l) = \langle l\rangle^s e^{a|l|}$ ($l \in \mathbb{Z}$, $\langle l\rangle = \max(1, |l|)$). The space $\ell^{a,s}_b$ of Definition 1.6 is realized as `lp (fun _ : Int => Complex) 2` through $c \mapsto (\omega^b_{a,s}(l)c_l)_l$, an isometry as in Lemma 3.2. For c with $\sum_i |c_i||d_{l-i}| < \infty$ for all l, $(c * d)_l = \sum_{i\in\mathbb{Z}} c_i d_{l-i}$ (Lean: a `tsum` guarded by the summability proved before each use).

Source: P96-W p. 279 and §6 p. 294.

Lemma 3.10 (Passage between one- and two-sided sequences)**.** *Let $a \geq 0$, $p \in \mathbb{R}$.*

- *(i) (Even and odd extensions.) For $q \in \ell^{a,p}_{\mathbb{C}}$ define $(E^\pm q)_l = (\pm 1)^{[l<0]} q_{|l|}/\sqrt{\lambda_{|l|}}$ for $l \neq 0$ and $(E^\pm q)_0 = 0$. Then $E^\pm : \ell^{a,p}_{\mathbb{C}} \to \ell^{a,p+1/2}_b$ is linear with $\|E^\pm q\|_{a,p+1/2} \leq \sqrt{2}\|q\|_{a,p}$, and the same holds for the majorant $l \mapsto |(E^+ q)_l|$. The even sequence $x_l = |q_{|l|}|/\sqrt{|l|}$ ($l \neq 0$), $x_0 = 0$, satisfies $\|x\|^2_{a,p+1/2} = 2\|q\|^2_{a,p}$.*
- *(ii) (Restriction with gain.) For $X \in \ell^{a,p+1/2}_b$, the sequence $(X_l/\sqrt{\lambda_l})_{l\geq 1}$ lies in $\ell^{a,p+1}_{\mathbb{C}}$ with norm at most $\big(\sum_{l\geq 1} l^{2p+1} e^{2al}|X_l|^2\big)^{1/2} \leq \|X\|_{a,p+1/2}$, and at most $\|X\|_{a,p+1/2}/\sqrt{2}$ if $|X|$ is even.*
- *(iii) (Pairing.) $J(z, \bar z)_j = z_j$, $J(z, \bar z)_{-j} = \bar z_j$ ($j \geq 1$), $J(z, \bar z)_0 = 0$ defines a linear isometry from $\ell^{a,p}_{\mathbb{C}} \times \ell^{a,p}_{\mathbb{C}}$ with the norm $(\|z\|^2 + \|\bar z\|^2)^{1/2}$ onto $\{w \in \ell^{a,p}_b : w_0 = 0\}$; in particular $\|J(z, \bar z)\|_{a,p} \leq \sqrt{2}\max(\|z\|_{a,p}, \|\bar z\|_{a,p})$. (Here $\bar z$ is an independent variable, as in P96-W p. 282.)*

Proof. **Step S3.10: Extensions.** $\|E^\pm q\|^2 = 2\sum_{j\geq 1} j^{2p+1} e^{2aj}|q_j|^2/\lambda_j \leq 2\sum_j j^{2p} e^{2aj}|q_j|^2$ since $j/\lambda_j < 1$; the majorant has the same norm; with $\sqrt{|l|}$ the factor j/λ_j becomes 1. This computation is self-contained (every $a \geq 0$, $p \in \mathbb{R}$) and analogous to Step S2.6, which is stated only for $p \geq \frac{1}{2}$.

Step S3.11: Restriction. $\sum_{l\ge1} l^{2p+2}e^{2al}|X_l|^2/\lambda_l \le \sum_{l\ge1} l^{2p+1}e^{2al}|X_l|^2$ since $l/\lambda_l < 1$; for even $|X|$ the sum over $l \ge 1$ is at most half of the full sum. This computation is self-contained (every $a \ge 0$, $p \in \mathbb{R}$, arbitrary X) and analogous to Step S2.8, which treats only $p \ge \frac12$ and $X = x * x' * x''$ for the map T.

Step S3.12: Pairing. $\|J(z,\bar z)\|^2 = \sum_{j\ge1} \omega_{a,p}(j)^2(|z_j|^2 + |\bar z_j|^2)$, since $\langle \pm j\rangle = j$ for $j \ge 1$. □

Source: P96-W p. 280 (proof of Lemma 3, with the corrected constant $\sqrt{2}$, errata item N-1, see Theorem 2.5) and p. 282 (the coordinates w).

Lemma 3.11 (Young and ℓ^1 control)**.** *(i) For $X \in \ell^2(\mathbb{Z})$ and $Y \in \ell^1(\mathbb{Z})$, each sum $\sum_i |X_i||Y_{l-i}|$ converges, and $\|X * Y\|_{\ell^2} \le \|X\|_{\ell^2}\|Y\|_{\ell^1}$.*

(ii) For $s > \frac12$, $\sum_{l\in\mathbb{Z}} \langle l\rangle^{-2s} = 1 + 2\zeta(2s) < \infty$, and for $a \ge 0$ and $Y \in \ell^{a,s}_b$, $\|Y\|_{\ell^1} \le (1 + 2\zeta(2s))^{1/2}\|Y\|_{0,s} \le (1 + 2\zeta(2s))^{1/2}\|Y\|_{a,s}$.

(iii) For nonnegative sequences X, Y, Z with $X, Y, Z \in \ell^{0,s}_b$, $s > \frac12$, convolution is commutative and associative, and monotone: $0 \le X \le X'$ pointwise implies $X * Y \le X' * Y$. For complex sequences, $|c * d| \le |c| * |d|$ pointwise.

Proof. **Step S3.13: Weighted Cauchy–Schwarz.**

$$\Big(\sum_i |X_i||Y_{l-i}|\Big)^2 \le \Big(\sum_i |X_i|^2|Y_{l-i}|\Big)\Big(\sum_i |Y_{l-i}|\Big) = \|Y\|_{\ell^1}\sum_i |X_i|^2|Y_{l-i}|,$$

finite since $|X_i|^2 \le \|X\|^2$.

Step S3.14: Summation. Summing over l and exchanging the nonnegative double sum ((W1-E2)) gives $\sum_l |(X * Y)_l|^2 \le \|Y\|_{\ell^1}\sum_i |X_i|^2 \sum_l |Y_{l-i}| = \|Y\|^2_{\ell^1}\|X\|^2_{\ell^2}$.

Step S3.15: ℓ^1 control. $\sum_l \langle l\rangle^{-2s} = 1 + 2\sum_{l\ge1} l^{-2s}$ ($\langle 0\rangle = 1$), finite by (W1-E2). Then $\sum_l |Y_l| = \sum_l \langle l\rangle^{-s}\cdot\langle l\rangle^s|Y_l| \le (1 + 2\zeta(2s))^{1/2}\|Y\|_{0,s}$, and $\|Y\|_{0,s} \le \|Y\|_{a,s}$ since $e^{a|l|} \ge 1$.

Step S3.16: Algebraic rules. By (ii) all sequences involved are in $\ell^1 \cap \ell^2$, so the double series defining $(X * Y) * Z$ and $X * (Y * Z)$ converge absolutely and may be rearranged (Fubini for series); commutativity is the substitution $i \mapsto l - i$; monotonicity and $|c * d| \le |c| * |d|$ are termwise. □

Source: standard; used in P96-W §6 p. 294 and KP96 App. A pp. 31–32.

Lemma 3.12 (Convolution algebra)**.** *For $a \ge 0$, $s \ge 1$ and $c, d \in \ell^{a,s}_b$: $\|c * d\|_{a,s} \le K_s\|c\|_{a,s}\|d\|_{a,s}$ with $K_s = 2^s(1 + 2\zeta(2s))^{1/2}$, and for three factors $\|c * d * e\|_{a,s} \le K_s^2\|c\|\|d\|\|e\|$.*

Proof. This is Lemma 2.4, whose Step S2.5 uses Lemma 3.11(i)–(ii); the three-factor form applies it twice, with associativity from Lemma 3.11(iii). In Lean the lemma is proved once, in the W1 module, and W0 imports it. □

Source: P96-W Lemma 2, p. 279, and §6, pp. 294–295 (the proof there is labelled "Appendix A" on p. 279 but printed as §6); KP96 Lemma 2 p. 12.

Remark 3.13 (The algebra for $\frac12 < s < 1$). P96-W §6 proves the algebra for all $s > \frac12$ by a different weight split: $\langle l\rangle/(\langle l - i\rangle\langle i\rangle) \le \langle l - i\rangle^{-1} + \langle i\rangle^{-1}$ and $(\alpha + \beta)^{2s} \le 2^{2s-1}(\alpha^{2s} + \beta^{2s})$ give the same constant K_s. No later milestone needs $s < 1$ (the cubic estimates use $s = p + \frac12 \ge 1$), so this extension is not part of the plan.

3.5. **The cubic field.** The next lemma isolates the only property of the quartic coefficients that the estimates use: the selection rule of Lemma 2.2 together with the decay $(ijkl)^{-1/2}$. It serves both the cubic gradient (here) and the Birkhoff generating field of milestone W3, whose coefficients are those of G divided by the divisors of P96-W Lemma 4.

Lemma 3.14 (Selection-rule trilinear maps)**.** *Let $a \geq 0$, $p \geq \frac{1}{2}$, $s = p + \frac{1}{2}$, $M \geq 0$.*

(a) (One-sided.) Let $c : \mathbb{N}_+^4 \to \mathbb{C}$ satisfy $|c_{ijkl}| \leq M(ijkl)^{-1/2} \sum_{\varepsilon \in \{\pm 1\}^3} [\, l = \varepsilon_1 i + \varepsilon_2 j + \varepsilon_3 k \,]$. For $q, q', q'' \in \ell^{a,p}_{\mathbb{C}}$ the sums $T_c(q, q', q'')_l = \sum_{i,j,k \geq 1} c_{ijkl} q_i q'_j q''_k$ converge absolutely, and

$$\|T_c(q, q', q'')\|_{a,p+1} \leq 2MK_s^2 \, \|q\|_{a,p} \|q'\|_{a,p} \|q''\|_{a,p}.$$

(b) (Two-sided.) Let $c : (\mathbb{Z} \setminus 0)^4 \to \mathbb{C}$ satisfy $|c_{ijkl}| \leq M|ijkl|^{-1/2} \sum_{\varepsilon \in \{\pm 1\}^3} [\, l = \varepsilon_1 i + \varepsilon_2 j + \varepsilon_3 k \,]$. For $w, w', w'' \in \ell^{a,p}_b$ the sums $T_c(w, w', w'')_l = \sum_{i,j,k \neq 0} c_{ijkl} w_i w'_j w''_k$ ($l \neq 0$) converge absolutely; with $T_c(\cdot)_0 = 0$,

$$\|T_c(w, w', w'')\|_{a,p+1} \leq 64MK_s^2 \, \|w\|_{a,p} \|w'\|_{a,p} \|w''\|_{a,p}.$$

In both cases T_c is trilinear; it is symmetric if c is symmetric in its first three indices.

Proof. **Step S3.17: One-sided majorant.** In (a) let $x_i = |q_{|i|}|/\sqrt{|i|}$ ($i \neq 0$), $x_0 = 0$, even; similarly x', x''. By Lemma 3.10(i), $\|x\|_{a,s} = \sqrt{2}\|q\|_{a,p}$.

Step S3.18: One-sided selection. $\sum_{i,j,k} |c_{ijkl} q_i q'_j q''_k| \leq M l^{-1/2} \sum_{i,j,k \geq 1} \sum_{\varepsilon} [l = \varepsilon_1 i + \varepsilon_2 j + \varepsilon_3 k] \, x_i x'_j x''_k \leq M l^{-1/2} X_l$, $X = x * x' * x''$, because $(i, j, k, \varepsilon) \mapsto (\varepsilon_1 i, \varepsilon_2 j, \varepsilon_3 k)$ is injective into $(\mathbb{Z} \setminus 0)^3$ and the x's are even (as in Step S2.7). X is finite and even by Lemmas 3.11 and 3.12.

Step S3.19: One-sided smoothing. $\|T_c\|^2_{a,p+1} \leq M^2 \sum_{l \geq 1} l^{2p+1} e^{2al} X_l^2 \leq \frac{M^2}{2} \|X\|^2_{a,s}$ (evenness), and $\|X\|_{a,s} \leq K_s^2 \|x\| \|x'\| \|x''\| = 2^{3/2} K_s^2 \prod \|q\|$. Hence the bound $2MK_s^2 \prod \|q\|$.

Step S3.20: Two-sided majorant. In (b) let $\tilde{x}_i = (|w_i| + |w_{-i}|)/\sqrt{|i|}$ ($i \neq 0$), $\tilde{x}_0 = 0$, even, and similarly $\tilde{x}', \tilde{x}''$. Then $\|\tilde{x}\|^2_{a,s} = \sum_{i \neq 0} |i|^{2p} e^{2a|i|} (|w_i| + |w_{-i}|)^2 \leq 4 \|w\|^2_{a,p}$.

Step S3.21: Two-sided selection. For fixed ε, the substitution $i' = \varepsilon_1 i$, $j' = \varepsilon_2 j$, $k' = \varepsilon_3 k$ is a bijection of $(\mathbb{Z} \setminus 0)^3$, and $|w_i|/\sqrt{|i|} \leq \tilde{x}_{i'}$. Hence each of the eight ε-terms is at most $(\tilde{x} * \tilde{x}' * \tilde{x}'')_l =: \tilde{X}_l$, and $|T_c(w, w', w'')_l| \leq 8M|l|^{-1/2} \tilde{X}_l$.

Step S3.22: Two-sided smoothing. $\|T_c\|^2_{a,p+1} \leq 64M^2 \sum_{l \neq 0} |l|^{2p+1} e^{2a|l|} \tilde{X}_l^2 \leq 64M^2 \|\tilde{X}\|^2_{a,s}$ and $\|\tilde{X}\|_{a,s} \leq K_s^2 \cdot 8 \prod \|w\|$, so $\|T_c\| \leq 64MK_s^2 \prod \|w\|$. Trilinearity and symmetry are termwise, the sums being absolutely convergent. □

Source: P96-W pp. 279–280 (Lemma 3) and p. 284 ((13)–(14), estimate of $\partial F/\partial w_l$, constants not made explicit); KP96 p. 16. The explicit constants are ours.

Theorem 3.15 (The cubic map as a continuous trilinear map)**.** *Let $a \geq 0$, $p \geq \frac{1}{2}$, $C_* = K^2_{p+1/2}/\pi$, and $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$.*

(i) There is a unique continuous symmetric $\mathbb{K}$-trilinear map $\mathcal{T}^{a,p}_{\mathbb{K}} : (\ell^2_{\mathbb{K}})^3 \to \ell^2_{\mathbb{K}}$ (Lean `ContinuousMultilinearMap K (fun _ : Fin 3 => lp ... 2) (lp ... 2))` with $\kappa_{a,p+1} \mathcal{T}^{a,p}_{\mathbb{K}}(y, y', y'') = T(\kappa_{a,p} y, \kappa_{a,p} y', \kappa_{a,p} y'')$, where T is the trilinear map of Theorem 2.5; and $\|\mathcal{T}^{a,p}_{\mathbb{K}}\| \leq C_*$.

(ii) (Real structure.) With ι and conjugation of Definition 3.22, $\mathcal{T}^{a,p}_{\mathbb{C}}(\iota y, \iota y', \iota y'') = \iota \, \mathcal{T}^{a,p}_{\mathbb{R}}(y, y', y'')$ and $\overline{\mathcal{T}^{a,p}_{\mathbb{C}}(z, z', z'')} = \mathcal{T}^{a,p}_{\mathbb{C}}(\bar{z}, \bar{z}', \bar{z}'')$.

(iii) (Compatibility.) For $a \geq a' \geq 0$, $p \geq p' \geq \frac{1}{2}$, with $\hat{\imath}$, $\hat{\imath}'$ the representatives of $\ell^{a,p} \subset \ell^{a',p'}$ and $\ell^{a,p+1} \subset \ell^{a',p'+1}$: $\mathcal{T}^{a',p'}(\hat{\imath} y, \hat{\imath} y', \hat{\imath} y'') = \hat{\imath}' \mathcal{T}^{a,p}(y, y', y'')$.

(iv) (Frozen identification.) For $y \in \ell^2_{\mathbb{R}}$ and $l \geq 1$,

$$\texttt{cubicGrad m hm y l} = \frac{\mathcal{T}^{1,1}_{\mathbb{R}}(y,y,y)_l}{\omega_{1,2}(l)}, \qquad \textit{i.e.} \qquad G_q(\kappa_{1,1} y) = \kappa_{1,2}\, \mathcal{T}^{1,1}_{\mathbb{R}}(y,y,y).$$

For $a > 0$ and $p \geq \frac{1}{2}$ likewise $G_q(\kappa_{a,p} y) = \kappa_{a,p+1} \mathcal{T}^{a,p}_{\mathbb{R}}(y,y,y)$.

Proof. **Step S3.23: Coordinates.** For $y, y', y'' \in \ell^2_{\mathbb{K}}$ put $q = \kappa_{a,p} y$ etc. By Theorem 2.5 (equivalently Lemma 3.14(a) with $c_{ijkl} = g_{ijkl}(\lambda_i\lambda_j\lambda_k\lambda_l)^{-1/2}$ and $M = 1/(2\pi)$, using $\lambda \geq j$ and the bound after (3)), each sum $T(q,q',q'')_l$ is an absolutely convergent `tsum` over $\mathbb{N}_+^3$ and $T(q,q',q'') \in \ell^{a,p+1}_{\mathbb{K}}$ with norm $\leq C_* \prod \|q\|_{a,p} = C_* \prod \|y\|$. Define $\mathcal{T}(y,y',y'') = \tau_{a,p+1} T(q,q',q'')$.

Step S3.24: Continuity. Trilinearity and symmetry hold termwise (symmetry of g, Lemma 2.2); the bound makes the multilinear map continuous with norm $\leq C_*$ (Lean: `MultilinearMap.mkContinuous`). Uniqueness: a map with the stated coordinates is determined, since $\kappa_{a,p+1}$ is injective.

Step S3.25: Real structure and compatibility. The coefficients $g_{ijkl}(\lambda_i\lambda_j\lambda_k\lambda_l)^{-1/2}$ are real, so the coordinate sums commute with ι and with conjugation (absolute convergence). (iii) holds because the inclusions do not change coefficients (Lemma 3.5(iii)), so both sides have coefficients $T(q,q',q'')$.

Step S3.26: Frozen identification. By the frozen definitions, `reconU m hm y` is $U_m(\kappa_{1,1} y)$ (Definition 1.7; the `tsum` is absolutely convergent by Lemma 2.1), the interval integral over $[0,\pi]$ is the integral of Definition 1.9, and $\sqrt{\lambda_l} > 0$. So `cubicGrad m hm y l` $= G_q(\kappa_{1,1} y)_l$, which is $T(q,q,q)_l$ by Lemma 2.3 ($a = 1 > 0$). The general case is the same with $a > 0$. □

Source: P96-W Lemma 3, pp. 279–280; Theorem 2.5 for the bound. The Lean packaging is ours.

Corollary 3.16 (Analyticity and Lipschitz bounds of the cubic gradient)**.** *Let $a \geq 0$, $p \geq \frac{1}{2}$, $\mathcal{T} = \mathcal{T}^{a,p}_{\mathbb{K}}$, C_* as above, and $\mathcal{N}(y) = \mathcal{T}(y,y,y)$.*

(i) $\mathcal{N} : \ell^2_{\mathbb{K}} \to \ell^2_{\mathbb{K}}$ is $\mathbb{K}$-analytic on all of $\ell^2_{\mathbb{K}}$ (Lean `AnalyticOnNhd K N Set.univ`), with $\|\mathcal{N}(y)\| \leq C_* \|y\|^3$.

(ii) $D\mathcal{N}(y)h = 3\mathcal{T}(y,y,h)$, $D^2\mathcal{N}(y)(h,k) = 6\mathcal{T}(y,h,k)$, $D^3\mathcal{N}(y) = 6\mathcal{T}$; so $\|D\mathcal{N}(y)\| \leq 3C_*\|y\|^2$ and $\|D^2\mathcal{N}(y)\| \leq 6C_*\|y\|$.

(iii) $\|\mathcal{N}(y) - \mathcal{N}(y')\| \leq 3C_* \|y - y'\| \max(\|y\|, \|y'\|)^2$.

(iv) (Frozen form.) The map $\mathcal{G}^w : \ell^2_{\mathbb{R}} \to \ell^2_{\mathbb{R}}$, $\mathcal{G}^w(y)_l = \omega_{1,2}(l) \cdot$ `cubicGrad m hm y l`, is $\mathcal{N}$ for $(a,p) = (1,1)$, $\mathbb{K} = \mathbb{R}$; it is real analytic everywhere and $\|\mathcal{G}^w(y)\| \leq C_m \|y\|^3$, $C_m = 8(1+2\zeta(3))/\pi$. The coefficient sequence $G_q(\kappa_{1,1} y)$ has weighted coordinates $\mathcal{G}^w(y)$ at weight $(1,2)$, $(l^{-1}\mathcal{G}^w(y)_l)$ at weight $(1,1)$ and $(l^{-2}\mathcal{G}^w(y)_l)$ at weight $(1,0)$.

Proof. **Step S3.27: Analyticity.** $\mathcal{N} = \mathcal{T} \circ \Delta$ with $\Delta(y) = (y,y,y)$ continuous linear; (W1-E3).

Step S3.28: Derivatives. Expanding $\mathcal{T}(y+h, y+h, y+h)$ by trilinearity, the linear term is $\mathcal{T}(h,y,y) + \mathcal{T}(y,h,y) + \mathcal{T}(y,y,h) = 3\mathcal{T}(y,y,h)$ by symmetry and the rest is $O(\|h\|^2)$ with constant $\leq 3C_*(\|y\| + \|h\|)\|h\|^2$; the higher derivatives are the same computation (Lean: `ContinuousMultilinearMap.hasFDerivAt` composed with Δ).

Step S3.29: Lipschitz. $\mathcal{N}(y) - \mathcal{N}(y') = \mathcal{T}(y-y', y, y) + \mathcal{T}(y', y-y', y) + \mathcal{T}(y', y', y-y')$, so the difference is at most $C_*\|y - y'\|(\|y\|^2 + \|y\|\|y'\| + \|y'\|^2) \leq 3C_*\|y - y'\| \max^2$.

Step S3.30: Frozen form. Theorem 3.15(iv) and $\omega_{1,1}(l)/\omega_{1,2}(l) = l^{-1}$, $\omega_{1,0}(l)/\omega_{1,2}(l) = l^{-2}$; $C_* = K^2_{3/2}/\pi = C_m$ (Theorem 2.5). □

Source: P96-W Lemma 3, p. 279 (analyticity near 0, with $O(\|q\|^3)$); the global polynomial form and the constants are ours.

Proposition 3.17 (Gradient identity)**.** *Let $a > 0$, $p \geq \frac{1}{2}$, and for $q \in \ell^{a,p}$ let $G(q) = \frac{1}{4}\int_0^\pi U_m(q)^4\,dx$ (Definition 1.8). Let $\langle u, v\rangle_0 = \sum_{j\geq 1} u_j v_j$ be the* unweighted *pairing, absolutely convergent for $u \in \ell^{a,p+1}$, $v \in \ell^{a,p}$.*

(i) $G(q) = \frac{1}{4}\langle G_q(q), q\rangle_0 = \frac{1}{4}\mathcal{Q}(q,q,q,q)$ with the symmetric continuous 4-linear form $\mathcal{Q}(q_1,q_2,q_3,q_4) = \langle T(q_1,q_2,q_3), q_4\rangle_0$, $|\mathcal{Q}| \leq C_* \prod \|q_i\|_{a,p}$.

(ii) G is real analytic on $\ell^{a,p}$ and $DG(q)h = \langle G_q(q), h\rangle_0$ for all $h \in \ell^{a,p}$.

(iii) $G_q(q)$ is not *the Riesz gradient of G for the inner product of $\ell^{a,p}$; that gradient is $(\omega_{a,p}(j)^{-2} G_q(q)_j)_j$. In frozen weighted coordinates: for $G^w(y) = G(\kappa_{1,1} y)$,*

$$DG^w(y)h = \sum_{j\geq 1} \texttt{cubicGrad m hm y j} \cdot \texttt{coeff h j}, \qquad \nabla^{\ell^2} G^w(y) = \Big(\frac{\mathcal{G}^w(y)_j}{\omega_{1,1}(j)\,\omega_{1,2}(j)}\Big)_j.$$

Proof. **Step S3.31: Quartic form.** By Lemma 2.3, $G(q) = \frac{1}{4}\sum_{i,j,k,l} g_{ijkl}\hat{q}_i\hat{q}_j\hat{q}_k\hat{q}_l$ absolutely; grouping the sum over l last gives $\frac{1}{4}\sum_l q_l\, T(q,q,q)_l$. The same absolute convergence, with four different arguments, defines $\mathcal{Q}$; it is symmetric because g is. Bound: $|\langle u, v\rangle_0| \leq \|u\|_{0,0}\|v\|_{0,0} \leq \|u\|_{a,p+1}\|v\|_{a,p}$ (Cauchy–Schwarz and Lemma 3.5), and Theorem 2.5.

Step S3.32: Derivative. $G = \frac{1}{4}\mathcal{Q} \circ (q \mapsto (q,q,q,q))$ is analytic ((W1-E3)), and by symmetry $DG(q)h = \frac{1}{4}\cdot 4\mathcal{Q}(q,q,q,h) = \langle T(q,q,q), h\rangle_0 = \langle G_q(q), h\rangle_0$ (Theorem 3.15(iv)).

Step S3.33: Weighted coordinates. $\langle G_q(q), h\rangle_0 = \sum_j \omega_{a,p}(j)^2\big(\omega_{a,p}(j)^{-2} G_q(q)_j\big)h_j$ identifies the Riesz gradient. For the frozen form substitute $q = \kappa_{1,1} y$, $h \mapsto \kappa_{1,1} h$ and Corollary 3.16(iv); the ℓ^2 gradient is read off from $\sum_j G_q(\kappa_{1,1}y)_j\, h_j/\omega_{1,1}(j)$. □

Source: P96-W (7) p. 278 and (8)–(9) p. 280 (where G_q is "the gradient of G" without naming the pairing).

Proposition 3.18 (The Hamiltonian field)**.** *Let $a > 0$ and $p \geq \frac{1}{2}$.*

(i) The linear field $X_\Lambda(q,p) = (\lambda p, -\lambda q)$ is bounded $E_{a,p} \to E_{a,p-1}$ with norm $\leq \sqrt{1+m}$, and is not bounded $E_{a,p} \to E_{a,p}$.

(ii) The nonlinear field $X_G(q,p) = (0, -G_q(q))$ is a real analytic map $E_{a,p} \to E_{a,p+1}$ (smoothing of order one), $\|X_G(q,p)\|_{E_{a,p+1}} \leq C_*\|q\|_{a,p}^3$, with the Lipschitz and derivative bounds of Corollary 3.16. Its complex extension $(z,\bar{z}) \mapsto$ (same formula with $\mathcal{T}_\mathbb{C}$) is $\mathbb{C}$-analytic $E_{a,p}^\mathbb{C} \to E_{a,p+1}^\mathbb{C}$.

(iii) $X_H = X_\Lambda + X_G$ is real analytic $E_{a,p} \to E_{a,p-1}$.

(iv) (Hamiltonian structure.) $H = \Lambda + G$ is real analytic on $E_{a,p}$, and for every j, $DH(q,p)(0,e_j) = \lambda_j p_j$, $DH(q,p)(e_j,0) = \lambda_j q_j + G_q(q)_j$; that is, $X_H = (\partial_p H, -\partial_q H)$ coordinatewise, which is the system (2).

(v) (Frozen form, $a = p = 1$.) In weighted coordinates, with $D = D_\mu$, $\mu_j = \lambda_j/j$, and $S = D_{(1/j)}$, the map X^w : `WaveSpace` $\to$ `WaveSpace`,

$$X^w(y,y') = \big(Dy',\ -Dy - S^2\mathcal{G}^w(y)\big),$$

represents $X_H : E_{1,1} \to E_{1,0}$; it is real analytic on all of `WaveSpace`, and its j-th coordinates are $\omega_{1,0}(j)\big(\lambda_j$ `coeff y' j`$,\ -\lambda_j$ `coeff y j` $-$ `cubicGrad m hm y j`$\big)$, the right sides of the frozen conclusion (4) times $\omega_{1,0}(j) = e^j$.

Proof. **Step S3.34: Linear part.** Lemma 3.6(i)–(ii) with $r = 1$ in each component; the max norm is preserved under componentwise bounds.

Step S3.35: Nonlinear part. Corollary 3.16 with Theorem 3.15(iv) in the second component, composed with the continuous linear projection $(q,p) \mapsto q$; complex version from Theorem 3.15(ii).

Step S3.36: Sum. X_G maps into $E_{a,p+1} \subset E_{a,p-1}$ (Lemma 3.5); a sum of a bounded linear and an analytic map is analytic.

Step S3.37: Hamiltonian. $\Lambda(q,p) = \frac{1}{2}(\langle \lambda q, q\rangle_0 + \langle \lambda p, p\rangle_0)$ is the diagonal of the symmetric bilinear form $\beta(q,q') = \sum_j \lambda_j q_j q'_j$, and $|\beta(q,q')| \le \sqrt{1+m}\sum_j j|q_j||q'_j| \le \sqrt{1+m}\|q\|_{0,1/2}\|q'\|_{0,1/2} \le \sqrt{1+m}\|q\|_{a,p}\|q'\|_{a,p}$ (Lemma 3.5, $p \ge \frac{1}{2}$), so Λ is a bounded quadratic form on $E_{a,p}$. Its derivative is $\langle \lambda q, h\rangle_0 + \langle \lambda p, k\rangle_0$; add Proposition 3.17(ii) and evaluate at $(e_j, 0)$, $(0, e_j)$.

Step S3.38: Frozen form. Coordinates of $X_H(\kappa_{1,1}y, \kappa_{1,1}y')$ times $\omega_{1,0}(j) = e^j$: $e^j \lambda_j y'_j/(je^j) = (\lambda_j/j)y'_j$, and $e^j\,\texttt{cubicGrad}_j = j^{-2}\mathcal{G}^w(y)_j$ by Corollary 3.16(iv). D and S are bounded by Lemma 3.4 ($\lambda_j/j \le \sqrt{1+m}$, $1/j \le 1$), so X^w is a polynomial map on `WaveSpace`, analytic by (W1-E3). □

Source: P96-W p. 280 ("X_G defines a real analytic map from some neighbourhood of the origin in $\ell^{a,s}\times\ell^{a,s}$ into $\ell^{a,s+1}\times\ell^{a,s+1}$ … X_Λ is unbounded of order 1"); P96-K Assumption C p. 121 ($X_P : \mathcal{P}^{a,p} \to \mathcal{P}^{a,\bar p}$, $\bar p > p$ for $d = 1$). Global analyticity is ours.

3.6. The sine system and analytic reconstruction.

Lemma 3.19 (Completeness of the sine system)**.** *(i) If $f \in L^2(0,\pi)$ and $\int_0^\pi f\phi_l\,dx = 0$ for all $l \ge 1$, then $f = 0$ almost everywhere on $(0,\pi)$. (ii) If moreover f is continuous on $[0,\pi]$, then $f = 0$ on $[0,\pi]$.*

Proof. **Step S3.39: Odd extension.** Define g on $(-\pi,\pi]$ by $g(x) = f(x)$ for $0 < x \le \pi$, $g(x) = -f(-x)$ for $-\pi < x < 0$, $g(0) = 0$, and regard it as a function on $\mathbb{R}/2\pi\mathbb{Z}$. Then $g \in L^2$, with $\|g\|^2 = 2\|f\|^2$ for the Lebesgue measure on $(-\pi,\pi]$.

Step S3.40: Fourier coefficients vanish. For $n \in \mathbb{Z}$, $\int_{-\pi}^{\pi} g(x)e^{-inx}dx = \int_0^\pi f(x)(e^{-inx} - e^{inx})dx = -2i\int_0^\pi f(x)\sin(nx)\,dx$ (substitute $x \mapsto -x$ on $(-\pi,0)$). This vanishes for $n = 0$, for $n \ge 1$ by hypothesis ($\sin(nx) = \sqrt{\pi/2}\,\phi_n(x)$), and for $n \le -1$ since $\sin(nx) = -\sin(|n|x)$.

Step S3.41: Hilbert basis. The Fourier coefficients of g for the normalized Haar measure are $(2\pi)^{-1}$ times the integrals above, hence all zero; by (W1-E8) (the coefficient map of a Hilbert basis is injective), $g = 0$ in L^2, so $g = 0$ a.e. and $f = 0$ a.e. on $(0,\pi)$.

Step S3.42: Continuity. If f is continuous and $f(x_0) \ne 0$ for some $x_0 \in [0,\pi]$, then $|f| > |f(x_0)|/2$ on a relatively open subinterval of $[0,\pi]$ of positive length, contradicting (i). □

Source: standard (P96-W p. 279, first paragraph, asserted); used in Step S2.10. Lean: transport of g to `AddCircle (2 * Real.pi)` and `fourierBasis`; the measure normalization is the factor $(2\pi)^{-1}$.

Lemma 3.20 (Analytic sine features)**.** *Let $a > 0$, $p \in \mathbb{R}$, $k \in \mathbb{N}$, and let $\beta : \mathbb{N}_+ \to \mathbb{R}$ satisfy $|\beta_j| \le Bj^\nu$ for some $B, \nu \ge 0$. For $x \in \mathbb{C}$ with $|\operatorname{Im} x| < a$ define $\Psi(x) \in \ell^2_{\mathbb{C}}$ by*

$$\Psi(x)_j = \frac{\beta_j\,\phi_j^{(k)}(x)}{\omega_{a,p}(j)}, \qquad \phi_j^{(k)}(x) = \sqrt{2/\pi}\,j^k \sin^{(k)}(jx).$$

(i) For $0 < a' < a$ put $b_j = \sqrt{2/\pi}\,Bj^{\nu+k-p}e^{-(a-a')j}$. Then $|\Psi(x)_j| \le b_j$ whenever $|\operatorname{Im} x| \le a'$, and $\sum_j b_j < \infty$.

(ii) Ψ is holomorphic on the strip $\{|\operatorname{Im} x| < a\}$, hence $\mathbb{C}$-analytic there; $\Psi(x)$ is real for real x, and $\Psi_{\mathbb{R}} = \operatorname{re}\circ\Psi|_{\mathbb{R}} : \mathbb{R} \to \ell^2_{\mathbb{R}}$ is real analytic.

(iii) For real x and $y \in \ell^2_{\mathbb{R}}$, $\langle \Psi_{\mathbb{R}}(x), y\rangle = \sum_j \beta_j\,(\kappa_{a,p}y)_j\,\phi_j^{(k)}(x)$, absolutely convergent.

Proof. **Step S3.43: Growth on the strip.** $|\sin^{(k)}(jx)| \le \cosh(j\operatorname{Im} x) \le e^{j|\operatorname{Im} x|}$, so $|\Psi(x)_j| \le \sqrt{2/\pi}Bj^{\nu+k-p}e^{-(a-a')j}$ on $|\operatorname{Im} x| \le a'$; the series converges since $a - a' > 0$.

Step S3.44: Normal convergence. The partial sums $\Psi^N(x) = \sum_{j\le N}\Psi(x)_j e_j$ are holomorphic (finite sums of entire functions times fixed vectors), and by (i) $\|\Psi(x) - \Psi^N(x)\| \le \sum_{j>N} b_j \to 0$

uniformly on $|\operatorname{Im} x| \le a'$. Every compact subset of the strip lies in some $\{|\operatorname{Im} x| \le a'\}$, so $\Psi^N \to \Psi$ locally uniformly and Ψ is holomorphic and analytic by (W1-E4).

Step S3.45: Real restriction. For real x all coordinates are real, so $\Psi(x) = \iota\Psi_{\mathbb{R}}(x)$; $\Psi_{\mathbb{R}} = \mathrm{re} \circ \Psi \circ (\mathbb{R} \hookrightarrow \mathbb{C})$ is real analytic by Lemma 3.23(ii). (iii) is `lp.inner_eq_tsum` and $y_j/\omega_{a,p}(j) = (\kappa_{a,p} y)_j$. □

Source: ours (P96-W Lemma 1, p. 278, uses complex neighbourhoods in x in the same way).

Corollary 3.21 (Analytic reconstruction)**.** *For $k \in \mathbb{N}$ let $\Psi_{\mathbb{R}}^{U,k}$, $\Psi_{\mathbb{R}}^{V,k} : \mathbb{R} \to \ell^2_{\mathbb{R}}$ be the real feature maps of Lemma 3.20 with $a = p = 1$ and $\beta_j = \lambda_j^{-1/2}$, respectively $\beta_j = \lambda_j^{1/2}$, and for $y \in \ell^2_{\mathbb{R}}$, $x \in \mathbb{R}$ define the* derived reconstructions

$$U_y^{(k)}(x) = \langle \Psi_{\mathbb{R}}^{U,k}(x), y\rangle = \sum_{j\ge 1} \frac{\texttt{coeff y j}}{\sqrt{\lambda_j}}\, \phi_j^{(k)}(x),$$

$$V_y^{(k)}(x) = \langle \Psi_{\mathbb{R}}^{V,k}(x), y\rangle = \sum_{j\ge 1} \sqrt{\lambda_j}\, \texttt{coeff y j}\, \phi_j^{(k)}(x),$$

the series converging absolutely.

(i) $U_y^{(0)}(x) =$ `reconU m hm y x` *and* $V_y^{(0)}(x) =$ `reconV m hm y x` *for all real x.*

(ii) For all k, y and real x: $x \mapsto U_y^{(k)}(x)$ has derivative $U_y^{(k+1)}(x)$ at x (Lean `HasDerivAt`*), and likewise for V. Hence $U_y^{(k)}$ is the k-th derivative of* `reconU m hm y` *(Lean* `iteratedDeriv k`*), and for $x \in [0,\pi]$,* `derivWithin (reconU m hm y) (Set.Icc 0 Real.pi) x` $= U_y^{(1)}(x)$.

(iii) Let S be a real normed space (in W2, $S = \mathbb{R}$; Lean `[NormedAddCommGroup S] [NormedSpace Real S]`*) and $Y : S \to \ell^2_{\mathbb{R}}$ real analytic at s_0 (Lean* `AnalyticAt Real Y s0`*). Then $(s,x) \mapsto U^{(k)}_{Y(s)}(x)$ and $(s,x) \mapsto V^{(k)}_{Y(s)}(x)$ are real analytic on $S \times \mathbb{R}$ at (s_0, x_0) for every $x_0 \in \mathbb{R}$ and every k.*

In particular (with $S = \mathbb{R}^n$, as in the first version of this corollary), for $Y : \mathbb{R}^n \to \ell^2_{\mathbb{R}}$ analytic at s_0, the maps $(s,x) \mapsto \partial_x^k$ `reconU m hm (Y s)`(x) *and $(s,x) \mapsto \partial_x^k$* `reconV m hm (Y s)`$(x)$*, defined as $U^{(k)}_{Y(s)}(x)$ and $V^{(k)}_{Y(s)}(x)$ and identified with the iterated x-derivatives by (ii), are real analytic at every (s_0, x_0).*

Proof. **Step S3.46: Bilinear form.** With $a = p = 1$, $|\beta_j| \le (1+m)^{1/4} j^{1/2}$, so Lemma 3.20 applies; its part (iii) gives the series forms and their absolute convergence (`coeff y j` $= (\kappa_{1,1} y)_j$). For $k = 0$ the series is the frozen `tsum` term by term, which is (i).

Step S3.47: Analyticity. The real inner product $\langle\cdot,\cdot\rangle : \ell^2_{\mathbb{R}} \times \ell^2_{\mathbb{R}} \to \mathbb{R}$ is a continuous bilinear map (`innerSL Real`), analytic at every pair (`ContinuousLinearMap.analyticAt_bilinear`). The maps $(s,x) \mapsto \Psi_{\mathbb{R}}^{U,k}(x)$ and $(s,x) \mapsto Y(s)$ are analytic at (s_0,x_0) (compositions of the analytic $\Psi_{\mathbb{R}}^{U,k}$, Y with the continuous linear projections of $S \times \mathbb{R}$), so their pair is analytic (`AnalyticAt.prod`) and the composite is analytic (`AnalyticAt.comp`); this is (iii).

Step S3.48: Derivatives. $\Psi_{\mathbb{R}}^{U,k}$ is analytic, hence differentiable, and $x \mapsto \langle \Psi_{\mathbb{R}}^{U,k}(x), y\rangle$ has derivative $\langle (\Psi_{\mathbb{R}}^{U,k})'(x), y\rangle$ (the pairing with fixed y is continuous linear). Coordinate evaluation is continuous linear, so the j-th coordinate of $(\Psi_{\mathbb{R}}^{U,k})'(x)$ is the derivative of $\beta_j \phi_j^{(k)}/\omega_{1,1}(j)$, which is $\beta_j \phi_j^{(k+1)}/\omega_{1,1}(j)$; thus $(\Psi_{\mathbb{R}}^{U,k})' = \Psi_{\mathbb{R}}^{U,k+1}$ and $\frac{d}{dx} U_y^{(k)} = U_y^{(k+1)}$. Iterating from (i) identifies $U_y^{(k)}$ with the k-th derivative of `reconU m hm y`. At $x \in [0,\pi]$ the derivative within $[0,\pi]$ equals the two-sided derivative, since the function is differentiable on $\mathbb{R}$ and $[0,\pi]$ has unique derivatives ($0 < \pi$, `uniqueDiffOn_Icc`). The same for V. This is (ii). □

Source: P96-W Lemma 1, p. 278 (real analyticity of u in t and x, argued via complex neighbourhoods); ours in this form. Consumed by W2 (frozen conclusion (5)).

3.7. **Analytic maps between weighted spaces.**

Definition 3.22 (Complexification)**.** $\iota : \ell^2_{\mathbb{R}} \to \ell^2_{\mathbb{C}}$, $(\iota y)_j = y_j$; $\mathrm{re} : \ell^2_{\mathbb{C}} \to \ell^2_{\mathbb{R}}$, $(\mathrm{re}\, z)_j = \mathrm{Re}\, z_j$; conjugation $(\bar z)_j = \overline{z_j}$; and componentwise on products, e.g. $E^{\mathbb{C}}_{a,p}$ for $E_{a,p}$, represented by $(\ell^2_{\mathbb{C}})^2$. A map f from an open subset of $\ell^2_{\mathbb{C}}$ (or a product) is *real on reals* if $f(\iota y) \in \iota(\ell^2_{\mathbb{R}})$ whenever defined.

Source: ours (P96-W p. 281: "real analytic means that H is a function of z and $\bar z$, real analytic in the real and imaginary part of z").

Lemma 3.23 (Complexification and real analyticity)**.** *(i) ι and* re *are continuous $\mathbb{R}$-linear, $\|\iota y\| = \|y\|$, $\|\mathrm{re}\, z\| \le \|z\|$, $\mathrm{re} \circ \iota = \mathrm{id}$, $\|\iota y + i\iota y'\|^2 = \|y\|^2 + \|y'\|^2$, and conjugation is a conjugate-linear isometry. All three commute with $\tau_{a,p}$, $\kappa_{a,p}$ and every real multiplier D_μ.*

(ii) If F_1, F_2 are complex Banach spaces, V a real normed space, $\ell : V \to F_1$ and $\pi : F_2 \to W$ continuous $\mathbb{R}$-linear, and f is $\mathbb{C}$-analytic at $\ell(x)$, then $\pi \circ f \circ \ell$ is $\mathbb{R}$-analytic at x.

(iii) In particular, if f is defined near ιy in $\ell^2_{\mathbb{C}}$ (or a product), $\mathbb{C}$-analytic at ιy, and real on reals near ιy, then $g = \mathrm{re} \circ f \circ \iota$ is $\mathbb{R}$-analytic at y and $\iota \circ g = f \circ \iota$ near y; its real derivative is $Dg(y)h = \mathrm{re}\, Df(\iota y)\iota h$.

Proof. (i) is coordinatewise. (ii): f is $\mathbb{R}$-analytic at $\ell(x)$ (`AnalyticAt.restrictScalars`), and composition with the $\mathbb{R}$-linear continuous maps ℓ, π preserves analyticity ((W1-E3)). (iii): (ii) with $\ell = \iota$, $\pi = \mathrm{re}$, and $\iota \circ \mathrm{re} = \mathrm{id}$ on $\iota(\ell^2_{\mathbb{R}})$; the derivative by the chain rule. □

Source: standard.

Definition 3.24 (Holomorphic maps on Banach domains)**.** Let F_1, F_2 be complex Banach spaces and $U \subset F_1$ open. A map $f : U \to F_2$ is *holomorphic* if it is complex Fréchet differentiable at every point of U (Lean `DifferentiableOn Complex f U`). Bounds on f are always hypotheses of the form $\|f(z)\| \le M$ for all $z \in U$, with $M \ge 0$ (Lean `forall z in U, norm (f z) <= M`); no unguarded supremum is used as a norm. Where the source writes $\|f\|_U$ we read such an M.

Source: P96-K p. 121 ("real analytic in $D(s,r)$") and App. A p. 146; P96-W p. 281.

Lemma 3.25 (Cauchy estimate)**.** *Let $r > 0$, $M \ge 0$, and let f be holomorphic on the open ball $B(v,r) \subset F_1$ with $\|f(z)\| \le M$ for $z \in B(v,r)$. Then $\|Df(v)\| \le M/r$ in the operator norm. Consequently, if $\delta > 0$, f is holomorphic on an open V with $\|f(z)\| \le M$ for $z \in V$, and $B(w,\delta) \subset V$, then $\|Df(w)\| \le M/\delta$; and f is (M/δ)-Lipschitz on every convex set C with $C + B(0,\delta) \subset V$.*

Proof. **Step S3.49: Complex lines.** Fix $u \in F_1$ with $\|u\| = 1$ and $0 < r' < r$. The map $\varphi(\zeta) = f(v + \zeta u)$ is holomorphic on $|\zeta| < r$ (chain rule), in particular continuous on $|\zeta| \le r'$, with $\varphi'(0) = Df(v)u$ and $\|\varphi\| \le M$.

Step S3.50: One-variable estimate. By (W1-E4), $\|Df(v)u\| \le M/r'$; let $r' \uparrow r$ and take the supremum over u.

Step S3.51: Margins. Apply this on $B(w,\delta)$; the Lipschitz bound is the mean value inequality ((W1-E5)) on C. □

Source: P96-K Lemma A.3, p. 147, and its proof.

Lemma 3.26 (Derivatives of holomorphic maps are continuous)**.** *Let $U \subset F_1$ be open and $f : U \to F_2$ holomorphic. Let $w_0 \in U$, $\rho > 0$ and $B \ge 0$ with $B(w_0, 2\rho) \subset U$ and $\|f(z)\| \le B$ on $B(w_0, 2\rho)$. Then $\|Df(w) - Df(w')\| \le 2B\rho^{-2}\|w - w'\|$ for $w, w' \in B(w_0, \rho/2)$. Such ρ, B exist for every w_0, so $Df : U \to \mathcal{L}(F_1, F_2)$ is continuous (Lean `ContinuousOn (fderiv Complex f) U`).*

Proof. **Step S3.52: Existence of ρ, B.** f is continuous at w_0, so $\|f(z)\| \le \|f(w_0)\| + 1 =: B$ on some ball $B(w_0, 2\rho) \subset U$.

Step S3.53: Cauchy integral along lines. Let $w \in B(w_0, \rho/2)$ and $\|u\| = 1$. For $|\zeta| < 3\rho/2$, $w + \zeta u \in B(w_0, 2\rho)$, so $\varphi(\zeta) = f(w + \zeta u)$ is holomorphic there and (W1-E4) gives $Df(w)u = \varphi'(0) = \frac{1}{2\pi i} \oint_{|\zeta|=\rho/2} f(w + \zeta u)\zeta^{-2} d\zeta$.

Step S3.54: Lipschitz difference. For $|\zeta| = \rho/2$ the points $w + \zeta u$, $w' + \zeta u$ lie in the convex ball $B(w_0, \rho)$, on which $\|Df\| \le B/\rho$ by Lemma 3.25 (margin ρ inside $B(w_0, 2\rho)$), so $\|f(w + \zeta u) - f(w' + \zeta u)\| \le (B/\rho)\|w - w'\|$. Subtracting the two Cauchy integrals, $\|(Df(w) - Df(w'))u\| \le \frac{1}{2\pi} \cdot \pi\rho \cdot (\rho/2)^{-2}(B/\rho)\|w - w'\| = 2B\rho^{-2}\|w - w'\|$; take the supremum over u (`ContinuousLinearMap.opNorm_le_bound`). □

Source: standard (P96-K App. A uses the Cauchy integral the same way).

Lemma 3.27 (Uniform limits)**.** *Let $U \subset F_1$ be open, $f_k : U \to F_2$ holomorphic, and $f_k \to f$ uniformly on U.*

(i) (Qualitative.) f is holomorphic on U, and $Df_k \to Df$ uniformly on every ball $B(w, \delta)$ with $\delta > 0$ and $B(w, 2\delta) \subset U$. If every f_k is real on reals, so is f. No boundedness of f is assumed.

(ii) (Quantitative.) If moreover $M \ge 0$ and $\|f(z)\| \le M$ for all $z \in U$, then $\|Df(w)\| \le M/\delta$ whenever $\delta > 0$ and $B(w, \delta) \subset U$.

Proof. **Step S3.55: Cauchy in derivatives.** Given $\varepsilon > 0$, uniform convergence gives K with $\|f_k(z) - f_l(z)\| \le \varepsilon$ for $k, l \ge K$ and all $z \in U$. For $w' \in B(w, \delta)$, $B(w', \delta) \subset U$, so Lemma 3.25 applied to $f_k - f_l$ (bound ε) gives $\|Df_k(w') - Df_l(w')\| \le \varepsilon/\delta$. Thus (Df_k) is uniformly Cauchy on $B(w, \delta)$ in the complete space of bounded $\mathbb{C}$-linear maps, with a uniform limit g.

Step S3.56: Limit. By (W1-E5) on the open ball, f is $\mathbb{C}$-differentiable there with $Df = g$. Since w is arbitrary, f is holomorphic on U. Real on reals: $\iota(\ell^2_{\mathbb{R}})$ is closed. (ii) is Lemma 3.25 applied to f itself. □

Source: standard (used implicitly in P96-K §4, convergence of the iteration).

Lemma 3.28 (Holomorphic maps of finitely many variables are analytic)**.** *Let $U \subset \mathbb{C}^n$ be open (max norm), F a complex Banach space and $f : U \to F$ holomorphic. Then f is $\mathbb{C}$-analytic on U (Lean `AnalyticOnNhd Complex f U`). (For $n = 0$, $\mathbb{C}^0$ is a singleton, U is empty or that point, and f is constant, hence analytic.) Quantitatively, for $n \ge 1$: let $r > 0$ with $\bar{P}(z_0, r) \subset U$; then there are $r_1 > r$ and $M_1 \ge 0$ with $\bar{P}(z_0, r_1) \subset U$ and $\|f\| \le M_1$ on $\bar{P}(z_0, r_1)$, and f is the sum on $P(z_0, r_1) \supset \bar{P}(z_0, r)$ of an absolutely convergent power series $\sum_\alpha c_\alpha (z - z_0)^\alpha$ with $\|c_\alpha\| \le M_1 r_1^{-|\alpha|}$; grouped by degree it is a `FormalMultilinearSeries` (p_m) with $\|p_m\| \le \binom{m+n-1}{n-1} M_1 r_1^{-m}$ and `HasFPowerSeriesOnBall f p z0 r1`. If $F = E^{\mathbb{C}}_{a,p}$ and f is real on reals at the real points of U, then $\mathrm{re} \circ f|_{U \cap \mathbb{R}^n}$ is real analytic into $E_{a,p}$.*

Proof. **Step S3.57: Margin.** $\bar{P}(z_0, r)$ is compact and U open, so $\bar{P}(z_0, r_1) \subset U$ for some $r_1 > r$; f is continuous on the compact $\bar{P}(z_0, r_1)$, hence bounded there by some M_1 (true in any normed F).

Step S3.58: Coefficients. Track K's Lemma `bp:p1:holomorphic-analytic` (its steps `han-cauchy` through `han-bounds`, label prefix `step:p1:`), with radius r_1 and W replaced by the complete space F, gives $c_\alpha \in F$ with $\|c_\alpha\| \le M_1 r_1^{-|\alpha|}$ and $f(z) = \sum_\alpha c_\alpha (z - z_0)^\alpha$ absolutely on $P(z_0, r_1)$. Those steps use only the one-variable Cauchy formula (valid for complete F, (W1-E4)) and Fubini and term-by-term integration of uniformly convergent series of continuous F-valued maps on a compact torus (Bochner integral, (W1-E6)); compactness is that of the domain torus, not of the target.

Step S3.59: Degree grouping. For $|\alpha| = m$ let $\iota_\alpha : \{1,\dots,m\} \to \{1,\dots,n\}$ be the nondecreasing map taking the value i exactly α_i times, and $p_m(h_1,\dots,h_m) = \sum_{|\alpha|=m} \big(\prod_{l=1}^m (h_l)_{\iota_\alpha(l)}\big) c_\alpha$, a continuous m-linear map $(\mathbb{C}^n)^m \to F$ (each product of coordinate functionals has norm ≤ 1 for the max norm; Lean: `ContinuousMultilinearMap.mkPiAlgebraFin` composed with coordinate projections, then `smulRight`). Then $p_m(h,\dots,h) = \sum_{|\alpha|=m} c_\alpha h^\alpha$ and $\|p_m\| \le \#\{\alpha : |\alpha| = m\} M_1 r_1^{-m} = \binom{m+n-1}{n-1} M_1 r_1^{-m}$ ($n \ge 1$, stars and bars). For $0 \le \rho < r_1$, $\sum_m \|p_m\| \rho^m < \infty$ (polynomial times geometric), so the radius of (p_m) is $\ge r_1$, and regrouping the absolutely convergent multi-index series by degree gives $f(z_0 + h) = \sum_m p_m(h,\dots,h)$ for $|h| < r_1$: this is `HasFPowerSeriesOnBall f p z0 r1`, so f is analytic at z_0, and $z_0 \in U$ is arbitrary (every point of U is the centre of a closed polydisc of positive radius in U).

Step S3.60: Real points. The real statement is Lemma 3.23(iii). □

Source: standard; track K blueprint `bp:p1:holomorphic-analytic`.

Corollary 3.29 (Real-analytic tori from complex strips)**.** *Let $s > 0$, $T_s = \{\theta \in \mathbb{C}^2 : \max_i |\operatorname{Im}\theta_i| < s\}$, and let $\tilde K : T_s \to E_{1,1}^{\mathbb{C}}$ be holomorphic, real on reals, and 1-periodic. Then $K = \operatorname{re} \circ \tilde K|_{\mathbb{R}^2} : \mathbb{R}^2 \to$ `WaveSpace` satisfies `AnalyticOnNhd Real K Set.univ`, is $\mathbb{Z}^2$-periodic, and `fderiv Real K θ h` $=$ $\operatorname{re}\, D\tilde K(\theta)h$ for $h \in \mathbb{R}^2$.*

Proof. Lemma 3.28 with $U = T_s$ and Lemma 3.23(iii); periodicity restricts. □

Source: ours; the analyticity clause of the frozen `IsAnalyticTorusEmbedding` (consumed by W6, W8).

Lemma 3.30 (Holomorphic flows)**.** *Let F be a complex Banach space, $V \subset F$ open, $X : V \to F$ holomorphic with $\|X(z)\| \le M$ for all $z \in V$, and $U \subset V$ open with $B(w, 2\delta) \subset V$ for all $w \in U$, where $0 < M < \delta$ (so $\delta > 0$). Put $L = M/\delta < 1$.*

(i) For every $w \in U$ there is a unique C^1 curve $t \mapsto \phi^t(w)$, $t \in [-1,1]$, with $\phi^0(w) = w$, $\frac{d}{dt}\phi^t(w) = X(\phi^t(w))$ and values in V; it satisfies $\|\phi^t(w) - w\| \le |t|M \le M$ for all $w \in U$.

(ii) For each $t \in [-1,1]$, $w \mapsto \phi^t(w)$ is holomorphic on U and $\|D\phi^t(w)\| \le e^{L|t|}$.

(iii) If $F = E^{\mathbb{C}}$ for a real space E and X is real on reals, then ϕ^t is real on reals.

(iv) (Parameters.) Let $X(\cdot,\xi)$, $\xi \in \Pi$, each satisfy the hypotheses with the same V, U, M, δ, let $\xi, \zeta \in \Pi$, and let $N \ge 0$ with $\|X(z,\xi) - X(z,\zeta)\| \le N$ for all $z \in V$. Then for $w \in U$ and $|t| \le 1$, $\|\phi^t(w,\xi) - \phi^t(w,\zeta)\| \le |t|\, e^{L|t|} N$.

Proof. **Step S3.61: Lipschitz region.** For $w \in U$ and $w' \in B(w,\delta)$, $B(w',\delta) \subset B(w,2\delta) \subset V$, so $\|DX(w')\| \le L$ by Lemma 3.25, and X is L-Lipschitz on the convex ball $B(w,\delta)$.

Step S3.62: Picard iterates. Let $\phi_0(w,t) = w$ and $\phi_{k+1}(w,t) = w + \int_0^t X(\phi_k(w,\sigma))\, d\sigma$. By induction, $\|\phi_k(w,t) - w\| \le |t|M < \delta$, so the iterates stay in $B(w,\delta)$ and are continuous in (w,t), and $\|\phi_{k+1}(w,t) - \phi_k(w,t)\| \le ML^k|t|^{k+1}/(k+1)!$ uniformly in $w \in U$.

Step S3.63: Holomorphy of the iterates. By induction on k: $w \mapsto \phi_k(w,t)$ is holomorphic on U, $\|D_w\phi_k(w,t)\| \le \sum_{i \le k} (L|t|)^i/i! \le e^{L|t|}$, and $t \mapsto D_w\phi_k(w,t)$ is continuous. For the step, fix $w_0 \in U$ and a ball $B(w_0,\varepsilon) \subset U$. For w in it and $\sigma \in [-1,1]$, the integrand $X(\phi_k(w,\sigma))$ has the w-derivative $DX(\phi_k(w,\sigma)) \circ D_w\phi_k(w,\sigma)$ (chain rule), which is continuous in σ because DX is continuous on V (Lemma 3.26), $\sigma \mapsto \phi_k(w,\sigma)$ is continuous, and $\sigma \mapsto D_w\phi_k(w,\sigma)$ is continuous by induction; continuity gives the measurability required by (W1-E6). It is dominated by Le^L uniformly for $w \in B(w_0,\varepsilon)$, because $\phi_k(w,\sigma) \in B(w,\delta)$ where $\|DX\| \le L$ (Step S3.61, applied at $w \in U$). Hence `hasFDerivAt_integral_of_dominated_of_fderiv_le` gives $D_w\phi_{k+1}(w,t) = I + \int_0^t DX(\phi_k) D_w\phi_k\, d\sigma$, which is continuous in t; by the induction bound inside the integral its norm is at most $1 + \int_0^{|t|} L \sum_{i=0}^k (L\sigma)^i/i!\, d\sigma = \sum_{i=0}^{k+1} (L|t|)^i/i!$. (The constant Le^L serves only as the uniform domination bound for differentiating under the integral.)

Step S3.64: Limit. The iterates converge uniformly on $U\times[-1,1]$ to ϕ, which satisfies the integral equation (pass to the limit under the integral), hence is a C^1 solution; uniqueness among solutions in V is (W1-E7) on $B(w,\delta)$ (a solution with values in V stays in $B(w,\delta)$ for $|t|\le 1$, since $\|\dot\phi\|\le M$). By Lemma 3.27, ϕ^t is holomorphic; the derivative bound passes to the limit. This proves (i), (ii).

Step S3.65: Reality. If X is real on reals, every Picard iterate of a real initial point is real, and the real subspace is closed.

Step S3.66: Parameters. Let $\psi^t=\phi^t(w,\xi)-\phi^t(w,\zeta)$; both curves lie in the convex ball $B(w,\delta)$, on which $X(\cdot,\zeta)$ is L-Lipschitz. Then $\dot\psi^t=[X(\phi^t_\xi,\xi)-X(\phi^t_\xi,\zeta)]+[X(\phi^t_\xi,\zeta)-X(\phi^t_\zeta,\zeta)]$, so $\|\dot\psi^t\|\le N+L\|\psi^t\|$, $\psi^0=0$, and Gronwall ((W1-E7)) gives $\|\psi^t\|\le N(e^{L|t|}-1)/L\le |t|e^{L|t|}N$. □

Source: P96-K Lemma A.4, p. 147 (its two displayed estimates: $\|\phi^t-\mathrm{id}\|_U\le\|X\|_V$, which is (i), and the Lipschitz estimate in the parameter, which is (iv)), and P96-K §3, (10), p. 130 (existence of the flow on a smaller domain).
SOURCE_GAP: In the proof of the second estimate of P96-K Lemma A.4, the bound $\|X(\phi^s(\cdot,\xi),\zeta)-X(\phi^s(\cdot,\zeta),\zeta)\|\le\|DX\|_V\|\Delta\phi^s\|_U$ is a mean value estimate, which needs the segment between the two trajectory points to lie in V; V is only an open domain, not convex, so this is unjustified as stated. (The lemma also does not assume $\|DX\|_V<\infty$; that only makes the estimate possibly vacuous and is not by itself a defect.) Here both trajectories lie in the convex ball $B(w,\delta)$ (ball margin 2δ, $M<\delta$), which repairs the step; in the KAM step W5 must instantiate these margins from (9)–(10) of P96-K pp. 129–130 in its chosen norm.

3.8. **Support for the KAM step.**

Definition 3.31 (KAM phase space, domains, norms)**.** Let $a\ge 0$, $p\in\mathbb{R}$. The complex phase space is $\mathcal{P}^{a,p}_{\mathbb{C}}=\mathbb{C}^2\times\mathbb{C}^2\times\ell^{a,p}_{\mathbb{C}}\times\ell^{a,p}_{\mathbb{C}}$ with points (θ,y,u,v) (Lean: `(Fin 2 -> Complex) × (Fin 2 -> Complex) × lp ... 2 × lp ... 2` in weighted coordinates); $|\cdot|$ is the max norm on $\mathbb{C}^2$. For $s,r>0$,

$$D(s,r)=\{(\theta,y,u,v):\max_i|\mathrm{Im}\,\theta_i|<s,\ |y|<r^2,\ \|u\|_{a,p}+\|v\|_{a,p}<r\},$$

and for $W=(\Theta,Y,U,V)\in\mathcal{P}^{a,\bar p}_{\mathbb{C}}$,

$$|W|_{\bar p,r}=|\Theta|+\frac{|Y|}{r^2}+\frac{\|U\|_{a,\bar p}}{r}+\frac{\|V\|_{a,\bar p}}{r}.$$

For a linear map $L:\mathcal{P}^{a,p}_{\mathbb{C}}\to\mathcal{P}^{a,\bar p}_{\mathbb{C}}$, $|L|_{r,s}$ is the least $C\ge 0$ with $|LW|_{\bar p,r}\le C|W|_{p,s}$ for all W (the operator norm; used only for bounded L, Lean `ContinuousLinearMap.opNorm`). The angle θ has period one: $x=2\pi\theta$ is P96-K's angle, a P96-K strip $|\mathrm{Im}\,x|<s_K$ is $|\mathrm{Im}\,\theta|<s_K/(2\pi)$, and the θ-component of a vector field is $(2\pi)^{-1}$ times its x-component.

Source: P96-K p. 120 ($\mathcal{P}^{a,p}$), p. 121 ((1) and $D(s,r)$), p. 129 (operator norm); period conversion ours.

Remark 3.32 (Conventions to carry into W4–W6). P96-K writes $z=(u-iv)/\sqrt{2}$, $\bar z=(u+iv)/\sqrt{2}$ (p. 124), the complex conjugate of the W0 convention $z=(q+ip)/\sqrt{2}$ (Definition 1.11). Since the normal form $\langle\omega,y\rangle+\langle\Omega,z\bar z\rangle$ and the smallness norms are invariant under $z\leftrightarrow\bar z$, this is a relabelling, not a defect; W4–W6 should keep the W0 convention and state the swap once. The domain $D(s,r)$ uses the sum $\|u\|+\|v\|$, not the max norm of `WaveSpace`; the two differ by a factor at most 2.

Lemma 3.33 (Domains)**.** *Let $0<\sigma\le 1$, $0<2\sigma<s$, $r>0$. Then $D(s,r)$ is open and convex, $D(s',r')\subset D(s,r)$ for $s'\le s$, $r'\le r$, and every $W_0\in D(s-2\sigma,r/2)$ satisfies $W_0+W\in D(s-\sigma,r)$ whenever $|W|_{p,r}<\sigma/2$. Also $|W|_{\bar p,r}\le|W|_{p,r}$ for $\bar p\le p$ and $|W|_{p,r}\le|W|_{p,r'}$ for $r'\le r$.*

Proof. Openness and convexity: each defining condition is a strict sublevel set of a continuous convex function ($\theta\mapsto|\mathrm{Im}\,\theta_i|$ is a seminorm). For the margin, $|\Theta|<\sigma/2$ gives $|\mathrm{Im}(\theta+\Theta)|<s-2\sigma+\sigma/2<s-\sigma$; $|Y|<r^2\sigma/2\le r^2/2$ gives $|y+Y|<r^2/4+r^2/2<r^2$; $\|U\|+\|V\|<r\sigma/2\le r/2$ gives

$\|u+U\|+\|v+V\| < r/2+r/2 = r$. The norm comparisons are Lemma 3.5(i) and $1/r^2 \le 1/r'^2$, $1/r \le 1/r'$. □

Source: P96-K p. 129 ("every point in $D(s-2\sigma, r/2)$ has at least $|\cdot|_{p,r}$-distance $\sigma/2$ to the boundary of $D(s-\sigma, r)$", with $\sigma \le 1$ from (6)); checked here.

Lemma 3.34 (Fourier coefficients under an imaginary shift)**.** *Let F be a complex Banach space, $n \ge 1$, $s > 0$, $T_s = \{\theta \in \mathbb{C}^n : \max_i |\mathrm{Im}\,\theta_i| < s\}$, and $u : T_s \to F$ holomorphic and 1-periodic ($u(\theta + e_r) = u(\theta)$ for $1 \le r \le n$). Let $\hat u_k = \int_{[0,1]^n} u(x) e^{-2\pi i k\cdot x} dx$. Then for every $\eta \in \mathbb{R}^n$ with $\max_i|\eta_i| < s$ and every $k \in \mathbb{Z}^n$,*

$$\int_{[0,1]^n} u(x+i\eta)\, e^{-2\pi i k\cdot x}\, dx = e^{-2\pi k\cdot\eta}\, \hat u_k.$$

Proof. **Step S3.67: One periodic variable.** Let $f : \mathbb{C} \to F$ be complex differentiable on the open strip $S = \{|\mathrm{Im}\, z| < s\}$ (values outside S are irrelevant) with $f(z+1) = f(z)$ for $z \in S$, and let $|\tau| < s$. The closed rectangle with corners 0 and $1+i\tau$ lies in S, so (W1-E9) gives $\int_0^1 f(x)\,dx - \int_0^1 f(x+i\tau)\,dx + i\int_0^\tau f(1+iy)\,dy - i\int_0^\tau f(iy)\,dy = 0$. The two vertical integrals are equal by periodicity, hence $\int_0^1 f(x+i\tau)\,dx = \int_0^1 f(x)\,dx$.

Step S3.68: Shift one coordinate at a time. For $0 \le r \le n$ let $\eta^{(r)} = (\eta_1, \dots, \eta_r, 0, \dots, 0)$ and $I_r = \int_{[0,1]^n} u(x+i\eta^{(r)}) e^{-2\pi i k\cdot(x+i\eta^{(r)})} dx$; the integrands are continuous since $x + i\eta^{(r)} \in T_s$. Let $1 \le r \le n$ and fix $x_t \in [0,1]$, $t \ne r$. Let $y(z) \in \mathbb{C}^n$ have coordinates $x_t + i\eta_t^{(r)}$ for $t \ne r$ and z for $t = r$ (note $\eta_t^{(r)} = \eta_t^{(r-1)}$ for $t \ne r$), and $f(z) = u(y(z)) e^{-2\pi i k\cdot y(z)}$. For $z \in S$, $y(z) \in T_s$, so f is complex differentiable on S (chain rule with the affine map y, Definition 3.24), and $f(z+1) = f(z)$ because u is 1-periodic in θ_r and $k_r \in \mathbb{Z}$. Step S3.67 with $\tau = \eta_r$ shows that the integrals over $x_r \in [0,1]$ of the integrands of I_r and I_{r-1} agree; integrating over the remaining x_t (Fubini, (W1-E9)) gives $I_r = I_{r-1}$.

Step S3.69: Conclusion. $I_0 = \hat u_k$, and $I_n = e^{2\pi k\cdot\eta} \int_{[0,1]^n} u(x+i\eta) e^{-2\pi i k\cdot x} dx$ since $e^{-2\pi i k\cdot(x+i\eta)} = e^{2\pi k\cdot\eta} e^{-2\pi i k\cdot x}$. Check: $n=1$, $u(z) = e^{2\pi i z}$, $k=1$: $u(x+i\eta) = e^{-2\pi\eta} e^{2\pi i x}$ has first coefficient $e^{-2\pi\eta} = e^{-2\pi k\eta}\hat u_1$. □

Source: standard (Cauchy's theorem); the argument is that of track K's Step `step:p1:dec-shift`, there for the particular shift $t_j = -\rho'\,\mathrm{sgn}(k_j)$ and here for an arbitrary η and Banach values.

Lemma 3.35 (Hilbert-valued Fourier series on strips)**.** *Let $\mathcal{H}$ be a complex Hilbert space (in the applications $\ell^2_{\mathbb{C}}$, representing $\ell^{a,p}_{\mathbb{C}}$), $n \ge 1$, $s > 0$, $T_s = \{\theta \in \mathbb{C}^n : \max_i|\mathrm{Im}\,\theta_i| < s\}$, and $u : T_s \to \mathcal{H}$ holomorphic, 1-periodic, with $\|u(\theta)\| \le M$ for all $\theta \in T_s$ ($M \ge 0$). Let $\hat u_k = \int_{[0,1]^n} u(x) e^{-2\pi i k\cdot x} dx$ ($k \in \mathbb{Z}^n$).*

- *(i)* $\sum_k \|\hat u_k\|^2 e^{4\pi|k|_1 s} \le 2^n M^2$.
- *(ii) Conversely, if $c_k \in \mathcal{H}$ satisfy $\sum_k \|c_k\| e^{2\pi|k|_1 s} < \infty$, then $v(\theta) = \sum_k c_k e^{2\pi i k\cdot\theta}$ converges absolutely and uniformly on T_s, v is holomorphic and 1-periodic on T_s, $\hat v_k = c_k$, and $\|v(\theta)\| \le \sum_k \|c_k\| e^{2\pi|k|_1 s}$ on T_s.*
- *(iii) Every u as in (i) is the sum of its Fourier series on $T_{s'}$ for $s' < s$.*

Proof. **Step S3.70: Shifted coefficients.** For $\eta \in \mathbb{R}^n$ with $\max|\eta_i| < s$, $\hat u_k\, e^{-2\pi k\cdot\eta}$ is the k-th Fourier coefficient of the continuous periodic map $x \mapsto u(x+i\eta)$: Lemma 3.34 with $F = \mathcal{H}$.

Step S3.71: Parseval. For a continuous periodic $\mathcal{H}$-valued g, $\sum_k \|\hat g_k\|^2 = \int_{[0,1]^n} \|g\|^2$, which is $\le M^2$ if $\|g(x)\| \le M$ for all x: expand in an orthonormal basis (e_α) of $\mathcal{H}$ (for $\ell^2_{\mathbb{C}}$, the

coordinate vectors), apply scalar Parseval ((W1-E8)) to each $\langle e_\alpha, g\rangle$ and sum (Tonelli). Hence $\sum_k \|\hat{u}_k\|^2 e^{-4\pi k\cdot\eta} \le M^2$.

Step S3.72: Orthants. For $s' < s$ and $\varepsilon \in \{\pm 1\}^n$ take $\eta = -s'\varepsilon$, so $e^{-4\pi k\cdot\eta} = e^{4\pi s'\varepsilon\cdot k}$. Since $e^{4\pi s'|k|_1} = e^{4\pi s'\varepsilon\cdot k}$ for $\varepsilon_i = \operatorname{sgn} k_i$ (any sign when $k_i = 0$), $e^{4\pi s'|k|_1} \le \sum_\varepsilon e^{4\pi s'\varepsilon\cdot k}$; summing over the 2^n choices gives (i) with s', and monotone convergence gives s.

Step S3.73: Summation. (ii): $|e^{2\pi ik\cdot\theta}| \le e^{2\pi|k|_1 s}$ on T_s; uniform convergence of holomorphic partial sums and Lemma 3.27; coefficients by term-by-term integration. (iii): by (i) and Cauchy–Schwarz, $\sum_k \|\hat{u}_k\| e^{2\pi|k|_1 s'} \le 2^{n/2} M (\sum_k e^{-4\pi|k|_1(s-s')})^{1/2} < \infty$; the sum v of (ii) has the same coefficients as u on $\mathbb{R}^n$, so $\langle e, u - v\rangle$ vanishes on $\mathbb{R}^n$ by scalar uniqueness (`bp:p1:fourier-basic`) and on $T_{s'}$ by the identity theorem (`bp:p1:strip-identity`). □

Source: P96-K p. 126 ("the usual L^2-bound $\sum_k \|\hat{R}_k\|^2 e^{2|k||s|} \le 2^n \|\hat{R}\|^2_{D(s)}$", period 2π; with $\theta = x/2\pi$, $e^{2|k|s_K} = e^{4\pi|k|_1 s}$ for $s = s_K/2\pi$). Valid for *Hilbert*-valued maps only (the term F^{10} of P96-K Lemma 1); for operator-valued maps see Lemma 3.36 and Remark 3.38.

Lemma 3.36 (Banach-valued Fourier coefficients on strips)**.** *Let F be a complex Banach space (in the applications the operator space $\mathcal{L}(\ell^2_{\mathbb{C}}, \ell^2_{\mathbb{C}})$ with the operator norm, representing operators $\ell^{a,p}_{\mathbb{C}} \to \ell^{a,\bar{p}}_{\mathbb{C}}$), $n \ge 1$, $s > 0$, $M \ge 0$, and $u : T_s \to F$ holomorphic and 1-periodic with $\|u(\theta)\| \le M$ for all $\theta \in T_s$. Let $\hat{u}_k = \int_{[0,1]^n} u(x) e^{-2\pi ik\cdot x} dx \in F$ (Bochner integral of a continuous map).*

- *(i) $\|\hat{u}_k\| \le M e^{-2\pi|k|_1 s}$ for every $k \in \mathbb{Z}^n$.*
- *(ii) For $0 < \sigma < s$: $\sum_k \|\hat{u}_k\| e^{2\pi|k|_1(s-\sigma)} \le M \coth(\pi\sigma)^n$; more generally, for $c \ge 0$, $\sum_k (1 + |k|_1)^c \|\hat{u}_k\| e^{2\pi|k|_1(s-\sigma)} \le M\, S_{n,c}(\sigma)$ with $S_{n,c}$ of Lemma 3.37.*
- *(iii) (Summation.) If $c_k \in F$ and $\sum_k \|c_k\| e^{2\pi|k|_1\rho} < \infty$ for some $\rho > 0$, then $v(\theta) = \sum_k c_k e^{2\pi ik\cdot\theta}$ converges absolutely and uniformly on T_ρ, v is holomorphic and 1-periodic there, $\hat{v}_k = c_k$, and $\|v(\theta)\| \le \sum_k \|c_k\| e^{2\pi|k|_1\rho}$ on T_ρ.*
- *(iv) Everything is linear in u; applied to $u(\cdot,\xi) - u(\cdot,\zeta)$ it gives the parameter-difference (Lipschitz) versions with M replaced by a bound for the difference.*

Proof. **Step S3.74: Scalar reduction.** Let $\varphi \in F^*$ with $\|\varphi\| \le 1$. Then $\varphi \circ u$ is holomorphic, 1-periodic and bounded by M on T_s, and $\varphi(\hat{u}_k) = \widehat{(\varphi \circ u)}_k$ (`ContinuousLinearMap.integral_comp_comm`, (W1-E4)).

Step S3.75: Scalar decay. Track K's `bp:p1:decay` gives $|\widehat{(\varphi \circ u)}_k| \le e^{-2\pi|k|_1 s'} M$ for $0 \le s' < s$; let $s' \uparrow s$. Since this holds for all norm-one φ, $\|\hat{u}_k\| \le M e^{-2\pi|k|_1 s}$ (`NormedSpace.norm_le_dual_bound`). This is (i).

Step S3.76: Lattice sum. By (i), $\|\hat{u}_k\| e^{2\pi|k|_1(s-\sigma)} \le M e^{-2\pi|k|_1\sigma}$; sum with Lemma 3.37.

Step S3.77: Summation. (iii): $|e^{2\pi ik\cdot\theta}| \le e^{2\pi|k|_1\rho}$ on T_ρ, so the partial sums (holomorphic trigonometric polynomials) converge uniformly; Lemma 3.27(i) gives holomorphy, and term-by-term integration ((W1-E6)) the coefficients. (iv) is linearity of $u \mapsto \hat{u}_k$. □

Source: ours; replaces the operator-valued step of P96-K p. 127 (see Remark 3.38); the scalar ingredient is track K's `bp:p1:decay` and `bp:p1:summation`.

Lemma 3.37 (Weighted lattice sums)**.** *Let $n \ge 1$, $\sigma > 0$, $\varepsilon = 2\pi\sigma$, and $|k|_1 = \sum_i |k_i|$ for $k \in \mathbb{Z}^n$.*

- *(i) $\sum_{k\in\mathbb{Z}^n} e^{-\varepsilon|k|_1} = \coth(\varepsilon/2)^n = \coth(\pi\sigma)^n$.*
- *(ii) $\#\{k \in \mathbb{Z}^n : |k|_1 = m\} \le 2^n \binom{m+n-1}{n-1} \le 2^n (m+1)^{n-1}$ for $m \ge 0$.*

(iii) For $c \ge 0$ and $b = c + n - 1$,

$$\sum_{k\in\mathbb{Z}^n} (1+|k|_1)^c e^{-\varepsilon|k|_1} \le S_{n,c}(\sigma) := 2^n \Big(\frac{2b}{e\varepsilon}\Big)^b e^{\varepsilon/2}\Big(1+\frac{2}{\varepsilon}\Big),$$

with $(2b/(e\varepsilon))^b := 1$ when $b = 0$.

(iv) For $\tau \ge 1$ the small-divisor weight $A_k = 1 + |k|_1^\tau$ of P96-K p. 121 satisfies $A_k \le (1+|k|_1)^\tau$, so $\sum_k A_k^\gamma (1+|k|_1)^\beta e^{-\varepsilon|k|_1} \le S_{n,\gamma\tau+\beta}(\sigma)$ for $\gamma, \beta \ge 0$. In particular P96-K's $B_\sigma^2 = 2^n \sum_k (1+|k|)^2 A_k^4 e^{-2|k|\sigma}$ (period 2π) is at most $2^n S_{n,4\tau+2}(\sigma/\pi)$ in the period-one units used here.

Proof. **Step S3.78: Product.** $e^{-\varepsilon|k|_1} = \prod_i e^{-\varepsilon|k_i|}$ and the sum factorizes (Tonelli, (W1-E2)): $\sum_{m\in\mathbb{Z}} e^{-\varepsilon|m|} = 1 + 2e^{-\varepsilon}/(1-e^{-\varepsilon}) = (1+e^{-\varepsilon})/(1-e^{-\varepsilon}) = \coth(\varepsilon/2)$.

Step S3.79: Counting. A k with $|k|_1 = m$ is determined by $(|k_1|, \dots, |k_n|)$, a composition of m into n nonnegative parts ($\binom{m+n-1}{n-1}$ choices), and at most 2^n signs; $\binom{m+n-1}{n-1} = \prod_{i=1}^{n-1} \frac{m+i}{i} \le (m+1)^{n-1}$.

Step S3.80: Weighted sum. By (ii), the sum in (iii) is at most $2^n \sum_{m\ge0} (m+1)^b e^{-\varepsilon m}$. Since $t^b e^{-\varepsilon t/2} \le (2b/(e\varepsilon))^b$ for $t > 0$ (maximum at $t = 2b/\varepsilon$), $(m+1)^b \le (2b/(e\varepsilon))^b e^{\varepsilon(m+1)/2}$, and $\sum_{m\ge0} e^{\varepsilon(m+1)/2 - \varepsilon m} = e^{\varepsilon/2}/(1-e^{-\varepsilon/2}) \le e^{\varepsilon/2}(1+2/\varepsilon)$, using $1 - e^{-x} \ge x/(1+x)$ for $x > 0$.

Step S3.81: Divisor weights. $1 + x^\tau \le (1+x)^\tau$ for $x \ge 0$, $\tau \ge 1$ (convexity: $(1+x)^\tau \ge 1 + x^\tau$). The B_σ statement is the substitution $|k| = |k|_1$, $e^{-2|k|\sigma_K} = e^{-2\pi|k|_1(\sigma_K/\pi)}$. □

Source: ours (P96-K (4) p. 128 asserts $B_\sigma \le b\,\sigma^{-(2\tau+n+1)}$ without proof).

Remark 3.38 (The operator-valued Fourier step of P96-K Lemma 1). For the term F^{11} P96-K p. 127 writes: "Expanding $\tilde R$ into its Fourier series with operator valued coefficient $\tilde R_k$ we have, as before, $\sum_k |||\tilde R_k|||^2 e^{2|k|s} \le 2^n |||\tilde R|||_{D(s)}$." (page image; period 2π, operator norms $|||\cdot|||$ of the weight-conjugated matrix $\tilde R = (v_i \ddot R_{ij} w_j)$ on ℓ^2; the right side is printed unsquared, evidently for $|||\tilde R|||_{D(s)}^2$ as on p. 126). This is false in either reading: the operator space is not a Hilbert space, and Lemma 3.35 does not apply. *Counterexample* (verified): fix $s_K > 0$, $N \in \mathbb{N}$, and on $|\mathrm{Im}\, x| < s_K$ (any $n \ge 1$, depending on x_1 only) let $\tilde R(x) e_j = e^{-js_K} e^{ijx_1} e_j$ for $j \le N$ and $\tilde R(x) e_j = 0$ for $j > N$. This is an entire, 2π-periodic, finite-rank diagonal family with $|||\tilde R(x)||| = \max_{j\le N} e^{-j(s_K + \mathrm{Im}\, x_1)} \le 1$ on the strip (supremum 1). Its nonzero coefficients are $\tilde R_{(j,0,\dots,0)} = e^{-js_K} P_j$ (P_j the coordinate projection), so $\sum_k |||\tilde R_k|||^2 e^{2|k|s_K} = \sum_{j\le N} 1 = N$, which exceeds $2^n \cdot 1$ (the right side in either reading, the supremum being 1) once $N > 2^n$ (for $n = 2$, $N = 8$). The family is the weight-conjugated mixed normal Hessian of the quadratic Hamiltonian $R = \sum_{j\le N} j^{p-\bar p} e^{-js_K} e^{ijx_1} z_j \bar z_j$, so it has the form of the term R^{11} of P96-K Lemma 1: the raw Hessian $\ddot R = \partial_z \partial_{\bar z} R$, as an operator $\ell^{a,p}_{\mathbb{C}} \to \ell^{a,\bar p}_{\mathbb{C}}$, is the diagonal $\ddot R_{jj} = j^{p-\bar p} e^{-js_K} e^{ijx_1}$, and its representative $\tau_{a,\bar p} \ddot R \kappa_{a,p}$ (Definition 3.1) has entries $\omega_{a,\bar p}(j) \ddot R_{jj} / \omega_{a,p}(j) = j^{\bar p - p} \ddot R_{jj} = e^{-js_K} e^{ijx_1} = \tilde R_{jj}$. If normal modes are labelled by shifted indices, the factor $j^{p-\bar p}$ is replaced by the corresponding ratio $\omega_{a,p}/\omega_{a,\bar p}$ of the weights actually carried by that mode. Replacing e^{ijx_1} by $\cos(jx_1)$ (raw coefficient $j^{p-\bar p} e^{-js_K} \cos(jx_1)$) gives a version real on reals whose representative has coefficients $\frac12 e^{-js_K} P_j$ at $\pm j e_1$, strip bound ≤ 1 (as $|\cos(jx_1)| \le e^{j|\mathrm{Im}\, x_1|}$) and square sum $N/2$, so the contradiction persists for $N > 2^{n+1}$.

What W4 must use instead. Bound the coefficients individually and sum in ℓ^1 after a strip loss: by Lemma 3.36(i), $|||\tilde R_k||| \le M e^{-2\pi|k|_1 s}$; by Lemma 3.39(iii)–(iv) and the divisor bound of P96-K p. 127, $|||\tilde F_k||| \le 3(A_k/\alpha)|||\tilde R_k|||$; then by Lemma 3.36(iii) and Lemma 3.37(iv),

$$|||\tilde F(\theta)||| \le \sum_k |||\tilde F_k||| e^{2\pi|k|_1(s-\sigma)} \le \frac{3M}{\alpha} \sum_k A_k e^{-2\pi|k|_1\sigma} \le \frac{3M}{\alpha} S_{n,\tau}(\sigma) \qquad (\theta \in T_{s-\sigma}).$$

The parameter differences (with $|\Delta\delta^{-1}_{k,ij}| \lesssim A_k^2\alpha^{-2}(|k||\Delta\omega| + \dots)/|i-j|$) are handled the same way with weights $A_k(1+|k|_1)$ and $A_k^2(1+|k|_1)$, i.e. $S_{n,\tau+1}$ and $S_{n,2\tau+1}$. The resulting powers of σ differ from the source's B_σ (they are no worse: $S_{n,c}(\sigma) \lesssim \sigma^{-(c+n)}$); W4 must recompute its constants and the smallness exponents of P96-K (6) with them. W1 makes no claim that the published exponents survive unchanged.

SOURCE_GAP: P96-K p. 127 (PDF p. 10), display after "Expanding $\tilde{R}$ into its Fourier series with operator valued coefficient $\tilde{R}_k$ we have, as before": the square-sum bound with constant 2^n fails for operator-valued families (Remark 3.38). Replacement: Lemmas 3.36 and 3.37. This does not by itself refute Theorem A; the homological estimate must be redone in W4.

Lemma 3.39 (Hilbert-matrix bound)**.** *Let A be a bounded linear operator on $\ell^2_{\mathbb{K}}$ with matrix $A_{ij} = (Ae_j)_i$ $(i,j \geq 1)$.*

(i) The matrix $B_{ij} = |A_{ij}|/|i-j|$ $(i \neq j)$, $B_{ii} = 0$, defines a bounded operator, $(Bv)_i = \sum_j B_{ij}v_j$ absolutely convergent, with $\|B\| \leq \frac{\pi}{\sqrt{3}}\|A\|$.

(ii) The diagonal (A_{ii}) defines a multiplier with norm $\leq \|A\|$.

(iii) If F_{ij} satisfies $|F_{ij}| \leq c|A_{ij}|/|i-j|$ for $i \neq j$ and $|F_{ii}| \leq c|A_{ii}|$, then F defines a bounded operator with $\|F\| \leq c(1+\pi/\sqrt{3})\|A\| \leq 3c\|A\|$.

(iv) The same holds for $A : \ell^{a,p}_{\mathbb{K}} \to \ell^{a,\bar{p}}_{\mathbb{K}}$ with the matrix of the representative $\hat{A}$ (Definition 3.1): the ratios $|F_{ij}|/|A_{ij}|$ are unchanged by the weights.

Proof. **Step S3.82: Row and column sums.** $\sum_j |A_{ij}|^2 = \|A^*e_i\|^2 \leq \|A\|^2$ and $\sum_i |A_{ij}|^2 = \|Ae_j\|^2 \leq \|A\|^2$. By Cauchy–Schwarz and $\sum_{k\in\mathbb{Z}\setminus 0} k^{-2} = \pi^2/3$ ((W1-E2)), $\sum_j B_{ij} \leq \Lambda\|A\|$ and $\sum_i B_{ij} \leq \Lambda\|A\|$ with $\Lambda = \pi/\sqrt{3}$.

Step S3.83: Schur test. $\|Bv\|^2 \leq \sum_i \big(\sum_j B_{ij}\big)\big(\sum_j B_{ij}|v_j|^2\big) \leq \Lambda\|A\| \sum_j \big(\sum_i B_{ij}\big)|v_j|^2 \leq \Lambda^2\|A\|^2\|v\|^2$, via finite partial sums ((W1-E1)).

Step S3.84: Diagonal and combination. $|A_{ii}| = |\langle e_i, Ae_i\rangle| \leq \|A\|$ and Lemma 3.4. For (iii), split F into off-diagonal and diagonal parts, dominated entrywise by cB and $c\,\mathrm{diag}(|A_{ii}|)$; an entrywise dominated matrix has operator norm at most that of the nonnegative dominating matrix (apply it to $|v|$). $1+\pi/\sqrt{3} < 2.82$.

Step S3.85: Weights. $\hat{A}_{ij} = \omega_{a,\bar{p}}(i)A_{ij}/\omega_{a,p}(j)$ and likewise for F, so the hypotheses of (iii) transfer. □

Source: P96-K Lemma A.1, pp. 145–146, and its use on p. 127 (the factor $3(A_k/\alpha)$).

3.9. **Ledger.**

PLAN nodes. `bpW:w1:space` → Lemma 3.2 (with Definition 3.1 and Lemma 3.3); `embed` → Lemma 3.5 (with Lemma 3.4); `diag` → Lemma 3.6; `twosided` → Lemma 3.10 (with Definition 3.9); `young` → Lemma 3.11; `algebra` → Lemma 3.12 (relabels Lemma 2.4); `trilinear` → Lemma 3.14 and Theorem 3.15; `cubic-analytic` → Corollary 3.16; `hamfield` → Proposition 3.18; `gradient-identity` → Proposition 3.17; `sine-completeness` → Lemma 3.19.

Added nodes and consumers. W1-internal: 3.34 (used by 3.35). W2: 3.3, 3.20, 3.21, 3.18. W3: 3.10(iii), 3.14(b), 3.23, 3.25, 3.30. W4: 3.35 (Hilbert-valued terms only), 3.36, 3.37, 3.39, 3.8, 3.31, and Remark 3.38 (the operator-valued Fourier step must be redone; not supplied by W1). W5: 3.25, 3.26, 3.30, 3.33. W6, W8: 3.27, 3.28, 3.29, 3.3. The algebra extension to $\frac{1}{2} < s < 1$ (Remark 3.13) is not planned.

Source gaps and conventions. P96-K p. 127 operator-valued Fourier square-sum bound (false; Remark 3.38); P96-K Lemma A.4 (segment hypothesis in the mean value step, Lemma 3.30); P96-W p. 280 constant (N-1, used corrected in Lemma 3.10); P96-K's $z = (u-iv)/\sqrt{2}$ versus W0 (Remark 3.32, a relabelling); P96-W §6 printed as "Appendix A" on p. 279 (label only).

4. Milestone W2: PDE reconstruction

This part delivers conclusion (5) of the frozen headline `KAM.Wave.nlw2` from its conclusions (3) and (4). Given an analytic map $K : \mathbb{R}^2 \to$ `WaveSpace` whose coefficients obey the coefficient system (2) along the linear flow $\theta \mapsto \theta + t\nu$ (frozen conclusion (4)), and a bound $\|K(\theta)\| \le B$, we show that for every θ_0 the function

$$u(t,x) = \texttt{reconU m hm (K (}\theta_0 \texttt{ + t • }\nu\texttt{)).1 x}$$

is real analytic at every point of $\mathbb{R} \times [0,\pi]$, is a classical solution of (1) on all of $\mathbb{R}$ in the exact sense of the frozen `KAM.Wave.IsClassicalSolution`, has $u_t = V_m(P)$, and obeys $|u| + |u_x| + |u_t| \le M_m B$ with an explicit M_m depending only on m (Theorem 4.9). Corollary 4.11 records how milestone W8 chooses the single constant C that the frozen statement uses in both (3) and (5).

Route. The source (P96-W Lemma 1, p. 278) argues with absolutely convergent series on complex neighbourhoods in t and x and differentiates termwise. We avoid termwise differentiation in t altogether. For fixed x, the maps $y \mapsto U_m(y)(x)$, $y \mapsto \partial_x U_m(y)(x)$, $y \mapsto \partial_x^2 U_m(y)(x)$ and $y \mapsto V_m(y)(x)$ are continuous linear functionals $\langle \Psi(x), \cdot \rangle$ on $\ell^2_{\mathbb{R}}$, with $x \mapsto \Psi(x)$ real analytic (Lemma 3.20). Hence $u(t,x) = \langle \Psi^0(x), Y_1(t) \rangle$ with $Y(t) = K(\theta_0 + t\nu)$; joint analyticity is the analyticity of a continuous bilinear map evaluated on two analytic maps, and the time derivative is $\langle \Psi^0(x), Y_1'(t) \rangle$, where $Y_1'(t) \in \ell^2_{\mathbb{R}}$ is the Fréchet derivative of the analytic curve. The coefficients of $Y'(t)$ are identified by the frozen conclusion (4) (Lemma 4.6); this gives $u_t = V_m(Y_2)$ and $u_{tt} = V_m(Y_2')$, and the force identity (4) in frozen form (Lemma 4.3) turns $V_m(Y_2')$ into $u_{xx} - mu - u^3$.

Standing data. Standing assumption 1.1 ($m > 0$, Lean `hm : 0 < m`) and the conventions of §1 are in force; $a = p = 1$ (Convention 1.5). We write $\ell^2_{\mathbb{R}}$ for `lp (fun _ : PNat => Real) 2`, $\omega(j) = \omega_{1,1}(j) = je^j$ (frozen `ellWeight 1 1 j`), $\kappa = \kappa_{1,1}$, so that `coeff y j` $= y_j/\omega(j)$ (Lemma 3.2(iv)), $\lambda_j =$ `lam m hm j`, $\phi_j =$ `sineMode j`, $U_m(y) =$ `reconU m hm y`, $V_m(y) =$ `reconV m hm y` and $\mathcal{G}(y)_j =$ `cubicGrad m hm y j` for $y \in \ell^2_{\mathbb{R}}$. A point of `WaveSpace` is $w = (w.1, w.2)$ with the max norm, so $\|w.1\|, \|w.2\| \le \|w\|$. Time derivatives are two-sided; x-derivatives in the frozen definition are within $[0,\pi]$, but all x-derivatives proved below are two-sided at every $x \in \mathbb{R}$ and are restricted to $[0,\pi]$ only at the end.

4.1. External facts.

Besides the facts (W1-E1)–(W1-E3) of W1, the proofs use the following.

(W2-E1) *One-variable calculus.* A continuous linear map composed with a curve differentiable at t is differentiable at t with the image derivative (`ContinuousLinearMap.hasFDerivAt`, `HasFDerivAt.comp_hasDerivAt`); the derivative of $s \mapsto \langle \Psi(s), y \rangle$ is $\langle \Psi'(s), y \rangle$ (`HasDerivAt.inner`); derivatives are unique (`HasDerivAt.unique`); translation: if F has derivative d at 0 then $s \mapsto F(s-t)$ has derivative d at t (`HasDerivAt.comp_sub_const`); a two-sided derivative is a derivative within any set (`HasDerivAt.hasDerivWithinAt`); on $[0,\pi]$, $\pi > 0$, derivatives within are unique, so `derivWithin f (Set.Icc 0 Real.pi) x` equals any f' with `HasDerivWithinAt f f' (Set.Icc 0 Real.pi) x`, $x \in [0,\pi]$ (`uniqueDiffOn_Icc`, `HasDerivWithinAt.derivWithin`).

(W2-E2) *Analytic maps on $\mathbb{R} \times \mathbb{R}$.* The projections $z \mapsto z.1$, $z \mapsto z.2$ and affine maps $t \mapsto \theta_0 + t\nu$ are analytic (`analyticAt_fst`, `analyticAt_snd`, `analyticAt_const`, `AnalyticAt.add`, `AnalyticAt.smul`); sums, products and powers of real analytic functions are analytic; an analytic map is differentiable (`AnalyticAt.differentiableAt`, `DifferentiableAt.hasDerivAt`) and continuous (`AnalyticAt.continuousAt`); continuity at every point of a set gives `ContinuousOn` (`ContinuousAt.continuousWithinAt`).

(W2-E3) *Series and norms.* `tsum_congr` (termwise equal series have equal `tsum`, no summability needed); `Summable.tsum_sub`, `Summable.tsum_add` and `tsum_mul_left` for summable series; `tsum_zero`; comparison (`Summable.of_norm_bounded`); summability of $j^k e^{-j}$ over $j \ge 1$ for $k \in \mathbb{N}$ and $\sum_{j \ge 1} e^{-2j} = (e^2 - 1)^{-1}$ (geometric series,

`tsum_geometric_of_lt_one`); Cauchy–Schwarz $|\langle \Psi, y\rangle| \le \|\Psi\|\|y\|$ (`abs_real_inner_le_norm`); on a product, `norm_fst_le`, `norm_snd_le`.

(W2-E4) *Trigonometry.* $\sin 0 = 0$, $\sin(n\pi) = 0$ for $n \in \mathbb{N}$ (`Real.sin_nat_mul_pi`), $|\sin|, |\cos| \le 1$, $\frac{d}{dx}\sin(jx) = j\cos(jx)$, $\frac{d}{dx}\cos(jx) = -j\sin(jx)$; $\sqrt{\lambda}\cdot\sqrt{\lambda} = \lambda$ and $\lambda/\sqrt{\lambda} = \sqrt{\lambda}$ for $\lambda > 0$ (`Real.mul_self_sqrt`, `Real.div_sqrt`), $\lambda_j^2 = j^2 + m$ (`Real.sq_sqrt`).

4.2. **Reconstruction functionals.**

Definition 4.1 (Derived reconstructions)**.** For $y \in \ell^2_{\mathbb{R}}$ and $x \in \mathbb{R}$ put $D^0 y(x) = U_m(y)(x)$ (frozen `reconU m hm y x`) and

$$D^1 y(x) = \sum_{j\ge 1} \frac{\kappa(y)_j}{\sqrt{\lambda_j}} \sqrt{\tfrac{2}{\pi}}\, j\cos(jx), \qquad D^2 y(x) = -\sum_{j\ge 1} \frac{\kappa(y)_j}{\sqrt{\lambda_j}} \sqrt{\tfrac{2}{\pi}}\, j^2 \sin(jx),$$

as `tsum` over $j \in \mathbb{N}_+$. (Lean: non-frozen helpers, e.g. `KAM.Wave.reconUx m hm y x` and `KAM.Wave.reconUxx m hm y x`; distinct names, not substitutes for frozen declarations.)

Source: ours; D^k is the termwise k-th x-derivative of the series of Definition 1.7 (P96-W p. 277), i.e. the function $\partial_x^k U_m$ of Corollary 3.21 written out for $k \le 2$. Lean guard: $\sqrt{\lambda_j} > 0$.

Lemma 4.2 (Reconstruction functionals)**.** *Put $c_0 = \sqrt{2/\pi}\,(e^2-1)^{-1/2}$. There are real analytic maps $\Psi^0, \Psi^1, \Psi^2, \Psi^V : \mathbb{R} \to \ell^2_{\mathbb{R}}$ (Lean `AnalyticOnNhd Real Ψ Set.univ`) such that for all $y \in \ell^2_{\mathbb{R}}$ and $x \in \mathbb{R}$:*

(i) $D^k y(x) = \langle \Psi^k(x), y\rangle$ for $k = 0,1,2$ and $V_m(y)(x) = \langle \Psi^V(x), y\rangle$; each of the four series is absolutely convergent;

(ii) $\|\Psi^0(x)\| \le c_0$, $\|\Psi^1(x)\| \le c_0$, $\|\Psi^V(x)\| \le (1+m)^{1/4} c_0$; hence $|D^0 y(x)|, |D^1 y(x)| \le c_0\|y\|$ and $|V_m(y)(x)| \le (1+m)^{1/4} c_0 \|y\|$;

(iii) $D^k y$ has derivative $D^{k+1}y(x)$ at x for $k = 0,1$ (Lean `HasDerivAt`);

(iv) $D^0 y(0) = D^0 y(\pi) = 0$.

Proof. **Step S4.1: Choice of the features.** Apply Lemma 3.20 with $a = p = 1$: for Ψ^k ($k = 0,1,2$) take $\beta_j = \lambda_j^{-1/2}$, which satisfies $|\beta_j| \le 1 = Bj^\nu$ with $B = 1$, $\nu = 0$, since $\lambda_j > j \ge 1$; for Ψ^V take $k = 0$ and $\beta_j = \lambda_j^{1/2} \le (1+m)^{1/4} j^{1/2}$ (as $\lambda_j^2 = j^2 + m \le (1+m)j^2$), so $B = (1+m)^{1/4}$, $\nu = \frac{1}{2}$. Let $\Psi^k = \Psi_{\mathbb{R}}$, $\Psi^V = \Psi_{\mathbb{R}}$ be the resulting real restrictions; they are real analytic on $\mathbb{R}$ by Lemma 3.20(ii). Their coordinates are $\Psi^k(x)_j = \lambda_j^{-1/2}\phi_j^{(k)}(x)/\omega(j)$ and $\Psi^V(x)_j = \lambda_j^{1/2}\phi_j(x)/\omega(j)$ with $\phi_j^{(0)} = \phi_j$, $\phi_j^{(1)}(x) = \sqrt{2/\pi}\, j\cos(jx)$, $\phi_j^{(2)}(x) = -\sqrt{2/\pi}\, j^2 \sin(jx)$.

Step S4.2: Pairings. By Lemma 3.20(iii), $\langle \Psi^k(x), y\rangle = \sum_j \lambda_j^{-1/2}\kappa(y)_j \phi_j^{(k)}(x)$ and $\langle \Psi^V(x), y\rangle = \sum_j \lambda_j^{1/2}\kappa(y)_j\phi_j(x)$, absolutely convergent. Termwise these are the summands of the frozen `reconU`, of Definition 4.1, and of the frozen `reconV` (`coeff y j` $= \kappa(y)_j$), so the `tsum`s agree by `tsum_congr` (W2-E3). This is (i).

Step S4.3: Norms. $\|\Psi^k(x)\|^2 = \sum_j \lambda_j^{-1}\phi_j^{(k)}(x)^2/(j^2 e^{2j}) \le \frac{2}{\pi}\sum_j \lambda_j^{-1} j^{2k-2} e^{-2j}$. For $k = 0,1$ the factor $\lambda_j^{-1} j^{2k-2}$ is at most $\lambda_j^{-1} < 1$, so $\|\Psi^k(x)\|^2 \le \frac{2}{\pi}\sum_{j\ge1} e^{-2j} = c_0^2$. For Ψ^V the factor is $\lambda_j j^{-2} \le (1+m)^{1/2} j^{-1} \le (1+m)^{1/2}$, giving $(1+m)^{1/2} c_0^2$. The bounds on the norm in ℓ^2 follow from these bounds on all finite partial sums ((W1-E1)); the pointwise bounds are Cauchy–Schwarz (W2-E3) applied to (i).

Step S4.4: x-derivatives. By Step S3.48 of Corollary 3.21, the derivative of the analytic map Ψ^k at x has the coordinates $\frac{d}{dx}\Psi^k(x)_j$, which by (W2-E4) are $\lambda_j^{-1/2}\phi_j^{(k+1)}(x)/\omega(j) = \Psi^{k+1}(x)_j$ for

$k = 0, 1$; two vectors of $\ell^2_{\mathbb{R}}$ with the same coordinates are equal, so $(\Psi^k)'(x) = \Psi^{k+1}(x)$. Then $x \mapsto \langle \Psi^k(x), y\rangle$ has derivative $\langle \Psi^{k+1}(x), y\rangle$ by (W2-E1), and (i) gives (iii).

Step S4.5: Boundary values. $\phi_j(0) = \sqrt{2/\pi}\sin 0 = 0$ and $\phi_j(\pi) = \sqrt{2/\pi}\sin(j\pi) = 0$ for $j \in \mathbb{N}_+$ ((W2-E4), after casting j to $\mathbb{N}$), so every summand of `reconU m hm y 0` and of `reconU m hm y Real.pi` vanishes and the `tsum` is 0 (`tsum_zero`). □

Source: ours; the functional form of P96-W Lemma 1, p. 278 ("absolutely convergent in some complex neighbourhood of the x-interval $[0, \pi]$"). Boundary values: Lemma 2.1.

Lemma 4.3 (Force identity in frozen form)**.** *Let $y \in \ell^2_{\mathbb{R}}$. For every $x \in \mathbb{R}$ the series*

$$\Sigma_{\mathrm{lin}}(x) = \sum_{j\ge 1} \sqrt{\lambda_j}\,\lambda_j\,\kappa(y)_j\,\phi_j(x), \qquad \Sigma_{\mathrm{cub}}(x) = \sum_{j\ge 1} \sqrt{\lambda_j}\,\mathcal{G}(y)_j\,\phi_j(x)$$

converge absolutely, and

(i) $\Sigma_{\mathrm{lin}}(x) = -D^2y(x) + m\,D^0y(x)$ *for every* $x \in \mathbb{R}$*;*
(ii) $\Sigma_{\mathrm{cub}}(x) = U_m(y)(x)^3$ *for every* $x \in [0, \pi]$*.*

Proof. **Step S4.6: Summability.** By (W1-E1), $|\kappa(y)_j| \le \|y\|/(je^j)$; with $\sqrt{\lambda_j}\lambda_j \le (1+m)^{3/4}j^{3/2}$ and $|\phi_j| \le \sqrt{2/\pi}$ the summands of Σ_{lin} are at most $\sqrt{2/\pi}(1+m)^{3/4}\|y\|\,je^{-j}$. By Corollary 3.16(iv), $\mathcal{G}(y)_j = \mathcal{G}^w(y)_j/\omega_{1,2}(j)$ with $\mathcal{G}^w(y) \in \ell^2_{\mathbb{R}}$, so $|\mathcal{G}(y)_j| \le \|\mathcal{G}^w(y)\|/(j^2e^j)$ and the summands of Σ_{cub} are at most $\sqrt{2/\pi}(1+m)^{1/4}\|\mathcal{G}^w(y)\|e^{-j}$. Both majorants are summable ((W2-E3)). The series $D^0y(x)$ and $D^2y(x)$ converge absolutely by Lemma 4.2(i).

Step S4.7: Linear part. Termwise, $\sqrt{\lambda_j}\lambda_j = \lambda_j^2/\sqrt{\lambda_j} = (j^2+m)/\sqrt{\lambda_j}$ ((W2-E4)), so the j-th summand of $\Sigma_{\mathrm{lin}}(x)$ is $\frac{\kappa(y)_j}{\sqrt{\lambda_j}}\big(j^2\phi_j(x) + m\phi_j(x)\big)$, which is minus the j-th summand of $D^2y(x)$ plus m times that of $D^0y(x)$. Sum, using absolute convergence of both series (`Summable.tsum_add`, `tsum_mul_left`). This is the second identity of (4) in frozen coordinates.

Step S4.8: Cubic part. By Step S3.26 of Theorem 3.15, $U_m(y) = U_m(q)$ and $\mathcal{G}(y) = G_q(q)$ for $q = \kappa(y) \in \ell^{1,1}$ (here $U_m(q)$, G_q are the objects of Definitions 1.7, 1.9). The first identity of (4) (Proposition 2.6, Step S2.10; it uses the completeness Lemma 3.19(ii)) states $U_m(q)^3 = \sum_l \sqrt{\lambda_l}G_q(q)_l\phi_l$ on $[0, \pi]$, which is (ii). □

Source: P96-W Lemma 1, p. 278 (the computation of u_{tt} and the completeness argument); Proposition 2.6 for the proof. Only $x \in [0, \pi]$ is claimed in (ii), which is all the frozen definition needs.

4.3. The coefficient system along an orbit.

Definition 4.4 (Analytic coefficient solution)**.** A curve $Y : \mathbb{R} \to$ `WaveSpace` is an *analytic coefficient solution* if

(C1) Y is real analytic at every $t \in \mathbb{R}$ (Lean `∀ t, AnalyticAt Real Y t`);
(C2) for all $t \in \mathbb{R}$ and $j \in \mathbb{N}_+$, the function $s \mapsto$ `coeff (Y s).1 j` has derivative λ_j `coeff (Y t).2 j` at t, and $s \mapsto$ `coeff (Y s).2 j` has derivative $-\lambda_j$ `coeff (Y t).1 j − cubicGrad m hm (Y t).1 j` at t (Lean `HasDerivAt`).

Source: P96-W Lemma 1, p. 278 ("a real analytic solution of (6)", curve $I \to \ell^{a,s} \times \ell^{a,s}$), with $I = \mathbb{R}$, $a = s = 1$, and (6) read coordinatewise in frozen coordinates (system (2)).

Lemma 4.5 (Torus orbits are analytic coefficient solutions)**.** *Let $K : (\mathrm{Fin}\,2 \to \mathbb{R}) \to$ `WaveSpace` satisfy `AnalyticOnNhd Real K Set.univ`, and let $\nu \in \mathrm{Fin}\,2 \to \mathbb{R}$ be such that for all θ and $j \in \mathbb{N}_+$*

$$\begin{aligned} \frac{d}{dt}\Big|_{t=0} \kappa\big(K(\theta + t\nu).1\big)_j &= \lambda_j\,\kappa\big(K(\theta).2\big)_j, \\ \frac{d}{dt}\Big|_{t=0} \kappa\big(K(\theta + t\nu).2\big)_j &= -\lambda_j\,\kappa\big(K(\theta).1\big)_j - \mathcal{G}\big(K(\theta).1\big)_j, \end{aligned} \tag{7}$$

each as a `HasDerivAt` *statement at* 0 *about* `fun t => coeff (K (θ + t • ν)).i j` *(the frozen conclusion (4) of* `nlw2` *for one* I*, with* $\nu = \nu(I)$*,* $K = K_I$*). Then for every* θ_0 *the curve* $Y(t) = K(\theta_0 + t\nu)$ *is an analytic coefficient solution.*

Proof. **Step S4.9: Analyticity.** $t \mapsto \theta_0 + t\nu$ is analytic and K is analytic at its value, so Y is analytic at t ((W2-E2), `AnalyticAt.comp`).

Step S4.10: Shift. Fix t and apply (7) at $\theta = \theta_0 + t\nu$: $F(h) =$ `coeff (K (θ + h • ν)).1 j` has the stated derivative at $h = 0$. By the translation rule (W2-E1), $s \mapsto F(s - t)$ has the same derivative at $s = t$, and $F(s-t) =$ `coeff (Y s).1 j` because $\theta_0 + t\nu + (s-t)\nu = \theta_0 + s\nu$ (`add_smul`, `add_assoc`, `add_sub_cancel`). The value of the derivative is λ_j `coeff (K θ).2 j` $= \lambda_j$ `coeff (Y t).2 j`. The second component is identical. □

Source: ours (the frozen (4) is stated at $t = 0$ for every θ; the shift is the group property of the linear flow on the torus).

Lemma 4.6 (Coefficients of the curve derivative)**.** *Let* Y *be an analytic coefficient solution and* $t \in \mathbb{R}$*. Then* $Y_1 = (Y\cdot).1$ *and* $Y_2 = (Y\cdot).2$ *are differentiable at* t *as maps* $\mathbb{R} \to \ell^2_{\mathbb{R}}$*; with* $Y_i'(t)$ *their derivatives, for every* $j \in \mathbb{N}_+$*,*

$$\kappa(Y_1'(t))_j = \lambda_j\, \kappa(Y_2(t))_j, \qquad \kappa(Y_2'(t))_j = -\lambda_j\, \kappa(Y_1(t))_j - \mathcal{G}(Y_1(t))_j.$$

Proof. **Step S4.11: Differentiability.** Y_i is the composite of Y with the continuous linear projection, so it is analytic at t, hence differentiable ((W2-E2)).

Step S4.12: Coordinates. By Lemma 3.3 with $(a, p) = (a', p') = (1, 1)$ (so $\hat{\imath} = \mathrm{id}$) and $v = Y_i'(t)$, the function $s \mapsto \kappa(Y_i(s))_j$ has derivative $\kappa(Y_i'(t))_j$ at t. By (C2) it also has the derivative stated there; uniqueness of derivatives ((W2-E1)) gives the identities. □

Source: ours.

Remark 4.7 (Regularity hidden in the hypotheses). Lemma 4.6 shows $\lambda\, \kappa(Y_2(t)) = \kappa(Y_1'(t)) \in \ell^{1,1}$, i.e. the momentum coefficients of a point of an invariant torus satisfying (3)–(4) lie in $\ell^{1,2}$. This is not an extra hypothesis but a consequence of differentiability into $E_{1,1}$; it is the frozen counterpart of the statement of Lemma 3.3 that (4) means $\hat{\imath}(DK(\theta)\nu) = X^w(K(\theta))$ with X^w the field of Proposition 3.18(v), the left side lying in $E_{1,1}$. The proof below uses only the coordinate form.

4.4. Classical solutions from analytic coefficient solutions.

Theorem 4.8 (P96-W Lemma 1 in frozen form)**.** *Let* Y *be an analytic coefficient solution, and define* $u, u_t, u_x, u_{xx}, u_{tt} : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ *by*

$$u(t,x) = U_m(Y_1(t))(x), \qquad u_t(t,x) = V_m(Y_2(t))(x),$$

$$u_x(t,x) = D^1 Y_1(t)(x), \qquad u_{xx}(t,x) = D^2 Y_1(t)(x), \qquad u_{tt}(t,x) = u_{xx}(t,x) - m\, u(t,x) - u(t,x)^3,$$

where $Y_i(t) = (Y\, t).i$*; in Lean* u *is* `fun t x => reconU m hm (Y t).1 x`*. Put* $M_m = c_0\big(2 + (1+m)^{1/4}\big)$ *with* c_0 *as in Lemma 4.2. Then:*

- *(i) each of the five functions, uncurried, is real analytic at every point of* $\mathbb{R} \times \mathbb{R}$ *(Lean* `AnalyticAt Real (fun z : Real × Real => reconU m hm (Y z.1).1 z.2) z` *for all* z*, and likewise);*
- *(ii) for all* $t \in \mathbb{R}$ *and* $x \in \mathbb{R}$*,* $s \mapsto u(s,x)$ *has derivative* $u_t(t,x)$ *at* t*;*
- *(iii) for all* $t \in \mathbb{R}$ *and* $x \in [0,\pi]$*,* $s \mapsto u_t(s,x)$ *has derivative* $u_{tt}(t,x)$ *at* t*;*
- *(iv) for all* $t, x \in \mathbb{R}$*,* $u(t,\cdot)$ *has derivative* $u_x(t,x)$ *and* $u_x(t,\cdot)$ *has derivative* $u_{xx}(t,x)$ *at* x*;*
- *(v)* `IsClassicalSolution m hm u`*, with witnesses* $(u_t, u_x, u_{tt}, u_{xx})$*;*
- *(vi) for all* $t \in \mathbb{R}$ *and* $x \in [0,\pi]$*,* `derivWithin (u t) (Set.Icc 0 Real.pi) x` $= u_x(t,x)$ *and*

$$|u(t,x)| + |\texttt{derivWithin (u t) (Set.Icc 0 Real.pi) x}| + |V_m(Y_2(t))(x)| \le M_m \|Y(t)\|.$$

Proof. **Step S4.13: Joint analyticity.** Y_1, Y_2 are analytic at every t (Step S4.11). By Corollary 3.21, applied with the parameter domain $\mathbb{R}$ and $Y = Y_1$ ($k = 0, 1, 2$) or $Y = Y_2$ (V_m, $k = 0$), the maps $(s, x) \mapsto \langle \Psi(x), Y_i(s) \rangle$ are real analytic at every $(s_0, x_0) \in \mathbb{R}^2$. Concretely, in Mathlib terms: $z \mapsto (\Psi(z.2), Y_i(z.1))$ is analytic into $\ell^2_{\mathbb{R}} \times \ell^2_{\mathbb{R}}$ at z (`AnalyticAt.prod` of $\Psi \circ$ snd and $Y_i \circ$ fst, (W2-E2)), and the real inner product is a continuous bilinear map, analytic on the product (`ContinuousLinearMap.analyticAt_bilinear` with `innerSL Real`, (W1-E3)); compose (`AnalyticAt.comp`). By Lemma 4.2(i) these functions are exactly u, u_x, u_{xx} (with Ψ^0, Ψ^1, Ψ^2 and Y_1) and u_t (with Ψ^V and Y_2), equal as functions on $\mathbb{R}^2$, so analyticity transfers by rewriting. Finally u_{tt} is a polynomial in u, u_{xx}, analytic by (W2-E2). This is (i).

Step S4.14: First time derivative. Fix x. The map $y \mapsto D^0 y(x) = \langle \Psi^0(x), y \rangle$ is a continuous linear functional on $\ell^2_{\mathbb{R}}$ (Lemma 4.2(i)), so $s \mapsto u(s, x)$ has derivative $\langle \Psi^0(x), Y_1'(t) \rangle = U_m(Y_1'(t))(x)$ at t ((W2-E1)). The j-th summand of this `tsum` is, by Lemma 4.6 and $\lambda_j / \sqrt{\lambda_j} = \sqrt{\lambda_j}$ ((W2-E4)),

$$\frac{\kappa(Y_1'(t))_j}{\sqrt{\lambda_j}} \phi_j(x) = \frac{\lambda_j \kappa(Y_2(t))_j}{\sqrt{\lambda_j}} \phi_j(x) = \sqrt{\lambda_j}\, \kappa(Y_2(t))_j\, \phi_j(x),$$

the j-th summand of $V_m(Y_2(t))(x)$; `tsum_congr` gives $U_m(Y_1'(t))(x) = u_t(t, x)$. This is (ii).

Step S4.15: Second time derivative. Likewise $y \mapsto V_m(y)(x) = \langle \Psi^V(x), y \rangle$ is continuous linear, so $s \mapsto u_t(s, x)$ has derivative $V_m(Y_2'(t))(x)$ at t. By Lemma 4.6 its j-th summand is $\sqrt{\lambda_j}\big(-\lambda_j \kappa(Y_1(t))_j - \mathcal{G}(Y_1(t))_j\big)\phi_j(x)$. Both resulting series are absolutely convergent (Lemma 4.3, Step S4.6, with $y = Y_1(t)$), so `Summable.tsum_sub` gives $V_m(Y_2'(t))(x) = -\Sigma_{\mathrm{lin}}(x) - \Sigma_{\mathrm{cub}}(x)$. For $x \in [0, \pi]$, Lemma 4.3 turns this into $-(-u_{xx}(t, x) + m\, u(t, x)) - u(t, x)^3 = u_{tt}(t, x)$. This is (iii).

Step S4.16: Space derivatives. (iv) is Lemma 4.2(iii) with $y = Y_1(t)$.

Step S4.17: The frozen definition. Unfold `IsClassicalSolution m hm u` and supply $(u_t, u_x, u_{tt}, u_{xx})$. Continuity: each of the five uncurried functions is analytic, hence continuous, at every point (Step S4.13, (W2-E2)), so it is `ContinuousOn` on `Set.univ` $\times^s$ `Set.Icc 0 Real.pi`; the quantifier over the five-element set is discharged by case analysis on set membership (`Set.mem_insert_iff`, `Set.mem_singleton_iff`). Derivatives, for $t \in \mathbb{R}$, $x \in [0, \pi]$: the two time derivatives are (ii) and (iii); the two derivatives within $[0, \pi]$ follow from (iv) by `HasDerivAt.hasDerivWithinAt`. The equation $u_{tt} = u_{xx} - mu - u^3$ holds by the definition of u_{tt}. The boundary values $u(t, 0) = u(t, \pi) = 0$ are Lemma 4.2(iv). This is (v).

Step S4.18: Bound. For $x \in [0, \pi]$, $u(t, \cdot)$ has derivative within $[0, \pi]$ equal to $u_x(t, x)$ (Step S4.17), so `derivWithin` equals $u_x(t, x)$ by uniqueness on $[0, \pi]$ ((W2-E1), $\pi > 0$); this includes the endpoints. By Lemma 4.2(ii), $|u(t, x)| \le c_0 \|Y_1(t)\|$, $|u_x(t, x)| \le c_0 \|Y_1(t)\|$, $|V_m(Y_2(t))(x)| \le (1+m)^{1/4} c_0 \|Y_2(t)\|$, and $\|Y_i(t)\| \le \|Y(t)\|$ ((W2-E3)). Adding gives (vi). □

Source: P96-W Lemma 1, p. 278, with $I = \mathbb{R}$, $a = s = 1$, $f(u) = u^3$. Strengthened: the source asserts real analyticity and the classical equation; it does not state $u_t = V_m(p)$ or the bound (vi). The proof is reorganized (continuous linear functionals instead of termwise t-differentiation on complex discs).

4.5. The W8 form: conclusion (5) from conclusions (3)–(4).

Theorem 4.9 (PDE solutions on invariant tori)**.** *Let $m > 0$, let M_m be as in Theorem 4.8, and let $K : (\mathrm{Fin}\, 2 \to \mathbb{R}) \to$ `WaveSpace`, $\nu : \mathrm{Fin}\, 2 \to \mathbb{R}$ and $B \in \mathbb{R}$ satisfy*

(H1) `IsAnalyticTorusEmbedding K`*;*
(H2) the invariance (7) for every θ and every $j \in \mathbb{N}_+$;
(H3) $\|K(\theta)\| \le B$ for every θ.

Then for every $\theta_0 : \mathrm{Fin}\, 2 \to \mathbb{R}$, with $u(t, x) =$ `reconU m hm (K (θ₀ + t • ν)).1 x`*,*

(a) for every $z \in$ `Set.univ` $\times^s$ `Set.Icc 0 Real.pi`*,* `AnalyticAt Real (fun z : Real × Real => reconU m hm (K (θ₀ + z.1 • ν)).1 z.2) z`*;*

(b) *IsClassicalSolution m hm (fun t x => reconU m hm (K (θ₀ + t • ν)).1 x);*
(c) *for all* $t \in \mathbb{R}$, $x \in [0,\pi]$*:* *HasDerivAt (fun s => reconU m hm (K (θ₀ + s • ν)).1 x) (reconV m hm (K (θ₀ + t • ν)).2 x) t;*
(d) *for all* $t \in \mathbb{R}$, $x \in [0,\pi]$*:* $|u(t,x)| + |$*derivWithin (u t) (Set.Icc 0 Real.pi) x*$| + |V_m(K(\theta_0 + t\nu).2)(x)| \le M_m B$.

Proof. **Step S4.19: Reduction to a curve.** The second conjunct of (H1) is `AnalyticOnNhd Real K Set.univ`; with (H2), Lemma 4.5 shows that $Y(t) = K(\theta_0 + t\nu)$ is an analytic coefficient solution. The functions u of Theorems 4.8 and 4.9 coincide.

Step S4.20: Conclusions. (a) is Theorem 4.8(i) restricted to $z \in \mathbb{R} \times [0,\pi]$; (b) is (v); (c) is (ii) restricted to $x \in [0,\pi]$, since $u_t(t,x) = V_m(Y_2(t))(x)$ is the stated derivative; (d) is (vi) with $\|Y(t)\| \le B$ by (H3) and $M_m > 0$. □

Source: P96-W Lemma 1, p. 278, applied to the quasi-periodic solutions $t \mapsto K(\theta_0 + t\nu)$ of the Main Theorem (P96-W pp. 272–273 and the assembly pp. 290–291, where Lemma 1 is invoked); the bound (d) is ours.

Remark 4.10 (Hypotheses). (H1)–(H3) are, for one $I \in \Gamma$ with $K = K_I$, $\nu = \nu(I)$ and $B = C\sqrt{I_0 + I_1}$, the frozen conclusions (3) (first and third conjuncts) and (4) of `nlw2`, which W8 obtains from the KAM construction independently of (5); none of them mentions u, so no hypothesis is conclusion-shaped. Of (H1) only analyticity is used; periodicity, injectivity and the injective derivative play no role in W2. A proof that holds only `AnalyticOnNhd Real K Set.univ` (e.g. from Corollary 3.29) may call Lemma 4.5 and Theorem 4.8 directly. The Lean statement of Theorem 4.9 may therefore take `AnalyticOnNhd Real K Set.univ` in place of (H1); W8 then passes the projection `h.2.1` of its embedding proof.

Corollary 4.11 (Clause (5) with the common constant)**.** *Let* $m > 0$, $C_3 > 0$, $\Gamma \subset \mathrm{Fin}\,2 \to \mathbb{R}$, $\nu : (\mathrm{Fin}\,2 \to \mathbb{R}) \to \mathrm{Fin}\,2 \to \mathbb{R}$ *and* $K : (\mathrm{Fin}\,2 \to \mathbb{R}) \to (\mathrm{Fin}\,2 \to \mathbb{R}) \to$ *WaveSpace, and suppose that for every* $I \in \Gamma$*:* K_I *satisfies IsAnalyticTorusEmbedding,* $\|K_I(\theta)\| \le C_3\sqrt{I_0 + I_1}$ *for all* θ*, and the invariance (4) of nlw2 holds. Put* $C = \max(1, M_m)\, C_3$*. Then* $C > 0$, $C \ge C_3$*, the bound* $\|K_I(\theta)\| \le C\sqrt{I_0 + I_1}$ *holds, and conclusion (5) of nlw2 holds verbatim with this* C*.*

Proof. $C \ge C_3 > 0$ since $\max(1, M_m) \ge 1$. As $\sqrt{I_0 + I_1} \ge 0$ (`Real.sqrt_nonneg`), $C_3\sqrt{I_0 + I_1} \le C\sqrt{I_0 + I_1}$, which gives the bound in (3) with C. For $I \in \Gamma$ and θ_0, apply Theorem 4.9 with $K = K_I$, $\nu = \nu(I)$ and $B = C_3\sqrt{I_0 + I_1}$: (a)–(c) are the first three conjuncts of (5), and (d) gives the fourth with $M_m C_3\sqrt{I_0 + I_1} \le C\sqrt{I_0 + I_1}$. □

Source: ours (constant management for the frozen statement).

Remark 4.12 (Order of constants). In `nlw2` the single existential C serves (3) and (5). The construction (W6, W8) produces C_3 depending only on m, before any I or θ_0; then $C = \max(1, M_m)C_3$ with

$$M_m = \sqrt{\tfrac{2}{\pi}}\,\frac{2 + (1+m)^{1/4}}{\sqrt{e^2 - 1}} \approx 0.316\,\big(2 + (1+m)^{1/4}\big)$$

also depends only on m. Since $M_m < 1$ exactly when $(1+m)^{1/4} < \sqrt{\pi(e^2-1)/2} - 2 \approx 1.17$, i.e. for small m, the maximum with 1 cannot be dropped in general. The constant M_m is not optimal (the factor λ_j^{-1} in Step S4.3 is discarded); only its independence of I, θ_0, t, x matters.

4.6. Lean totality, sources and summary.

Lean totality. The divisions by $\sqrt{\lambda_j}$ (in `reconU`, `cubicGrad` and Definition 4.1) are by a positive number. Every `tsum` used is absolutely convergent (Lemmas 4.2(i), 4.3); the identities obtained by `tsum_congr` need no summability. The curve derivatives $Y_i'(t)$ are taken only at points where Y_i is differentiable (Step S4.11). `derivWithin` in (5) is identified, not bounded through its junk value, by uniqueness of derivatives within $[0,\pi]$, endpoints included. `Real.sqrt (I 0 + I 1)` appears only as the nonnegative factor in B.

Comparison with the source. P96-W Lemma 1 (p. 278; read on the page image) assumes a real analytic solution $I \to \ell^{a,s} \times \ell^{a,s}$ of (6) and concludes that $u = \sum_j q_j \lambda_j^{-1/2} \phi_j$ is a classical solution of (1), real analytic on $I \times [0, \pi]$. Theorem 4.8 is this lemma for $I = \mathbb{R}$, $a = s = 1$, $f(u) = u^3$, in frozen coordinates, strengthened by $u_t = V_m(p)$ and the bound (vi). Deviations: (6) is read coordinatewise (Definition 4.4) and supplied by the torus invariance (4) through Lemma 4.5; the source's termwise differentiation in t on complex discs is replaced by continuous linear functionals; the source's completeness of (ϕ_j) among odd L^2 functions on $[-\pi, \pi]$ enters through Lemma 3.19 inside Proposition 2.6. The sign slip in the second line of the source's u_{tt} display (printed $+\lambda_j^2 q_j$; errata item N-4) is already recorded in Part 0; the proof above uses the correct sign. No new source defect was found on pp. 277–280.
Results. Definitions 4.1, 4.4; Lemmas 4.2, 4.3, 4.5, 4.6; Theorems 4.8, 4.9; Corollary 4.11; Remarks 4.7, 4.10, 4.12. Inputs from Part 0: Lemma 2.1, Proposition 2.6 with (4). Inputs from W1: Lemmas 3.2, 3.3, 3.20, 3.19; Theorem 3.15(iv); Corollaries 3.16(iv), 3.21; Proposition 3.18(v) (Remark 4.7 only). External: (W2-E1)–(W2-E4) and (W1-E1)–(W1-E3). Used by W8 (Corollary 4.11).

5. Milestone W3: the analytic partial Birkhoff normal form

Part 0 computed the partial quartic normal form formally (Proposition 2.11): the generating Hamiltonian F divides each nonresonant quartic monomial with at least one tangential index by its divisor, and the time-one map of X_F removes these monomials up to order six. This part makes that computation analytic. It proves (a) a quantitative lower bound for the divisors (Lemma 5.13, P96-W Lemma 4); (b) that F is a bounded quartic polynomial whose Hamiltonian field gains one weight (Theorem 5.16); (c) that the time-one map Γ of X_F is an analytic, real, symplectic, near-identity change of coordinates between balls of the weighted phase space (Theorems 5.21, 5.23, Corollary 5.27); (d) that $H \circ \Gamma = \Lambda + \bar{G} + G^{\mathrm{nor}} + K$ with $K = O(\|z\|^6)$ and a one-smoothing holomorphic field X_K (Theorem 5.25), in the form stated in §5.8 for milestones W4–W6.
Route. The flow of X_F is not constructed by an ODE existence theorem. Since X_F is a bounded cubic polynomial, its time-t map is written as an explicit power series $\Gamma_t = \sum_k t^k \varphi_k$ whose terms are continuous multilinear maps (the Lie series of the identity). The coefficient bounds follow by induction, the flow equation follows from a Leibniz identity, and analyticity is immediate from the power series. Symplecticity is the classical argument (the derivative of the pulled-back form vanishes). The remainder K is split as $(G \circ \Gamma - G) + (\Lambda \circ \Gamma - \Lambda - Z')$; the first part is handled by the gradient transport formula, the second by the homological identity, which cancels the unbounded part of X_Λ before any estimate.
Standing data. Standing assumption 1.1 ($m > 0$) and the conventions of §1 are in force. Throughout this part $a \geq 0$ and $p \geq \frac{1}{2}$ are fixed; the headline uses $a = p = 1$ (Convention 1.5). The tangential modes are $J = \{1, 2\}$ and the normal modes are $j \geq 3$. All constants are listed with their dependence on m, a, p in §5.9. The source fixes a general finite $J = \{1, \dots, n\}$; we keep n general only in Lemma 5.13.
Inputs from W1. The weighted spaces are those of Lemma 3.2 with the embedding $\ell^{a,p+1} \subset \ell^{a,p}$ of Lemma 3.5; the multiplier bound is Lemma 3.6; "holomorphic" is Definition 3.24. Facts needed here in a two-sided, complexified form are stated and proved below (Lemmas 5.3, 5.4, 5.8, 5.10; Remark 5.9).

5.1. External facts.

Each item names its Mathlib counterpart (checkout of §7 of the scoping report). "Banach" means a complete normed space over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$.

(E1) *Multilinear maps.* Continuous multilinear maps between Banach spaces form a Banach space under the operator norm; inserting a continuous multilinear map into one slot of another, or composing with continuous linear maps, gives a continuous multilinear map of norm at most

the product of the norms. The diagonal $P(w) = M(w, \dots, w)$ of a continuous d-linear M is C^∞, with $DP(w)v = \sum_{i=1}^{d} M(w, \dots, w, \underset{i}{v}, w, \dots, w)$. (Mathlib: `ContinuousMultilinearMap`, `ContinuousMultilinearMap.opNorm_le_bound`, `ContinuousMultilinearMap.hasFDerivAt`, `ContinuousMultilinearMap.compContinuousLinearMap`.)

(E2) *Power series.* If $(M_n)_{n\geq 0}$ are continuous n-linear maps with $\sum_n \|M_n\| r^n < \infty$, then $f(w) = \sum_n M_n(w, \dots, w)$ has this formal power series on the open ball of radius r, is analytic at every point of that ball, and its derivative is obtained termwise. Finite sums, compositions and continuous multilinear images of analytic maps are analytic; so is $A \mapsto A^{-1}$ on the units of a Banach algebra; and the derivative of an analytic map into a Banach space is analytic. (Mathlib: `FormalMultilinearSeries`, `HasFPowerSeriesOnBall`, `HasFPowerSeriesOnBall.analyticAt_of_mem`, `HasFPowerSeriesOnBall.fderiv`, `AnalyticAt.comp`, `analyticAt_inverse`, `AnalyticOnNhd.fderiv`, `ContinuousMultilinearMap.analyticOnNhd` and `ContinuousLinearMap.analyticOnNhd_uncurry_of_multilinear`.) A $\mathbb{C}$-analytic map is $\mathbb{R}$-analytic for the underlying real structures (`AnalyticOnNhd.restrictScalars`). A `FormalMultilinearSeries` indexed by degree carries explicit zero coefficients in the degrees that do not occur.

(E3) *Termwise differentiation.* If f_k are differentiable on an open convex set U of a normed space, $\sum_k f_k(x_0)$ converges and $\sup_U \|Df_k\| \leq u_k$ with $\sum u_k < \infty$, then $\sum_k f_k$ is differentiable on U with derivative $\sum_k Df_k$. Absolutely summable families in a Banach space may be regrouped along any partition of the index set; for a continuous multilinear B and absolutely summable families $(a_i^1), \dots, (a_i^r)$, $B(\sum a^1, \dots, \sum a^r) = \sum_{i_1,\dots,i_r} B(a_{i_1}^1, \dots, a_{i_r}^r)$, the right side absolutely summable (Cauchy products are the case of regrouping by total degree). (Mathlib: `hasFDerivAt_tsum_of_isPreconnected`, `hasDerivAt_tsum_of_isPreconnected`, `HasSum.sigma`, `Summable.sigma`, `tsum_mul_tsum_eq_tsum_sum_antidiagonal_of_summable_norm` as the ring prototype; the multilinear case follows by continuity of B and iterated regrouping.)

(E4) *Mean value inequality; zero derivative.* If f is differentiable on a convex set with $\|Df\| \leq M$, then $\|f(y) - f(x)\| \leq M\|y - x\|$; a differentiable function on an interval with vanishing derivative is constant. (Mathlib: `Convex.norm_image_sub_le_of_norm_hasFDerivWithin_le`, `is_const_of_deriv_eq_zero`.)

(E5) *Integrals in a parameter.* For $Y : [0,1] \times U \to V$ (V Banach, U open in a Banach space) continuous, differentiable in the second variable with $D_w Y$ continuous and bounded on $[0,1] \times U$: $w \mapsto \int_0^1 Y(t,w)\, dt$ is differentiable with derivative $\int_0^1 D_w Y(t,w)\, dt$; the fundamental theorem of calculus holds for C^1 functions $[0,1] \to V$; continuous linear maps commute with the Bochner integral; $\|\int Y\| \leq \int \|Y\|$; and a series $\sum_j g_j$ of continuous functions on $[0,1]$ with $\sup\|g_j\| \leq c_j$, $\sum c_j < \infty$, may be integrated termwise. Over $\mathbb{K} = \mathbb{C}$ the first statement yields complex differentiability. (Mathlib: `hasFDerivAt_integral_of_dominated_of_fderiv_le`, `intervalIntegral.integral_eq_sub_of_hasDerivAt`, `ContinuousLinearMap.intervalIntegral_comp_comm`, `intervalIntegral.norm_integral_le_of_norm_le`, `intervalIntegral.hasSum_integral_of_dominated_convergence`.)

(E6) *Neumann series.* If A is a bounded operator on a Banach space with $\|A - I\| \leq \frac{1}{2}$, then A is invertible, $\|A^{-1}\| \leq 2$ and $\|A^{-1} - I\| \leq 2\|A - I\|$. (Mathlib: `NormedRing.inverse_one_sub`, `Units.oneSub`, `geom_series_eq_inverse`, `tsum_geometric_le_of_norm_lt_one`.) In particular $A^{-1} = \sum_{n\geq 0}(I - A)^n$, absolutely convergent.

Remark 5.1 (Flows of analytic vector fields: what Mathlib provides). Mathlib provides local existence (Picard–Lindelöf, `IsPicardLindelof`, `IsPicardLindelof.exists_forall_mem_closedBall_eq_hasDerivWithinAt_lipschitzOnWith`)

and uniqueness (`ODE_solution_unique`) of solutions of Lipschitz ODEs in Banach spaces, and Gronwall's inequality. It does *not* provide analytic (or C^1) dependence of the flow on initial data, symplecticity of Hamiltonian flows, or the theorem that complex-differentiable maps on open subsets of a complex Banach space are analytic (only $f : \mathbb{C} \to E$ is covered: `DifferentiableOn.analyticOnNhd`). W1 constructs flows of general bounded holomorphic fields by Picard iteration and uniform limits (Lemma 3.30). This part does not use that lemma: the time-t map of the polynomial field X_F is constructed as an explicit power series (Definition 5.19), and symplecticity and invertibility are proved for it directly (Theorems 5.21, 5.23). This gives analyticity (not only holomorphy), the gain of one weight in $\Gamma_t - \mathrm{id}$, and explicit constants. Likewise the field X_K of the remainder is proved analytic by an explicit power series (Lemma 5.24); the holomorphic-implies-analytic theorem on Banach domains is never invoked. Uniqueness of ODE solutions is not used.

5.2. **Coordinates, symplectic form, Hamiltonian fields.**

Definition 5.2 (Two-sided coordinate space)**.** For $a \geq 0$ and $r \in \mathbb{R}$ let $\mathfrak{W}_{a,r}$ be the space of complex families $w = (w_l)_{l \in \mathbb{Z} \setminus 0}$ with

$$\|w\|_{a,r}^2 = \sum_{l \neq 0} |l|^{2r} e^{2a|l|} |w_l|^2 < \infty.$$

Write $\mathfrak{W} = \mathfrak{W}_{a,p}$ and $\mathfrak{W}^+ = \mathfrak{W}_{a,p+1}$, with norms $\|\cdot\|$ and $\|\cdot\|_+$. For $w \in \mathfrak{W}$ write $\tilde{w} = (w_l)_{|l| \leq 2}$ (tangential part) and $\hat{w} = (w_l)_{|l| \geq 3}$ (normal part), extended by zero. Put $\lambda_l' = \mathrm{sgn}(l)\lambda_{|l|}$. In Lean the carrier is `lp (fun _ : Bool × PNat => Complex) 2` in weighted coordinates (the pair $(\mathrm{sign}, |l|)$ indexes l), so that the norm is exactly the one displayed.

Source: ours; P96-W p. 282 (the coordinates $w_{\pm j}$ and the space $\ell_b^{a,s}$ restricted to $l \neq 0$).

Lemma 5.3 (Complex coordinates and real structure)**.** *Let $E^{\mathbb{C}} = \ell_{\mathbb{C}}^{a,p} \times \ell_{\mathbb{C}}^{a,p}$ with $\|(q,p)\|_E = \max(\|q\|_{a,p}, \|p\|_{a,p})$, and define $\mathcal{Z} : E^{\mathbb{C}} \to \mathfrak{W}$ by*

$$(\mathcal{Z}x)_j = \frac{q_j + ip_j}{\sqrt{2}}, \qquad (\mathcal{Z}x)_{-j} = \frac{q_j - ip_j}{\sqrt{2}} \qquad (j \geq 1).$$

Then $\mathcal{Z}$ is a $\mathbb{C}$-linear bijection with $\|x\|_E \leq \|\mathcal{Z}x\| \leq \sqrt{2}\|x\|_E$; it maps $E_{a,p+1}^{\mathbb{C}}$ onto $\mathfrak{W}^+$ with the same bounds. The map $\varrho(w)_l = \overline{w_{-l}}$ is a conjugate-linear isometric involution of $\mathfrak{W}$ and of $\mathfrak{W}^+$, and $\mathcal{Z}$ maps the real space $E_{a,p}$ bijectively onto its fixed set $\mathfrak{W}_{\mathbb{R}} = \{\varrho w = w\}$. On real points, $w_j = z_j$ and $w_{-j} = \bar{z}_j$ (Definition 1.11) and $w_j w_{-j} = |z_j|^2 = I_j$.

Proof. For complex q_j, p_j the parallelogram identity gives $|q_j + ip_j|^2 + |q_j - ip_j|^2 = 2(|q_j|^2 + |p_j|^2)$, so $\|\mathcal{Z}x\|^2 = \|q\|^2 + \|p\|^2 \in [\|x\|_E^2, 2\|x\|_E^2]$. The inverse is $q_j = (w_j + w_{-j})/\sqrt{2}$, $p_j = (w_j - w_{-j})/(i\sqrt{2})$. If q, p are real then $w_{-j} = \overline{w_j}$, i.e. $\varrho w = w$; conversely $\varrho w = w$ makes both formulas real. The weights depend on $|l|$ only, so ϱ is isometric. □

Source: ours; P96-W p. 281 (definition of z) and p. 282 ($z_j = w_j$, $\bar{z}_j = w_{-j}$).

Lemma 5.4 (Symplectic form)**.** $\Omega(w, w') = i \sum_{l \neq 0} \mathrm{sgn}(l)\, w_l w'_{-l}$ *is a continuous antisymmetric bilinear form on $\mathfrak{W}$ with $|\Omega(w,w')| \leq \|w\|\|w'\|$. It is nondegenerate: $\Omega(\zeta, e_{-l}) = i\,\mathrm{sgn}(l)\zeta_l$ for the unit vectors e_l, so $\Omega(\zeta, \cdot) = 0$ implies $\zeta = 0$. For real $x, x' \in E$, $\Omega(\mathcal{Z}x, \mathcal{Z}x') = \sum_j (q_j p'_j - p_j q'_j)$, the form $\sum_j dq_j \wedge dp_j$ of Definition 1.10. Moreover $\Omega(\varrho w, \varrho w') = \overline{\Omega(w, w')}$.*

Proof. Cauchy–Schwarz and $|l|^p e^{a|l|} \geq 1$ give the bound. Substituting the inverse formulas of Lemma 5.3, $q_j p'_j - p_j q'_j = i(w_j w'_{-j} - w_{-j} w'_j)$. The last identity follows from $\overline{\mathrm{sgn}(l)} = \mathrm{sgn}(l) = -\mathrm{sgn}(-l)$ after the substitution $l \mapsto -l$. □

Source: ours; P96-W p. 281 (symplectic structure $i \sum dz_j \wedge d\bar{z}_j$).

Definition 5.5 (Hamiltonian field; one-smoothing holomorphic class)**.** Let $U \subset \mathfrak{W}$ be open and $P : U \to \mathbb{C}$ complex differentiable. A map $X_P : U \to \mathfrak{W}$ is the *Hamiltonian field* of P if $DP(w)v = \Omega(X_P(w), v)$ for all $w \in U$, $v \in \mathfrak{W}$; by Lemma 5.4 it is unique when it exists, and $X_P(w)_l = -i \operatorname{sgn}(l)\, DP(w)e_{-l}$. P is *real* if $P(\varrho w) = \overline{P(w)}$ for $w \in U = \varrho U$; then $X_P \circ \varrho = \varrho \circ X_P$. The class $\mathcal{A}(U)$ consists of the real P that are analytic on U (Mathlib `AnalyticOnNhd Complex`) and whose Hamiltonian field exists, takes values in $\mathfrak{W}^+$, and is analytic as a map $U \to \mathfrak{W}^+$ (`AnalyticOnNhd Complex`). Analytic maps are holomorphic in the sense of Definition 3.24.

The equations $\dot w = X_P(w)$ are Hamilton's equations: on real points, by Lemma 5.4, $\mathcal{Z}^{-1}X_P = (\partial_p P, -\partial_q P)$, the convention of Definition 1.10. The reality statement holds because $DP(\varrho w)\varrho v = \overline{DP(w)v}$ and Lemma 5.4 give $\Omega(X_P(\varrho w), \varrho v) = \Omega(\varrho X_P(w), \varrho v)$ for all v.
Source: ours; P96-W p. 281 (the class $A(\ell^{a,s}, \ell^{a,s+1})$ of real analytic maps from a neighbourhood of the origin; here with an explicit complex domain, and real analyticity obtained through Lemma 5.6).

Lemma 5.6 (Real restrictions)**.** *Let $U \subset \mathfrak{W}$ be open with $\varrho U = U$, $V \in \{\mathfrak{W}, \mathfrak{W}^+\}$ with real form $E_V \in \{E_{a,p}, E_{a,p+1}\}$, and $f : U \to V$ analytic with $f \circ \varrho = \varrho \circ f$. Then $f^{\mathbb{R}}(x) = \mathcal{Z}^{-1} f(\mathcal{Z}x)$, $x \in \mathcal{Z}^{-1}(U) \cap E_{a,p}$, takes values in E_V and is real analytic (`AnalyticOnNhd Real`). Likewise, if $P : U \to \mathbb{C}$ is analytic and real, $x \mapsto P(\mathcal{Z}x)$ is real valued and real analytic.*

Proof. f is $\mathbb{R}$-analytic by (E2) (`AnalyticOnNhd.restrictScalars`). The maps $x \mapsto \mathcal{Z}x$ ($E_{a,p} \to \mathfrak{W}$) and $\Pi(y) = \mathcal{Z}^{-1}\big(\frac{1}{2}(y + \varrho y)\big)$ ($V \to E_V$) are continuous and $\mathbb{R}$-linear (Lemma 5.3), hence $\mathbb{R}$-analytic, and $\Pi(y) = \mathcal{Z}^{-1}y$ when $\varrho y = y$. Since $\varrho \mathcal{Z}x = \mathcal{Z}x$, $\varrho f(\mathcal{Z}x) = f(\mathcal{Z}x)$, so $f^{\mathbb{R}} = \Pi \circ f \circ \mathcal{Z}$, a composition of $\mathbb{R}$-analytic maps. For P use Re in place of Π. □

Source: ours.

Definition 5.7 (Smoothing symmetric forms)**.** A continuous symmetric d-linear form $\mathfrak{P} : \mathfrak{W}^d \to \mathbb{C}$ ($d \geq 2$) is *smoothing with constant c* if there is a continuous $(d-1)$-linear map $\nabla\mathfrak{P} : \mathfrak{W}^{d-1} \to \mathfrak{W}^+$ with $\mathfrak{P}(w^1, \dots, w^{d-1}, v) = \Omega(\nabla\mathfrak{P}(w^1, \dots, w^{d-1}), v)$ and $\|\nabla\mathfrak{P}(w^1, \dots, w^{d-1})\|_+ \leq c \prod \|w^i\|$. Its diagonal $P(w) = \mathfrak{P}(w, \dots, w)$ then has $X_P(w) = d\,\nabla\mathfrak{P}(w, \dots, w)$ by (E1), and on every ball:

$$|P(w)| \leq c\|w\|^d, \qquad \|X_P(w)\|_+ \leq dc\|w\|^{d-1},$$
$$\|X_P(y) - X_P(w)\|_+ \leq d(d-1)c \max(\|y\|, \|w\|)^{d-2}\|y - w\|.$$

Source: ours.

Lemma 5.8 (The linear part)**.** $\Lambda(w) = \sum_{j \geq 1} \lambda_j w_j w_{-j}$ *is a continuous quadratic form on $\mathfrak{W}$ with $|\Lambda(w)| \leq \sqrt{1+m}\,\|w\|^2$, and $\Lambda(\mathcal{Z}x) = \frac{1}{2}\sum \lambda_j (q_j^2 + p_j^2)$ for real x (Definition 1.8). Its derivative is $D\Lambda(w)v = \sum_{l \neq 0} \lambda_{|l|} w_{-l} v_l$ for all $w, v \in \mathfrak{W}$. For $w \in \mathfrak{W}^+$ its Hamiltonian field is $X_\Lambda(w)_l = -i\lambda'_l w_l$, a bounded linear map $\mathfrak{W}^+ \to \mathfrak{W}$ of norm $\leq \sqrt{1+m}$; X_Λ does not map $\mathfrak{W}$ into itself.*

Proof. $\lambda_j \leq \sqrt{1+m}\, j \leq \sqrt{1+m}\, j^{2p} e^{2aj}$ since $p \geq \frac{1}{2}$, and Cauchy–Schwarz gives the bound. The field is computed from Definition 5.5: $\Omega(X_\Lambda(w), v) = \sum_l \lambda_{|l|} w_l v_{-l}$. The operator bound is Lemma 3.6(i) with $r = 1$. □

Source: P96-W p. 280 (X_Λ unbounded of order one) and p. 281 ($\Lambda = \sum \lambda_j |z_j|^2$).

Remark 5.9 (Requests to W1). Lemmas 5.3, 5.4, 5.8 and 5.10 are not in the W1 plan. They are proved here; if W1 adopts them, the labels here become citations. Lemma 5.3 is the pairing of Lemma 3.10(iii) composed with $(q, p) \mapsto (z, \bar z)$. Proposition 3.18 is stated for real $E_{a,p}$; this part uses its complexified, two-sided form (Lemma 5.11).

5.3. **The quartic Hamiltonian in two-sided coordinates.** For $x=(x_1,x_2,x_3,x_4)\in(\mathbb{Z}\setminus 0)^4$ write $|x|=(|x_1|,\dots,|x_4|)$, $w_x=w_{x_1}w_{x_2}w_{x_3}w_{x_4}$, and

$$\mathfrak{g}_x=\frac{g_{|x|}}{\sqrt{\lambda_{|x_1|}\lambda_{|x_2|}\lambda_{|x_3|}\lambda_{|x_4|}}},\qquad \delta_x=\lambda'_{x_1}+\lambda'_{x_2}+\lambda'_{x_3}+\lambda'_{x_4},$$

with g from Lemma 2.2. Call x *paired* if the multiset $\{x_t : x_t>0\}$ equals $\{-x_t : x_t<0\}$, i.e. x is a permutation of $(r,-r,s,-s)$; paired x have $\delta_x=0$. Let $\mathcal{T}=\{x : \min_t|x_t|\le 2\}$ (at least one tangential index).

Lemma 5.10 (Majorant for the cubic sums)**.** *For nonnegative $u,u',u''\in\ell^{a,p}$ put $\mathcal{M}(u,u',u'')_l=\lambda_l^{-1/2}\sum_{i,j,k\ge1}|g_{ijkl}|\frac{u_iu'_ju''_k}{\sqrt{\lambda_i\lambda_j\lambda_k}}$. The sums converge and*

$$\|\mathcal{M}(u,u',u'')\|_{a,p+1}\le C_*\|u\|_{a,p}\|u'\|_{a,p}\|u''\|_{a,p},\qquad C_*=\frac{K_{p+1/2}^2}{\pi},$$

with K_s from Lemma 2.4. C_* does not depend on m; for $a=p=1$, $C_*=8(1+2\zeta(3))/\pi$.

Proof. Steps S2.6–S2.8 of the proof of Theorem 2.5 bound exactly the absolute series $\sum|g_{ijkl}||\hat q_i\hat q'_j\hat q''_k|$; apply them with $q=u$, $q'=u'$, $q''=u''$. □

Source: P96-W (13)–(14), p. 284 (constant "c"); the constant is that of Theorem 2.5.

Lemma 5.11 (Quartic forms)**.** *For $\mathcal{S}\subset(\mathbb{Z}\setminus 0)^4$ invariant under permutations of the four entries and under $x\mapsto -x$, and coefficients (c_x) with $c_x=c_{\sigma x}=\overline{c_{-x}}$ and $|c_x|\le|\mathfrak{g}_x|$, the form*

$$\mathfrak{P}(w^1,w^2,w^3,w^4)=\frac{1}{16}\sum_{x\in\mathcal{S}}c_x\,w^1_{x_1}w^2_{x_2}w^3_{x_3}w^4_{x_4}$$

converges absolutely, is continuous, symmetric, and smoothing with constant $C_*/4$, with $\nabla\mathfrak{P}(w^1,w^2,w^3)_l=-\frac{i\,\mathrm{sgn}(l)}{16}\sum_{y:(y,-l)\in\mathcal{S}}c_{(y,-l)}w^1_{y_1}w^2_{y_2}w^3_{y_3}$; its diagonal P is real. In particular, with $c=\mathfrak{g}$ restricted to the sets $(\mathbb{Z}\setminus0)^4$, $\mathcal{T}$, $\mathcal{T}\cap\{\text{paired}\}$, $\mathcal{T}^c$, we obtain the forms $\mathfrak{G}$, $\mathfrak{G}^{\tan}$, $\bar{\mathfrak{G}}$, $\mathfrak{G}^{\mathrm{nor}}$ and their diagonals $G,G^{\tan},\bar G,G^{\mathrm{nor}}$; $G=G^{\tan}+G^{\mathrm{nor}}$; and

$$\|X_P(w)\|_+\le C_*\|w\|^3,\qquad |P(w)|\le\tfrac{C_*}{4}\|w\|^4\qquad(P\in\{G,G^{\tan},\bar G,G^{\mathrm{nor}}\}).$$

Proof. **Step S5.1: Majorize.** Given w^1,w^2,w^3 let $u_i^t=|w_i^t|+|w_{-i}^t|$ $(i\ge1)$, so $\|u^t\|_{a,p}\le\sqrt{2}\|w^t\|$. Summing over the signs of $y\in(\mathbb{Z}\setminus0)^3$, $\sum_y|\mathfrak{g}_{(y,-l)}||w^1_{y_1}w^2_{y_2}w^3_{y_3}|\le\mathcal{M}(u^1,u^2,u^3)_{|l|}$.

Step S5.2: Bound the gradient. Hence, by Lemma 5.10, $\|\nabla\mathfrak{P}\|_+^2\le2\cdot16^{-2}\|\mathcal{M}(u^1,u^2,u^3)\|_{a,p+1}^2$, so $\|\nabla\mathfrak{P}\|_+\le\frac{\sqrt2}{16}C_*(\sqrt2)^3\prod\|w^t\|=\frac{C_*}{4}\prod\|w^t\|$.

Step S5.3: Identify. By Lemma 5.4, $\Omega(\nabla\mathfrak{P}(w^1,w^2,w^3),v)=i\sum_l\mathrm{sgn}(l)\nabla\mathfrak{P}(\cdot)_lv_{-l}=\frac{1}{16}\sum_l\sum_y c_{(y,-l)}w^1_{y_1}w^2_{y_2}w^3_{y_3}v_{-l}$, which is $\mathfrak{P}(w^1,w^2,w^3,v)$ after $l\mapsto-l$; the four-fold series converges absolutely by Step S5.1 and Cauchy–Schwarz against v. Symmetry follows from the permutation invariance of $\mathcal{S}$ and c. Reality: $P(\varrho w)=\frac{1}{16}\sum c_x\overline{w_{-x}}=\frac{1}{16}\sum_x c_{-x}\overline{w_x}=\overline{P(w)}$. The bounds are Definition 5.7 with $d=4$, $c=C_*/4$. $\mathfrak{g}$ is invariant under permutations and $x\mapsto-x$ and real; the four sets are invariant; $(\mathbb{Z}\setminus0)^4=\mathcal{T}\sqcup\mathcal{T}^c$. □

Source: P96-W p. 282 (expansion of H in the w-coordinates) and (13)–(14), p. 284. The prefactor $\frac{1}{16}$ is the corrected one (E-PDE-W-1, Remark 2.12).

Lemma 5.12 (Identification with Part 0)**.** *For real $x=(q,p)\in E_{a,p}$: $G(\mathcal{Z}x)=G(q)$ of Definition 1.8 when $a>0$ (for $a=0$ this is the definition of G on $E_{0,p}$); $\mathcal{Z}^{-1}X_G(\mathcal{Z}x)=(0,-T(q,q,q))$ with T from Theorem 2.5, which is $(0,-G_q(q))$ for $a>0$; $G^{\mathrm{nor}}(w)=G^{\mathrm{nor}}(\hat w)$ and $X_{G^{\mathrm{nor}}}(w)$ has*

vanishing tangential components, so $|G^{\mathrm{nor}}(w)| \le \frac{C_*}{4}\|\hat{w}\|^4$ *and* $\|X_{G^{\mathrm{nor}}}(w)\|_+ \le C_*\|\hat{w}\|^3$*; and for all* $w \in \mathfrak{W}$*, with* $I_j := w_j w_{-j}$*,*

$$\bar{G}(w) = \tfrac{1}{2}\textstyle\sum_{i,j\in J} A_{ij} I_i I_j + \sum_{j\ge 3,\, i\in J} B_{ji} I_i I_j,$$

with the corrected A*,* B *of Theorem 2.10. Finally* $x \in \mathcal{T}$ *with* $\mathfrak{g}_x \neq 0$ *has* $\delta_x = 0$ *if and only if* x *is paired.*

Proof. **Step S5.4: The quartic.** Expanding $\prod_t (z_{i_t} + \bar{z}_{i_t})$ in (5) into the 16 sign choices gives $G(\mathcal{Z}x) = \frac{1}{16}\sum_x \mathfrak{g}_x w_x$; absolute convergence is Lemma 5.11, and $G(q)$ agrees with (5) by Lemma 2.3.

Step S5.5: The field. On real points $\mathcal{Z}^{-1} X_G = (\partial_p G, -\partial_q G)$ (paragraph after Definition 5.5), and $\partial_p G = 0$. For every $a \ge 0$ (including $a = 0$), Step S5.4 gives $G(\mathcal{Z}x) = \frac{1}{4}\mathcal{Q}(q,q,q,q)$ with $\mathcal{Q}(q^1,q^2,q^3,q^4) = \sum_{i,j,k,l\ge 1} g_{ijkl}\hat{q}_i^1\hat{q}_j^2\hat{q}_k^3\hat{q}_l^4$ ($\hat{q}_j = q_j/\sqrt{\lambda_j}$). This series converges absolutely on $(\ell^{a,p})^4$: summing over i,j,k first, its absolute series is $\sum_l |q_l^4|\,\mathcal{M}(|q^1|,|q^2|,|q^3|)_l \le \|q^4\|_{a,p}\|\mathcal{M}(|q^1|,|q^2|,|q^3|)\|_{a,p+1} \le C_*\prod_t \|q^t\|_{a,p}$ (Cauchy–Schwarz, weights ≥ 1, Lemma 5.10). Hence $\mathcal{Q}$ is 4-linear, and symmetric by the symmetry of g (Lemma 2.2). Multilinearity gives $G(\mathcal{Z}(q + te_l, p)) = G(\mathcal{Z}x) + t\,\mathcal{Q}(q,q,q,e_l) + t^2(\cdots)$, a polynomial in t, so $\partial_{q_l} G = \mathcal{Q}(q,q,q,e_l) = T(q,q,q)_l$ by the definition of T in Theorem 2.5. For $a > 0$ this is also the formula of Lemma 2.3 (equivalently Proposition 3.17).

Step S5.6: Normal part. Every $x \in \mathcal{T}^c$ has all $|x_t| \ge 3$.

Step S5.7: Count the paired terms. For integers $1 \le r < s$ with $r \in J$: the ordered x that are permutations of $(r,-r,s,-s)$ are 24 in number, with $\mathfrak{g}_x = g_{rrss}/(\lambda_r\lambda_s) = 1/(\pi\lambda_r\lambda_s)$ and $w_x = I_r I_s$, contributing $\frac{24}{16\pi\lambda_r\lambda_s} I_r I_s = \frac{3}{2\pi\lambda_r\lambda_s} I_r I_s$. For $r = s \in J$ there are 6 orderings, $\mathfrak{g}_x = \frac{3}{2\pi\lambda_r^2}$, contributing $\frac{9}{16\pi\lambda_r^2} I_r^2 = \frac{1}{2} A_{rr} I_r^2$. These are the coefficients of Theorem 2.10.

Step S5.8: Resonance. $\mathfrak{g}_x \neq 0$ and $\delta_x = 0$ force pairing by Lemma 2.7 (applied to the indices $|x|$ with signs $\operatorname{sgn} x_t$). □

Source: P96-W pp. 282–284; Theorem 2.10 for the coefficients.

5.4. **Quantitative small divisors.**

Lemma 5.13 (Divisor lower bound; P96-W Lemma 4)**.** *Let* $x \in (\mathbb{Z}\setminus 0)^4$ *satisfy the selection rule* $\varepsilon_1|x_1| + \cdots + \varepsilon_4|x_4| = 0$ *for some* $\varepsilon \in \{\pm 1\}^4$*, and let* x *be not paired. With* $n = \min_t |x_t|$*,*

$$|\delta_x| \ge \frac{m}{27\,(n^2+m)^{3/2}}.$$

Consequently, if $x \in \mathcal{T}$*,* $\mathfrak{g}_x \neq 0$ *and* x *is not paired, then* $|\delta_x| \ge \delta_*(m) := \dfrac{m}{27(4+m)^{3/2}}$.

Proof. Let $a \le b \le c \le d$ be $|x_1|,\dots,|x_4|$ sorted, so $n = a$, and write $f(s) = \sqrt{s^2+m}$, $\lambda_k = f(k)$. Recall $f'(s) = s/f(s)$ is increasing and $f''(s) = m/f(s)^3$ is positive and decreasing on $s \ge 0$. Put $\beta = m/(27\lambda_a^3)$ and note $\beta \le \frac{1}{27\lambda_a}$ because $m \le \lambda_a^2$.

Step S5.9: Selection rule. We show $d = a+b+c$ or $d = b+c-a$. Solving the relation for d gives $d = \pm a \pm b \pm c$. Since $1 \le a \le b \le c \le d$: $d = a-b+c \le c$ forces $a = b$, $d = c$, hence $d = b+c-a$; $d = a+b-c \le b$ forces $a = b = c = d$, hence $d = b+c-a$; every choice with at least two minus signs gives $d \le c-a < c$ or $d \le 0$, impossible.

Step S5.10: Signs. $|\delta_x|$ is unchanged by $x \mapsto -x$, so assume at most two of the x_t are negative. Pairing and δ_x depend only on the signed multiset, so it suffices to bound, for each sign pattern on (a,b,c,d), the corresponding divisor.

Step S5.11: No minus sign. $\delta = \lambda_a + \lambda_b + \lambda_c + \lambda_d \ge 4 > \beta$.

Step S5.12: One minus sign. Each such divisor is $\geq \lambda_a + \lambda_b + \lambda_c - \lambda_d$. Since $f(k) - k = m/(f(k)+k)$,

$$\lambda_a + \lambda_b + \lambda_c - \lambda_d = (a+b+c-d) + m\Big(\tfrac{1}{\lambda_a+a} + \tfrac{1}{\lambda_b+b} + \tfrac{1}{\lambda_c+c} - \tfrac{1}{\lambda_d+d}\Big) \geq \frac{m}{2\lambda_a} \geq \beta,$$

using $a+b+c \geq d$ (Step S5.9), $\lambda_c + c \leq \lambda_d + d$ and $\lambda_a \geq 1$.

Step S5.13: Two minus signs, outer pairs. $\delta_3 = \lambda_c + \lambda_d - \lambda_a - \lambda_b$ vanishes identically iff $a = b = c = d$ (then x is paired); otherwise $c > a$ or $d > b$, and since $\lambda_{k+1} - \lambda_k \geq f'(k) \geq f'(a) = a/\lambda_a$ for $k \geq a$, $\delta_3 \geq a/\lambda_a \geq \beta$. Likewise $\delta_2 = (\lambda_b - \lambda_a) + (\lambda_d - \lambda_c)$ is paired iff $a = b$ and $c = d$, and otherwise $\delta_2 \geq a/\lambda_a \geq \beta$.

Step S5.14: Two minus signs, inner pair. $\delta_1 = \lambda_a + \lambda_d - \lambda_b - \lambda_c$ is paired iff $a = b$ and $c = d$. Otherwise:

(i) if $d = b + c - a$, put $t = b - a = d - c \geq 1$ (if $t = 0$ the tuple is paired). As f' increases and $c \geq b$, $\lambda_d - \lambda_c = \int_c^{c+t} f' \geq \int_b^{b+t} f'$, so $\delta_1 \geq \int_a^{a+t} \big(f'(s+t) - f'(s)\big)\,ds \geq \int_a^{a+1}\int_s^{s+1} f''(r)\,dr\,ds \geq f''(a+2)$, and $f''(a+2) = m/((a+2)^2+m)^{3/2} \geq m/(27\lambda_a^3)$ because $(a+2)^2 \leq 9a^2$;

(ii) if $d = a+b+c$, then $\lambda_d - \lambda_c \geq \lambda_{d-(c-a)} - \lambda_a = \lambda_{2a+b} - \lambda_a$ (shift both endpoints down by $c-a$), so $\delta_1 \geq \lambda_{b+2a} - \lambda_b \geq 2af'(b) \geq 2/\sqrt{1+m} \geq \beta$, the last since $m/\lambda_a^3 \leq m/(\sqrt{1+m}\,\lambda_a^2) \leq 1/\sqrt{1+m}$.

Step S5.15: Conclusion. Patterns with two minus signs are exactly $\pm\delta_1, \pm\delta_2, \pm\delta_3$, and pairing of x coincides with the identical vanishing in each case. The corollary: $\mathfrak{g}_x \neq 0$ implies the selection rule (Lemma 2.2), $n \leq 2$ on $\mathcal{T}$, and $n \mapsto m/(n^2+m)^{3/2}$ is decreasing. □

Source: P96-W Lemma 4, p. 283; proof pp. 285–286 ("some absolute constant c"; here $c = \frac{1}{27}$).
SOURCE_GAP: Two steps of the printed proof are incorrect, although the lemma is true. (1) P96-W p. 285, Case 1: "$\delta'(m) = \frac{1}{2}(\frac{1}{\lambda_i} + \frac{1}{\lambda_j} + \frac{1}{\lambda_k} - \frac{1}{\lambda_l}) \geq \frac{1}{\lambda_i}$" is false: for $(i,j,k,l) = (1,10,10,19)$ (selection rule $1-10-10+19 = 0$) and $m = 1$ the left side is ≈ 0.427 while $1/\lambda_1 \approx 0.707$. The correct bound is $\geq \frac{1}{2\lambda_i}$, which halves the constant (Step S5.12 avoids calculus). (2) P96-W p. 286, Case 2: "it suffices to study $\delta = \delta_{+--+}$" because $\delta_{-+-+}, \delta_{--++} \geq \delta_{+--+}$. When $i = j < k = l$ the divisor δ_{+--+} is resonant (identically 0) while $\delta_{--++} = 2(\lambda_k - \lambda_i)$ is not, so the reduction gives only $\delta_{--++} \geq 0$; the missing bound is Step S5.13. Also on p. 286, "$\lambda_{j+2i} - \lambda_j \geq 2(\lambda_{j+1} - \lambda_j) \geq 2if'(j)$" should read $\geq 2i(\lambda_{j+1} - \lambda_j)$; the final bound is unaffected.

Remark 5.14. The bound is attained in order: $\lambda_{n+1} - 2\lambda_n + \lambda_{n-1} \sim m/n^3$ (P96-W p. 286). For the headline, $\delta_*(m) \sim m/216$ as $m \to 0$ and $\delta_*(m) \sim 1/(27\sqrt{m})$ as $m \to \infty$; no uniformity in m is claimed (Standing assumption 1.1).

5.5. The generating Hamiltonian.

Definition 5.15 (Generating Hamiltonian)**.** For $x \in \mathcal{T}$ not paired put $f_x = \mathfrak{g}_x/\delta_x$, and $f_x = 0$ otherwise. (In Lean, division by 0 returns 0; Lemma 5.13 guarantees $\delta_x \neq 0$ wherever $\mathfrak{g}_x \neq 0$ and f_x is used.) Let

$$\mathfrak{F}(w^1, \dots, w^4) = \frac{i}{16} \sum_x f_x\, w^1_{x_1} w^2_{x_2} w^3_{x_3} w^4_{x_4}, \qquad F(w) = \mathfrak{F}(w,w,w,w),$$

and $S = 4\nabla\mathfrak{F} : \mathfrak{W}^3 \to \mathfrak{W}^+$, so that $X_F(w) = S(w,w,w)$.

Source: P96-W (12), p. 282, with the corrected prefactor $\frac{1}{16}$ and the time convention of Definition 1.11 (the printed $iF_{ijkl} = G_{ijkl}/\delta$ becomes $F = \frac{i}{16}\sum \mathfrak{g}/\delta\, w^4$); the same F as in Proposition 2.11.

Theorem 5.16 (F is bounded and one-smoothing)**.** *$\mathfrak{F}$ is a continuous symmetric 4-linear form, smoothing with constant $C_*/(4\delta_*)$, and F is real. Hence S is a continuous symmetric trilinear map*

$\mathfrak{W}^3 \to \mathfrak{W}^+$ *with*

$$\|S(w^1,w^2,w^3)\|_+ \le C_F\|w^1\|\|w^2\|\|w^3\|, \qquad C_F = C_F(m) = \frac{C_*}{\delta_*(m)} = \frac{27\,C_*\,(4+m)^{3/2}}{m},$$

$\Omega(S(w^1,w^2,w^3),v) = 4\mathfrak{F}(w^1,w^2,w^3,v)$ *is symmetric in all four arguments,* $S(\varrho w^1,\varrho w^2,\varrho w^3) = \varrho S(w^1,w^2,w^3)$*, and* $|F(w)| \le \frac{C_F}{4}\|w\|^4$*.* X_F *is a cubic polynomial, analytic on* $\mathfrak{W}$ *with values in* $\mathfrak{W}^+$*.*

Proof. **Step S5.16: Coefficients.** f is invariant under permutations; $f_{-x} = \mathfrak{g}_x/(-\delta_x) = -f_x$; and $|f_x| \le |\mathfrak{g}_x|/\delta_*$ by Lemma 5.13. So $c_x := i\delta_* f_x$ satisfies the hypotheses of Lemma 5.11 ($c_{-x} = i\delta_* f_{-x} = -i\delta_* f_x$, so $\overline{c_{-x}} = i\delta_* f_x = c_x$), and $\mathfrak{F} = \delta_*^{-1}\mathfrak{P}_c$.

Step S5.17: Conclusions. Lemma 5.11 gives smoothing with constant $C_*/(4\delta_*)$ and reality; $S = 4\nabla\mathfrak{F}$ gives C_F. The reality of S follows from the reality of $\mathfrak{F}$ as in Definition 5.5, applied to the multilinear form. Analyticity: (E2). □

Source: P96-W (13)–(14), p. 284.

Proposition 5.17 (Homological identity)**.** *Let* $Z' = \bar G - G^{\tan}$*, the diagonal of* $\mathfrak{z} = \bar{\mathfrak{G}} - \mathfrak{G}^{\tan}$ *(a form as in Lemma 5.11, smoothing with constant* $C_*/4$*, so* $\|X_{Z'}(w)\|_+ \le C_*\|w\|^3$*). For every* $w \in \mathfrak{W}$*,*

$$D\Lambda(w)\,X_F(w) = Z'(w), \qquad \textit{equivalently} \qquad G + D\Lambda\cdot X_F = \bar G + G^{\mathrm{nor}}.$$

Proof. **Step S5.18: Field coordinates.** By Definition 5.15 and Lemma 5.11, $X_F(w)_l = 4\nabla\mathfrak{F}(w,w,w)_l = \frac{\operatorname{sgn}(l)}{4}\sum_y f_{(y,-l)}w_y$.

Step S5.19: Euler identity. By Lemma 5.8, $D\Lambda(w)X_F(w) = \sum_l \lambda_{|l|}w_{-l}X_F(w)_l$. Substituting $l = -r$ and $\lambda_{|r|}\operatorname{sgn}(-r) = -\lambda'_r$, this equals $-\frac14\sum_r \lambda'_r w_r\sum_y f_{(y,r)}w_y = -\frac{1}{16}\sum_x \delta_x f_x w_x$, by symmetrizing over the position of r in x (f symmetric). All series converge absolutely: $\lambda_{|l|} \le \sqrt{1+m}|l|^{p+1}e^{a|l|}$ and $\|X_F(w)\|_+ < \infty$ with Cauchy–Schwarz.

Step S5.20: Divide. $\delta_x f_x = \mathfrak{g}_x$ for $x \in \mathcal{T}$ not paired (both vanish if $\mathfrak{g}_x = 0$; Lemma 5.13 otherwise) and $\delta_x f_x = 0$ else. So the sum is $-\frac{1}{16}\sum_{\mathcal{T}\setminus\text{paired}}\mathfrak{g}_x w_x = \bar G - G^{\tan}$ by Lemma 5.11. The second form uses $G = G^{\tan} + G^{\mathrm{nor}}$. □

Source: P96-W p. 283 ("$G + \{\Lambda, F\} = \bar G + \hat G$"); the identity of Proposition 2.11 with $\{\Lambda,F\} = D\Lambda\cdot X_F$, now on all of $\mathfrak{W}$.

Remark 5.18. The identity holds on $\mathfrak{W}$ although X_Λ is defined only on $\mathfrak{W}^+$: it is stated with $D\Lambda(w)$, which is a bounded functional on $\mathfrak{W}$ (Lemma 5.8). This is what lets the remainder estimate below avoid the unbounded field.

5.6. The Lie series of the flow.

Definition 5.19 (Lie coefficients)**.** Define continuous $(2k+1)$-linear maps $\Phi_k : \mathfrak{W}^{2k+1} \to \mathfrak{W}$ by $\Phi_0(w) = w$ and

$$\Phi_{k+1}(w^1,\dots,w^{2k+3}) = \frac{1}{k+1}\sum_{i=1}^{2k+1}\Phi_k\big(w^1,\dots,w^{i-1},S(w^i,w^{i+1},w^{i+2}),w^{i+3},\dots,w^{2k+3}\big),$$

and let $\varphi_k(w) = \Phi_k(w,\dots,w)$. For a C^1 map Q from an open subset of $\mathfrak{W}$ to a Banach space, let $(LQ)(w) = DQ(w)\,X_F(w)$.

Source: ours (the Lie series $\Gamma_t = \sum_k t^k L^k(\mathrm{id})/k!$; P96-W p. 283 uses the time-one map of X_F without constructing it).

Lemma 5.20 (Coefficient bounds and the Leibniz identity)**.** *For all* $k \ge 0$*:*

(i) $\varphi_{k+1} = \frac{1}{k+1} L\varphi_k$, *hence* $\varphi_k = L^k(\mathrm{id})/k!$;
(ii) $\|\Phi_k\| \le (2C_F)^k$; *for* $k \ge 1$, Φ_k *takes values in* $\mathfrak{W}^+$ *and* $\|\Phi_k\|_{\mathfrak{W}^{2k+1}\to\mathfrak{W}^+} \le (2C_F)^k$;
(iii) $\Phi_k(\varrho w^1, \dots) = \varrho\Phi_k(w^1, \dots)$;
(iv) $(k+1)\,\varphi_{k+1} = \sum_{a+b+c=k} S(\varphi_a, \varphi_b, \varphi_c)$ *and*

$$(k+1)\,D\varphi_{k+1}(w)v = \sum_{a+b+c=k} 3\,S\big(D\varphi_a(w)v, \varphi_b(w), \varphi_c(w)\big).$$

Proof. **Step S5.21: Recursion.** (i) is (E1): $D\varphi_k(w)v = \sum_i \Phi_k(w, \dots, \underset{i}{v}, \dots, w)$ with $v = S(w,w,w)$.

Step S5.22: Norms. By (E1), $\|\Phi_{k+1}\| \le \frac{2k+1}{k+1} C_F\|\Phi_k\| \le 2C_F\|\Phi_k\|$ (as $\|\cdot\| \le \|\cdot\|_+$). For values in $\mathfrak{W}^+$: $\Phi_1 = S$, and for $k \ge 1$ each summand is Φ_k (into $\mathfrak{W}^+$) applied to arguments in $\mathfrak{W}$, so the same recursion holds for the $\mathfrak{W}^+$-valued norms.

Step S5.23: Reality. Induction from Theorem 5.16.

Step S5.24: Leibniz rule. For a continuous multilinear B and C^∞ maps $Q_1, \dots, Q_r$, the product and chain rules give $L\,B(Q_1, \dots, Q_r) = \sum_i B(Q_1, \dots, LQ_i, \dots, Q_r)$, and by induction on n, $L^n B(Q_1, \dots, Q_r) = \sum_{n_1+\dots+n_r=n} \frac{n!}{n_1!\cdots n_r!} B(L^{n_1}Q_1, \dots, L^{n_r}Q_r)$.

Step S5.25: The identity. $L\,\mathrm{id} = X_F = S(\mathrm{id}, \mathrm{id}, \mathrm{id})$, so by (i) and Step S5.24, $(k+1)!\,\varphi_{k+1} = L^k(S(\mathrm{id}, \mathrm{id}, \mathrm{id})) = \sum_{a+b+c=k} \frac{k!}{a!b!c!} S(a!\varphi_a, b!\varphi_b, c!\varphi_c)$. Differentiating and using the symmetry of S gives the second identity. □

Source: ours.

Theorem 5.21 (The flow of X_F)**.** *Let* $\rho_0 = \rho_0(m) = (40C_F)^{-1/2}$, $B_\rho = \{w \in \mathfrak{W} : \|w\| < \rho\}$, *and for real* $|t| < 2$ *and* $w \in B_{\rho_0}$

$$\Gamma_t(w) = \sum_{k\ge 0} t^k \varphi_k(w).$$

Then, with $r = 2|t|C_F\|w\|^2 \le \frac{1}{10}$*:*

(i) (Analyticity.) *The series converges absolutely;* $w \mapsto \Gamma_t(w)$ *has the power series* $(t^k\Phi_k)$ *on* B_{ρ_0} *and is analytic there;* $\Gamma_t - \mathrm{id}$ *is analytic* $B_{\rho_0} \to \mathfrak{W}^+$; $(t, w) \mapsto \Gamma_t(w)$ *is* C^1 *on* $(-2, 2) \times B_{\rho_0}$.
(ii) (Flow and transport equations.) $\Gamma_0 = \mathrm{id}$, $\partial_t\Gamma_t(w) = X_F(\Gamma_t(w))$ *and* $\partial_t\Gamma_t(w) = D\Gamma_t(w)X_F(w)$.
(iii) (Estimates.) $\|\Gamma_t(w)\| \le \frac{10}{9}\|w\|$, $\|\Gamma_t(w) - w\|_+ \le 3|t|C_F\|w\|^3$, *and* $\|D\Gamma_t(w) - I\|_{\mathfrak{W}\to\mathfrak{W}^+} \le 10|t|C_F\|w\|^2 \le \frac{1}{2}$. *Consequently* $D\Gamma_t(w)$ *maps* $\mathfrak{W}^+$ *into itself and is invertible on* $\mathfrak{W}$ *and on* $\mathfrak{W}^+$, *with inverses of norm* ≤ 2 *and* $\|D\Gamma_t(w)^{-1} - I\|_{\mathfrak{W}\to\mathfrak{W}^+} \le 20|t|C_F\|w\|^2$.
(iv) (Reality.) $\Gamma_t(\varrho w) = \varrho\Gamma_t(w)$.
(v) (Variational equation.) $\partial_t D\Gamma_t(w)v = DX_F(\Gamma_t(w))\,D\Gamma_t(w)v$, *where* $DX_F(y)\xi = 3S(y, y, \xi)$.
(vi) (Inverse.) *With* $\rho_1 = \rho_0/2$*: for* $|t| \le 1$ *and* $w \in B_{\rho_1}$, $\Gamma_{-t}(\Gamma_t(w)) = w$. *Hence* $\Gamma = \Gamma_1$ *is injective on* B_{ρ_1}, $\Gamma(B_{\rho_1}) \subset B_{10\rho_1/9}$, *and* $\Gamma(B_{\rho_1}) \supset B_{9\rho_1/10}$.

Proof. **Step S5.26: Convergence and analyticity.** By Lemma 5.20(ii), $|t|^k\|\varphi_k(w)\| \le r^k\|w\|$ and $r \le 4C_F\rho_0^2 = \frac{1}{10}$; the formal multilinear series with coefficient $t^k\Phi_k$ in degree $2k+1$ and 0 in degree 0 and every even degree has $\sum_k |t|^k\|\Phi_k\|\rho^{2k+1} < \infty$ for $\rho < \rho_0$, so (E2) applies, in $\mathfrak{W}$ and (for $k \ge 1$) in $\mathfrak{W}^+$. The partial derivatives of $t^k\varphi_k(w)$ in t and w are bounded by $k|t|^{k-1}(2C_F)^k\rho^{2k+1}$ and $(2k+1)|t|^k(2C_F)^k\rho^{2k}$ on $|t| < 2$, $\|w\| < \rho < \rho_0$; these are summable, so (E3) gives joint C^1 with termwise derivatives.

Step S5.27: Flow equation. Termwise, $\partial_t\Gamma_t = \sum_k (k+1)t^k\varphi_{k+1} = \sum_k t^k \sum_{a+b+c=k} S(\varphi_a,\varphi_b,\varphi_c)$ (Lemma 5.20(iv)), which is $S(\Gamma_t,\Gamma_t,\Gamma_t)$ by the Cauchy product of absolutely convergent series by (E3) and continuity of S.

Step S5.28: Transport equation. Termwise, $\partial_t\Gamma_t = \sum_k t^k L\varphi_k = \big(\sum_k t^k D\varphi_k\big)X_F = D\Gamma_t\, X_F$ by Lemma 5.20(i) and Step S5.26.

Step S5.29: Estimates. $\sum_k r^k \le \frac{10}{9}$ and $\sum_{k\ge1} r^k \le \frac{10}{9}r$ give the first two bounds ($\frac{10}{9}\cdot 2 \le 3$). For the third, $\|D\varphi_k(w)\|_{\mathfrak{W}\to\mathfrak{W}^+} \le (2k+1)(2C_F)^k\|w\|^{2k}$ and $\sum_{k\ge1}(2k+1)r^k = \frac{2r}{(1-r)^2} + \frac{r}{1-r} \le 5r$ for $r \le \frac14$. Since $\|\cdot\| \le \|\cdot\|_+$, the same operator bounds hold on $\mathfrak{W}$ and on $\mathfrak{W}^+$, and (E6) gives the inverses; $D\Gamma_t^{-1} - I = -D\Gamma_t^{-1}(D\Gamma_t - I)$ with $\|D\Gamma_t^{-1}\|_{\mathfrak{W}^+\to\mathfrak{W}^+} \le 2$.

Step S5.30: Reality. Lemma 5.20(iii), termwise; t is real.

Step S5.31: Variational equation. Termwise, by Lemma 5.20(iv) and the Cauchy product,

$$\partial_t D\Gamma_t v = \sum_k (k+1)t^k D\varphi_{k+1}v = \sum_k t^k \sum_{a+b+c=k} 3S(D\varphi_a v,\varphi_b,\varphi_c) = 3S(D\Gamma_t v,\Gamma_t,\Gamma_t).$$

Step S5.32: Inverse. For $w \in B_{\rho_1}$ and s between 0 and t, $\|\Gamma_s(w)\| \le \frac{10}{9}\|w\| < \rho_0$, so $\psi(s) = \Gamma_{-s}(\Gamma_s(w))$ is defined and C^1 (chain rule, Step S5.26). Its derivative is $-(\partial_\tau\Gamma_\tau)|_{\tau=-s}(\Gamma_s w) + D\Gamma_{-s}(\Gamma_s w)\,\partial_s\Gamma_s(w) = -D\Gamma_{-s}(\Gamma_s w)X_F(\Gamma_s w) + D\Gamma_{-s}(\Gamma_s w)X_F(\Gamma_s w) = 0$, by the transport equation at the first term and the flow equation at the second. By (E4), $\psi(t) = \psi(0) = w$. For the image: if $\|y\| < \frac{9}{10}\rho_1$ then $w = \Gamma_{-1}(y) \in B_{\rho_1}$ and $\Gamma(w) = y$ (the same argument with $t=-1$). □

Source: P96-W p. 285 ("in a sufficiently small neighbourhood of the origin the time-1-map is well defined and gives rise to a real analytic, symplectic change of coordinates Γ with the estimates $\|\Gamma - \mathrm{id}\|_{a,s+1} = O(\|w\|_{a,s}^3)$, $\|D\Gamma - I\|^{\mathrm{op}}_{a,s+1,s} = O(\|w\|_{a,s}^2)$"), asserted there without proof; constructed and proved here. Lean: every `tsum` above is absolutely convergent by Step S5.26.

Remark 5.22. X_F is Lipschitz on every ball, so local uniqueness (`ODE_solution_unique_of_mem_Icc`, with the Lipschitz bound on a ball containing both curves) identifies $t \mapsto \Gamma_t(w)$ with the solution of $\dot y = X_F(y)$, $y(0) = w$. This identification is not used.

Theorem 5.23 (Symplecticity and gradient transport)**.** *For $|t| < 2$, $w \in B_{\rho_0}$ and $\xi,\eta \in \mathfrak{W}$:*

$$\Omega\big(D\Gamma_t(w)\xi, D\Gamma_t(w)\eta\big) = \Omega(\xi,\eta).$$

Consequently, if P is complex differentiable near $\Gamma_t(B_\rho)$ ($\rho \le \rho_0$) with Hamiltonian field X_P, then $P\circ\Gamma_t$ has Hamiltonian field

$$X_{P\circ\Gamma_t}(w) = D\Gamma_t(w)^{-1}X_P(\Gamma_t(w)) \qquad (w \in B_\rho),$$

which takes values in $\mathfrak{W}^+$ when X_P does.

Proof. **Step S5.33: Skewness of DX_F.** $\Omega(DX_F(y)\xi,\eta) = 3\Omega(S(y,y,\xi),\eta) = 12\mathfrak{F}(y,y,\xi,\eta)$ is symmetric in (ξ,η) (Theorem 5.16); with antisymmetry of Ω, $\Omega(DX_F(y)\xi,\eta) + \Omega(\xi, DX_F(y)\eta) = 0$.

Step S5.34: Constant form. $g(t) = \Omega(D\Gamma_t\xi, D\Gamma_t\eta)$ is differentiable with $g'(t) = \Omega(DX_F(\Gamma_t)D\Gamma_t\xi, D\Gamma_t\eta) + \Omega(D\Gamma_t\xi, DX_F(\Gamma_t)D\Gamma_t\eta) = 0$ by Theorem 5.21(v), bilinearity and continuity of Ω, and Step S5.33. As $D\Gamma_0 = I$, $g \equiv \Omega(\xi,\eta)$ by (E4).

Step S5.35: Gradient transport.

$$D(P\circ\Gamma_t)(w)v = \Omega(X_P(\Gamma_t w), D\Gamma_t v) = \Omega\big(D\Gamma_t\, D\Gamma_t^{-1}X_P(\Gamma_t w), D\Gamma_t v\big) = \Omega(D\Gamma_t^{-1}X_P(\Gamma_t w), v),$$

using invertibility on $\mathfrak{W}$ (Theorem 5.21(iii)) and Step S5.34. Invertibility on $\mathfrak{W}^+$ gives the last claim. □

Source: P96-W p. 285 (“symplectic”; “$\Gamma^* X_H = D\Gamma^{-1} X_H \circ \Gamma = X_{H\circ\Gamma}$”), asserted there.

5.7. **The normal form.** Let $H = \Lambda + G$ on $\mathfrak{W}$ (finite by Lemmas 5.8, 5.11); on real points it is the Hamiltonian of Definition 1.8. Its Hamiltonian field $X_H = X_\Lambda + X_G$ is defined on $\mathfrak{W}^+$, with values in $\mathfrak{W}$.

The field of the remainder will be an integral over $t \in [0,1]$ of

$$Y(t,w) = D\Gamma_t(w)^{-1} X_{Z'}(\Gamma_t w) - X_{Z'}(w) \in \mathfrak{W}^+ \qquad (0 \le t \le 1,\ w \in B_{\rho_1}), \tag{8}$$

with $Z' = \bar{G} - G^{\tan}$ (Proposition 5.17), $\rho_1 = \frac{1}{2}(40C_F)^{-1/2}$, and $D\Gamma_t(w)^{-1}$ the inverse on $\mathfrak{W}^+$ (Theorem 5.21(iii)). Holomorphy of $w \mapsto \int_0^1 Y\,dt$ would follow by differentiating under the integral; analyticity needs more, and the next lemma provides it by expanding Y into a power series in (t,w) whose coefficients can be integrated in t one at a time.

Lemma 5.24 (Power series of the integrand)**.** *There are continuous $(2j+3)$-linear maps $Y_j : \mathfrak{W}^{2j+3} \to \mathfrak{W}^+$, $j \ge 0$, with*

$$\sum_{j\ge0} \|Y_j\|\, \rho_1^{2j+3} \le 4C_*\rho_1^3, \qquad Y(t,w) = \sum_{j\ge0} t^j Y_j(w,\dots,w) \quad (0 \le t \le 1,\ w \in B_{\rho_1}),$$

the series converging absolutely. Consequently

$$\int_0^1 Y(t,w)\,dt = \sum_{j\ge0} \frac{Y_j(w,\dots,w)}{j+1} \qquad (w \in B_{\rho_1}),$$

and $w \mapsto \int_0^1 Y(t,w)\,dt$ has on B_{ρ_1} the formal power series q with $q_{2j+3} = Y_j/(j+1)$ and $q_n = 0$ for all other n; in particular it is analytic $B_{\rho_1} \to \mathfrak{W}^+$ (`AnalyticOnNhd Complex`). Moreover $\int_0^1 Y(t,\varrho w)\,dt = \varrho \int_0^1 Y(t,w)\,dt$.

Proof. Throughout, $0 \le s \le 1$, $0 \le \rho \le \rho_1$ and $r = 2C_F s\rho^2 \le 2C_F\rho_1^2 = \frac{1}{80}$.

Step S5.36: Majorant calculus. For a Banach space V, a (t,w)-*series* with values in V is a family $P = (P_{j,d})_{j,d\ge0}$ of continuous d-linear maps $\mathfrak{W}^d \to V$; its majorant is $\mu_P(s,\rho) = \sum_{j,d} \|P_{j,d}\| s^j \rho^d \in [0,\infty]$. If $\mu_P(|t|,\|w\|) < \infty$, the family $(t^j P_{j,d}(w,\dots,w))_{j,d}$ is absolutely summable; its sum is denoted $P(t,w)$. For a continuous multilinear $B : V_1 \times \cdots \times V_n \to V$ and (t,w)-series $P^1,\dots,P^n$, the composite series $B(P^1,\dots,P^n)$ has coefficients $\sum B\big(P^1_{j_1,d_1},\dots,P^n_{j_n,d_n}\big)$ (arguments concatenated in order), the sum over $j_1+\cdots+j_n = j$, $d_1+\cdots+d_n = d$, which is finite. By (E1), $\mu_{B(P^1,\dots,P^n)} \le \|B\| \prod_i \mu_{P^i}$, and by (E3) its sum is $B(P^1(t,w),\dots,P^n(t,w))$ wherever all $\mu_{P^i}(|t|,\|w\|) < \infty$. Sums of series add coefficientwise. Call P *graded* if $P_{j,d} = 0$ unless $d = 2j+e$ for a fixed e (its degree shift); composites of graded series are graded, the shifts adding.

Step S5.37: The flow. The series Γ with $\Gamma_{k,2k+1} = \Phi_k$ (shift 1) sums to $\Gamma_t(w)$ (Theorem 5.21), and $\mu_\Gamma(s,\rho) \le \sum_k (2C_F)^k s^k \rho^{2k+1} = \rho/(1-r)$ by Lemma 5.20(ii).

Step S5.38: The derivative. For $k \ge 1$ let $A_{k,2k}$ be the continuous $2k$-linear map $\mathfrak{W}^{2k} \to L(\mathfrak{W}^+)$ sending $(w^1,\dots,w^{2k})$ to $v \mapsto \sum_{i=1}^{2k+1} \Phi_k(w^1,\dots,w^{i-1},v,w^i,\dots,w^{2k})$; all other coefficients vanish (shift 0, no $j=0$ term). By Lemma 5.20(ii) and $\|v\| \le \|v\|_+$, $\|A_{k,2k}\| \le (2k+1)(2C_F)^k$, so $\mu_A \le \sum_{k\ge1}(2k+1)r^k \le 5r \le \frac{1}{16}$ (as in Step S5.29). Its diagonal is $D\varphi_k(w)$ restricted to $\mathfrak{W}^+$ by (E1), so A sums to $D\Gamma_t(w) - I$ on $\mathfrak{W}^+$ (Theorem 5.21(i), termwise derivative).

Step S5.39: The inverse. Composition is a bilinear map of norm 1 on the Banach algebra $L(\mathfrak{W}^+)$. The powers $(-A)^n$ are composite series with $\mu_{(-A)^n} \le \mu_A^n$, and $(-A)^n$ has no coefficient with $j < n$. Let $N_{j,d} = \sum_{n\le j} ((-A)^n)_{j,d}$, a finite sum. Then $\mu_N \le \sum_n \mu_A^n \le \frac{16}{15}$, the family indexed by n and the coefficients of $(-A)^n$ is absolutely summable, and regrouping by (E3) gives

$N(t,w)=\sum_n(-A(t,w))^n=(I+A(t,w))^{-1}=D\Gamma_t(w)^{-1}$ on $\mathfrak{W}^+$ by (E6). N is graded with shift 0.

Step S5.40: The cubic field along the flow. $X_{Z'}(y)=4\nabla\mathfrak{z}(y,y,y)$ with $4\nabla\mathfrak{z}$ trilinear of norm $\le C_*$ (Lemma 5.11). The composite $4\nabla\mathfrak{z}(\Gamma,\Gamma,\Gamma)$ has majorant $\le C_*\rho^3(1-r)^{-3}$, sums to $X_{Z'}(\Gamma_t w)$, and has shift 3.

Step S5.41: The integrand. Application $L(\mathfrak{W}^+)\times\mathfrak{W}^+\to\mathfrak{W}^+$ is bilinear of norm 1. Let $\mathcal{Y}$ be the composite $N\big(4\nabla\mathfrak{z}(\Gamma,\Gamma,\Gamma)\big)$ minus the series whose only coefficient is $4\nabla\mathfrak{z}$ in position $(0,3)$. Then $\mathcal{Y}$ sums to $Y(t,w)$, has shift 3, and $\mu_{\mathcal{Y}}(1,\rho_1)\le\frac{16}{15}C_*\rho_1^3(\frac{80}{79})^3+C_*\rho_1^3\le 4C_*\rho_1^3$. Put $Y_j=\mathcal{Y}_{j,2j+3}$; all other coefficients vanish.

Step S5.42: Integrate in t. For fixed $w\in B_{\rho_1}$ the functions $t\mapsto t^jY_j(w,\dots,w)$ are continuous on $[0,1]$ with $\sup_t\|\cdot\|_+\le\|Y_j\|\rho_1^{2j+3}$, summable in j. By (E5) the series is integrated termwise, and $\int_0^1 t^j\,dt=\frac{1}{j+1}$.

Step S5.43: Power series. $\sum_n\|q_n\|\rho_1^n\le\sum_j\|Y_j\|\rho_1^{2j+3}<\infty$, so (E2) gives the formal power series on B_{ρ_1} and analyticity.

Step S5.44: Reality. For real t, Γ_t, $D\Gamma_t$ and $X_{Z'}$ commute with ϱ (Theorem 5.21(iv), Definition 5.5), so $Y(t,\varrho w)=\varrho Y(t,w)$; ϱ is continuous and $\mathbb{R}$-linear and therefore commutes with the integral over $t\in[0,1]$ by (E5). □

Source: ours. P96-W p. 284 obtains analyticity of X_F from coordinatewise analyticity and local boundedness ([9, Appendix A] of P96-W) and does not discuss the analyticity of X_K separately.

Theorem 5.25 (Analytic partial Birkhoff normal form; P96-W Main Proposition for $J=\{1,2\}$)**.** *Let $\Gamma=\Gamma_1$ and $\rho_1=\rho_1(m)=\frac12(40C_F(m))^{-1/2}$. Define on B_{ρ_1}*

$$K=H\circ\Gamma-\Lambda-\bar G-G^{\mathrm{nor}}.$$

Then:

- *(i) K is analytic on B_{ρ_1} and real, and $|K(w)|\le C_K\|w\|^6$ with $C_K=10\,C_*C_F$;*
- *(ii) $K\in\mathcal{A}(B_{\rho_1})$ (Definition 5.5): its Hamiltonian field X_K is an analytic map $B_{\rho_1}\to\mathfrak{W}^+$ (`AnalyticOnNhd Complex`), real, with $\|X_K(w)\|_+\le C_K'\|w\|^5$, $C_K'=66\,C_*C_F$. Its real restriction $X_K^{\mathbb{R}}=\mathcal{Z}^{-1}X_K\circ\mathcal{Z}$ is a real analytic map $B_{r_1}^E\to E_{a,p+1}$ ($r_1=\rho_1/\sqrt2$, B_r^E the real ball of $E_{a,p}$), equal to $(\partial_pK^{\mathbb{R}},-\partial_qK^{\mathbb{R}})$ for the real analytic $K^{\mathbb{R}}=K\circ\mathcal{Z}$;*
- *(iii) $\bar G$ and G^{nor} are as in Lemma 5.12: $\bar G=\frac12\langle AI,I\rangle+\langle BI,Z\rangle$ with the corrected coefficients, G^{nor} depends only on $\hat w$ and $|G^{\mathrm{nor}}|\le\frac{C_*}{4}\|\hat w\|^4$; both are in $\mathcal{A}(\mathfrak{W})$ with cubic fields bounded by $C_*\|w\|^3$ (resp. $C_*\|\hat w\|^3$);*
- *(iv)* (Conjugacy.) *For $w\in B_{\rho_1}\cap\mathfrak{W}^+$, $\Gamma(w)\in\mathfrak{W}^+$ and*

$$D\Gamma(w)\big[X_\Lambda(w)+X_{\bar G}(w)+X_{G^{\mathrm{nor}}}(w)+X_K(w)\big]=X_H(\Gamma(w)).$$

Proof. Write $Z'=\bar G-G^{\tan}$ (Proposition 5.17). Since $Z'+G-\bar G-G^{\mathrm{nor}}=0$,

$$K=K_1+K_2,\qquad K_1=G\circ\Gamma-G,\qquad K_2=\Lambda\circ\Gamma-\Lambda-Z'.$$

Step S5.45: Analyticity and reality. Λ, G, $\bar G$, G^{nor} are continuous polynomials and Γ is analytic on B_{ρ_1} (Theorem 5.21(i)); compose by (E2). Reality: each polynomial is real and $\Gamma\circ\varrho=\varrho\circ\Gamma$.

Step S5.46: The bound for K_1. On the segment from w to $\Gamma(w)$ every point y has $\|y\|\le\frac{10}{9}\|w\|$, and $|DG(y)v|=|\Omega(X_G(y),v)|\le C_*\|y\|^3\|v\|$. By (E4) and Theorem 5.21(iii), $|K_1(w)|\le C_*(\frac{10}{9})^3\|w\|^3\cdot 3C_F\|w\|^3\le 5C_*C_F\|w\|^6$.

Step S5.47: The bound for K_2. Let $h(t)=\Lambda(\Gamma_tw)-\Lambda(w)-tZ'(w)$ for $t\in[0,1]$. Then $h(0)=0$ and, by the flow equation and Proposition 5.17 at $y=\Gamma_tw$, $h'(t)=D\Lambda(\Gamma_tw)X_F(\Gamma_tw)-$

$Z'(w) = Z'(\Gamma_t w) - Z'(w)$. As in Step S5.46 (with $\|X_{Z'}(y)\|_+ \le C_*\|y\|^3$), $|h'(t)| \le 5C_*C_F\, t\|w\|^6$, so $|K_2(w)| = |h(1)| \le 5C_*C_F\|w\|^6$. Together with Step S5.46, (i) holds.

Step S5.48: The field of K_1. By Theorem 5.23, $X_{K_1}(w) = D\Gamma(w)^{-1}X_G(\Gamma w) - X_G(w) = D\Gamma^{-1}\big[X_G(\Gamma w) - X_G(w)\big] + (D\Gamma^{-1} - I)X_G(w)$. By Definition 5.7 (with $d = 4$, $c = C_*/4$) and Theorem 5.21(iii), $\|X_{K_1}(w)\|_+ \le 2\cdot 3C_*(\frac{10}{9})^2\|w\|^2\cdot 3C_F\|w\|^3 + 20C_F\|w\|^2\cdot C_*\|w\|^3 \le 44C_*C_F\|w\|^5$. It is analytic into $\mathfrak{W}^+$: $w \mapsto D(\Gamma - \mathrm{id})(w) \in L(\mathfrak{W}, \mathfrak{W}^+)$ is analytic by (E2), restriction $L(\mathfrak{W}, \mathfrak{W}^+) \to L(\mathfrak{W}^+)$ is bounded linear, inversion is analytic on the units of $L(\mathfrak{W}^+)$, and $X_G \circ \Gamma : B_{\rho_1} \to \mathfrak{W}^+$ is analytic.

Step S5.49: The field of K_2. By the fundamental theorem of calculus (E5) and Step S5.47, $K_2(w) = \int_0^1 \big(Z'(\Gamma_t w) - Z'(w)\big)dt$. The integrand is C^1 in (t, w) and, by Theorem 5.23, $D_w\big(Z'(\Gamma_t w) - Z'(w)\big)v = \Omega(Y(t,w), v)$ with Y as in (8). As in Step S5.48, $\|Y(t,w)\|_+ \le 44C_*C_F\, t\|w\|^5$; Y is continuous in (t, w) and $D_w Y$ is bounded on $[0,1] \times B_\rho$ for each $\rho < \rho_1$ (Lemma 5.24: the termwise derivative of the series). By (E5), $DK_2(w)v = \int_0^1 \Omega(Y(t,w), v)dt = \Omega\big(\int_0^1 Y(t,w)dt, v\big)$, so $X_{K_2}(w) = \int_0^1 Y(t,w)\,dt$, with $\|X_{K_2}(w)\|_+ \le 22C_*C_F\|w\|^5$. By Lemma 5.24, X_{K_2} is analytic $B_{\rho_1} \to \mathfrak{W}^+$ with an explicit power series.

Step S5.50: Conclusion of (ii). $X_K = X_{K_1} + X_{K_2}$, with $44 + 22 = 66$, analytic by Steps S5.48–S5.49; reality from Definition 5.5 and (i). The real restriction is Lemma 5.6 applied to X_K (with $V = \mathfrak{W}^+$) and to K; on real points $\mathcal{Z}^{-1}X_K = (\partial_p K, -\partial_q K)$ (paragraph after Definition 5.5), and $\|\mathcal{Z}x\| \le \sqrt{2}\|x\|_E < \rho_1$ for $x \in B^E_{r_1}$.

Step S5.51: (iii). Lemmas 5.11 and 5.12.

Step S5.52: Conjugacy. Let $w \in B_{\rho_1} \cap \mathfrak{W}^+$. Then $\Gamma(w) = w + (\Gamma(w) - w) \in \mathfrak{W}^+$, so $X_H(\Gamma w) \in \mathfrak{W}$ is defined, and $DH(y)u = \Omega(X_H(y), u)$ for $y \in \mathfrak{W}^+$, $u \in \mathfrak{W}$ (Lemma 5.8). By the chain rule and Theorem 5.23, $D(H\circ\Gamma)(w)v = \Omega(X_H(\Gamma w), D\Gamma v) = \Omega(D\Gamma^{-1}X_H(\Gamma w), v)$. On the other hand $D(H\circ\Gamma)(w)v = D(\Lambda + \bar{G} + G^{\mathrm{nor}} + K)(w)v = \Omega(X_\Lambda(w) + X_{\bar{G}}(w) + X_{G^{\mathrm{nor}}}(w) + X_K(w), v)$. Nondegeneracy (Lemma 5.4) gives the identity after applying $D\Gamma(w)$. □

Source: P96-W Main Proposition, pp. 281–282; proof pp. 282–285, (12)–(14). Deviations: $J = \{1, 2\}$; fixed m (the source's uniformity on compact m-intervals and analytic dependence on m are not needed by W-NLW2 and are omitted); the corrected coefficients (E-PDE-W-1); time convention of Definition 1.11.
SOURCE_GAP: P96-W p. 285 states two conditional facts: "with $X_H \in A(\ell^{a,s}, \ell^{a,s+1})$, also $\Gamma^* X_H = D\Gamma^{-1}X_H \circ \Gamma = X_{H\circ\Gamma} \in A(\ell^{a,s}, \ell^{a,s+1})$", and the analogous statement for the bracket $[X_F, X_H] = X_{\{H,F\}}$; it concludes $X_K \in A(\ell^{a,s}, \ell^{a,s+1})$. The printed proof leaves implicit how these facts apply to the Taylor remainder when the original linear field X_Λ is unbounded (p. 280). The homological identity shows $\{\Lambda, F\} = D\Lambda \cdot X_F = Z'$ (Proposition 5.17), so $\{H, F\} = Z' + \{G, F\}$ has a smoothing field, and the source's route can be completed along these lines. Steps S5.47 and S5.49 give the expansion in detail, using only $D\Lambda \cdot X_F = Z'$. The conditional facts themselves are correct; this is an omitted application, not a false statement. Analyticity: the source appeals to coordinatewise analyticity and local boundedness ([9, Appendix A] of P96-W); here it comes from explicit power series (Theorem 5.21, Lemma 5.24).

Remark 5.26 (Regularity class). Γ, K and X_K are analytic on the complex ball B_{ρ_1} (`AnalyticOnNhd Complex`): Γ and K by composition of power series, X_{K_1} by composition and operator inversion, X_{K_2} by the explicit power series of Lemma 5.24. Their real restrictions are real analytic (Lemma 5.6). The theorem that complex-differentiable maps on open subsets of complex Banach spaces are analytic is not used. Analyticity implies holomorphy in the sense of Definition 3.24, so the Cauchy estimates of W1 also apply.

Corollary 5.27 (Real change of coordinates). *Let $r_1 = \rho_1/\sqrt{2}$ and $B^E_r = \{x \in E_{a,p} : \|x\|_E < r\}$ (real). The map $\Gamma^{\mathbb{R}} = \mathcal{Z}^{-1}\circ\Gamma\circ\mathcal{Z} : B^E_{r_1} \to E_{a,p}$ is well defined (real-valued) and:*

(i) it is real analytic: it has on $B^E_{r_1}$ the real power series whose $(2k+1)$-st term is $\mathcal{Z}^{-1}\Phi_k(\mathcal{Z}\cdot,\dots,\mathcal{Z}\cdot)$ restricted to $E^{2k+1}_{a,p}$ (Mathlib `AnalyticOnNhd Real`);
(ii) $\|\Gamma^{\mathbb{R}}(x)-x\|_{E_{a,p+1}} \le 6\sqrt{2}\,C_F\|x\|_E^3$ and $\Gamma^{\mathbb{R}}(B^E_{r_1}) \subset B^E_{2r_1}$;
(iii) it is symplectic for $\sum dq_j\wedge dp_j$, injective, with inverse $\mathcal{Z}^{-1}\Gamma_{-1}\mathcal{Z}$ on its image, and $\Gamma^{\mathbb{R}}(B^E_{r_1}) \supset B^E_{r_1/2}$;
(iv) for real $x \in B^E_{r_1}$: $H(\Gamma^{\mathbb{R}}x) = \Lambda(x)+\bar G(x)+G^{\mathrm{nor}}(x)+K(\mathcal{Z}x)$, where on real points $\bar G = \frac12\langle AI,I\rangle + \langle BI,Z\rangle$ with $I_j=(q_j^2+p_j^2)/2$.

Proof. $\|\mathcal{Z}x\| \le \sqrt{2}\|x\|_E < \rho_1$, and Γ preserves $\mathfrak{W}_{\mathbb{R}}$ (Theorem 5.21(iv)), so $\Gamma^{\mathbb{R}}$ is real-valued (Lemma 5.3). (i): each Φ_k commutes with ϱ (Lemma 5.20(iii)), so its restriction to real arguments is real; the real multilinear maps have norm $\le 2^{(2k+1)/2}\|\Phi_k\|$ ($\|\mathcal{Z}\| \le \sqrt{2}$, $\|\mathcal{Z}^{-1}\| \le 1$), so the radius is at least $\rho_0/\sqrt{2}$; apply (E2) over $\mathbb{R}$. (ii): Theorem 5.21(iii) and $\|\mathcal{Z}x\| \le \sqrt{2}\|x\|_E$ ($3\cdot 2^{3/2} \le 6\sqrt{2}$; $\frac{10}{9}\sqrt{2} < 2$). (iii): Lemma 5.4, Theorem 5.23, Theorem 5.21(vi): for $y \in B^E_{r_1/2}$, $\|\mathcal{Z}y\| < \rho_1/2$, so $w = \Gamma_{-1}(\mathcal{Z}y)$ is real with $\|\mathcal{Z}^{-1}w\|_E \le \|w\| \le \frac59\rho_1 < r_1$ and $\Gamma(w) = \mathcal{Z}y$. (iv): Theorem 5.25 and Lemma 5.12. □

Source: P96-W p. 281 ("real analytic, symplectic change of coordinates Γ in some neighbourhood of the origin in $\ell^{a,s}$") and p. 290 (use of Γ in real coordinates).

Corollary 5.28 (Invariant tori are transported)**.** *Let $\nu \in \mathbb{R}^2$ and $\Psi : \mathbb{R}^2 \to B_{\rho_1} \cap \mathfrak{W}^+$ be differentiable as a map into $\mathfrak{W}$ with $D\Psi(\theta)\nu = X_{H'}(\Psi(\theta))$ for all θ, where $X_{H'} = X_\Lambda + X_{\bar G} + X_{G^{\mathrm{nor}}} + X_K$. Then $\Gamma\circ\Psi$ takes values in $\mathfrak{W}^+$ and $D(\Gamma\circ\Psi)(\theta)\nu = X_H(\Gamma\circ\Psi(\theta))$. If Ψ is real ($\varrho\Psi = \Psi$), then $(Q,P) = \mathcal{Z}^{-1}\Gamma\circ\Psi$ satisfies, for every $j \ge 1$, $D_\nu Q_j = \lambda_j P_j$ and $D_\nu P_j = -\lambda_j Q_j - T(Q,Q,Q)_j$, which is the coefficient system (2) ($T = G_q$ for $a > 0$).*

Proof. Chain rule (Γ is differentiable on $\mathfrak{W}$) and Theorem 5.25(iv). Coordinates are continuous linear functionals, and on real points $\mathcal{Z}^{-1}X_H = (\lambda p, -\lambda q - T(q,q,q))$ by Lemmas 5.8 and 5.12. □

Source: P96-W p. 290 ("These tori carry the quasi-periodic motions $\gamma = \Gamma\circ\Psi(\dots)$").

The tori produced by the KAM iteration lie a priori only in $\mathfrak{W}_{a,p+1/2}$, and their invariance equation is an identity in the weaker space $\mathfrak{W}_{a,p-1}$ (Theorem 8.14(i),(iii)). The next corollary shows that this weak form suffices: the equation itself forces every torus point into $\mathfrak{W}^+$, where Theorem 5.25(iv) applies pointwise.

Corollary 5.29 (Weak invariance is transported)**.** *For $w \in \mathfrak{W}$ let $X_\Lambda(w) = (-i\lambda'_l w_l)_{l\neq 0} \in \mathfrak{W}_{a,p-1}$, extending Lemma 5.8. Let $\nu \in \mathbb{R}^2$ and $\Psi : \mathbb{R}^2 \to B_{\rho_1}$ be differentiable as a map into $\mathfrak{W}$, and suppose that for every θ*

$$D\Psi(\theta)\nu = X_\Lambda(\Psi(\theta)) + X_{\bar G}(\Psi(\theta)) + X_{G^{\mathrm{nor}}}(\Psi(\theta)) + X_K(\Psi(\theta)) \qquad \text{in } \mathfrak{W}_{a,p-1}.$$

Then $\Psi(\theta) \in \mathfrak{W}^+$ with $\|\Psi(\theta)\|_+ \le \|D\Psi(\theta)\nu\| + C_*\|\Psi(\theta)\|^3 + C'_K\|\Psi(\theta)\|^5$ for every θ, and all conclusions of Corollary 5.28 hold: $\Gamma\circ\Psi$ takes values in $\mathfrak{W}^+$, $D(\Gamma\circ\Psi)(\theta)\nu = X_H(\Gamma\circ\Psi(\theta))$ in $\mathfrak{W}$, and for real Ψ the coefficient system (2) holds for $(Q,P) = \mathcal{Z}^{-1}\Gamma\circ\Psi$.

Proof. **Step S5.53: The linear field on $\mathfrak{W}$.** By Lemma 3.6(i) (with $p-1$ in place of p and $r=1$), $X_\Lambda : \mathfrak{W} \to \mathfrak{W}_{a,p-1}$ is bounded with norm $\le \sqrt{1+m}$, so the hypothesis is meaningful; on $\mathfrak{W}^+$ it agrees with Lemma 5.8. The other three fields take values in $\mathfrak{W}^+ \subset \mathfrak{W} \subset \mathfrak{W}_{a,p-1}$ (Theorem 5.25(ii),(iii)), with the norm-one inclusions of Lemma 3.5.

Step S5.54: Regularity of the torus points. Fix θ and write $w = \Psi(\theta)$. The hypothesis gives $X_\Lambda(w) = D\Psi(\theta)\nu - X_{\bar G}(w) - X_{G^{\mathrm{nor}}}(w) - X_K(w)$, whose right side lies in $\mathfrak{W}$ with norm $\le \|D\Psi(\theta)\nu\| + C_*\|w\|^3 + C'_K\|w\|^5$ (Theorem 5.25; $\|X_{\bar G}(w) + X_{G^{\mathrm{nor}}}(w)\| \le C_*\|w\|^3$ by Lemma 5.11 applied to $\bar{\mathfrak{G}} + \mathfrak{G}^{\mathrm{nor}}$, whose coefficient sets are disjoint). Since $w_l = i\,X_\Lambda(w)_l/\lambda'_l$ and $|l| \le \lambda_{|l|}$,

each weight satisfies $|l|^{p+1}e^{a|l|}|w_l| \le |l|^p e^{a|l|}|X_\Lambda(w)_l|$ (Lemma 3.6(i) with $r=-1$), so $w \in \mathfrak{W}^+$ with $\|w\|_+ \le \|X_\Lambda(w)\|$.

Step S5.55: Transport. Since $w \in B_{\rho_1} \cap \mathfrak{W}^+$, both sides of the hypothesis lie in $\mathfrak{W}$ and agree there, i.e. $D\Psi(\theta)\nu = X_{H'}(w)$ in $\mathfrak{W}$ with $X_{H'}$ as in Corollary 5.28. The chain rule (Γ is differentiable on $\mathfrak{W}$, Ψ differentiable into $\mathfrak{W}$) and Theorem 5.25(iv) at w give $D(\Gamma\circ\Psi)(\theta)\nu = D\Gamma(w)X_{H'}(w) = X_H(\Gamma(w))$, and $\Gamma(w) \in \mathfrak{W}^+$. The coefficient system follows as in the proof of Corollary 5.28. The argument is pointwise in θ; no differentiability of Ψ into $\mathfrak{W}^+$ is needed. □

Source: ours. The KAM tori are invariant in a weaker space than the one P96-W p. 290 tacitly uses; see the return-trip item of Remark 5.30.

5.8. **Interface to milestones W4–W6.**

Remark 5.30 (What W3 supplies toward P96-K Theorem A). P96-K Theorem A receives its Hamiltonian $N+P$ only after the action-angle substitution and the choice of parameter and phase-space domains (KP96 p. 25, Step 1, and p. 26, Step 3; milestone W5, by the ownership ruling recorded in W4's `bpW:w4:from-w3`). W3 supplies the Cartesian input to that construction. After Theorem 5.25, the Hamiltonian is $H' = N_0 + \bar G + G^{\mathrm{nor}} + K$, $N_0 = \Lambda$, on the complex ball $B_{\rho_1} \subset \mathfrak{W}$, with:

- *Phase space and weights.* $\mathfrak{W} = \mathfrak{W}_{a,p}$ (complexification of $E_{a,p}$ via $\mathcal{Z}$; $a = p = 1$ for the headline). The variables w_l and w_{-l} are *independent* complex variables on B_{ρ_1}; reality is $\varrho w = w$. This is needed because the KAM domain complexifies the angles, so that $w_{-j} \ne \overline{w_j}$ there. In P96-K's notation (p. 120) the normal coordinates are $u_j = q_{j+2}$, $v_j = p_{j+2}$ ($j \ge 1$; KP96 p. 25, Step 1, with $q_j^{\mathrm{KP}} = \sqrt{2}z_j$), in $\ell^{a,p}$ with weights $(j+2)^p e^{a(j+2)}$. Since $j^p e^{aj} \le (j+2)^p e^{a(j+2)} \le 3^p e^{2a} j^p e^{aj}$, these are equivalent to P96-K's $j^p e^{aj}$ within the factor $3^p e^{2a}$ in the phase space and $3^{p+1}e^{2a}$ in the target weight $p+1$; P96-K's sum norm $\|u\| + \|v\|$ lies between the max norm and twice it; $\mathcal{Z}$ contributes $\sqrt{2}$ (Lemma 5.3). W5 must carry these fixed factors into its domains and perturbation norms (W4 works with the shifted wave weights throughout, so the factor $3^p e^{2a}$ does not arise there).
- *Smoothing.* P96-K Assumption C with $\bar p = p+1$ (so $\bar p - p = 1$, $\Delta = \min(\bar p - p, 1) = 1$): the fields of $\bar G$, G^{nor}, K map into $\mathfrak{W}^+ = \mathfrak{W}_{a,p+1}$; in particular their normal components $(\partial_{p_j}R, -\partial_{q_j}R)_{j\ge3}$ lie in $\ell^{a,p+1}$, which is P96-K's $(P_v, -P_u) \in \ell^{a,\bar p}$.
- *Regularity class: Assumption C in Cartesian form.* P96-K p. 121 requires that X_P define, near the torus, a real analytic map $\mathcal{P}^{a,p} \to \mathcal{P}^{a,\bar p}$, real analytic on the complex domain $D(s,r)$ with finite sup of $|X_P|_r$. W3 gives: $\bar G$, G^{nor} are real quartic polynomials with real cubic fields $\mathfrak{W} \to \mathfrak{W}^+$ (analytic everywhere); $K \in \mathcal{A}(B_{\rho_1})$, so X_K is analytic $B_{\rho_1} \to \mathfrak{W}^+$ on the *complex* ball, real, and its real restriction $X_K^{\mathbb{R}} : B_{r_1}^E \to E_{a,p+1}$ is real analytic (Theorem 5.25(ii)). This is the literal analytic hypothesis in the Cartesian coordinates (q,p); composing with the (analytic) action-angle substitution on the two tangential modes, and checking that $D(s,r)$ is mapped into B_{ρ_1}, is W5's step. No dependence on parameters: the parameter ξ of P96-K enters later, through the action-angle substitution (KP96 p. 25, Step 1), and its Lipschitz seminorms are then obtained by Cauchy estimates in ξ (KP96 p. 26, Step 3).
- *Size.* On B_{ρ_1}: $|K| \le C_K\|w\|^6$, $\|X_K\|_+ \le C_K'\|w\|^5$, $|G^{\mathrm{nor}}| \le \frac{C_*}{4}\|\hat w\|^4$, $\|X_{G^{\mathrm{nor}}}\|_+ \le C_*\|\hat w\|^3$. This is the hypothesis $|R| = O(\|\hat z\|^4) + O(\|z\|^g)$, $g = 6$, of the Cantor Manifold Theorem (P96-W p. 289) with $R = G^{\mathrm{nor}} + K$, strengthened by the field bounds that KP96 p. 26, Step 3 uses ("the hypotheses for R_u, R_v"). The exponent condition $g = 6 > 4 + (4-\Delta)/\kappa = \frac{11}{2}$ uses $\kappa = 2$ (Remark 2.14).
- *Normal form.* $\bar G = \frac12\langle AI, I\rangle + \langle BI, Z\rangle$ with $I_j = w_j w_{-j}$, the corrected A, B (Theorem 2.10); nondegeneracy is Theorem 2.13.
- *Return trip.* The KAM tori $\tilde\Psi_\xi$ of W6 take values in B_{ρ_1} with normal part only in $\mathfrak{W}_{a,p+1/2}$, and are invariant in the sense of an identity in $\mathfrak{W}_{a,p-1}$ (Theorem 8.14(i),(iii)); they are

not known a priori to lie in $\mathfrak{W}^+$, so the strong Corollary 5.28 does not apply directly (the $\bar{p}$-space embeddings of KP96 p. 28, Step 5, are not what W6 establishes). Corollary 5.29 covers exactly this case: the invariance equation forces each torus point into $\mathfrak{W}^+$, and $\Gamma \circ \tilde{\Psi}_\xi$ is then an invariant torus of H satisfying the coefficient system. Analyticity in θ of $\Gamma^{\mathbb{R}} \circ \Psi^{\mathbb{R}}$ follows from Corollary 5.27(i) and analyticity of $\Psi^{\mathbb{R}}$; injectivity of $\Gamma^{\mathbb{R}}$ transports the family injectivity (6) of W-NLW2; the closeness $\|\Gamma^{\mathbb{R}}(x) - x\| \le 6\sqrt{2} C_F \|x\|^3$ is $o(r)$ in (3).

Source: P96-K Assumption C and the norm (1), p. 121; P96-W Cantor Manifold Theorem, assumption C, pp. 288–289, and p. 290; KP96 §7, Steps 1–5, pp. 25–28.

5.9. **Constants and their dependence on m.** They are chosen in this order; each depends only on the listed data.

Constant	Value	Depends on
C_*	$K^2_{p+1/2}/\pi$; $= 8(1+2\zeta(3))/\pi \approx 8.67$ for $a = p = 1$	p only (not m, not a)
$\delta_*(m)$	$m/(27(4+m)^{3/2})$	m
$C_F(m)$	$C_*/\delta_* = 27C_*(4+m)^{3/2}/m$	m, p
$\rho_0(m)$, $\rho_1(m)$	$(40C_F)^{-1/2}$, $\frac{1}{2}(40C_F)^{-1/2}$	m, p
$C_K(m)$, $C'_K(m)$	$10C_*C_F$, $66C_*C_F$	m, p
Λ bound	$\sqrt{1+m}$	m

Asymptotically $C_F \sim 216 C_*/m$ as $m \to 0$ and $C_F \sim 27 C_* \sqrt{m}$ as $m \to \infty$; nothing is uniform in m, as Standing assumption 1.1 allows. Lean totality: the divisions in f_x are guarded by Lemma 5.13; $\sqrt{\lambda_j} > 0$; every `tsum` is absolutely convergent (Lemmas 5.10, 5.11, 5.24, Theorem 5.21); the integral in Step S5.49 has a continuous integrand on $[0,1]$; derivatives are stated as `HasFDerivAt` at points of differentiability.

5.10. **Summary of status.**

- *Proved here*: Lemmas 5.3, 5.4, 5.8, 5.10, 5.11, 5.12, 5.13, 5.20, 5.6, 5.24; Theorems 5.16, 5.21, 5.23, 5.25; Proposition 5.17; Corollaries 5.27, 5.28, 5.29.
- *External*: (E1)–(E6), all in Mathlib.
- *From W1*: Lemmas 3.2, 3.5, 3.6, Definition 3.24, Proposition 3.17 (the last also covered by Lemma 2.3).
- *Source defects*: P96-W p. 285 (Case 1 derivative inequality false; constant halves); P96-W p. 286 (Case 2 reduction omits the subcase $i = j < k = l$; a factor slip $2 \to 2i$); P96-W pp. 282, 284 factor 16 (E-PDE-W-1, already recorded). Not a false statement but an omitted application: P96-W p. 285, regularity of X_K (the SOURCE_GAP note after Theorem 5.25).

6. Milestone W4: the normal homological equation

One step of the KAM iteration (milestone W5) removes from the perturbation its part of order at most two in the normal variables by the time-one map of a Hamiltonian F that solves the linearized equation $\{F, N\} + \hat{N} = R$ of P96-K §2. This part solves that equation for the two-frequency wave setting ($n = 2$, $d = 1$, $\delta = -1$, $\bar{p} = p + 1$) and proves the estimates of P96-K Lemmas 1 and 2 with explicit constants, including the Lipschitz dependence on the parameter ξ.

Route. The equation splits into seven components (scalar, tangential-linear, two normal-vector and three normal-operator components). In every component the solution is a Fourier multiplier in the angle θ: the k-th Fourier coefficient is divided by a small divisor $\langle k, \omega\rangle + \langle l, \Omega\rangle$, entrywise for the vector and operator components. One abstract lemma (Lemma 6.15) handles all components. It uses only the Banach-valued coefficient bound $\|\hat{f}_k\| \le m\, e^{-2\pi|k|_1 s}$ and the explicit lattice sums of Lemma 6.12; the operator-valued division is bounded by the Hilbert-matrix (Schur) lemma of P96-K App. A. P96-K p. 127 instead sums operator-valued coefficients by a square-sum bound which is false (Remark 6.9); the replacement keeps the power $\sigma^{-(2\tau+n+1)}$ of P96-K (4), so the later

smallness exponents of Theorem A are unchanged (Remark 6.21). The size of the components of R is read off from the weighted field norm $|X_R|_r$ by evaluating the field at chosen points and extracting polynomial coefficients by averages over roots of unity (Lemmas 6.17, 6.18); no Cauchy integral is needed. Derivatives of F in θ are obtained by solving the same equations with the θ-derivatives of the components of R as data, so no factor $|k|$ is lost.
Inputs. From W1: the weighted spaces and transports (Definition 3.1, Lemma 3.2), the KAM phase space, domains and norms (Definition 3.31, Remark 3.32), the Hilbert-matrix bound (Lemma 3.39), the spectral tails (Lemma 3.8), the sup-weighted sequences (Definition 3.7), and the Banach-valued Fourier Lemma 3.36 with the scalar uniqueness of Lemma 3.35(iii) (the exact content used is listed in Remark 6.10). From W0: the action coefficients (Theorem 2.10). From W3: the form of the Hamiltonian (Remark 5.30); see §6.9.
Regularity used. W4 uses only complex differentiability (`DifferentiableOn Complex`) of the θ-dependent components of R on a strip, and boundedness. It does not use analyticity of X_R on an infinite-dimensional domain; P96-K Lemma 1 assumes "X_R real analytic", which is more than is used here. In particular W4 does not depend on the pending upgrade of the W3 interface from holomorphic to analytic fields (review finding W3-R1); that dependency, if any, sits in W5, where R is produced from P.

6.1. **External facts.** Besides the external facts of W1 (§3.2), this part uses:

(W4-E1) *Roots of unity.* For $m \in \mathbb{Z}$, $\sum_{u=0}^{3} i^{um} = 4$ if $4 \mid m$ and $= 0$ otherwise (finite geometric sum; Mathlib `Finset.sum_range` with `geom_sum_eq`).

(W4-E2) *Parallelogram law* in a complex inner product space: $\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$ (`parallelogram_law_with_norm`).

(W4-E3) *Exponential inequalities.* $e^t \geq 1+t$ for real t (`Real.add_one_le_exp`); hence $e^{t-1} \geq t$, and for $q > 0$, $x \geq 0$: $x^q e^{-x} \leq (q/e)^q$. Also $\sum_{N \geq 1} e^{-xN} = 1/(e^x - 1)$ for $x > 0$ (`tsum_geometric_of_lt_one`).

(W4-E4) *Periodic integration by parts.* If $g : \mathbb{R} \to E$ (E Banach) is C^1 and 1-periodic, then $\int_0^1 g' = 0$ (`intervalIntegral.integral_eq_sub_of_hasDerivAt`); continuous linear maps commute with Bochner integrals (`ContinuousLinearMap.integral_comp_comm`).

(W4-E5) *Riesz representation of bounded sesquilinear forms.* Let E be a complex Hilbert space (Mathlib: `[NormedAddCommGroup E] [InnerProductSpace Complex E] [CompleteSpace E]`; here $E = \ell^2_{\mathbb{C}}$, complete by (W1-E1)) and $B : E \to E \to \mathbb{C}$ a bounded form, conjugate-linear in the first and linear in the second argument (Mathlib type `E →L⋆[Complex] E →L[Complex] Complex`). Then $B^\sharp =$ `InnerProductSpace.continuousLinearMapOfBilin B` is a continuous linear map $E \to E$ with $\langle B^\sharp v, w\rangle = B(v,w)$ for all v, w (`InnerProductSpace.continuousLinearMapOfBilin_apply`), where $\langle\cdot,\cdot\rangle$ is the inner product, conjugate-linear in the first slot. Moreover $\|u\| = \sup_{\|w\|\leq 1} |\langle u, w\rangle|$ (`InnerProductSpace.toDual` is a linear isometry equivalence $E \simeq E^*$, and $\langle u, w\rangle =$ `toDual Complex E u w`, `InnerProductSpace.toDual_apply_apply`). (Checked in Mathlib, `Mathlib/Analysis/InnerProductSpace/Dual.lean`.)

The bound in (W4-E3) follows from $e^{x/q - 1} \geq x/q$ raised to the power q.

6.2. **Setting and hypotheses.**

Convention 6.1 (Normal modes, reindexing, coordinates)**.** Fix $a \geq 0$, $p \geq 0$ and put $\bar{p} = p + 1$ (the wave case of P96-K Assumption C, $\bar{p} - p = 1$). All results of this part hold for $a, p \geq 0$ except the identification of the weak bracket with a derivative of N, which needs the guard (G) ($a > 0$ or $p \geq \frac{1}{2}$; Lemma 6.16(iv)); the wave consumer $a = p = 1$ satisfies it. The tangential modes are the wave modes $1, 2$; the normal modes are the wave modes $J \geq 3$. Following P96-K (index $j \geq 1$) we

write

$$J = j + 2 \qquad (j \geq 1)$$

and index every normal quantity by the *P96-K index* j: Ω_j, z_j, $\bar{z}_j$, matrix entries M_{ij}. For $q \in \mathbb{R}$ let $\ell^N_{a,q}$ be the space of complex sequences $(z_j)_{j\geq 1}$ with

$$\|z\|^2_{a,q} = \sum_{j\geq 1} \omega_{a,q}(j+2)^2 |z_j|^2 < \infty, \qquad \omega_{a,q}(J) = J^q e^{aJ}.$$

It is realized in Lean as the closed subspace of `lp (fun _ : PNat => Complex) 2` (weighted coordinates at weight (a, q), Definition 3.1) of vectors whose entries at $J = 1, 2$ vanish, the entry at J being the coordinate $j = J - 2$; this is the convention of Lemma 3.8. The weights are the wave weights $\omega_{a,q}(J)$ at the wave index, not P96-K's $j^q e^{aj}$; see Remark 6.2. We write $\|\cdot\| = \|\cdot\|_{a,p}$ on the normal variables and $\|\cdot\|_{\bar{p}} = \|\cdot\|_{a,\bar{p}}$ on gradients. On $\mathbb{C}^2$ (angles θ, actions y, frequencies ω) $|\cdot|$ is the max norm; on $\mathbb{Z}^2$, $|k|_1 = |k_1| + |k_2|$.

The unweighted bilinear pairing is $\langle x, x' \rangle_0 = \sum_{j\geq 1} x_j x'_j$ (no conjugation), and on $\mathbb{C}^2$, $\langle b, y \rangle = b_1 y_1 + b_2 y_2$. Since $\omega_{a,q}(J)\, \omega_{a,q'}(J) = J^{q+q'} e^{2aJ} \geq 1$ whenever $q + q' \geq 0$, Cauchy–Schwarz gives

$$|\langle x, x' \rangle_0| \leq \|x\|_{a,q} \|x'\|_{a,q'} \qquad (q + q' \geq 0). \tag{9}$$

The normal phase variables are $(u, v) \in \ell^N_{a,p} \times \ell^N_{a,p}$ (P96-K's u, v; wave $u_j = q_{j+2}$, $v_j = p_{j+2}$), and

$$z = \frac{u + iv}{\sqrt{2}}, \qquad \bar{z} = \frac{u - iv}{\sqrt{2}}, \qquad u = \frac{z + \bar{z}}{\sqrt{2}}, \qquad v = \frac{z - \bar{z}}{i\sqrt{2}},$$

with $z, \bar{z}$ independent complex variables on complex domains. This is the W0 convention (Definition 1.11); P96-K p. 124 uses $z = (u - iv)/\sqrt{2}$, which exchanges z and $\bar{z}$ (Remark 3.32). From these formulas,

$$\|z\|, \|\bar{z}\| \leq \frac{\|u\| + \|v\|}{\sqrt{2}}, \qquad \|u\| + \|v\| \leq \sqrt{2}\, (\|z\| + \|\bar{z}\|). \tag{10}$$

The angle $\theta \in \mathbb{C}^2$ has period one ($x = 2\pi\theta$ is P96-K's angle); the strip is $T_s = \{\theta \in \mathbb{C}^2 : |\operatorname{Im} \theta| < s\}$ with $s = s_K/(2\pi)$, and the domain $D(s, r)$ is that of Definition 3.31 (with the normal norms above). The symplectic form is the pull-back $2\pi \sum_i d\theta_i \wedge dy_i + \sum_j du_j \wedge dv_j$, so Hamilton's equations are

$$\dot{\theta} = \frac{1}{2\pi} H_y, \quad \dot{y} = -\frac{1}{2\pi} H_\theta, \quad \dot{u} = H_v, \quad \dot{v} = -H_u, \qquad \text{equivalently} \quad \dot{z} = -iH_{\bar{z}}, \quad \dot{\bar{z}} = iH_z, \tag{11}$$

where $H_u, H_v, H_z, H_{\bar{z}}$ are gradients for $\langle \cdot, \cdot \rangle_0$ and $H_z = (H_u - iH_v)/\sqrt{2}$, $H_{\bar{z}} = (H_u + iH_v)/\sqrt{2}$, $H_u = (H_z + H_{\bar{z}})/\sqrt{2}$, $H_v = i(H_z - H_{\bar{z}})/\sqrt{2}$. Frequencies are angular: ω is P96-K's ω, and the period-one frequency of the frozen statement is $\nu = \omega/(2\pi)$.

Source: P96-K pp. 120, 124 (phase space $\mathcal{P}^{a,p}$, $z = (u - iv)/\sqrt{2}$); conversions ours (W1 Definition 3.31, Remark 3.32). Lean guards: no division; θ-periodicity is `Function.Periodic` in each coordinate.

Remark 6.2 (Why the wave weights are kept). P96-K weights the j-th normal coordinate by $j^p e^{aj}$; the wave space weights it by $(j + 2)^p e^{a(j+2)}$. The two norms are equivalent within the factor $3^p e^{2a}$ (Remark 5.30). Every estimate of this part is insensitive to the choice of positive weights (Lemma 6.14) except one: the diagonal of an operator $\ell^N_{a,p} \to \ell^N_{a,\bar{p}}$ is a multiplier μ_j with $|\mu_j|\, (j + 2)^{\bar{p}-p} \leq \|\cdot\|$, and $(j + 2) \geq j$ is all that P96-K needs there (Theorem 6.19(iv)). So W4 works with the wave weights throughout and never incurs the factor $3^p e^{2a}$.

Definition 6.3 (Normal operators and their norms)**.** $\mathcal{L}$ is the Banach space of bounded linear maps $\ell^N_{a,p} \to \ell^N_{a,\bar{p}}$ with the operator norm $\|\cdot\|$. For $M \in \mathcal{L}$ its matrix is $M_{ij} = \langle M e_j, e_i \rangle_0$, where e_j is the j-th unit sequence; M^T is the map with matrix (M_{ji}) when it lies in $\mathcal{L}$; M is *symmetric* if $M^T = M$. The *bilinear norm* of a bounded bilinear form β on $\ell^N_{a,p}$ is $\|\beta\|_{\text{bil}} = \sup\{|\beta(z, z')| : \|z\| = \|z'\| = 1\}$; $\mathcal{B}$ is the Banach space of these forms, and $\Lambda : \mathcal{L} \to \mathcal{B}$, $\Lambda M(z, z') = \langle Mz, z' \rangle_0$, satisfies $\|\Lambda M\|_{\text{bil}} \leq \|M\|$

by (9). The *dual norm* of c is $\|c\|_* = \sup\{|\langle c,z\rangle_0| : \|z\|=1\} = \big(\sum_j |c_j|^2\omega_{a,p}(j+2)^{-2}\big)^{1/2}$, and ℓ^N_* is the space of c with $\|c\|_*<\infty$ (a weighted ℓ^2, Lean: `lp` with weight $\omega_{a,p}(j+2)^{-1}$).

Source: ours (P96-K p. 127 speaks of "bounded linear operators $\ell^{a,p}\to\ell^{a,\bar p}$" and their "operator norm $\|\cdot\|$").

Definition 6.4 (Taylor Hamiltonians)**.** Let $s>0$. The class $\mathcal{T}_s$ consists of the functions

(12) $R(\theta,y,z,\bar z) = a(\theta)+\langle b(\theta),y\rangle+\langle c^+(\theta),z\rangle_0+\langle c^-(\theta),\bar z\rangle_0+\langle P(\theta)z,z\rangle_0+\langle Q(\theta)z,\bar z\rangle_0+\langle P'(\theta)\bar z,\bar z\rangle_0$

whose *components* $a:T_s\to\mathbb{C}$, $b:T_s\to\mathbb{C}^2$, $c^\pm:T_s\to\ell^N_{a,\bar p}$, $P,P',Q,Q^T:T_s\to\mathcal{L}$ are complex differentiable, 1-periodic in each θ_i, and bounded, with $P(\theta)$, $P'(\theta)$ symmetric. For $R\in\mathcal{T}_s$ and $(\theta,y,z,\bar z)\in T_s\times\mathbb{C}^2\times\ell^N_{a,p}\times\ell^N_{a,p}$ put

$$R_y = b,\quad R_\theta=(\partial_{\theta_1}R,\partial_{\theta_2}R),\quad R_z = c^+ + 2Pz+Q^T\bar z,\quad R_{\bar z}=c^-+Qz+2P'\bar z,$$

with R_u,R_v as in Convention 6.1; $R_z,R_{\bar z},R_u,R_v\in\ell^N_{a,\bar p}$. The *field* is $X_R=\big(\frac{1}{2\pi}R_y,-\frac{1}{2\pi}R_\theta,R_v,-R_u\big)$ and, for $r>0$,

$$|X_R|_r = \frac{1}{2\pi}|R_y|+\frac{1}{2\pi r^2}|R_\theta|+\frac{1}{r}\big(\|R_u\|_{\bar p}+\|R_v\|_{\bar p}\big),$$

the norm of Definition 3.31 in period-one coordinates. R is *real* if for real θ: $a(\theta)\in\mathbb{R}$, $b(\theta)\in\mathbb{R}^2$, $c^-(\theta)=\overline{c^+(\theta)}$, $P'(\theta)=\overline{P(\theta)}$ and $Q(\theta)^T=\overline{Q(\theta)}$ (entrywise conjugation); equivalently R is real at real points $\theta\in\mathbb{R}^2$, $y\in\mathbb{R}^2$, $\bar z=\overline{z}$. The *mean* of a component g is $[g]=\hat g_0=\int_{[0,1]^2}g(x)\,dx$ (Bochner integral over the real torus).

Source: P96-K (3) p. 124 ("$R=\sum_{2|m|+|q+\bar q|\le 2}\sum_k R_{kmq\bar q}e^{i\langle k,x\rangle}y^m z^q\bar z^{\bar q}$") and the decomposition $R=R^0+R^1+R^2$ on p. 125; here $R^{00}=a+\langle b,y\rangle$, $R^{10}=c^+$, $R^{01}=c^-$, $R^{20}=P$, $R^{11}=Q$, $R^{02}=P'$. The derivatives are the gradients of the polynomial (12): for symmetric P, $D_z\langle Pz,z\rangle_0 h=2\langle Pz,h\rangle_0$, and $D_z\langle Qz,\bar z\rangle_0 h=\langle Q^T\bar z,h\rangle_0$.

Definition 6.5 (Frequencies, divisors, Melnikov conditions)**.** Let $\Pi\subset\mathbb{R}^2$, and let $\omega:\Pi\to\mathbb{R}^2$, $\Omega=(\Omega_j)_{j\ge1}:\Pi\to\mathbb{R}^{\mathbb{N}_+}$ satisfy, for constants $M_0,M_\omega,M_\Omega\ge0$ and all $\xi,\zeta\in\Pi$,

(13)
$$\Omega_j = j+2+\tilde\Omega_j,\quad \sup_j j|\tilde\Omega_j(\xi)|\le M_0,\quad |\omega(\xi)-\omega(\zeta)|\le M_\omega|\xi-\zeta|,\quad \sup_j j|\Omega_j(\xi)-\Omega_j(\zeta)|\le M_\Omega|\xi-\zeta|,$$

and put $M=M_\omega+M_\Omega$ (P96-K's $M=|\omega|^{\mathcal{L}}+|\Omega|^{\mathcal{L}}_{-\delta}$, $-\delta=1$). Let $\tau\ge0$, $A_0=1$ and $A_k=1+|k|_1^\tau$ for $k\ne0$ (so $A_0=1$ also when $\tau=0$; this is the convention throughout this part), $|l|=\sum_j|l_j|$ and $\langle l\rangle=\max(1,|\sum_j jl_j|)$ for finitely supported $l\in\mathbb{Z}^{\mathbb{N}_+}$, and

$$\mathcal{Z}=\{(k,l)\in\mathbb{Z}^2\times\mathbb{Z}^{\mathbb{N}_+}:(k,l)\ne0,\ |l|\le2\}.$$

For $\alpha>0$, the *Melnikov conditions* (H_α) are

$$|\langle k,\omega(\xi)\rangle+\langle l,\Omega(\xi)\rangle|\ge\alpha\frac{\langle l\rangle}{A_k}\qquad\text{for all }\xi\in\Pi,\ (k,l)\in\mathcal{Z}.$$

The divisors used below are, for $k\in\mathbb{Z}^2$ and $i,j\ge1$ (with ξ suppressed):

$$\delta^0_k=\langle k,\omega\rangle\ (k\ne0),\qquad \delta^+_{k,j}=\langle k,\omega\rangle-\Omega_j,\qquad \delta^-_{k,j}=\langle k,\omega\rangle+\Omega_j,$$
$$\delta^P_{k,ij}=\langle k,\omega\rangle-\Omega_i-\Omega_j,\quad \delta^{P'}_{k,ij}=\langle k,\omega\rangle+\Omega_i+\Omega_j,\quad \delta^Q_{k,ij}=\langle k,\omega\rangle+\Omega_i-\Omega_j\ ((k,i)\ne(0,j)),$$

with the *weights* $w^0=1$, $w^\pm_j=j$, $w^P_{ij}=w^{P'}_{ij}=i+j$, $w^Q_{ij}=\max(1,|i-j|)$.

Source: P96-K p. 121 ("$\langle l\rangle_d=\max(1,|\sum j^d l_j|)$, $A_k=1+|k|^\tau$", Z), Lemma 1 p. 125 (the hypothesis), Assumption B p. 120; $d=1$. The divisor list is P96-K p. 125 ("$\langle k,\omega\rangle+\langle q-\bar q,\Omega\rangle$") in the W0 sign convention, see Lemma 6.16(iii). P96-K's $|k|$ for $k\in\mathbb{Z}^n$ is not specified; we use $|k|_1$, which is what the lattice sums of Lemma 6.12 need.

Lemma 6.6 (The wave frequencies)**.** *Let $\lambda_J = \sqrt{J^2+m}$, $b = (\lambda_1^{-1}, \lambda_2^{-1})$, and for $\xi \in \mathbb{R}^2$*

$$\omega(\xi) = (\lambda_1, \lambda_2) + A\xi, \qquad \Omega_j(\xi) = \lambda_{j+2} + \frac{3}{2\pi\lambda_{j+2}}\langle b, \xi\rangle \quad (j \geq 1),$$

with A of Theorem 2.10. Then for every $\Pi \subset \mathbb{R}^2$:

(i) (13) holds with $M_\omega = \frac{9}{8\pi\lambda_1^2} + \frac{3}{2\pi\lambda_1\lambda_2}$, $M_\Omega = \frac{3}{2\pi}(\lambda_1^{-1}+\lambda_2^{-1})$ and $M_0 = \frac{m}{2} + \frac{3}{2\pi}\sup_\Pi|\langle b,\xi\rangle|$ (when finite). In particular P96-K Assumption B holds with $d = 1$, $\delta = -1$ and the fixed sequence $\bar\Omega_j = j+2$, whose constant 2 is a lower-order term allowed by Assumption B.

(ii) (Reindexing.) Write $\langle l\rangle^{\mathrm{w}} = \max(1, |\sum_j (j+2)l_j|)$ for the wave index. For $1 \leq |l| \leq 2$, $\langle l\rangle \leq \langle l\rangle^{\mathrm{w}} \leq 3\langle l\rangle$ unless $\sum_j l_j = 0$, in which case $\langle l\rangle^{\mathrm{w}} = \langle l\rangle$. Hence (H_α) stated with $\langle l\rangle^{\mathrm{w}}$ implies (H_α), and (H_α) implies the conditions with $\langle l\rangle^{\mathrm{w}}$ and $\alpha/3$.

Proof. **Step S6.1: Lipschitz constants.** $|A(\xi-\zeta)| \leq \max_i \sum_{i'} A_{ii'}|\xi-\zeta|$, the maximal row sum being the first since $\lambda_1 < \lambda_2$ and all $A_{ii'} > 0$ (Theorem 2.10). For Ω, $j|\Omega_j(\xi) - \Omega_j(\zeta)| = \frac{3j}{2\pi\lambda_{j+2}}|\langle b, \xi-\zeta\rangle| \leq \frac{3}{2\pi}|b|_1|\xi-\zeta|$ since $\lambda_{j+2} > j$.

Step S6.2: Tails. $\tilde\Omega_j = (\lambda_{j+2} - (j+2)) + \frac{3\langle b,\xi\rangle}{2\pi\lambda_{j+2}}$ and $0 < \lambda_J - J = m/(\lambda_J + J) \leq m/(2J)$, so $j|\tilde\Omega_j| \leq \frac{m}{2} + \frac{3}{2\pi}|\langle b,\xi\rangle|$. This is Lemma 3.8 in the P96-K index.

Step S6.3: Index comparison. $\sum_j (j+2)l_j = \sum_j jl_j + 2\sum_j l_j$. If $\sum l_j = 0$ the two agree. Otherwise, up to a global sign, $l = e_j$ ($\sum jl_j = j$, $\sum(j+2)l_j = j+2 \leq 3j$) or $l = e_i + e_j$ ($i+j$ and $i+j+4 \leq 3(i+j)$), and the ratio lies in $[1,3]$. □

Source: ours; P96-W p. 290 (frequencies of the normal form) with the corrected A, B (E-PDE-W-1, Theorem 2.10); P96-K Assumption B p. 120. The source applies P96-K with normal index $j \geq 1$ without spelling out the shift $j \mapsto j+2$.

Lemma 6.7 (The conditions with $k = 0$ for the wave frequencies)**.** *In the setting of Lemma 6.6, let $\alpha_0(m) = 2/\sqrt{9+m}$. If $\xi \in \mathbb{R}^2$ satisfies $|\langle b,\xi\rangle| \leq 2\pi$ and $0 < \alpha \leq \alpha_0(m)$, then $|\langle l, \Omega(\xi)\rangle| \geq \alpha\langle l\rangle$ for every l with $1 \leq |l| \leq 2$. That is, the conditions (H_α) with $k = 0$ hold on $\Pi \subset \{|\langle b,\xi\rangle| \leq 2\pi\}$.*

Proof. Write $\beta = \frac{3}{2\pi}\langle b,\xi\rangle$, so $|\beta| \leq 3$, and $J = j+2 \geq 3$, $I = i+2$.

Step S6.4: One mode. $\Omega_j \geq \lambda_J - |\beta|/\lambda_J \geq J - 1 = j+1 > \alpha\langle e_j\rangle$, since $\lambda_J > J \geq 3$ and $\alpha \leq 1$. The same bound gives $\Omega_i + \Omega_j \geq i+j+2 > \alpha\langle e_i + e_j\rangle$.

Step S6.5: Differences. For $i > j$: $\Omega_i - \Omega_j = (\lambda_I - \lambda_J)\big(1 - \beta/(\lambda_I\lambda_J)\big) \geq \frac{2}{3}(\lambda_I - \lambda_J)$, using $\frac{1}{\lambda_I} - \frac{1}{\lambda_J} = -\frac{\lambda_I - \lambda_J}{\lambda_I\lambda_J}$ and $\lambda_I\lambda_J > 9$. Further $\lambda_I - \lambda_J = \frac{(I-J)(I+J)}{\lambda_I + \lambda_J} \geq \frac{I-J}{\sqrt{1+m/9}}$ because $\lambda_K \leq K\sqrt{1+m/9}$ for $K \geq 3$. Hence $\Omega_i - \Omega_j \geq \alpha_0(m)(i-j)$, and $\langle e_i - e_j\rangle = i-j$. Signs of l are symmetric. □

Source: ours. Superseded by Lemma 9.12 (W7) and used by no later result; kept for reference. A numerical check over $j \leq 150$ and $m \in \{10^{-3}, \frac{1}{2}, 1, 10, 10^3\}$ gives minimal ratio ≥ 1.12 (`.drafts/W/bp-scratch-w4/checks.py`). This settles the sets $\mathcal{R}^0_{0l}(\alpha)$ of P96-K Theorem A for W7: they are empty for $\alpha \leq \alpha_0(m)$.

Remark 6.8 (What is checked about the Melnikov conditions). The conditions with $k \neq 0$ cannot hold for all ξ; P96-K Theorem A removes the resonance zones and Theorem D (milestone W7) estimates their measure. What W4 checks is that the conditions of P96-K Lemma 1 are the right ones in the wave setting: (a) they are stated in the P96-K index j with $d = 1$, and the reindexing changes $\langle l\rangle$ by at most the factor 3 (Lemma 6.6(ii)); (b) the exchange $z \leftrightarrow \bar z$ between W0 and P96-K replaces l by $-l$, which maps $\mathcal{Z}$ to itself and preserves $\langle l\rangle$, so the conditions are the same; (c) every divisor of Definition 6.5 is of the form $\langle k,\omega\rangle + \langle l,\Omega\rangle$ with $(k,l) \in \mathcal{Z}$, and $\langle l\rangle \geq w$ (Lemma 6.13); (d) the conditions with $k = 0$ hold outright for small α (Lemma 6.7).

6.3. **Fourier series with Banach-space coefficients.**

Remark 6.9 (The operator-valued square-sum bound of P96-K p. 127 is false). P96-K p. 127 states, for the operator-valued coefficients $\tilde{R}_k$ of $\tilde{R} = \partial_{\bar{z}}\partial_z R$, "we have, as before, $\sum_k \|\tilde{R}_k\|^2 e^{2|k||s|} \le 2^n \|\tilde{R}\|_{D(s)}$" ("as before" refers to the L^2 bound for $\ell^{a,\bar{p}}$-valued coefficients on p. 126, printed there with $\|\dot{R}\|^2_{D(s)}$; on p. 127 the square on the right side is missing). This is false, even with the square. In period-one notation, fix $s > 0$, $N \ge 1$ and let $A_N(\theta)$ be the diagonal operator on ℓ^2 with

$$A_N(\theta)e_j = e^{-2\pi j s}e^{2\pi i j\theta_1}e_j \quad (1 \le j \le N), \qquad A_N(\theta)e_j = 0 \quad (j > N).$$

It is entire and 1-periodic, and on T_s its entries have modulus $e^{-2\pi j(s+\operatorname{Im}\theta_1)} < 1$, so $\sup_{T_s}\|A_N(\theta)\| = 1$ (approached as $\operatorname{Im}\theta_1 \to -s$). Its nonzero Fourier coefficients are $\hat{A}_{(j,0)} = e^{-2\pi j s}e_j \otimes e_j$, of norm $e^{-2\pi j s}$, so

$$\sum_k \|\hat{A}_k\|^2 e^{4\pi|k|_1 s} = \sum_{j=1}^{N} 1 = N,$$

which exceeds $2^n \sup\|A_N\|^2 = 4$ for $N \ge 5$ ($n = 2$; verified in `.drafts/W/bp-scratch-w4/checks.py`). Diagonal operators between weighted spaces are transported to this example by the diagonal maps of Definition 3.1, and the example is the mixed Hessian of the quadratic Hamiltonian $\sum_{j\le N}\mu_j(\theta)z_j\bar{z}_j$, so it lies in the class P96-K uses. The Hilbert-space bound (Lemma 3.35(i)) is true for $\mathcal{H}$-valued maps; it fails here because $\mathcal{L}$ with the operator norm is not a Hilbert space. The valid replacement is the Banach-valued bound $\|\hat{A}_k\| \le \sup_{T_s}\|A\|\, e^{-2\pi|k|_1 s}$ (attained here for every $k = (j, 0)$), followed by an ℓ^1 summation after a strip loss. This is review finding W1-R1, rechecked here.

Remark 6.10 (What W4 uses from W1). W4 uses exactly the following, for a complex Banach space E, $s > 0$ and $f : T_s \to E$ complex differentiable, 1-periodic, with $\|f(\theta)\| \le m$ on T_s, and with $\hat{f}_k = \int_{[0,1]^2} f(x)e^{-2\pi i\langle k,x\rangle}dx$:

- (F1) $\|\hat{f}_k\| \le m\, e^{-2\pi|k|_1 s}$ for all $k \in \mathbb{Z}^2$ (contour shift with the sign correction of W1-R3);
- (F2) if $c_k \in E$ and $\sum_k \|c_k\|e^{2\pi|k|_1 s'} < \infty$, then $g(\theta) = \sum_k c_k e^{2\pi i\langle k,\theta\rangle}$ converges absolutely and uniformly on $T_{s'}$, g is complex differentiable and 1-periodic there, $\hat{g}_k = c_k$, and $\|g(\theta)\| \le \sum_k \|c_k\|e^{2\pi|k|_1 s'}$;
- (F3) (uniqueness, scalar case $E = \mathbb{C}$) if f is as above and $\hat{f}_k = 0$ for all k, then $f = 0$ on $T_{s'}$ for every $s' < s$, hence on T_s.

(F1) is Lemma 3.36(i), (F2) is Lemma 3.36(iii), and (F3) is Lemma 3.35(iii) with $\mathcal{H} = \mathbb{C}$ (a Hilbert space; only scalar uniqueness is used). The Hilbert-valued square-sum bound Lemma 3.35(i) is not used for operators. The $n = 2$ lattice sums are proved below with sharper constants (Lemma 6.12); Lemma 3.37 gives the same powers of σ for general n.

Lemma 6.11 (Coefficients of linear images and derivatives). *Let $f : T_s \to E$ be as in Remark 6.10.*

- *(i) For a continuous linear $\Lambda : E \to E'$, $\widehat{(\Lambda f)}_k = \Lambda\hat{f}_k$.*
- *(ii) For $i \in \{1, 2\}$, $\partial_{\theta_i} f$ is complex differentiable and 1-periodic on T_s, and $\widehat{(\partial_{\theta_i} f)}_k = 2\pi i k_i \hat{f}_k$.*
- *(iii) If f depends on $\xi \in \Pi$ and $\|f(\theta,\xi) - f(\theta,\zeta)\| \le m^{\mathcal{L}}|\xi - \zeta|$ on T_s, then $\|\hat{f}_k(\xi) - \hat{f}_k(\zeta)\| \le m^{\mathcal{L}}|\xi - \zeta|e^{-2\pi|k|_1 s}$.*

Proof. **Step S6.6: Linear images.** (W4-E4): Λ commutes with the integral defining $\hat{f}_k$.

Step S6.7: Derivatives. The derivative of a complex differentiable map is complex differentiable ((W1-E4) in the variable θ_i; on the real torus it suffices that f is C^1), and periodicity passes to $\partial_{\theta_i} f$. On the real torus, $\partial_{x_i}\big(f(x)e^{-2\pi i\langle k,x\rangle}\big) = (\partial_{x_i} f)e^{-2\pi i\langle k,x\rangle} - 2\pi i k_i f(x)e^{-2\pi i\langle k,x\rangle}$ and the left

side integrates to zero over a period in x_i ((W4-E4)); integrate over the other variable (Fubini for continuous functions on $[0,1]^2$).

Step S6.8: Parameters. Apply (F1) to $f(\cdot,\xi)-f(\cdot,\zeta)$, whose coefficients are $\hat f_k(\xi)-\hat f_k(\zeta)$. □

Source: standard.

Lemma 6.12 (Lattice sums in $\mathbb{Z}^2$)**.** *Let $\sigma>0$ and $S_q(\sigma)=\sum_{k\in\mathbb{Z}^2}|k|_1^q e^{-2\pi|k|_1\sigma}$ for $q\ge 0$ (with $0^0=1$).*

(i) $\#\{k\in\mathbb{Z}^2:|k|_1=N\}=4N$ *for* $N\ge 1$.
(ii) $S_0(\sigma)=\coth(\pi\sigma)^2\le(1+(\pi\sigma)^{-1})^2$.
(iii) For $q>0$, $S_q(\sigma)\le\dfrac{4}{\pi\sigma}\Big(\dfrac{q+1}{e\pi\sigma}\Big)^{q+1}$.
(iv) Let $\tau\ge 0$, $A_0=1$ *and* $A_k=1+|k|_1^\tau$ *for* $k\ne 0$ *(as in Definition 6.5), and*

$$B_\sigma^{\mathrm W}=\sum_{k\in\mathbb{Z}^2}(1+|k|_1)A_k^2e^{-2\pi|k|_1\sigma}.$$

Then $\sum_k A_k e^{-2\pi|k|_1\sigma}\le B_\sigma^{\mathrm W}$, *and for* $0<\sigma\le 1$,

$$B_\sigma^{\mathrm W}\le\frac{b_{\mathrm W}(\tau)}{\sigma^{2\tau+3}},\qquad b_{\mathrm W}(\tau)=2\big(C_0+C_1+C_{2\tau}+C_{2\tau+1}\big),\qquad C_0=\big(1+\tfrac{1}{\pi}\big)^2,\quad C_q=\frac{4}{\pi}\Big(\frac{q+1}{e\pi}\Big)^{q+1}\ (q>0),$$

a sum of four terms counted with multiplicity (for $\tau=0$ the term $C_{2\tau}$ is C_0; for $\tau=\frac12$, $C_{2\tau}=C_1$ occurs twice). Since $n=2$, $2\tau+3=2\tau+n+1$.

Proof. **Step S6.9: Counting.** For $N\ge 1$, k_1 ranges over $-N,\dots,N$; $|k_1|=N$ gives one point each and $|k_1|<N$ gives two ($k_2=\pm(N-|k_1|)$), in total $2+2(2N-1)=4N$.

Step S6.10: Product. $\sum_k e^{-2\pi|k|_1\sigma}=\big(\sum_{m\in\mathbb{Z}}e^{-2\pi|m|\sigma}\big)^2$ and $\sum_{m\in\mathbb{Z}}e^{-x|m|}=1+\frac{2}{e^x-1}=\coth\frac{x}{2}$ ((W4-E3)); $\coth t=1+\frac{2}{e^{2t}-1}\le 1+\frac1t$.

Step S6.11: Weighted sums. For $q>0$ the term $k=0$ vanishes and, with $x=2\pi\sigma$, $S_q=4\sum_{N\ge1}N^{q+1}e^{-xN}$ by (i). By (W4-E3), $N^{q+1}e^{-xN/2}\le(2(q+1)/(ex))^{q+1}$ and $\sum_{N\ge1}e^{-xN/2}=1/(e^{x/2}-1)\le 2/x$.

Step S6.12: The constant. $A_k\ge 1$ gives the first claim. The term $k=0$ of $B_\sigma^{\mathrm W}$ is 1 ($A_0=1$). For $k\ne0$, $(1+|k|_1)A_k^2\le 2(1+|k|_1)(1+|k|_1^{2\tau})=2\big(1+|k|_1+|k|_1^{2\tau}+|k|_1^{2\tau+1}\big)$. Summing over $k\ne0$, the four sums are at most S_0-1, S_1, $S_{2\tau}$, $S_{2\tau+1}$ (for $\tau=0$ the third is again at most S_0-1), so $B_\sigma^{\mathrm W}\le 1+2(S_0-1)+2(S_1+S'_{2\tau}+S_{2\tau+1})\le 2(S_0+S_1+S'_{2\tau}+S_{2\tau+1})$, where $S'_{2\tau}=S_{2\tau}$ for $\tau>0$ and $S'_0=S_0$. By (ii)–(iii), $S_0\le C_0\sigma^{-2}$ and $S_q\le C_q\sigma^{-(q+2)}$ for $q>0$, each term counted separately even when two exponents coincide ($\tau=\frac12$), and $\sigma^{-(q+2)}\le\sigma^{-(2\tau+3)}$ for $\sigma\le1$ and $q\le 2\tau+1$. □

Source: ours. Checked numerically for $\tau\in\{0,0.5,1,3,3.5,5\}$, $\sigma\in[0.02,1]$ (`checks.py`); e.g. $b_{\mathrm W}(3)\approx 5.76$.

6.4. **Divisors and the multiplier solve.**

Lemma 6.13 (Divisor bounds)**.** *Assume (13) and* (H_α). *For every divisor δ of Definition 6.5 with weight w, all $\xi,\zeta\in\Pi$ and $k\in\mathbb{Z}^2$:*

$$|\delta(\xi)|\ge\frac{\alpha w}{A_k},\qquad\Big|\frac{1}{\delta(\xi)}-\frac{1}{\delta(\zeta)}\Big|\le\frac{2(1+|k|_1)A_k^2M}{\alpha^2 w}|\xi-\zeta|.$$

The weights satisfy $w_{ij}^Q\ge|i-j|$, $w_{ij}^P\ge|i-j|$ for $i\ne j$, and all weights are ≥ 1.

Proof. **Step S6.13: Membership.** Each divisor is $\langle k,\omega\rangle+\langle l,\Omega\rangle$ with $l\in\{0,\mp e_j,\mp(e_i+e_j),e_i-e_j\}$, $|l|\le 2$, and $(k,l)\neq 0$ by the exclusions of Definition 6.5; so $(k,l)\in\mathcal{Z}$. Then $\langle l\rangle = 1,\ j,\ i+j,\ \max(1,|i-j|)$ respectively, which is the weight, and (H_α) gives the lower bound.

Step S6.14: Lipschitz. $|\delta(\xi)-\delta(\zeta)|\le|k|_1 M_\omega|\xi-\zeta|+\sum_j|l_j||\Omega_j(\xi)-\Omega_j(\zeta)|\le(|k|_1+2)M|\xi-\zeta|$ by (13) ($|\Omega_j(\xi)-\Omega_j(\zeta)|\le M_\Omega|\xi-\zeta|/j$ and $|l|\le 2$). Divide by $|\delta(\xi)\delta(\zeta)|\ge\alpha^2 w^2/A_k^2\ge\alpha^2 w/A_k^2$ and use $|k|_1+2\le 2(1+|k|_1)$. □

Source: P96-K pp. 126–127. Deviations: (a) P96-K p. 127 asserts "$|\langle k,\omega\rangle+\Omega_i-\Omega_j|\ge\alpha(1+|i-j|)/A_k$, since $d\ge 1$"; from $\langle l\rangle_1=\max(1,|i-j|)$ only $\alpha\max(1,|i-j|)/A_k$ follows, a factor up to 2 smaller (suspected erratum; harmless, since Lemma A.1 needs only $|i-j|$). (b) For $d=1$ the interpolation $2|i^d-j^d|\ge|i-j|(i^{d-1}+j^{d-1})$ and the factor $(i^\delta+j^\delta)$ of p. 127 are not needed: $|\Omega_i(\xi)-\Omega_i(\zeta)|\le M_\Omega/i$ is used directly.

Lemma 6.14 (Hadamard division of operators)**.** *Let $K_S=1+\pi/\sqrt{3}$ (<2.82). Let $d=(d_{ij})_{i,j\ge1}$ be complex numbers with $|d_{ij}|\le c/|i-j|$ for $i\neq j$ and $|d_{ii}|\le c$. Let μ,ν be positive sequences and A a matrix whose weighted matrix $(\mu_i A_{ij}\nu_j)$ defines a bounded operator on ℓ^2 of norm N_A. Then $(\mu_i d_{ij}A_{ij}\nu_j)$ defines a bounded operator of norm $\le cK_S N_A$. In particular:*

- *(i) ($\mu_i=\omega_{a,\bar p}(i+2)$, $\nu_j=\omega_{a,p}(j+2)^{-1}$.) The Hadamard map $\mathcal{H}_d: M\mapsto(d_{ij}M_{ij})$ is a bounded linear map $\mathcal{L}\to\mathcal{L}$ of norm $\le cK_S$, and $\mathcal{H}_d(M)^T=\mathcal{H}_{d^T}(M^T)$.*
- *(ii) ($\mu_i=\nu_i=\omega_{a,p}(i+2)^{-1}$.) The same Hadamard map on bilinear forms, $\beta\mapsto\big((z,z')\mapsto\sum d_{ij}\beta(e_j,e_i)z_j z'_i\big)$, is bounded $\mathcal{B}\to\mathcal{B}$ with norm $\le cK_S$, and $\Lambda\circ\mathcal{H}_d=\mathcal{H}_d\circ\Lambda$.*
- *(iii) For diagonal d ($d_{ij}=0$ for $i\neq j$), $|d_{ii}|\le c$, the bound improves to c, for every μ,ν; in particular $|M_{ii}|\mu_i\nu_i\le\|M\|$ for $M\in\mathcal{L}$, i.e. $(i+2)|M_{ii}|\le\|M\|$.*

Proof. **Step S6.15: Transport.** Apply Lemma 3.39(iii) to the ℓ^2-matrix $\hat A=(\mu_i A_{ij}\nu_j)$ and $\hat F=(\mu_i d_{ij}A_{ij}\nu_j)$: $|\hat F_{ij}|\le c|\hat A_{ij}|/|i-j|$ off the diagonal and $|\hat F_{ii}|\le c|\hat A_{ii}|$.

Step S6.16: Representatives. In (i), the weighted matrix of $M\in\mathcal{L}$ is the matrix of its representative $\tau_{a,\bar p}\circ M\circ\kappa_{a,p}$ (Definition 3.1, restricted to the normal entries), so bounded weighted matrices are exactly elements of $\mathcal{L}$ with the same norm; the transpose identity is entrywise. In (ii), let $\hat\beta(x,y)=\beta(\kappa x,\kappa y)$ on $\ell^2_{\mathbb{C}}\times\ell^2_{\mathbb{C}}$, where $(\kappa x)_j=\mu_j x_j$ with $\mu_j=\omega_{a,p}(j+2)^{-1}$ (the coefficient map of Definition 3.1 on normal entries); it is complex bilinear with $\|\hat\beta\|=\|\beta\|_{\mathrm{bil}}$, since κ is an isometry onto $\ell^N_{a,p}$. Let C be coordinatewise conjugation on $\ell^2_{\mathbb{C}}$ (a conjugate-linear isometry, $\langle Cu,w\rangle=\langle u,w\rangle_0$ by (W1-E1)). The form $B(v,w)=\hat\beta(Cv,w)$ is conjugate-linear in v, linear in w, bounded by $\|\hat\beta\|$; put $T=C\circ B^\sharp\circ C$ with $B^\sharp$ of (W4-E5). Then T is complex linear, continuous, and $\langle Tx,y\rangle_0=\langle CTx,y\rangle=\langle B^\sharp Cx,y\rangle=B(Cx,y)=\hat\beta(x,y)$. By the norm formula of (W4-E5) (applied to CTx, with $\|CTx\|=\|Tx\|$), $\|Tx\|=\sup_{\|y\|\le1}|\hat\beta(x,y)|$, so $\|T\|=\|\hat\beta\|=\|\beta\|_{\mathrm{bil}}$. The matrix of T is $\langle Te_j,e_i\rangle_0=\hat\beta(e_j,e_i)=\mu_j\mu_i\beta(e_j,e_i)$, the weighted matrix. Conversely every bounded operator T on $\ell^2_{\mathbb{C}}$ defines the bounded form $(x,y)\mapsto\langle Tx,y\rangle_0$ of norm $\le\|T\|$ by (9)-type Cauchy–Schwarz. So bounded weighted matrices correspond to elements of $\mathcal{B}$ with the same norm, and Lemma 3.39(iii) transfers; Λ does not change entries.

Step S6.17: Diagonal. (iii) is Lemma 3.39(ii) and Lemma 3.4; $\mu_i\nu_i=(i+2)^{\bar p-p}=i+2$ in (i). □

Source: P96-K Lemma A.1, pp. 145–146, and its use on p. 127 ("$\|\tilde F_k\|\le 3(A_k/\alpha)\|\tilde R_k\|$"); Lemma 3.39. The bilinear version (ii) is ours; it is needed for the θ-derivatives of the quadratic components (Theorem 6.19).

Lemma 6.15 (Multiplier solve)**.** *Let E,E' be complex Banach spaces, $0<\sigma<s$, $\alpha>0$, $\kappa\ge1$, $M\ge0$, and for $k\in\mathbb{Z}^2$ and $\xi\in\Pi$ let $T_k(\xi)\in L(E,E')$ satisfy*

$$\|T_k(\xi)\|\le\frac{\kappa A_k}{\alpha},\qquad \|T_k(\xi)-T_k(\zeta)\|\le\frac{2\kappa(1+|k|_1)A_k^2 M}{\alpha^2}|\xi-\zeta|.$$

Let $f(\cdot,\xi) : T_s \to E$ be complex differentiable, 1-periodic, with $\|f(\theta,\xi)\| \le m$ and $\|f(\theta,\xi) - f(\theta,\zeta)\| \le m^{\mathcal{L}}|\xi-\zeta|$ on T_s. Define

$$(\mathcal{S}f)(\theta,\xi) = \sum_{k\in\mathbb{Z}^2} T_k(\xi)\hat{f}_k(\xi)\, e^{2\pi i\langle k,\theta\rangle}.$$

(i) The series converges absolutely and uniformly on $T_{s-\sigma}$; $\mathcal{S}f(\cdot,\xi)$ is complex differentiable and 1-periodic there, with coefficients $T_k\hat{f}_k$, and $\|\mathcal{S}f(\theta,\xi)\| \le \kappa B^{\mathrm{W}}_\sigma m/\alpha$.

(ii) $\|\mathcal{S}f(\theta,\xi) - \mathcal{S}f(\theta,\zeta)\| \le \frac{\kappa B^{\mathrm{W}}_\sigma}{\alpha}\big(m^{\mathcal{L}} + \frac{2Mm}{\alpha}\big)|\xi-\zeta|$ on $T_{s-\sigma}$.

(iii) If $\partial_{\theta_i} f$ is bounded on T_s, then $\partial_{\theta_i}\mathcal{S}f = \mathcal{S}(\partial_{\theta_i} f)$ on $T_{s-\sigma}$.

(iv) If $\Lambda : E' \to E''$ is continuous linear, then $\Lambda\mathcal{S}f$ is the solve of f with the operators ΛT_k. If $\Lambda_0 : E \to E_0$, $\Lambda_1 : E' \to E_1$ and $T'_k \in L(E_0,E_1)$ satisfy $\Lambda_1 T_k = T'_k\Lambda_0$, then $\Lambda_1\mathcal{S}f = \mathcal{S}'(\Lambda_0 f)$.

Proof. **Step S6.18: Coefficients.** By (F1), $\|T_k\hat{f}_k\| \le \frac{\kappa A_k}{\alpha} m e^{-2\pi|k|_1 s}$, so $\sum_k \|T_k\hat{f}_k\| e^{2\pi|k|_1(s-\sigma)} \le \frac{\kappa m}{\alpha}\sum_k A_k e^{-2\pi|k|_1\sigma} \le \frac{\kappa m}{\alpha} B^{\mathrm{W}}_\sigma$ (Lemma 6.12(iv)); (F2) with $s' = s-\sigma$ gives (i).

Step S6.19: Differences. $T_k(\xi)\hat{f}_k(\xi) - T_k(\zeta)\hat{f}_k(\zeta) = T_k(\xi)\big(\hat{f}_k(\xi) - \hat{f}_k(\zeta)\big) + \big(T_k(\xi) - T_k(\zeta)\big)\hat{f}_k(\zeta)$; by Lemma 6.11(iii) and (F1) its norm is at most $\big(\frac{\kappa A_k}{\alpha} m^{\mathcal{L}} + \frac{2\kappa(1+|k|_1)A_k^2 Mm}{\alpha^2}\big)|\xi-\zeta| e^{-2\pi|k|_1 s}$. Multiply by $e^{2\pi|k|_1(s-\sigma)}$, sum, and use $A_k \le (1+|k|_1)A_k^2$ and the definition of B^{W}_σ; the difference of the two series is the series of differences, bounded as in (F2).

Step S6.20: Commutation. (iii): by Lemma 6.11(ii), $T_k\widehat{(\partial_{\theta_i} f)}_k = 2\pi i k_i T_k\hat{f}_k$, and by Step S6.18 applied to the bounded map $\partial_{\theta_i} f$ these terms are summable with the weight $e^{2\pi|k|_1(s-\sigma)}$. So the termwise derivative of the series of $\mathcal{S}f$ converges uniformly on $T_{s-\sigma}$, and termwise differentiation ((W1-E5)) gives $\partial_{\theta_i}\mathcal{S}f = \sum_k 2\pi i k_i T_k\hat{f}_k e^{2\pi i\langle k,\theta\rangle} = \mathcal{S}(\partial_{\theta_i} f)$.

Step S6.21: Linear images. (iv): Λ commutes with absolutely convergent sums; and $\Lambda_1 T_k\hat{f}_k = T'_k\Lambda_0\hat{f}_k = T'_k\widehat{(\Lambda_0 f)}_k$ by Lemma 6.11(i). □

Source: P96-K pp. 126–127 ("It follows that $\|\dot{F}\|_{D(s-\sigma)} \le \sum_k \|\dot{F}_k\| e^{|k|(s-\sigma)} \le \dots$", "summing up as before"), in the Banach-valued form that replaces the square-sum step (Remark 6.9). Lean: the series is a `tsum` in E', summable in norm by Step S6.18.

6.5. **The Lie derivative along the normal form.** At a fixed real parameter ξ the normal form is formally $N = \langle\omega(\xi), y\rangle + \sum_j \Omega_j(\xi) z_j\bar{z}_j$. Its field is unbounded on $\ell^N_{a,p}$ ($\Omega_j \sim j$). For general $a,p \ge 0$ the series $\sum_j \Omega_j z_j\bar{z}_j$ need not even converge: for $a = p = 0$, $\Omega_j = j+2$ and $z_j = \bar{z}_j = 1/j$ it is $\sum_j (1/j + 2/j^2) = \infty$. So we define the bracket of N with $F \in \mathcal{T}$ as a *weak pairing* through the smoothing gradients of F, which exists for all $a,p \ge 0$, and identify it with an actual derivative of N only under the guard

$$\gamma_{a,p} := \sup_{j\ge 1} \frac{j+2}{\omega_{a,p}(j+2)^2} < \infty. \tag{G}$$

(G) holds if and only if $a > 0$ or $p \ge \frac12$: for $a > 0$, $J^{1-2p}e^{-2aJ} \to 0$; for $a = 0$, J^{1-2p} is bounded iff $p \ge \frac12$. The wave consumer $a = p = 1$ satisfies (G) with $\gamma_{1,1} = e^{-6}/3$ (at $J = 3$).

Lemma 6.16 (Lie derivative and weak bracket)**.** *Assume (13) and let $F \in \mathcal{T}_{s'}$ (Definition 6.4) with components $a_F, b_F, c^{\pm}_F, P_F, Q_F, P'_F$. For $(\theta, y, z, \bar{z}) \in T_{s'} \times \mathbb{C}^2 \times (\ell^N_{a,p})^2$ put*

$$L_N F = \frac{1}{2\pi}\langle\omega, F_\theta\rangle - i\langle F_z, \Omega z\rangle_0 + i\langle F_{\bar{z}}, \Omega\bar{z}\rangle_0, \qquad \langle\omega, F_\theta\rangle = \sum_i \omega_i\partial_{\theta_i} F,$$

and the weak bracket

$$\langle\!\langle N, F\rangle\!\rangle = -\frac{1}{2\pi}\langle\omega, F_\theta\rangle + \langle\Omega\bar{z}, \dot{Z}\rangle_0 + \langle\Omega z, \dot{\bar{Z}}\rangle_0, \qquad (\dot{Z}, \dot{\bar{Z}}) = (-iF_{\bar{z}}, iF_z),$$

which is the formal expression for $DN \cdot X_F$: $-\frac{1}{2\pi}F_\theta$ is the y-component and $(\dot Z, \dot{\bar Z})$ the normal components of X_F by (11).

(i) ($a, p \ge 0$.) $\Omega z \in \ell^N_{a,p-1}$ with $\|\Omega z\|_{a,p-1} \le (1+M_0)\|z\|$; the pairings converge absolutely, $L_N F$ and $\langle\!\langle N, F\rangle\!\rangle$ are continuous in $(z, \bar z)$ and polynomials of degree ≤ 2 in $(y, z, \bar z)$.

(ii) ($a, p \ge 0$.) $\langle\!\langle N, F\rangle\!\rangle = -L_N F$.

(iii) (Coefficients, $a, p \ge 0$.) For finitely supported $z, \bar z$, $L_N F$ is the finite expression obtained from (12) for F by replacing the coefficient of $e^{2\pi i\langle k,\theta\rangle}$ in a_F, b_F, $c^+_{F,j}$, $c^-_{F,j}$, $P_{F,ij}$, $Q_{F,ij}$, $P'_{F,ij}$ by $i\delta$ times it, with $\delta = \langle k, \omega\rangle$ (for a_F, b_F), $\delta^+_{k,j}$, $\delta^-_{k,j}$, $\delta^P_{k,ij}$, $\delta^Q_{k,ij}$ (with $\delta^Q_{0,jj} = 0$), $\delta^{P'}_{k,ij}$ respectively.

(iv) (Derivative, under (G).) If (G) holds, then $N_\Omega(z, \bar z) = \sum_j \Omega_j z_j \bar z_j$ converges absolutely and is a continuous bilinear form on $\ell^N_{a,p} \times \ell^N_{a,p}$ with $|N_\Omega(z, \bar z)| \le (1+M_0)\gamma_{a,p}\|z\|\|\bar z\|$; $N = \langle\omega, y\rangle + N_\Omega(z, \bar z)$ is a continuous polynomial on $\mathbb{C}^2 \times \mathbb{C}^2 \times (\ell^N_{a,p})^2$ (independent of θ), Fréchet differentiable with $DN(w)(\Theta, Y, Z, \bar Z) = \langle\omega, Y\rangle + N_\Omega(Z, \bar z) + N_\Omega(z, \bar Z)$, and

$$DN(w) \cdot X_F(w) = \langle\!\langle N, F\rangle\!\rangle(w) = -L_N F(w).$$

Without (G) the left side is not defined in general (example above), and only (ii) is available.

Proof. **Step S6.22: Bounds.** $|\Omega_j| \le j + 2 + M_0/j \le (1+M_0)(j+2)$, so $|\Omega_j|\omega_{a,p-1}(j+2) \le (1+M_0)\omega_{a,p}(j+2)$. The gradients $F_z, F_{\bar z}$ lie in $\ell^N_{a,\bar p} = \ell^N_{a,p+1}$, and (9) with $q = p-1$, $q' = p+1$ ($q + q' = 2p \ge 0$) bounds the pairings. Continuity in $(z, \bar z)$: F_z, $F_{\bar z}$ are affine continuous in $(z, \bar z)$ and the pairing is a bounded bilinear form on $\ell^N_{a,p-1} \times \ell^N_{a,p+1}$.

Step S6.23: Weak bracket. Substitute $(\dot Z, \dot{\bar Z})$: $\langle\!\langle N, F\rangle\!\rangle = -\frac{1}{2\pi}\langle\omega, F_\theta\rangle - i\langle\Omega\bar z, F_{\bar z}\rangle_0 + i\langle\Omega z, F_z\rangle_0$, which is $-L_N F$ by symmetry of $\langle\cdot,\cdot\rangle_0$. This is an identity of explicit expressions; no derivative of N is involved.

Step S6.24: Coefficients. For finitely supported $z, \bar z$ every pairing is a finite sum. Write $F_z = c^+_F + 2P_F z + Q^T_F \bar z$, $F_{\bar z} = c^-_F + Q_F z + 2P'_F \bar z$ (Definition 6.4) and collect: the coefficient of z_j is $\frac{1}{2\pi}\langle\omega, \partial_\theta c^+_{F,j}\rangle - i\Omega_j c^+_{F,j}$; of $z_i z_j$ (symmetrized), $\frac{1}{2\pi}\langle\omega, \partial_\theta P_{F,ij}\rangle - i(\Omega_i + \Omega_j)P_{F,ij}$; of $z_j \bar z_i$, $\frac{1}{2\pi}\langle\omega, \partial_\theta Q_{F,ij}\rangle - i(\Omega_j - \Omega_i)Q_{F,ij}$; and similarly for c^-_F, P'_F with $+i\Omega$. By Lemma 6.11, the k-th coefficient of $\frac{1}{2\pi}\langle\omega, \partial_\theta g\rangle$ is $i\langle k, \omega\rangle\hat g_k$, and the matrix entries are continuous linear functionals of the operators.

Step S6.25: The guarded derivative. Under (G), $|\Omega_j z_j \bar z_j| \le (1+M_0)(j+2)|z_j||\bar z_j| \le (1+M_0)\gamma_{a,p}\,\omega_{a,p}(j+2)^2|z_j||\bar z_j|$, and Cauchy–Schwarz gives absolute convergence and the bound; so N_Ω is a bounded bilinear form and N a continuous polynomial, whose derivative is the displayed one (Mathlib: `ContinuousLinearMap.hasFDerivAt` for the linear part and the bilinear-map derivative `IsBoundedBilinearMap.hasFDerivAt`). The direction $X_F(w)$ has normal components $\dot Z, \dot{\bar Z} \in \ell^N_{a,\bar p} \subset \ell^N_{a,p}$ (Lemma 3.5) and y-component $-\frac{1}{2\pi}F_\theta$, so $DN(w)\cdot X_F(w) = -\frac{1}{2\pi}\langle\omega, F_\theta\rangle + N_\Omega(\dot Z, \bar z) + N_\Omega(z, \dot{\bar Z})$, and $N_\Omega(\dot Z, \bar z) = \langle\Omega\bar z, \dot Z\rangle_0$, $N_\Omega(z, \dot{\bar Z}) = \langle\Omega z, \dot{\bar Z}\rangle_0$ termwise (both series converge absolutely). This is $\langle\!\langle N, F\rangle\!\rangle$, and (ii) concludes. □

Source: P96-K p. 125 ("by comparison of coefficients … $iF_{kmq\bar q} = R_{kmq\bar q}/(\langle k, \omega\rangle + \langle q - \bar q, \Omega\rangle)$"), in the W0 sign convention: our $\langle\cdot, \Omega\rangle$-terms are those of P96-K with $q \leftrightarrow \bar q$ (Remark 6.8(b)). P96-K writes the bracket as $\{F, N\}$; here $\{F, N\} := L_N F = -\langle\!\langle N, F\rangle\!\rangle$, which equals $-DN \cdot X_F$ under (G). P96-K takes $a \ge 0$, $p \ge 0$ (p. 120) and does not address the convergence of N; the guard (G) is ours. The identity (iv) is the input W5 needs to expand $N \circ X^t_F$; W5 must use it in this form because X_N is unbounded.

6.6. **Reading the components off the field norm.**

Lemma 6.17 (Coefficient extraction)**.** *Let E be a complex Banach space, $\rho_1,\rho_2>0$, and $g(\zeta,\eta)=\sum_{p+q\le 2} g_{pq}\zeta^p\eta^q$ with $g_{pq}\in E$. If $\|g(\zeta,\eta)\|\le K$ whenever $|\zeta|<\rho_1$, $|\eta|<\rho_2$, then $\|g_{pq}\|\le K\rho_1^{-p}\rho_2^{-q}$ for all $p+q\le 2$. The same holds for one variable.*

Proof. For $0<\rho_i'<\rho_i$, $g_{pq}\rho_1'^p\rho_2'^q=\frac{1}{16}\sum_{u,v=0}^{3} i^{-up-vq}g(\rho_1' i^u,\rho_2' i^v)$: expanding g, the monomial (p',q') contributes $\frac{1}{16}\sum_{u,v} i^{u(p'-p)+v(q'-q)}g_{p'q'}\rho_1'^{p'}\rho_2'^{q'}$, and since $|p'-p|,|q'-q|\le 2<4$ this vanishes unless $(p',q')=(p,q)$ ((W4-E1)). The triangle inequality gives $\|g_{pq}\|\le K\rho_1'^{-p}\rho_2'^{-q}$; let $\rho_i'\uparrow\rho_i$. □

Source: ours; it replaces the generalized Cauchy estimate of P96-K Lemma A.3 (p. 147) for the polynomial dependence on $(y,z,\bar z)$.

Lemma 6.18 (Component bounds)**.** *Let $s,r>0$, $R\in\mathcal{T}_s$, and suppose $|X_R(\theta,y,z,\bar z)|_r\le\varepsilon$ on $D(s,r)$. Then for all $\theta\in T_s$ and $i\in\{1,2\}$:*

(i) $|b(\theta)|\le 2\pi\varepsilon$, $\|c^{\pm}(\theta)\|_{\bar p}\le r\varepsilon/\sqrt{2}$, $\|2P(\theta)\|,\|Q(\theta)\|,\|Q(\theta)^T\|,\|2P'(\theta)\|\le\varepsilon$;

(ii) $|\partial_{\theta_i}a|\le 2\pi r^2\varepsilon$, $|\partial_{\theta_i}b|_1\le 2\pi\varepsilon$, $\|\partial_{\theta_i}c^{\pm}\|_*\le 2\sqrt{2}\pi r\varepsilon$, $\|\Lambda\partial_{\theta_i}P\|_{\mathrm{bil}},\|\Lambda\partial_{\theta_i}P'\|_{\mathrm{bil}}\le 4\pi\varepsilon$, $\|\Lambda\partial_{\theta_i}Q\|_{\mathrm{bil}}\le 16\pi\varepsilon$.

Proof. **Step S6.26: Test points.** By (10), the points $(\theta,y,0,0)$ with $|y|<r^2$; $(\theta,0,z,0)$ and $(\theta,0,0,\bar z)$ with $\|z\|,\|\bar z\|<r/\sqrt{2}$; and $(\theta,0,z,\bar z)$ with $\|z\|,\|\bar z\|<r/(2\sqrt{2})$ lie in $D(s,r)$ (for the second, $u=z/\sqrt{2}$, $v=z/(i\sqrt{2})$ and $\|u\|+\|v\|=\sqrt{2}\|z\|$). On $D(s,r)$ the hypothesis gives $|R_y|\le 2\pi\varepsilon$, $|R_\theta|\le 2\pi r^2\varepsilon$, and $\|R_z\|_{\bar p},\|R_{\bar z}\|_{\bar p}\le(\|R_u\|_{\bar p}+\|R_v\|_{\bar p})/\sqrt{2}\le r\varepsilon/\sqrt{2}$.

Step S6.27: Zeroth and first order. $R_y=b$ and $R_z(\theta,0,0,0)=c^+$, $R_{\bar z}(\theta,0,0,0)=c^-$. For $\|z\|=1$, $\zeta\mapsto R_z(\theta,0,\zeta z,0)=c^++2\zeta Pz$ is bounded by $r\varepsilon/\sqrt{2}$ on $|\zeta|<r/\sqrt{2}$, so $\|2Pz\|_{\bar p}\le\varepsilon$ by Lemma 6.17. Likewise $R_z(\theta,0,0,\eta\bar z)=c^++\eta Q^T\bar z$, $R_{\bar z}(\theta,0,\zeta z,0)=c^-+\zeta Qz$, $R_{\bar z}(\theta,0,0,\eta\bar z)=c^-+2\eta P'\bar z$.

Step S6.28: Angle derivatives. $\partial_{\theta_i}R(\theta,y,0,0)=\partial_{\theta_i}a+\langle\partial_{\theta_i}b,y\rangle$: its value at $y=0$ and, for $|y|=1$, the coefficient of ζ in $\zeta\mapsto\partial_{\theta_i}R(\theta,\zeta y,0,0)$ ($|\zeta|<r^2$) give the first two bounds ($|\cdot|_1$ is the dual of the max norm). For $\|z\|=1$, $\zeta\mapsto\partial_{\theta_i}R(\theta,0,\zeta z,0)=\partial_{\theta_i}a+\zeta\langle\partial_{\theta_i}c^+,z\rangle_0+\zeta^2\langle\partial_{\theta_i}Pz,z\rangle_0$ on $|\zeta|<r/\sqrt{2}$ gives $|\langle\partial_{\theta_i}c^+,z\rangle_0|\le 2\pi r^2\varepsilon\sqrt{2}/r$ and $|\langle\partial_{\theta_i}Pz,z\rangle_0|\le 2\pi r^2\varepsilon\cdot 2/r^2$. For $\|z\|=\|\bar z\|=1$, the coefficient of $\zeta\eta$ in $\partial_{\theta_i}R(\theta,0,\zeta z,\eta\bar z)$ on $|\zeta|,|\eta|<r/(2\sqrt{2})$ is $\langle\partial_{\theta_i}Qz,\bar z\rangle_0$, bounded by $2\pi r^2\varepsilon\cdot 8/r^2$. The c^-, P' cases are symmetric.

Step S6.29: Polarization. For symmetric $\beta=\Lambda\partial_{\theta_i}P$ and unit z,z': $\beta(z,z')=\frac{1}{4}\big(\beta(z+z',z+z')-\beta(z-z',z-z')\big)$, so $|\beta(z,z')|\le\frac{1}{4}\cdot 4\pi\varepsilon\,(\|z+z'\|^2+\|z-z'\|^2)=4\pi\varepsilon$ by (W4-E2). ($\partial_{\theta_i}P$ is symmetric because P is.) □

Source: P96-K p. 126 ("$\|\dot R\|_{D(s)}\le r|X_R|_r$"), p. 127 ("$\|\tilde R\|_{D(s)}\le\frac{1}{r}\|R_z\|_{D(s,r)}\le|X_R|_r$") and p. 128 ("All the other components F^{ij} admit the same estimates"); the bounds for the angle derivatives are not in the source, which leaves the y-component of X_F implicit.

6.7. **The homological equation.**

Theorem 6.19 (P96-K Lemma 1 for the wave setting)**.** *Assume Convention 6.1, (13) on $\Pi\subset\mathbb{R}^2$ with $M=M_\omega+M_\Omega$, $\tau\ge 0$, and (H_α) for some $\alpha>0$. Let $0<\sigma<s$, $r>0$, $\varepsilon,\varepsilon^{\mathcal{L}}\ge 0$, and for each $\xi\in\Pi$ let $R(\cdot,\xi)\in\mathcal{T}_s$ satisfy, at every point of $D(s,r)$ and for all $\xi,\zeta\in\Pi$,*

$$|X_{R(\xi)}|_r\le\varepsilon,\qquad |X_{R(\xi)}-X_{R(\zeta)}|_r\le\varepsilon^{\mathcal{L}}|\xi-\zeta|.$$

Define $F(\cdot,\xi)$ by the components (all series in θ, $e_k=e^{2\pi i\langle k,\theta\rangle}$)

$$a_F=\sum_{k\neq 0}\frac{\hat a_k}{i\delta_k^0}e_k,\quad b_F=\sum_{k\neq 0}\frac{\hat b_k}{i\delta_k^0}e_k,\quad (c_F^{\pm})_j=\sum_k\frac{(\hat c_k^{\pm})_j}{i\delta_{k,j}^{\pm}}e_k,$$

$$(P_F)_{ij}=\sum_k \frac{(\hat P_k)_{ij}}{i\delta^P_{k,ij}}e_k,\quad (Q_F)_{ij}=\sum_{(k,i)\neq(0,j)} \frac{(\hat Q_k)_{ij}}{i\delta^Q_{k,ij}}e_k,\quad (P'_F)_{ij}=\sum_k \frac{(\hat P'_k)_{ij}}{i\delta^{P'}_{k,ij}}e_k,$$

(the vector and operator series converging in $\ell^N_{a,\bar p}$ and $\mathcal{L}$, entries as displayed), and

$$\hat N=\langle\hat\omega,y\rangle+\sum_j \hat\Omega_j z_j\bar z_j,\qquad \hat\omega=[b],\quad \hat\Omega_j=([Q])_{jj}.$$

Let $C_1=7+10K_S$ (<36) and $B=B^{\mathrm W}_\sigma$ (Lemma 6.12). Then for every $\xi\in\Pi$:

- *(i) $F(\cdot,\xi)\in\mathcal{T}_{s-\sigma}$, with $\hat a_{F,0}=0$, $\hat b_{F,0}=0$, $(\hat Q_{F,0})_{jj}=0$; if R is real, so are F and $\hat N$ ($\hat\omega\in\mathbb{R}^2$, $\hat\Omega_j\in\mathbb{R}$).*
- *(ii) $L_NF+\hat N=R-\hat a_0$ on $T_{s-\sigma}\times\mathbb{C}^2\times(\ell^N_{a,p})^2$, with L_N of Lemma 6.16; equivalently $R+\langle\!\langle N,F\rangle\!\rangle=\hat N+\hat a_0$ (weak bracket of Lemma 6.16), and, under the guard (G) (in particular for $a=p=1$), $R+DN\cdot X_F=\hat N+\hat a_0$ with the Fréchet derivative of N.*
- *(iii) On $D(s-\sigma,r)$: $|X_F|_r\le\frac{C_1B}{\alpha}\varepsilon$ and $|X_{F(\xi)}-X_{F(\zeta)}|_r\le\frac{C_1B}{\alpha}\big(\varepsilon^{\mathcal{L}}+\frac{2M}{\alpha}\varepsilon\big)|\xi-\zeta|$.*
- *(iv) $|\hat\omega|\le2\pi\varepsilon$, $\sup_j(j+2)|\hat\Omega_j|\le\varepsilon$, and the same with $\varepsilon^{\mathcal{L}}|\xi-\zeta|$ for the differences; on $D(s,r)$, $|X_{\hat N}|_r\le2\varepsilon$ and $|X_{\hat N(\xi)}-X_{\hat N(\zeta)}|_r\le2\varepsilon^{\mathcal{L}}|\xi-\zeta|$.*

Proof. **Step S6.30: Division operators.** For each component define $T_k(\xi)$: division by $i\delta^0_k$ on $\mathbb{C}$ and $\mathbb{C}^2$ (zero for $k=0$); the diagonal multipliers $x\mapsto(x_j/(i\delta^\pm_{k,j}))_j$ on $\ell^N_{a,\bar p}$ and on ℓ^N_*; the Hadamard maps $\mathcal{H}_d$ with $d_{ij}=1/(i\delta_{k,ij})$ for $\delta\in\{\delta^P,\delta^Q,\delta^{P'}\}$ ($d_{jj}=0$ when $k=0$ for δ^Q) on $\mathcal{L}$ and on $\mathcal{B}$. By Lemma 6.13, $|d_{ij}|\le A_k/(\alpha w_{ij})$ with $w_{ij}\ge\max(1,|i-j|)$, and the differences $d(\xi)-d(\zeta)$ satisfy the same with $2(1+|k|_1)A_k^2M|\xi-\zeta|/\alpha^2$ in place of A_k/α. Lemma 6.14 (and Lemma 3.4 for the diagonal cases) gives the hypotheses of Lemma 6.15 with $\kappa=1$ for $a,b,c^\pm$ and $\kappa=K_S$ for P,Q,Q^T,P'.

Step S6.31: The solution is in the class. Each component of F is $\mathcal{S}g$ for the corresponding component g of R (Lemma 6.15(i)); for Q_F^T use $\mathcal{H}_d(M)^T=\mathcal{H}_{d^T}(M^T)$ (Lemma 6.14(i)), where d^T is the division by $\delta^Q_{k,ji}$, which obeys the same bounds. P_F, P'_F are symmetric since $\delta^P_{k,ij}$, $\delta^{P'}_{k,ij}$ are symmetric in (i,j). The vanishing coefficients are the excluded terms.

Step S6.32: The equation. Fix y and finitely supported $z,\bar z$. Then $\theta\mapsto L_NF+\hat N-R+\hat a_0$ is complex differentiable, periodic and bounded on $T_{s-\sigma}$ (the θ-derivatives of the components of F are bounded there by Lemma 6.15(iii) and Lemma 6.18(ii)). By Lemma 6.16(iii) and the definition of F, its k-th coefficient is a finite sum of terms $i\delta\cdot\hat g_k/(i\delta)-\hat g_k=0$ for the nonresonant terms, while the resonant terms are $-\hat a_0+\hat a_0$, $\langle\hat\omega-\hat b_0,y\rangle$ and $(\hat\Omega_j-(\hat Q_0)_{jj})z_j\bar z_j$, all zero. By (F3) the function vanishes. Both sides are continuous in $(z,\bar z)\in(\ell^N_{a,p})^2$ (Lemma 6.16(i); $\hat N$ by (iv) below), and finitely supported sequences are dense (Lemma 3.2(iii)). The second form is Lemma 6.16(ii), the third Lemma 6.16(iv).

Step S6.33: Component bounds of F. By Lemma 6.18, Step S6.30 and Lemma 6.15(i),(iii),(iv), on $T_{s-\sigma}$: $|b_F|\le2\pi\varepsilon B/\alpha$; $\|c^\pm_F\|_{\bar p}\le\frac{B}{\alpha}\frac{r\varepsilon}{\sqrt2}$; $\|2P_F\|,\|Q_F\|,\|Q_F^T\|,\|2P'_F\|\le K_S\frac{B}{\alpha}\varepsilon$; and for $i=1,2$, $|\partial_{\theta_i}a_F|\le\frac{B}{\alpha}2\pi r^2\varepsilon$, $|\partial_{\theta_i}b_F|_1\le\frac{B}{\alpha}2\pi\varepsilon$, $\|\partial_{\theta_i}c^\pm_F\|_*\le\frac{B}{\alpha}2\sqrt2\pi r\varepsilon$, $\|\Lambda\partial_{\theta_i}P_F\|_{\mathrm{bil}},\|\Lambda\partial_{\theta_i}P'_F\|_{\mathrm{bil}}\le K_S\frac{B}{\alpha}4\pi\varepsilon$, $\|\Lambda\partial_{\theta_i}Q_F\|_{\mathrm{bil}}\le K_S\frac{B}{\alpha}16\pi\varepsilon$. (For the bilinear bounds first apply Lemma 6.15(iv) with $\Lambda_0=\Lambda_1=\Lambda$ and the Hadamard maps on $\mathcal{B}$ (Lemma 6.14(ii)), so ΛP_F is the $\mathcal{B}$-valued solve of ΛP, then Lemma 6.15(iii) in $\mathcal{B}$, where $\partial_{\theta_i}\Lambda P=\Lambda\partial_{\theta_i}P$ is bounded by Lemma 6.18(ii). For $c^\pm$ likewise with the inclusion $\ell^N_{a,\bar p}\subset\ell^N_*$, $\|c\|_*\le\|c\|_{\bar p}$ by (9). For $|\cdot|_1$ on $\mathbb{C}^2$, division by a scalar $i\delta^0_k$ multiplies the norm by at most A_k/α.)

Step S6.34: Assembly. Let $(\theta,y,z,\bar z)\in D(s-\sigma,r)$, so $|y|<r^2$ and $\|z\|,\|\bar z\|<r/\sqrt2$ by (10). *θ-component:* $\frac{1}{2\pi}|F_y|=\frac{1}{2\pi}|b_F|\le\frac{B}{\alpha}\varepsilon$. *$y$-component:* $\partial_{\theta_i}F=\partial_{\theta_i}a_F+\langle\partial_{\theta_i}b_F,y\rangle+\sum_\pm\langle\partial_{\theta_i}c^\pm_F,\cdot\rangle_0+$

$\Lambda\partial_{\theta_i}P_F(z,z)+\Lambda\partial_{\theta_i}Q_F(z,\bar z)+\Lambda\partial_{\theta_i}P'_F(\bar z,\bar z)$, so

$$|\partial_{\theta_i}F|\le\frac{B}{\alpha}\varepsilon\,2\pi r^2\Big(1+1+2\cdot1+K_S\big(1+1+4\big)\Big)=\frac{B}{\alpha}\varepsilon\,2\pi r^2(4+6K_S),$$

using $|\langle\beta,y\rangle|\le|\beta|_1|y|$, $\|z\|^2<r^2/2$ and $\|z\|\|\bar z\|<r^2/2$; hence $\frac{1}{2\pi r^2}|F_\theta|\le(4+6K_S)\frac{B}{\alpha}\varepsilon$. *Normal components:* $F_z=c_F^+ +2P_Fz+Q_F^T\bar z$ gives $\|F_z\|_{\bar p}\le\frac{B}{\alpha}\frac{r\varepsilon}{\sqrt2}(1+2K_S)$, and the same for $F_{\bar z}$; by the formulas for F_u,F_v, $\frac1r(\|F_u\|_{\bar p}+\|F_v\|_{\bar p})\le\frac{\sqrt2}{r}(\|F_z\|_{\bar p}+\|F_{\bar z}\|_{\bar p})\le2(1+2K_S)\frac{B}{\alpha}\varepsilon$. Summing, $|X_F|_r\le(1+4+6K_S+2+4K_S)\frac{B}{\alpha}\varepsilon=C_1\frac{B}{\alpha}\varepsilon$.

Step S6.35: Lipschitz. $R(\xi)-R(\zeta)\in\mathcal{T}_s$ and $X_{R(\xi)-R(\zeta)}=X_{R(\xi)}-X_{R(\zeta)}$, so Lemma 6.18 applied to it gives every bound of Step S6.33's inputs with $\varepsilon^{\mathcal{L}}|\xi-\zeta|$ in place of ε. Lemma 6.15(ii) then gives each component difference of F with the factor $\varepsilon^{\mathcal{L}}+2M\varepsilon/\alpha$ in place of ε, and the assembly of Step S6.34, which is linear in the components, gives (iii).

Step S6.36: The normal form correction. $|\hat\omega|=|[b]|\le\sup_{\mathbb{R}^2}|b|\le2\pi\varepsilon$. $\hat\Omega_j=([Q])_{jj}$ and $\|[Q]\|\le\sup\|Q\|\le\varepsilon$ (the mean is an average), so $(j+2)|\hat\Omega_j|\le\varepsilon$ by Lemma 6.14(iii). For the field: since $z_j\bar z_j=\frac12(u_j^2+v_j^2)$, $\hat N_u=\hat\Omega u$, $\hat N_v=\hat\Omega v$ and $\|\hat\Omega u\|_{\bar p}\le\sup_j(j+2)|\hat\Omega_j|\|u\|$, so $X_{\hat N}=(\frac{1}{2\pi}\hat\omega,0,\hat\Omega v,-\hat\Omega u)$ has $|X_{\hat N}|_r\le\varepsilon+\varepsilon(\|u\|+\|v\|)/r\le2\varepsilon$ on $D(s,r)$. Differences: the same with $R(\xi)-R(\zeta)$.

Step S6.37: Reality. For real R and real ξ, the reality relations of Definition 6.4 read $\hat a_{-k}=\overline{\hat a_k}$, $\hat b_{-k}=\overline{\hat b_k}$, $\hat c_k^-=\overline{\hat c_{-k}^+}$, $\hat P'_k=\overline{\hat P_{-k}}$, $(\hat Q_k)_{ji}=\overline{(\hat Q_{-k})_{ij}}$. The divisors are real and satisfy $\delta^0_{-k}=-\delta^0_k$, $\delta^-_{k,j}=-\delta^+_{-k,j}$, $\delta^{P'}_{k,ij}=-\delta^P_{-k,ij}$, $\delta^Q_{k,ji}=-\delta^Q_{-k,ij}$; since $\overline{1/(i\delta)}=-1/(i\delta)$ for real δ, the coefficients of F satisfy the same relations. $\hat\omega$ and $\hat\Omega_j$ are means of functions real on $\mathbb{R}^2$ (Q_{jj} is real there since $Q^T=\overline{Q}$). □

Source: P96-K Lemma 1, p. 125, and its proof pp. 125–128. Deviations, all reported: (a) the operator-valued summation of p. 127 is replaced (Remark 6.9); (b) the constant B_σ with $B_\sigma^2=2^n\sum_k(1+|k|)^2A_k^4e^{-2|k|\sigma}$ is replaced by $B_\sigma^{\mathrm W}$; (c) constants $C_1<36$ and $2C_1$ instead of 2^6, 2^8; (d) $|X_{\hat N}|_r\le2|X_R|_r$ instead of $\le|X_R|_r$; (e) the equation holds modulo the constant $\hat a_0$ (P96-K's $\hat N=[R]$, summed over $|m|+|q|=1$, omits the constant term of R, so $\{F,N\}+\hat N=R$ holds only up to that constant; harmless, constants do not enter X); (f) θ has period one and the θ-component of X carries the factor $(2\pi)^{-1}$.
SOURCE_GAP: P96-K p. 128 states that the components other than F^{10}, F^{11} "admit the same estimates, or even better ones" and that "the estimates of X_F thus follow", without the bookkeeping of the y-component of X_F (which contains $\partial_\theta F^{ij}$ for all components) or of the constants; Steps S6.33 and S6.34 supply it.

Corollary 6.20 (P96-K Lemma 2)**.** *In the setting of Theorem 6.19 with $M>0$, let $0\le\lambda\le\alpha/M$ and write $|X|^\lambda=\varepsilon_X+\lambda\varepsilon_X^{\mathcal{L}}$ for a field with uniform bound ε_X and Lipschitz constant $\varepsilon_X^{\mathcal{L}}$ on the relevant domain. Then*

$$|X_{\hat N}|^\lambda_{r,D(s,r)}\le2|X_R|^\lambda_{r,D(s,r)},\qquad |X_F|^\lambda_{r,D(s-\sigma,r)}\le\frac{3C_1B_\sigma^{\mathrm W}}{\alpha}|X_R|^\lambda_{r,D(s,r)},$$

and for $0<\sigma\le1$, $B_\sigma^{\mathrm W}\le b_{\mathrm W}(\tau)\,\sigma^{-(2\tau+n+1)}$ ($n=2$), the bound (4) of P96-K with $b=b_{\mathrm W}(\tau)$; the constant $3C_1<108$ plays the role of P96-K's absolute constant a.

Proof. By Theorem 6.19(iii), $|X_F|^\lambda\le\frac{C_1B}{\alpha}\big(\varepsilon+\lambda\varepsilon^{\mathcal{L}}+\frac{2\lambda M}{\alpha}\varepsilon\big)\le\frac{3C_1B}{\alpha}(\varepsilon+\lambda\varepsilon^{\mathcal{L}})$ since $\lambda M/\alpha\le1$; (iv) gives the first bound; Lemma 6.12(iv) the last. □

Source: P96-K Lemma 2 and (4), p. 128.

6.8. **Comparison with P96-K and consequences for W5–W6.**

Remark 6.21 (Constants, strip powers, smallness exponents).

Quantity	P96-K (pp. 125, 128)	Here (period-one units, $n=2$)
Coefficient summation	L^2 bound and Cauchy–Schwarz	Banach bound (F1) and ℓ^1 sum
Strip constant	$B_\sigma^2 = 2^n \sum_k (1+\lvert k\rvert)^2 A_k^4 e^{-2\lvert k\rvert\sigma_K}$	$B_\sigma^{\mathrm{W}} = \sum_k (1+\lvert k\rvert_1) A_k^2 e^{-2\pi\lvert k\rvert_1\sigma}$
Power of σ	$B_\sigma \le b\,\sigma^{-(2\tau+n+1)}$	$B_\sigma^{\mathrm{W}} \le b_{\mathrm{W}}(\tau)\sigma^{-(2\tau+n+1)}$ $(\sigma\le 1)$
$\lvert X_F\rvert_{r,D(s-\sigma,r)}$	$2^6 B_\sigma \lvert X_R\rvert_r/\alpha$	$C_1 B_\sigma^{\mathrm{W}}\varepsilon/\alpha$, $C_1<36$
$\lvert X_F\rvert^{\mathcal{L}}$	$2^8 \frac{B_\sigma}{\alpha}(\lvert X_R\rvert^{\mathcal{L}} + \frac{M}{\alpha}\lvert X_R\rvert)$	$C_1 \frac{B_\sigma^{\mathrm{W}}}{\alpha}(\varepsilon^{\mathcal{L}} + \frac{2M}{\alpha}\varepsilon)$
$\lvert X_{\hat N}\rvert$	$\le \lvert X_R\rvert$	$\le 2\varepsilon$
Lemma 2 constant a	absolute	$3C_1 < 108$

(i) The Banach route costs σ^{-n} in the lattice sum where the L^2 route costs $\sigma^{-n/2}$; but P96-K gains nothing from this, since it bounds B_σ by $\sigma^{-(2\tau+n+1)}$ while its own definition gives $\sigma^{-(2\tau+1+n/2)}$. Our sum needs only the weight $(1+\lvert k\rvert_1)A_k^2$ (not $(1+\lvert k\rvert)^2A_k^4$) because the θ-derivatives of F are solved from the θ-derivatives of R (Step S6.33), so it attains the same power $\sigma^{-(2\tau+n+1)}$. (ii) Hence (8) of P96-K p. 129 holds with the same exponent $t=2\tau+n+2$, $\lvert X_F\rvert \prec \lvert X_P\rvert^\lambda/(\alpha\sigma^{t-1})$, and the smallness condition (6), the choice of σ_ν,η_ν and the iteration estimates of §4 are formally unchanged; only the constant c_0 of (6) and hence γ of Theorem A change, through $b_{\mathrm{W}}(\tau)$, C_1 and the factor 2 in $\lvert X_{\hat N}\rvert$. **The smallness exponents of Theorem A used by W5–W6 are not affected**, and the measure exponent of Theorem D (W7), which depends on κ and $\bar p - p$ only, is not affected either. (iii) Unit conversions for W5: $\sigma = \sigma_K/(2\pi)$; for $\sigma_K \le 1$ one has $\sigma\le 1$, so the bound of Lemma 6.12(iv) applies, with $\sigma^{-(2\tau+3)} = (2\pi)^{2\tau+3}\sigma_K^{-(2\tau+3)}$. The norm $\lvert\cdot\rvert_r$ of Definition 6.4 differs from P96-K's in the θ-component (factor $(2\pi)^{-1}$); the smallness conditions are homogeneous in these norms, so W5 may use either consistently.

6.9. **Interfaces.**

Remark 6.22 (From W3). After the Birkhoff map of Theorem 5.25, the normal coordinates are $z_j = w_{j+2}$, $\bar z_j = w_{-(j+2)}$ $(j\ge 1)$ in the notation of Definition 5.2, and on real points $u_j = q_{j+2}$, $v_j = p_{j+2}$ (Lemma 5.3). The restriction of the norm of $\mathfrak{W}$ to the normal part is $\|\hat w\|^2 = \|z\|^2 + \|\bar z\|^2 = \|u\|^2 + \|v\|^2$, with exactly the wave weights $\omega_{a,p}(j+2)$ of Convention 6.1. Hence (a) no weight conversion is needed (Remark 6.2); (b) the sum norm of $D(s,r)$ and the $\mathfrak{W}$-norm compare by $\|\hat w\| \le \|u\| + \|v\| \le \sqrt{2}\|\hat w\|$; (c) the smoothing class of W3 (X_K with values in $\mathfrak{W}^+$) gives normal gradients in $\ell^N_{a,\bar p}$, $\bar p = p+1$, as required by Definition 6.4; (d) the symplectic form Ω of Lemma 5.4 restricted to the normal part is $\sum_j du_j\wedge dv_j$, and $\dot z = -iH_{\bar z}$ agrees with $X_P(w)_l = -i\,\mathrm{sgn}(l)DP(w)e_{-l}$ (Definition 5.5). The exponent $g=6$ of Remark 5.30 is not used in W4.

Ownership (coordinator ruling on review finding W4-R3). W5, not W4, owns: the action–angle substitution of the tangential modes (KP96 p. 25, Step 1), which produces θ, y, the parameter ξ and the form $2\pi d\theta\wedge dy$; the check that the image of $D(s,r)$ lies in W3's ball B_{ρ_1} (the domain check); and every norm comparison between W3's coordinates and the KAM norms, including the sum-versus-$\mathfrak{W}$ comparison (b) above. W4 keeps the shifted wave weights $\omega_{a,p}(j+2)$, so the weight-equivalence factor $3^p e^{2a}$ mentioned in Remark 5.30 never arises; only the explicit phase-norm comparisons (b) remain, for W5. On this point Remark 5.30, which says "W4 must carry these two conversions", is superseded.

Remark 6.23 (To W5). W5 receives: Theorem 6.19 and Corollary 6.20 for arbitrary frequencies (ω_ν,Ω_ν) obeying (13) and $(\mathrm{H}_{\alpha_\nu})$ (no wave-specific input is used in the proof); the bracket identity $R + \langle\!\langle N,F\rangle\!\rangle = \hat N + \hat a_0$ for the weak bracket (Lemma 6.16(ii)), valid for all $a,p\ge 0$, and, under the guard (G), its identification $\langle\!\langle N,F\rangle\!\rangle = DN\cdot X_F$ with the Fréchet derivative of the continuous polynomial N (Lemma 6.16(iv)); the wave case $a=p=1$ satisfies (G), and W5 must work under

(G) whenever it differentiates N; and $\hat\Omega \in \ell^1_\infty$ with $|\hat\Omega|_1 \le \varepsilon$, so that $\Omega_{\nu+1} = \Omega_\nu + \hat\Omega$ keeps (13) with M_0 increased by ε. W5 must supply: the Taylor approximation R of P in the class $\mathcal{T}_s$ (P96-K (7), p. 129), including holomorphy and boundedness of the components and of Q^T; the action–angle substitution, the domain check and the norm comparisons (Remark 6.22, ownership); the parameter regularity of R in the form of the hypothesis $|X_{R(\xi)} - X_{R(\zeta)}|_r \le \varepsilon^{\mathcal{L}}|\xi-\zeta|$; and, if it needs analyticity of the field of P beyond Cauchy-type estimates, the revised W3 interface (finding W3-R1).

6.10. **Ledger.**
Results. Setting: Convention 6.1, Definitions 6.3, 6.4, 6.5; wave data: Lemmas 6.6, 6.7, Remark 6.8; Fourier: Remark 6.9, Lemmas 6.11, 6.12; divisors and solve: Lemmas 6.13, 6.14, 6.15; bracket: Lemma 6.16; extraction: Lemmas 6.17, 6.18; main: Theorem 6.19, Corollary 6.20; comparison and interfaces: Remarks 6.21, 6.22, 6.23. PLAN.md lists no PLANNED node labels for W4; the milestone content "operator solve, Lipschitz" is Theorem 6.19.
W1 inputs. Lemma 3.36(i),(iii) and Lemma 3.35(iii) (scalar), with the content (F1)–(F3) of Remark 6.10; Lemmas 3.39, 3.4, 3.2, 3.8; Definitions 3.1, 3.31. No forward references remain.
Suspected source errata (P96-K).. (1) p. 127, operator-valued square-sum bound: false (Remark 6.9); (2) p. 127, "$|\langle k,\omega\rangle + \Omega_i - \Omega_j| \ge \alpha(1+|i-j|)/A_k$": off by a factor up to 2 (Lemma 6.13); (3) p. 127, the (false) square-sum bound is printed with $\|\tilde R\|_{D(s)}$ unsquared on the right (typo; p. 126 has the square); (4) p. 125, $\{F,N\} + \hat N = R$ with $\hat N = [R]$ holds only modulo the constant term of R. Source gap: p. 128, the remaining components and the y-component of X_F (Theorem 6.19).

7. Milestone W5: one KAM step

This part performs one step of the KAM iteration of P96-K §3 (pp. 128–132) in the wave setting ($n=2$, $d=1$, $\delta=-1$) and constructs the Hamiltonian to which the iteration is applied. A step takes $H = N + P$, with N a normal form and P small, and returns $H\circ\Phi = N_+ + P_+$ with a corrected normal form N_+ and a perturbation P_+ of size $\epsilon^{4/3}$, where Φ is the time-one map of the field X_F of the solution F of the homological equation of W4.
Route. (1) The Taylor polynomial R of P of weighted degree ≤ 2 is defined through the scaling $S_t(\theta,y,u,v) = (\theta, t^2 y, tu, tv)$: $R(w)$ is the second-order Taylor polynomial of $t \mapsto P(S_t w)$ evaluated at $t=1$. Its field is a Cauchy integral in the single variable t, which gives P96-K (7) without a strip loss in θ (Lemma 7.8). (2) W4 solves the homological equation for R; W5 needs W4 with a general smoothing exponent and, for the preparatory step of W6, with modified divisors (Lemmas 7.5, 7.6). (3) The flow of X_F is W1's holomorphic flow with the ball margins of P96-K (9)–(10) made explicit; this instantiates the repair D-W-Pos96K-A4 of P96-K Lemma A.4 (Lemma 7.11). (4) The new Hamiltonian is computed at the level of functions, using W4's bracket identity $DN\cdot X_F = -L_N F$, which is the only place where the unbounded field of N enters (Lemma 7.13); its field is then the P96-K formula $X_{P_+} = \Phi^*(X_P - X_R) + \int_0^1 (\phi^t)^*[X_{R(t)}, X_F]\,dt$. (5) Estimates with explicit constants give Theorem 7.16. (6) The action–angle substitution carries W3's normal form into the form $N+P$ and the initial perturbation is estimated (Lemma 7.18, Proposition 7.19), as ruled in W4-R3.
Inputs. W1: holomorphic maps and the Cauchy estimate (Definition 3.24, Lemmas 3.25, 3.26), the flow (Lemma 3.30), the KAM space and domains (Definition 3.31, Remark 3.32, Lemma 3.33), analyticity in finitely many variables (Lemma 3.28). W3: the symplectic form, Hamiltonian fields and the normal form (Lemma 5.4, Definition 5.5, Theorem 5.25, Remark 5.30). W4: Theorem 6.19, Corollary 6.20, Lemma 6.16, Remarks 6.21, 6.23.
Weights. Throughout $a = p = 1$ (Convention 1.5). This satisfies the weight guard of W4's bracket identity (review W4-R1: $a > 0$ or $a = p = 1$), and $p \ge \frac12$ as W3 requires. All normal quantities carry the wave weights $\omega_{a,q}(J) = J^q e^{aJ}$ at the Dirichlet label $J = j+2$ (Convention 6.1); the P96-K weights $j^q e^{aj}$ are never used, so the equivalence factors of Remark 5.30 do not occur (Lemma 7.18(v)).

7.1. **External facts.** Besides W1 §3.2 and W3 §5.1:

(W5-E1) *Taylor series in one complex variable, Banach values.* If g is holomorphic on $|t|<\rho$ with values in a complex Banach space and $\|g(t)\|\le B$ there, then $g(t)=\sum_{k\ge0} g_k t^k$ on $|t|<\rho$ with $g_k=\frac{1}{2\pi i}\oint_{|t|=\rho'} g(t)t^{-k-1}dt$ $(\rho'<\rho)$, $\|g_k\|\le B\rho^{-k}$, and for $|t|\le 1<\rho$: $\|g(t)-\sum_{k\le 2} g_k t^k\|\le B\rho^{-3}/(1-\rho^{-1})$. (`DifferentiableOn.hasFPowerSeriesOnBall` on a slightly smaller closed disc; the coefficient bound is the Cauchy estimate (W1-E4), and the remainder bound is the sum of the geometric tail $\sum_{k\ge3} B\rho^{-k}$. Mathlib's `HasFPowerSeriesWithinOnBall.uniform_geometric_approx'` gives a bound of this type with unspecified constants and is not needed.)

(W5-E2) *Calculus along curves.* For C^1 maps $c:[0,1]\to V$ into an open subset of a Banach space and G C^1 on V: $G(c(1))-G(c(0))=\int_0^1 DG(c(t))c'(t)\,dt$; differentiation under the integral sign ((E5)); Neumann series ((E6)); Gronwall ((W1-E7)).

(W5-E3) *Symmetry of second derivatives.* Over $\mathbb{K}\in\{\mathbb{R},\mathbb{C}\}$: if f has derivative $f'(y)$ at every y in a neighbourhood of x, and f' has derivative f'' at x, then $f''(v)(w)=f''(w)(v)$ for all v,w. (Mathlib `second_derivative_symmetric_of_eventually` with `[IsRCLikeNormedField K]` ($K=\mathbb{K}$), `(hf : ∀ᶠ y in nhds x, HasFDerivAt f (f' y) y) (hx : HasFDerivAt f' f'' x)`, in `Mathlib/Analysis/Calculus/FDeriv/Symmetric.lean`.)

(W5-E4) *Principal square root.* $\sqrt{\zeta}:=\zeta^{1/2}$ (Mathlib `Complex.cpow` with exponent $(2:\mathbb{C})^{-1}$) satisfies $(\sqrt{\zeta})^2=\zeta$ (`Complex.cpow_ofNat_inv_pow`) and is complex differentiable at every ζ with $\operatorname{Re}\zeta>0$: such ζ lie in `Complex.slitPlane` (`Complex.mem_slitPlane_iff`: $0<\operatorname{Re}\zeta\vee\operatorname{Im}\zeta\ne0$), the hypothesis `f x ∈ slitPlane` of `DifferentiableAt.cpow_const`. Moreover $|\sqrt{\zeta}|=|\zeta|^{1/2}$, and $\sqrt{\zeta}>0$ for $\zeta>0$.

(W5-E5) *A triangular double integral.* For continuous $g:[0,1]\to V$ (Banach V), $\int_0^1\int_0^t g(\tau)\,d\tau\,dt=\int_0^1(1-\tau)g(\tau)\,d\tau$. No Fubini is needed: $h(t)=\int_0^t g$ is C^1 with $h'=g$ (`intervalIntegral.integral_hasDerivAt_right`), and integrating $\frac{d}{dt}\big((t-1)h(t)\big)=h(t)+(t-1)g(t)$ over $[0,1]$ by (W5-E2) gives $0=\int_0^1 h+\int_0^1(t-1)g$.

7.2. **Phase space, symplectic form, smoothing Hamiltonians.**

Definition 7.1 (Phase space and symplectic form)**.** For $q\in\mathbb{R}$ let $\mathcal{P}^q=\mathbb{C}^2\times\mathbb{C}^2\times\ell^N_{a,q}\times\ell^N_{a,q}$ with points $W=(\Theta,Y,U,V)$ and, for $r>0$, the norm $|W|_{q,r}=|\Theta|+|Y|/r^2+(\|U\|_{a,q}+\|V\|_{a,q})/r$ of Definition 3.31 (normal spaces of Convention 6.1). Write $\mathcal{P}=\mathcal{P}^p$, and $D(s,r)\subset\mathcal{P}$ for the domain of Definition 3.31. A point is *real* if $\Theta,Y\in\mathbb{R}^2$ and U,V are real; $\mathcal{P}^q_{\mathbb{R}}$ is the real subspace. On $\mathcal{P}$ define the complex bilinear form

$$\varpi_{\mathcal{P}}(W,W')=2\pi\big(\langle\Theta,Y'\rangle-\langle Y,\Theta'\rangle\big)+\langle U,V'\rangle_0-\langle V,U'\rangle_0.$$

For a linear $L:\mathcal{P}^q\to\mathcal{P}^{q'}$ we write $|L|_{q\to q',r}$ for the operator norm from $|\cdot|_{q,r}$ to $|\cdot|_{q',r}$; $|L|_{r,r}=|L|_{p\to p_+,r}$ when the target exponent p_+ is fixed.

Source: P96-K pp. 120–121, (1); W4 Convention 6.1 and (11) (the form $2\pi d\theta\wedge dy+du\wedge dv$ in period-one angles).

Lemma 7.2 (The form)**.** *$\varpi_{\mathcal{P}}$ is continuous, antisymmetric and nondegenerate on $\mathcal{P}$: $|\varpi_{\mathcal{P}}(W,W')|\le(4\pi r^2+2r^2)|W|_{p,r}|W'|_{p,r}$ for every $r>0$, and $\varpi_{\mathcal{P}}(W,W')=0$ for all W' implies $W=0$. For $q\ge p$ and $r'\le r$: $|W|_{p,r}\le|W|_{q,r}$ and $|W|_{q,r}\le|W|_{q,r'}\le(r/r')^2|W|_{q,r}$.*

Proof. **Step S7.1: Bound.** $2\pi|\langle\Theta,Y'\rangle|\le4\pi|\Theta||Y'|\le4\pi r^2|W|_{p,r}|W'|_{p,r}$ (max norms on $\mathbb{C}^2$), and $|\langle U,V'\rangle_0|\le\|U\|\|V'\|$ by (9).

Step S7.2: Nondegeneracy. Test with the unit vectors of $\mathbb{C}^2$ and the coordinate vectors e_j of $\ell^N_{a,p}$: $\varpi_{\mathcal{P}}(W,(0,e_i,0,0))=2\pi\Theta_i$, $\varpi_{\mathcal{P}}(W,(e_i,0,0,0))=-2\pi Y_i$, $\varpi_{\mathcal{P}}(W,(0,0,0,e_j))=U_j$, $\varpi_{\mathcal{P}}(W,(0,0,e_j,0))=-V_j$.

Step S7.3: Norms. Weights increase with q (Lemma 3.5); $1/r \le 1/r' \le (r/r')^2/r$ for $r' \le r$, and $(r/r')^2 \ge 1$ bounds the Θ-term. □

Source: ours.

Definition 7.3 (Smoothing Hamiltonians; parameter norms)**.** Let $p_+ \in [p+\frac12, p+1]$, $V \subset \mathcal{P}$ open, and $\Pi \subset \mathbb{R}^2$ (max norm). The class $\mathcal{S}_{p_+}(V)$ consists of the holomorphic $P : V \to \mathbb{C}$ (Definition 3.24) for which there is a holomorphic $X_P : V \to \mathcal{P}^{p_+}$ with

$$DP(w)W = \varpi_{\mathcal{P}}(X_P(w), W) \qquad (w \in V,\ W \in \mathcal{P});$$

X_P is unique by Lemma 7.2 and is the field $X_P = (\frac{1}{2\pi}P_y, -\frac{1}{2\pi}P_\theta, P_v, -P_u)$ of Definition 6.4. P is *real* if V is invariant under complex conjugation of all coordinates and P is real at real points. A *family* is $P(\cdot,\xi) \in \mathcal{S}_{p_+}(V)$, $\xi \in \Pi$. For $\lambda \ge 0$ and $r > 0$,

$$|X_P|^{\lambda}_{r,V} = \sup_{\xi\in\Pi,\, w\in V} |X_{P(\xi)}(w)|_{p_+,r} + \lambda \sup_{\xi\neq\zeta} \sup_{w\in V} \frac{|X_{P(\xi)}(w) - X_{P(\zeta)}(w)|_{p_+,r}}{|\xi-\zeta|},$$

read in Lean as the conjunction of the two bounds $|X_{P(\xi)}(w)| \le \epsilon_0$, $|X_{P(\xi)}(w) - X_{P(\zeta)}(w)| \le \epsilon_1|\xi-\zeta|$ with $\epsilon_0 + \lambda\epsilon_1 \le \epsilon$; no supremum is taken as a value. The same notation is used for maps $V \times \Pi \to \mathcal{P}^q$ other than fields.

Source: P96-K Assumption C and (1), p. 121, and $|X|^\lambda = |X| + \lambda|X|^{\mathcal{L}}$, p. 128; W3 Definition 5.5 (the same notion for the Cartesian form).

Remark 7.4 (Products). If $|a(\xi)| \le A_0$, $|a(\xi) - a(\zeta)| \le A_1|\xi-\zeta|$ and similarly for b, and $a \cdot b$ is a product of norm ≤ 1 (composition of operators, application, a bilinear form of norm ≤ 1), then $A_0B_0 + \lambda(A_1B_0 + A_0B_1) \le (A_0 + \lambda A_1)(B_0 + \lambda B_1)$: the λ-norm is submultiplicative. This is used without comment below.

7.3. **The homological equation in the generality needed.** The step of W6 runs in two regimes: the preparatory step, whose input field gains one weight ($p_+ = p+1$), and the iteration after it, whose fields gain only $p_+ = p + \frac12$ (P96-K §6, p. 140, $\check p = \bar p - \varpi$ with $\varpi = \frac12$). Moreover after the preparatory step the normal frequency drifts decay only like $j^{-1/2}$ (W7, Definition 9.9(H2)). W4 is written for $\bar p = p+1$ and tails $\sup_j j|\tilde\Omega_j| < \infty$.

Lemma 7.5 (W4 with a general smoothing exponent)**.** *Let $p_+ \in [p+\frac12, p+1]$. Replace in W4 the exponent $\bar p = p+1$ by p_+ (in Definitions 6.3, 6.4) and the frequency hypothesis (13) by*

$$\begin{aligned} &\Omega_j = j + 2 + \tilde\Omega_j, \qquad \sup_j j^{1/2}|\tilde\Omega_j(\xi)| \le M_0, \\ &|\omega(\xi) - \omega(\zeta)| \le M_\omega|\xi-\zeta|, \qquad \sup_j j^{1/2}|\Omega_j(\xi) - \Omega_j(\zeta)| \le M_\Omega|\xi-\zeta|. \end{aligned} \tag{14}$$

Then Lemma 6.16, Theorem 6.19 and Corollary 6.20 hold verbatim, with the same constants C_1, B^{W}_σ, $3C_1$, except that Theorem 6.19(iv) reads $\sup_j (j+2)^{p_+-p}|\hat\Omega_j| \le \varepsilon$ (and likewise for differences).

Proof. We list every use of $\bar p - p = 1$ or of the tail weight j in W4 and check it under the new hypotheses.

Step S7.4: Pairings. (9) needs $q + q' \ge 0$. In Definition 6.3 the pairs are (p_+, p) and (p,p); in Lemma 6.16(i) the pair is $(p-1, p_+)$ with $p - 1 + p_+ \ge 2p - \frac12 \ge 0$ since $p = 1$.

Step S7.5: Tails. The tails enter only Step S6.22, through $|\Omega_j| \le (1 + M_0)(j+2)$, which holds since $|\tilde\Omega_j| \le M_0 j^{-1/2} \le M_0$.

Step S7.6: Divisors. Step S6.14 uses $|\Omega_j(\xi) - \Omega_j(\zeta)| \le M_\Omega|\xi-\zeta|/j$ only through $\sum_j |l_j||\Omega_j(\xi) - \Omega_j(\zeta)| \le 2M_\Omega|\xi-\zeta|$, which holds with $j^{-1/2} \le 1$.

Step S7.7: Diagonals. Lemma 6.14(iii) gives $\mu_i\nu_i|M_{ii}| \le \|M\|$ with $\mu_i\nu_i = (i+2)^{p_+-p}$; this is used only in Step S6.36, where $\|\hat\Omega u\|_{p_+} \le \sup_j (j+2)^{p_+-p}|\hat\Omega_j|\,\|u\|$ replaces the case $p_+ - p = 1$.

Step S7.8: Everything else. Lemmas 6.11, 6.12, 6.14(i)–(ii), 6.15, 6.17, 6.18 are stated for arbitrary Banach spaces or arbitrary positive weights, and the remaining steps of Theorem 6.19 use only these. □

Source: ours. P96-K applies Lemma 1 after the preparatory step with $\check p$ in place of $\bar p$ (p. 141: "with parameters $\check\alpha$ and $\check p$ instead of α and $\bar p$") without comment. *Report to W4:* in Lean, W4 should be parametrized by p_+ and use (14); the proof is unchanged.

Lemma 7.6 (Modified divisors for the preparatory step)**.** *Assume (14) with $M = M_\omega + M_\Omega > 0$, $\hat\alpha > 0$, $\beta \in (0, \frac12]$, $p_+ \in [p+\frac12, p+1]$ with $\beta \le p+1-p_+$, and suppose every divisor of Definition 6.5 satisfies the bound of $(\mathrm{H}_{\hat\alpha})$ except the divisors $\delta^Q_{k,ij}$, $i \ne j$, for which only*

$$|\delta^Q_{k,ij}(\xi)| \ge \frac{\hat\alpha}{A_k}\,\frac{|i-j|}{(\min(i,j)+2)^\beta} \qquad (\xi \in \Pi) \tag{15}$$

is assumed. Let R be as in Theorem 6.19 with $\bar p = p+1$, and define F, $\hat N$ by the same formulas. Then:

(i) $F(\cdot,\xi) \in \mathcal{T}_{s-\sigma}$ with Q_F, Q_F^T bounded $\ell^N_{a,p} \to \ell^N_{a,p_+}$ and all other components as in W4; the equation (ii) and the bounds (iv) of Theorem 6.19 hold, and $|X_F|_{p_+,r} \le C_1 B^{\mathrm W}_\sigma \varepsilon/\hat\alpha$ on $D(s-\sigma,r)$.

(ii) If moreover $2\beta \le p+1-p_+$, then on $D(s-\sigma,r)$

$$|X_{F(\xi)} - X_{F(\zeta)}|_{p_+,r} \le \frac{C_1 B^\natural_\sigma}{\hat\alpha}\Big(\varepsilon^{\mathcal L} + \frac{2M}{\hat\alpha}\varepsilon\Big)|\xi-\zeta|, \qquad B^\natural_\sigma = \sum_{k\in\mathbb Z^2}(1+|k|_1)^2 A_k^2 e^{-2\pi|k|_1\sigma} \quad (\ge B^{\mathrm W}_\sigma),$$

so for $0 \le \lambda \le \hat\alpha/M$: $|X_F|^\lambda_{r,D(s-\sigma,r)} \le \frac{3C_1B^\natural_\sigma}{\hat\alpha}|X_R|^\lambda_{r,D(s,r)}$.

(iii) For $0 < \sigma \le 1$, $B^\natural_\sigma \le b^\natural(\tau)\sigma^{-(2\tau+4)}$ with $b^\natural(\tau) = 4\big(C_0 + C_2 + C_{2\tau} + C_{2\tau+2}\big)$ (four terms, counted with multiplicity), where $C_0 = (1+\frac1\pi)^2$ and $C_q = \frac4\pi\big(\frac{q+1}{e\pi}\big)^{q+1}$ for $q > 0$ (so $C_{2\tau} = C_0$ when $\tau = 0$).

Proof. Write $I = i+2$, $J = j+2$ (Dirichlet labels) and $c = p+1-p_+$.

Step S7.9: The Q-division. $(Q_F)_{ij} = d_{ij}(\hat Q_k)_{ij}$ with $d_{ij} = 1/(i\delta^Q_{k,ij})$, $|d_{ij}| \le A_k \min(I,J)^\beta/(\hat\alpha|i-j|)$ by (15). The p_+-weighted matrix of Q_F is $\big(\omega_{a,p+1}(I)(\hat Q_k)_{ij}\omega_{a,p}(J)^{-1}\big)\cdot I^{-c}d_{ij}$, and $I^{-c}\min(I,J)^\beta \le 1$ since $\beta \le c$. Lemma 6.14 with the factor $I^{-c}d_{ij}$, which is bounded by $A_k/(\hat\alpha|i-j|)$, gives the operator bound with $\kappa = K_S$; Q_F^T likewise (the factor is $I^{-c}d_{ji}$).

Step S7.10: Other components. The divisors of $c^\pm$, P, P' obey $(\mathrm{H}_{\hat\alpha})$, so W4's bounds hold in $\ell^N_{a,p+1} \subset \ell^N_{a,p_+}$ (norm ≤ 1).

Step S7.11: Lipschitz. By Step S6.14, $|d_{ij}(\xi) - d_{ij}(\zeta)| \le 2(1+|k|_1)M|\xi-\zeta|/|\delta(\xi)\delta(\zeta)| \le 2(1+|k|_1)A_k^2 M\min(I,J)^{2\beta}|\xi-\zeta|/(\hat\alpha^2|i-j|^2)$, and $I^{-c}\min(I,J)^{2\beta} \le 1$ when $2\beta \le c$. Lemma 6.15(ii) then applies as in Step S6.35.

Step S7.12: The action component. W4 bounds the term $\Lambda\partial_{\theta_i}Q_F(z,\bar z)$ of $\partial_{\theta_i}F$ in the bilinear space $\mathcal B$ (Step S6.33); this route is not available here, because the Hadamard factor d_{ij} grows like $\min(I,J)^\beta$ and the norm of $\mathcal B$ (two copies of $\ell^N_{a,p}$) supplies no compensating weight. Instead we bound $\partial_{\theta_i}Q_F$ as an operator $\ell^N_{a,p} \to \ell^N_{a,p_+}$. By Lemma 6.18(i), $\|Q(\theta)\| \le \varepsilon$ in $\mathcal L$ (target $\ell^N_{a,p+1}$), so $\|\hat Q_k\| \le \varepsilon e^{-2\pi|k|_1 s}$ ((F1)). By Step S7.9 the coefficient $2\pi i k_i(\hat Q_F)_k = 2\pi i k_i\, d\circ\hat Q_k$ has $\ell^N_{a,p} \to \ell^N_{a,p_+}$

norm $\le 2\pi|k_i|K_S A_k\varepsilon e^{-2\pi|k|_1 s}/\hat\alpha$, and $|k_i|A_k \le (1+|k|_1)A_k^2$. By (F2) and termwise differentiation ((W1-E5)), on $T_{s-\sigma}$

$$\|\partial_{\theta_i} Q_F(\theta)\|_{\ell^N_{a,p}\to\ell^N_{a,p_+}} \le \frac{2\pi K_S B_\sigma^{\mathrm{W}}}{\hat\alpha}\varepsilon, \qquad |\langle \partial_{\theta_i} Q_F\, z,\bar z\rangle_0| \le \frac{2\pi K_S B_\sigma^{\mathrm{W}}}{\hat\alpha}\varepsilon\|z\|\|\bar z\|$$

(pairing of ℓ^N_{a,p_+} with $\ell^N_{a,p}$, (9)). This is smaller than W4's bound $16\pi K_S(B_\sigma^{\mathrm{W}}/\hat\alpha)\varepsilon$ for $\|\Lambda\partial_{\theta_i}Q_F\|_{\mathrm{bil}}$, so W4's Step S6.34 goes through with the same C_1. For differences, the coefficient difference is $2\pi i k_i\big(d(\xi)\circ(\hat Q_k(\xi)-\hat Q_k(\zeta))+(d(\xi)-d(\zeta))\circ\hat Q_k(\zeta)\big)$; by Steps S7.9 and S7.11 ($2\beta\le c$; for $i=j$, $k\neq 0$ the divisor is $\langle k,\omega\rangle$ and $|\Delta(1/\delta)|\le|k|_1 M_\omega A_k^2|\xi-\zeta|/\hat\alpha^2$) its norm is at most $2\pi K_S\big(|k|_1 A_k\varepsilon^{\mathcal L}/\hat\alpha + 2|k|_1(1+|k|_1)A_k^2 M\varepsilon/\hat\alpha^2\big)e^{-2\pi|k|_1 s}|\xi-\zeta|$ (Lemma 6.11(iii)). Summing with $e^{2\pi|k|_1(s-\sigma)}$ and $|k|_1(1+|k|_1)A_k^2 \le (1+|k|_1)^2A_k^2$: $\|\Delta\partial_{\theta_i}Q_F\| \le \frac{2\pi K_S B_\sigma^\natural}{\hat\alpha}(\varepsilon^{\mathcal L}+\frac{2M}{\hat\alpha}\varepsilon)|\xi-\zeta|$. The extra factor $|k|_1$ is the price of differentiating the solved series instead of solving for the derivative data; it is why $B_\sigma^\natural$ replaces B_σ^{W} in (ii).

Step S7.13: Assembly. W4's Steps S6.33–S6.36, with the Q_F-contribution to $\partial_\theta F$ taken from Step S7.12 instead of the bilinear route, and all other components unchanged (their divisors obey ($\mathrm{H}_{\hat\alpha}$)). In (ii) every component difference is bounded with $B_\sigma^{\mathrm{W}} \le B_\sigma^\natural$ or, for the Q_F-part of $\partial_\theta F$, with $B_\sigma^\natural$; the assembly is linear in the component bounds. The λ-form follows as in Corollary 6.20.

Step S7.14: The constant. $(1+x)^2(1+x^\tau)^2 \le 4(1+x^2)(1+x^{2\tau})$ for $x\ge 0$ (Cauchy–Schwarz), so $B_\sigma^\natural \le 4(S_0+S_2+S_{2\tau}+S_{2\tau+2})$ (four terms, with multiplicity) in the notation of Lemma 6.12. By (ii) there, $S_0 \le (1+\frac{1}{\pi\sigma})^2 \le C_0\sigma^{-2}$, and by (iii), $S_q \le C_q\sigma^{-(q+2)}$ for $q>0$; for $\sigma\le 1$ every $\sigma^{-(q+2)}$ with $q\le 2\tau+2$ is $\le \sigma^{-(2\tau+4)}$. Summing the four terms gives (iii). □

Source: P96-K §6, p. 140, (23) ("$|\langle k,\omega\rangle+\Omega_i-\Omega_j| \ge \frac{\hat\alpha}{A_k}\cdot\frac{|i-j|}{\min(i^\varpi,j^\varpi)}$") and "$|X_F|^\lambda_{\check p,r,D(s-\sigma,r)} \le \frac{aB_\sigma}{\hat\alpha}|X_R|^\lambda_{\bar p,r,D(s,r)}$, $\check p=\bar p-\varpi$".

SOURCE_GAP: P96-K p. 140 asserts the λ-norm (that is, including the Lipschitz part) for the weight exponent $\beta=\varpi$ and the loss $\bar p-\check p=\varpi$. The termwise argument (Step S7.11) needs $2\beta\le\bar p-\check p$: the product of two modified divisors is only $\ge\hat\alpha^2|i-j|^2\min^{-2\beta}/A_k^2$, so the Lipschitz constant of $1/\delta$ carries $\min(I,J)^{2\beta}$ while the weight gain is only I^ϖ. The printed justification is therefore insufficient (suspected parameter-Lipschitz gap); no complete counterexample satisfying all the source's hypotheses is claimed. (A locally adverse multiplier exists: near-resonant adjacent divisors give entries whose weighted ξ-derivative is large; whether an admissible frequency family realizes this uniformly is open.) Hence with P96-K's choice $\beta=\varpi=\frac12$ and $p_+=p+\frac12$, the termwise proof of part (ii) is not available. Moreover the action component needs the separate argument of Step S7.12, which costs one power of σ in the Lipschitz part ($B_\sigma^\natural$); P96-K states the bound with B_σ. *Resolution adopted:* W7 separates the weight exponent from the loss, $\beta=\frac14$ with loss $\varpi=\frac12$ (Definition 9.1(vii), $\hat w=(i-j)j^{-1/4}$), so $2\beta\le p+1-p_+$ and part (ii) applies. See Remark 7.7.

Remark 7.7 (The exponents used by W6–W7). With $\beta=\frac14$ and loss $\frac12$ P96-K's Lemma 9 gives $\lambda=\kappa/(\kappa+1-\frac14)=\frac{8}{11}$ and the optimizing w of p. 142 becomes $w=\frac14/(4\kappa+4-\frac14)=\frac{1}{47}$, so $\hat\alpha=\alpha^{44/47}$, $\check\alpha=\alpha^{48/47}$ and $\tilde\mu=\frac{32}{47}>\frac23$; W7's density argument survives ($\frac{32}{47}\cdot 2>\frac43$; the admissible window is $t^{3/2}\lesssim\alpha=o(t^{47/32})$, and W7 uses $\alpha(t)=t^{95/64}$). W7 has adopted these values (Definition 9.6), and W6 states the preparatory step with them, the Lipschitz part included (Lemma 8.8(v)).

7.4. **Taylor approximation.**

Lemma 7.8 (Taylor approximation; P96-K (7))**.** *Let $s,r>0$, $p_+\in[p+\frac12,p+1]$, $P(\cdot,\xi)\in\mathcal S_{p_+}(D(s,r))$ real, 1-periodic in each θ_i, with $|X_P|^\lambda_{r,D(s,r)}\le\epsilon$. Put $S_t(\theta,y,u,v)=(\theta,t^2y,tu,tv)$ and $T_t(\Theta,Y,U,V)=(t^2\Theta,Y,tU,tV)$ ($t\in\mathbb C$). Define*

$$a(\theta)=P(\theta,0,0,0),\quad b(\theta)=2\pi X_P^\theta(\theta,0,0,0),\quad g(\theta)=(-X_P^v,X_P^u)(\theta,0,0,0),\quad \mathsf H(\theta)=D_{(u,v)}g$$

(the last the (u,v)-derivative of $(-X_P^v, X_P^u)$ at $(\theta,0,0,0)$), and $R = a + \langle b, y\rangle + \langle g,(u,v)\rangle_0 + \frac{1}{2}\langle \mathsf{H}(u,v),(u,v)\rangle_0$. Then:

- *(i) $R(\cdot,\xi)$ belongs to W4's class $\mathcal{T}_s$ (with $\bar{p} = p_+$): rewritten in $z,\bar{z}$ by Convention 6.1 its components are $c^+ = (g_u - ig_v)/\sqrt{2}$, $c^- = (g_u + ig_v)/\sqrt{2}$ and the blocks $P = \frac{1}{4}(\mathsf{H}_{uu} - \mathsf{H}_{vv} - i\mathsf{H}_{uv} - i\mathsf{H}_{vu})$, $Q = \frac{1}{2}(\mathsf{H}_{uu} + \mathsf{H}_{vv} - i\mathsf{H}_{uv} + i\mathsf{H}_{vu})$, $P' = \frac{1}{4}(\mathsf{H}_{uu} - \mathsf{H}_{vv} + i\mathsf{H}_{uv} + i\mathsf{H}_{vu})$ (H_{xy} the block from the y-variable to the x-gradient), all holomorphic, periodic and bounded, P, P' symmetric; R is real; and $R(w) = \sum_{k\le 2}\frac{1}{k!}\partial_t^k P(S_t w)|_{t=0}$.*
- *(ii) $X_R(w) = \sum_{k\le 2}\frac{1}{k!}\partial_t^k G(t,w)|_{t=0}$ with $G(t,w) = T_t X_P(S_t w)$.*
- *(iii) $|X_R|^\lambda_{r,D(s,r)} \le 3\epsilon$.*
- *(iv) For $0 < \eta \le \frac{1}{8}$: $|X_P - X_R|^\lambda_{\eta r, D(s,4\eta r)} \le 128\,\eta\,\epsilon$.*

Proof. **Step S7.15: Scaling identity.** For all W, W': $\varpi_{\mathcal{P}}(X, S_t W') = \varpi_{\mathcal{P}}(T_t X, W')$ (expand: the ΘY-terms acquire t^2, the UV-terms t). Hence for $t \neq 0$ and $S_t w \in D(s,r)$, $D(P\circ S_t)(w)W' = \varpi_{\mathcal{P}}(X_P(S_t w), S_t W') = \varpi_{\mathcal{P}}(G(t,w), W')$, so $X_{P\circ S_t} = G(t,\cdot)$; G is holomorphic in (t,w) on the open set $\{S_t w \in D(s,r)\}$, including $t = 0$.

Step S7.16: Discs. If $w \in D(s,r')$, $r' \le r$, then $S_t w \in D(s,r)$ for $|t| < \rho := r/r'$, and $|T_t X|_{p_+,r'} \le \rho^2 |X|_{p_+,r}$ for $|t| \le \rho$: the Θ-component gains $|t|^2 \le \rho^2$, the Y-component $r^2/r'^2 = \rho^2$, and the normal components $|t|/r' \le \rho/r' = \rho^2/r$.

Step S7.17: Taylor polynomial. Expanding $P(S_t w)$ at $t = 0$ by the chain rule: the coefficient of t^0 is a, of t^1 is $DP(\theta,0,0,0)(0,0,u,v) = \langle g,(u,v)\rangle_0$, of t^2 is $DP(\theta,0,0,0)(0,y,0,0) + \frac{1}{2}D^2P(\theta,0,0,0)[(0,0,u,v)^2] = \langle b, y\rangle + \frac{1}{2}\langle \mathsf{H}(u,v),(u,v)\rangle_0$ (using $DP\,W = \varpi_{\mathcal{P}}(X_P, W)$ and differentiating X_P once more, which exists since X_P is holomorphic). This is the last claim of (i).

Step S7.18: Components. a, b, g are compositions of holomorphic maps with $\theta \mapsto (\theta,0,0,0)$. $\mathsf{H}(\theta) = \frac{1}{2\pi i}\oint_{|\zeta|=\rho'}(-X_P^v, X_P^u)(\theta,0,\zeta U,\zeta V)\zeta^{-2}d\zeta$ (as an operator applied to (U,V), Lemma 3.26), holomorphic in θ by differentiation under the integral, bounded $\ell^{N\,2}_{a,p} \to \ell^{N\,2}_{a,p_+}$ by Lemma 3.25 (in fact $\|\mathsf{H}\| \le \epsilon$, Step S7.20). H is symmetric for $\langle\cdot,\cdot\rangle_0$ because D^2P is symmetric: $DP(w) = \varpi_{\mathcal{P}}(X_P(w),\cdot)$ at every point, and this has derivative $W \mapsto \varpi_{\mathcal{P}}(DX_P(w)W,\cdot)$ (X_P holomorphic), so (W5-E3) applies with $\mathbb{K} = \mathbb{C}$; its blocks satisfy $\mathsf{H}_{uv}^T = \mathsf{H}_{vu}$, so the blocks P, Q, Q^T, P' are bounded and P, P' symmetric. The z-formulas are the substitution $u = (z+\bar{z})/\sqrt{2}$, $v = (z-\bar{z})/(i\sqrt{2})$. Reality and periodicity are inherited from P.

Step S7.19: Field. For fixed w and W', $\varepsilon \mapsto R(w + \varepsilon W')$ is the sum of the Taylor coefficients of order ≤ 2 of $t \mapsto P(S_t(w + \varepsilon W'))$; differentiate in ε under the Cauchy integrals defining these coefficients and use Step S7.15: $DR(w)W' = \sum_{k\le 2}\frac{1}{k!}\partial_t^k \varpi_{\mathcal{P}}(G(t,w), W')|_{t=0}$, which is $\varpi_{\mathcal{P}}(\sum_{k\le 2} g_k(w), W')$ with g_k the Taylor coefficients of $G(\cdot,w)$. The g_k lie in $\mathcal{P}^{p_+}$; uniqueness gives (ii), and W4's formula for X_R agrees by the same uniqueness.

Step S7.20: Estimates. (iii): for $w \in D(s,r)$, $|G(t,w)|_{p_+,r} \le \epsilon$ on $|t| \le 1$ ($|t|^2, |t| \le 1$) and $G(\cdot,w)$ is holomorphic on a disc of radius > 1 (Step S7.16 with r' = the size of w); (W5-E1) on $|t| < 1$ gives $|g_k| \le \epsilon$, so $|X_R| \le 3\epsilon$. (iv): for $w \in D(s,4\eta r)$ take $r' = 4\eta r$, $\rho = 1/(4\eta) \ge 2$; (W5-E1) with $B = \rho^2\epsilon$ in $|\cdot|_{p_+,r'}$ gives $|X_P(w) - X_R(w)|_{p_+,4\eta r} \le \epsilon/(\rho - 1) \le 8\eta\epsilon$, and $|\cdot|_{p_+,\eta r} \le 16|\cdot|_{p_+,4\eta r}$ (Lemma 7.2). The Lipschitz parts: everything is linear in P; apply the same to $P(\xi) - P(\zeta)$. □

Source: P96-K (7), p. 129 ("these approximations are given by certain Cauchy integrals, and the estimates are the same as in a finite dimensional setting"). Constants 3 and 128 ours. The scaling S_t replaces the multi-index Taylor expansion of P96-K (3) and avoids a strip loss in the y-component.

7.5. **The step data and the flow.**

Definition 7.9 (Step data)**.** *Step data* consist of: $p_+ \in [p+\frac12, p+1]$; $\tau \geq 0$ (W6 uses $\tau = 24$, Remark 9.7); $0 < s \leq 1$, $0 < \sigma \leq \min(\frac14, \frac{s}{6})$, $0 < \eta \leq \frac18$, $r > 0$; a parameter set $\Pi \subset \mathbb{R}^2$; frequencies (ω, Ω) on Π satisfying (14) with $M = M_\omega + M_\Omega > 0$; $\alpha > 0$ such that (H_α) holds on Π (Definition 6.5) or, in the modified case, the hypotheses of Lemma 7.6 with $\hat\alpha = \alpha$; $0 \leq \lambda \leq \alpha/M$; the normal form $N = \langle \omega(\xi), y\rangle + \sum_j \Omega_j(\xi) z_j \bar z_j$; and a real periodic family $P \in \mathcal{S}_{p'}(D(s,r))$ with $p' = p+1$ in the modified case and $p' = p_+$ otherwise, and a number $\epsilon > 0$ with $\epsilon \geq |X_P|^\lambda_{r,D(s,r)}$ (norm $|\cdot|_{p',r}$; a positive majorant, also when $P = 0$). Put, in the unmodified case,

$$t = 2\tau + 4, \qquad c_F = 9C_1 b_{\mathrm{W}}(\tau), \qquad c_0 = 16 c_F, \qquad M_F = \frac{c_F\,\epsilon}{\alpha\sigma^{t-1}},$$

with C_1, b_{W} of Theorem 6.19 and Lemma 6.12; in the modified case $t^\natural = 2\tau + 5$, $c_F^\natural = 9C_1 b^\natural(\tau)$, $c_0^\natural = 16 c_F^\natural$, $C_2^\natural = 896 + 168 c_F^\natural$ (Lemma 7.6(iii)), and every statement below about the modified case is read with (t, c_F, c_0, C_2) replaced by $(t^\natural, c_F^\natural, c_0^\natural, C_2^\natural)$; thus $M_F = c_F^\natural \epsilon/(\alpha\sigma^{2\tau+4})$ there. The *smallness condition* is

$$\epsilon \leq \frac{\alpha\sigma^t\eta^2}{c_0} \qquad \Big(\text{equivalently } M_F \leq \frac{\sigma\eta^2}{16}\Big). \tag{16}$$

Source: P96-K (5)–(6), p. 129 ("$0 < \eta < \frac18$ and $0 < \sigma < s$, $\sigma \leq 1$", $t = 2\tau + n + 2$, c_0 "sufficiently large constant"); $n = 2$. Period-one strips (Remark 6.21(iii)). c_0 explicit here.

Lemma 7.10 (The solution of the homological equation in the step)**.** *Under step data, let R be the Taylor polynomial of Lemma 7.8 and F, $\hat N$, $\hat a_0$ the objects of Theorem 6.19 for R (through Lemma 7.5, or Lemma 7.6 in the modified case). Then $F \in \mathcal{T}_{s-\sigma} \cap \mathcal{S}_{p_+}(D(s-\sigma, r))$, $\hat N \in \mathcal{S}_{p_+}(\mathcal{P})$ (indeed p'), both real, $R + DN \cdot X_F = \hat N + \hat a_0$ on $D(s-\sigma, r)$, and*

$$|X_F|^\lambda_{r,D(s-\sigma,r)} \leq M_F, \qquad |X_{\hat N}|^\lambda_{r,D(s,r)} \leq 6\epsilon, \qquad |\hat\omega| + \lambda|\hat\omega|^{\mathcal{L}} \leq 6\pi\epsilon,$$

$$\sup_j (j+2)^{p_+ - p}\big(|\hat\Omega_j| + \lambda|\hat\Omega_j|^{\mathcal{L}}\big) \leq 3\epsilon.$$

In the modified case the Lipschitz parts require $2\beta \leq p + 1 - p_+$, and the bound for X_F is read with $c_F^\natural$, $t^\natural$ (Lemma 7.6(ii)–(iii): $3C_1 B_\sigma^\natural \cdot 3\epsilon/\alpha \leq c_F^\natural \epsilon/(\alpha\sigma^{2\tau+4})$).

Proof. **Step S7.21: Input.** Lemma 7.8(i),(iii): $R \in \mathcal{T}_s$ and $|X_R|^\lambda_{r,D(s,r)} \leq 3\epsilon$.

Step S7.22: Output. Corollary 6.20 (via Lemma 7.5) with $\varepsilon + \lambda\varepsilon^{\mathcal{L}} \leq 3\epsilon$: $|X_F|^\lambda \leq 3C_1 B_\sigma^{\mathrm{W}} \cdot 3\epsilon/\alpha \leq 9C_1 b_{\mathrm{W}}\epsilon/(\alpha\sigma^{2\tau+3}) = M_F$ since $\sigma \leq 1$ (Lemma 6.12(iv)); and $|X_{\hat N}|^\lambda \leq 2 \cdot 3\epsilon$, Theorem 6.19(iv). The bracket identity is Theorem 6.19(ii) together with the derivative identification of Lemma 6.16(iv), valid under the weight guard (G) since $a = p = 1$.

Step S7.23: Class. A member of $\mathcal{T}$ is a polynomial of degree ≤ 2 in (y, u, v) whose coefficients are holomorphic in finitely many variables θ, hence analytic (Lemma 3.28); so it is C^2 on its domain, its field (Definition 6.4) is holomorphic into $\mathcal{P}^{p_+}$ and satisfies $DF\,W = \varpi_{\mathcal{P}}(X_F, W)$ (W4 Definition 6.4: the field is formed from the gradients). □

Source: P96-K (8), p. 129.

Lemma 7.11 (The flow with explicit margins; P96-K (9)–(11) and Lemma A.4)**.** *Under step data and (16), apply Lemma 3.30 in the Banach space $(\mathcal{P}, |\cdot|_{p,r})$ to $X = X_{F(\xi)}$ with*

$$V = D(s-\sigma, r), \qquad U = D(s - 2\sigma, \tfrac{r}{2}), \qquad \delta = \frac{\sigma}{4}, \qquad M = M_F \leq \frac{\sigma}{256} < \delta, \qquad L = \frac{M}{\delta} \leq \frac{1}{64}.$$

The hypotheses hold, and for $|t| \leq 1$, $\xi, \zeta \in \Pi$:

(i) the flow ϕ^t exists on U with values in V, $\phi^t-\mathrm{id}$ is holomorphic $U\to\mathcal{P}^{p_+}$, and $|\phi^t-\mathrm{id}|^\lambda_{p_+,r,U}\le 2M_F$;
(ii) on $D(s-3\sigma,\frac{r}{4})$: $|D\phi^t-I|_{p\to p_+,r}\le 8M_F/\sigma$ and $|D\phi^t-I|^\lambda_{p\to p_+,r}\le 16M_F/\sigma\le\eta^2$;
(iii) $\phi^t\big(D(s-5\sigma,\eta r)\big)\subset D(s-4\sigma,2\eta r)\subset D(s-3\sigma,\frac{r}{4})$;
(iv) ϕ^t is real on real points and commutes with integer angle translations: with $\tau_k=(k,0,0,0)$, $k\in\mathbb{Z}^2$, $\phi^t(w+\tau_k)=\phi^t(w)+\tau_k$ for $w\in U$ (note $U+\tau_k=U$).

Proof. **Step S7.24: Margins.** By Lemma 3.33 (with s,σ,r), every $w\in U$ has $B(w,\sigma/2)=B(w,2\delta)\subset V$ in $|\cdot|_{p,r}$. $|X_F|_{p,r}\le|X_F|_{p_+,r}\le M_F$ on V (Lemma 7.2), and $M_F\le\sigma\eta^2/16\le\sigma/1024$.

Step S7.25: Flow. Lemma 3.30(i)–(iii). The identity $\phi^t(w)-w=\int_0^t X_F(\phi^\tau w)\,d\tau$ (fundamental theorem for the C^1 curve) shows $\phi^t-\mathrm{id}\in\mathcal{P}^{p_+}$ with $|\cdot|_{p_+,r}\le|t|M_F$; holomorphy into $\mathcal{P}^{p_+}$: the integrand is holomorphic in w into $\mathcal{P}^{p_+}$ with a uniform bound on its derivative (Lemma 3.25 on $B(w,\delta)$), so differentiate under the integral ((E5)).

Step S7.26: Parameters. Let $N_1=\sup_V|X_{F(\xi)}-X_{F(\zeta)}|_{p_+,r}$. Lemma 3.30(iv) gives $|\phi^t_\xi-\phi^t_\zeta|_{p,r}\le e^{1/64}N_1$. In the integral identity, $|X_{F(\xi)}(\phi^\tau_\xi)-X_{F(\zeta)}(\phi^\tau_\zeta)|_{p_+,r}\le N_1+\frac{M_F}{\delta}e^{1/64}N_1\le 2N_1$ (both trajectory points lie in the convex ball $B(w,\delta)$, on which $|DX_F|_{p\to p_+,r}\le M_F/\delta$ by Lemma 3.25). Hence $|\phi^t-\mathrm{id}|^\lambda\le M_F+2\lambda N_1/|\xi-\zeta|\le 2M_F$ after taking the supremum.

Step S7.27: Derivative. By Lemma 3.33 with $(s-\sigma,\sigma,r/2)$, each point of $D(s-3\sigma,r/4)$ has a $|\cdot|_{p,r/2}$-ball of radius $\sigma/2$ in U; as $|W|_{p,r/2}\le 4|W|_{p,r}$, it has a $|\cdot|_{p,r}$-ball of radius $\sigma/8$ in U. Lemma 3.25 for $\phi^t-\mathrm{id}$ (bound M_F, resp. $2N_1$ for the difference in ξ) gives (ii); $16M_F/\sigma\le\eta^2$ by (16).

Step S7.28: Image. For $w\in D(s-5\sigma,\eta r)$ the increment $W=\phi^t w-w$ has $|\Theta|\le M_F<\sigma$, $|Y|\le r^2M_F<3\eta^2r^2$, $\|U\|+\|V\|\le rM_F<\eta r$; and $2\eta\le\frac14$.

Step S7.29: Reality and equivariance. Lemma 3.30(iii), with $\mathcal{P}=(\mathcal{P}_\mathbb{R})^\mathbb{C}$ and X_F real on reals (Theorem 6.19(i)). $X_F(w+\tau_k)=X_F(w)$ since the components of F are 1-periodic, and $V+\tau_k=V$; so $t\mapsto\phi^t(w)+\tau_k$ is a C^1 solution in V starting at $w+\tau_k$, and it equals $\phi^t(w+\tau_k)$ by the uniqueness of Lemma 3.30(i). □

Source: P96-K (9)–(11), pp. 129–130 ("Hence the flow X_F^t exists on $D(s-3\sigma,r/4)$ … takes this domain into $D(s-2\sigma,r/2)$", "any point in $D(s-4\sigma,r/8)$ has $|\cdot|_r$-distance greater than $\sigma/32$"), and Lemma A.4, p. 147. This is the concrete instance of the repaired Lemma A.4 (D-W-Pos96K-A4, Lemma 3.30 and its SOURCE_GAP): ball margin $2\delta=\sigma/2$ around $U=D(s-2\sigma,r/2)$ inside $V=D(s-\sigma,r)$, field bound $M_F<\delta$, both trajectories in the convex ball $B(w,\delta)$. Our margin $\sigma/8$ for the derivative replaces P96-K's $\sigma/32$ on a larger domain.

Lemma 7.12 (Flow calculus)**.** *In the setting of Lemma 7.11, for $|t|\le 1$:*

(i) $(t,w)\mapsto\phi^t(w)$ and $(t,w)\mapsto D\phi^t(w)$ are continuous on $[-1,1]\times U$, and $D\phi^t(w)=I+\int_0^t DX_F(\phi^\tau w)D\phi^\tau(w)\,d\tau$.
(ii) (Symplecticity.) $\varpi_\mathcal{P}(D\phi^t(w)W,D\phi^t(w)W')=\varpi_\mathcal{P}(W,W')$ for $w\in U$, $W,W'\in\mathcal{P}$.
(iii) On $D(s-3\sigma,r/4)$, $D\phi^t(w)$ is invertible on $\mathcal{P}$ and maps $\mathcal{P}^{p_+}$ onto $\mathcal{P}^{p_+}$.
(iv) (Transport.) If $G\in\mathcal{S}_{p_+}(V')$ and $\phi^t(O)\subset V'$ for an open $O\subset D(s-3\sigma,r/4)$, then $G\circ\phi^t\in\mathcal{S}_{p_+}(O)$ with $X_{G\circ\phi^t}=(\phi^t)^*X_G:=D(\phi^t)^{-1}X_G\circ\phi^t$.
(v) (Time derivative.) For every C^1 function G on an open set containing $\phi^{[0,1]}(w)$, $\frac{d}{dt}G(\phi^t w)=DG(\phi^t w)X_F(\phi^t w)$; in particular for $G=N$, by the bracket identity, $\frac{d}{dt}N(\phi^t w)=(\hat N+\hat a_0-R)(\phi^t w)$.

Proof. **Step S7.30: Continuity.** $|\phi^t(w)-\phi^{t'}(w')|_{p,r}\le M_F|t-t'|+e^L|w-w'|_{p,r}$ for w' near w (the second by Lemma 3.30(ii) and the mean value inequality on a convex ball in U). $D\phi^t(w)W=\frac{1}{2\pi i}\oint_{|\zeta|=\rho}\phi^t(w+\zeta W)\zeta^{-2}d\zeta$ for small ρ, which is continuous in (t,w) uniformly for $|W|_{p,r}\le 1$.

Step S7.31: Variational equation. Differentiate $\phi^t(w)=w+\int_0^t X_F(\phi^\tau w)d\tau$ in w under the integral ((E5); the integrand's derivative $DX_F(\phi^\tau w)D\phi^\tau(w)$ is continuous in (τ,w) by Step S7.30 and Lemma 3.26, and bounded by Le^L).

Step S7.32: Skewness. F is C^2 (Step S7.23) and $\varpi_{\mathcal P}(DX_F(w)A,B)=D^2F(w)[A,B]$ (differentiate $DF(w)B=\varpi_{\mathcal P}(X_F(w),B)$ in w), which is symmetric by (W5-E3) (F has derivative DF near w and DF is differentiable); by antisymmetry of $\varpi_{\mathcal P}$, $\varpi_{\mathcal P}(DX_FA,B)+\varpi_{\mathcal P}(A,DX_FB)=0$.

Step S7.33: Symplecticity. $g(t)=\varpi_{\mathcal P}(D\phi^tW,D\phi^tW')$ is C^1 with $g'=0$ by Steps S7.31–S7.32; $g(0)=\varpi_{\mathcal P}(W,W')$ ((E4)).

Step S7.34: Inverse. $|D\phi^t-I|_{p\to p,r}\le|D\phi^t-I|_{p\to p_+,r}\le\frac18$ and the same on $\mathcal P^{p_+}$ (as $|\cdot|_{p,r}\le|\cdot|_{p_+,r}$); Neumann series (E6).

Step S7.35: Transport. As in Theorem 5.23: $D(G\circ\phi^t)(w)W=\varpi_{\mathcal P}(X_G(\phi^tw),D\phi^tW)=\varpi_{\mathcal P}(D\phi^tD(\phi^t)^{-1}X_G(\phi^tw),D\phi^tW)=\varpi_{\mathcal P}(D(\phi^t)^{-1}X_G(\phi^tw),W)$; the field is holomorphic into $\mathcal P^{p_+}$ (inversion is holomorphic on invertible operators and $w\mapsto D\phi^t(w)$ is holomorphic, being a Cauchy integral of a holomorphic map).

Step S7.36: Time derivative. Chain rule. N is a continuous quadratic form on $\mathcal P$ (for $p=1$, $\sum_j|\Omega_j||z_j||\bar z_j|\le(1+M_0)\|z\|\|\bar z\|$ since $\omega_{1,1}(J)^2\ge J$), hence C^1, and $DN(w)X_F(w)=-L_NF(w)=(\hat N+\hat a_0-R)(w)$ by Lemma 6.16(ii) and Theorem 6.19(ii). □

Source: P96-K p. 124 ("Φ takes the hamiltonian $H=N+R$ into N_++R_+") and p. 130, asserted. The symplecticity argument is that of W3 Theorem 5.23.

7.6. **The new Hamiltonian.** For a C^1 function G and a field X_F put $\{G,F\}=DG\cdot X_F$, and for G,F in W4's class $\mathcal T$, $[X_G,X_F]=DX_G\cdot X_F-DX_F\cdot X_G$.

Lemma 7.13 (New Hamiltonian; conjugacy)**.** *In the setting of Lemma 7.11, let* $\Phi=\phi^1$, $D_+=D(s-5\sigma,\eta r)$, $R(t)=(1-t)\hat N+tR$, $H=N+P$, $N_+=N+\hat N=\langle\omega_+,y\rangle+\sum_j\Omega_{+,j}z_j\bar z_j$ ($\omega_+=\omega+\hat\omega$, $\Omega_+=\Omega+\hat\Omega$)*, and*

$$P_+=(P-R)\circ\Phi+\int_0^1\{R(t),F\}\circ\phi^t\,dt.$$

(i) $H\circ\Phi=N_++P_++\hat a_0$ *on* D_+.

(ii) $P_+\in\mathcal S_{p_+}(D_+)$*, real and periodic, with* $X_{P_+}=\Phi^*(X_P-X_R)+\int_0^1(\phi^t)^*[X_{R(t)},X_F]\,dt$.

(iii) (Weak conjugacy.) For $w\in D_+$ *and* $W\in\mathcal P^{p_+}$*:* $\varpi_{\mathcal P}\big(X_H(\Phi w),D\Phi(w)W\big)=\varpi_{\mathcal P}\big(X_{N_+}(w)+X_{P_+}(w),W\big)$*, where* $X_N(w')=(\frac{\omega}{2\pi},0,\Omega v',-\Omega u')$ *is the bounded linear map* $\mathcal P\to\mathcal P^{p-1}$ *(plus a constant) of Lemma 6.16(i),* $X_H=X_N+X_P$*, and* $\varpi_{\mathcal P}$ *is extended to* $\mathcal P^{p-1}\times\mathcal P^{p_+}$ *by the same formula (continuous, since* $(p-1)+p_+\ge0$*, (9)).*

Proof. **Step S7.37: Expansion.** Fix $w\in D_+$; $\phi^t(w)\in D(s-4\sigma,2\eta r)$ for $|t|\le1$ (Lemma 7.11(iii)). By Lemma 7.12(v) and (W5-E2), $N(\Phi w)=N(w)+\int_0^1(\hat N+\hat a_0-R)(\phi^tw)\,dt$ and, for $G\in\{\hat N,R\}$, $G(\phi^tw)=G(w)+\int_0^t\{G,F\}(\phi^\tau w)\,d\tau$. Substituting the second into the first, applying (W5-E5) to $g(\tau)=\{\hat N-R,F\}(\phi^\tau w)$ (continuous by Lemma 7.12(i)), and writing $P=R+(P-R)$:

$$H(\Phi w)=N(w)+\hat N(w)+\hat a_0+(P-R)(\Phi w)+\int_0^1\{(1-\tau)\hat N+\tau R,F\}(\phi^\tau w)\,d\tau,$$

using $R(\Phi w)=R(w)+\int_0^1\{R,F\}(\phi^\tau w)d\tau$. This is (i).

Step S7.38: Bracket field. For $G, F \in \mathcal{T}$, $\{G,F\}$ is C^1 and $D\{G,F\}W = D^2G[X_F,W] + DG\,DX_FW = \varpi_{\mathcal{P}}(DX_GX_F,W) + \varpi_{\mathcal{P}}(X_G,DX_FW) = \varpi_{\mathcal{P}}([X_G,X_F],W)$ by Step S7.32 for G and F. Hence $\{R(t),F\} \in \mathcal{S}_{p_+}$ with field $[X_{R(t)},X_F]$, holomorphic on $D(s-\sigma,r)$.

Step S7.39: Field of P_+. Lemma 7.12(iv) for $G = P - R$ (in $\mathcal{S}_{p'} \subset \mathcal{S}_{p_+}$ on $D(s,r)$; $R \in \mathcal{T}_s$) and for $G = \{R(t),F\}$; differentiate under the t-integral (continuity in (t,w) by Lemma 7.12(i)). Reality: all ingredients are real.

Step S7.40: Conjugacy. $H \circ \Phi$ is C^1 on D_+; for $W \in \mathcal{P}^{p_+}$, $D(H\circ\Phi)(w)W = DN(\Phi w)D\Phi W + DP(\Phi w)D\Phi W$. Here $D\Phi W \in \mathcal{P}^{p_+}$, and for $w' \in \mathcal{P}$, $W'' \in \mathcal{P}^{p_+}$, $DN(w')W'' = \varpi_{\mathcal{P}}(X_N(w'),W'')$ by direct computation of the derivative of the quadratic form N (the pairings converge by Lemma 6.16(i)). The other side is $D(N_+ + P_+ + \hat{a}_0)(w)W = \varpi_{\mathcal{P}}(X_{N_+}(w) + X_{P_+}(w),W)$ by the same computation for N_+ and (ii). □

Source: P96-K p. 130 ("we obtain the new hamiltonian $H\circ\Phi = N_+ + P_+ \dots\ P_+ = (P-R)\circ X_F^1 + \int_0^1 \{R(t),F\}\circ X_F^t\,dt$ with $R(t) = (1-t)\hat{N} + tR$" and the field formula). Deviations: the constant $\hat{a}_0$ (W4 Theorem 6.19, deviation (e)); the conjugacy is stated in the weak form (iii) because X_N is unbounded and, when $p_+ < p+1$, $X_N(\Phi w)$ need not lie in $\mathcal{P}$.
SOURCE_GAP: P96-K p. 130 writes the new field as the pull-back of $X_P - X_R$ plus the integral of commutators without deriving it from $H\circ\Phi = N_+ + P_+$; the derivation needs $\{N,F\}$ only through W4's identity $DN\cdot X_F = -L_NF$ (Step S7.37), never $X_N \circ \Phi$. Supplied above.

Lemma 7.14 (Pull-back; P96-K (12))**.** *In the setting of Lemma 7.11, let $Y(\cdot,\xi)$ be holomorphic fields $D(s-3\sigma,4\eta r) \to \mathcal{P}^{p_+}$. Then for $|t| \le 1$,*

$$|(\phi^t)^*Y|^\lambda_{\eta r,D_+} \le 7\,|Y|^\lambda_{\eta r,D(s-3\sigma,4\eta r)} \qquad (\textit{norm } |\cdot|_{p_+,\eta r}).$$

Proof. **Step S7.41: Inverse in the small scale.** For $W \in \mathcal{P}^{p_+}$, by Lemma 7.2 and Lemma 7.11(ii), $|(D\phi^t - I)W|_{p_+,\eta r} \le \eta^{-2}|(D\phi^t - I)W|_{p_+,r} \le \eta^{-2}\frac{8M_F}{\sigma}|W|_{p,r} \le \frac{1}{2}|W|_{p_+,\eta r}$, so $|D(\phi^t)^{-1}|_{p_+\to p_+,\eta r} \le 2$ on $D(s-3\sigma,r/4) \supset D_+$. Fix $\xi \ne \zeta$, write Δ for the difference between the values at ξ and ζ, and $N_1 = \sup_V |X_{F(\xi)} - X_{F(\zeta)}|_{p_+,r}$, so $\lambda N_1 \le M_F|\xi-\zeta|$. By Step S7.27, $|\Delta D\phi^t|_{p\to p_+,r} \le 16N_1/\sigma$, hence $\lambda|\Delta D\phi^t|_{p_+\to p_+,\eta r} \le \eta^{-2}\cdot 16\lambda N_1/\sigma \le |\xi-\zeta|$ (by (16)); and $\Delta(D\phi^t)^{-1} = -(D\phi^t_\xi)^{-1}(\Delta D\phi^t)(D\phi^t_\zeta)^{-1}$ gives $\lambda|\Delta(D\phi^t)^{-1}| \le 4|\xi-\zeta|$.

Step S7.42: Derivative of Y. By Lemma 3.33 with $(s-2\sigma,\sigma,4\eta r)$, every point of $D(s-4\sigma,2\eta r)$ has a $|\cdot|_{p,4\eta r}$-ball of radius $\sigma/2$ in $D(s-3\sigma,4\eta r)$, and $|\cdot|_{p,4\eta r} \le |\cdot|_{p,\eta r}$; so $|DY|_{p\to p_+,\eta r} \le 2|Y|/\sigma$ there (Lemma 3.25).

Step S7.43: Assemble. $|(\phi^t)^*Y| \le 2|Y|$. For the differences, $\Delta[(D\phi^t)^{-1}Y\circ\phi^t] = \Delta(D\phi^t)^{-1}\,Y_\xi\circ\phi^t_\xi + (D\phi^t_\zeta)^{-1}\big[(\Delta Y)\circ\phi^t_\xi + Y_\zeta\circ\phi^t_\xi - Y_\zeta\circ\phi^t_\zeta\big]$; the last bracket is $\le \frac{2|Y|}{\sigma}|\Delta\phi^t|_{p,\eta r}$ (segment in the convex $D(s-4\sigma,2\eta r)$), and $\lambda|\Delta\phi^t|_{p,\eta r} \le \eta^{-2}\lambda\cdot 2N_1 \le \frac{\sigma}{8}|\xi-\zeta|$. Hence $\lambda|\Delta[(\phi^t)^*Y]| \le \big(4|Y| + 2\lambda|Y|^{\mathcal{L}} + 2\cdot\frac{2}{\sigma}\cdot\frac{\sigma}{8}|Y|\big)|\xi-\zeta|$, where $|Y|^{\mathcal{L}}$ is the Lipschitz constant of Y; dividing by $|\xi-\zeta|$ and adding the sup bound $2|Y|$: $|(\phi^t)^*Y|^\lambda \le \frac{13}{2}|Y| + 2\lambda|Y|^{\mathcal{L}} \le 7|Y|^\lambda$. □

Source: P96-K (12), p. 130, and its proof pp. 131–132 (domains $U = D(s-5\sigma,\eta r)$, $V = D(s-4\sigma,2\eta r)$, $W = D(s-2\sigma,4\eta r)$). Constant ours; our W is $D(s-3\sigma,4\eta r)$.

Lemma 7.15 (Commutators)**.** *In the setting of Lemma 7.11, for $0 \le t \le 1$:*

$$|[X_{R(t)},X_F]|^\lambda_{\eta r,D(s-3\sigma,4\eta r)} \le 24\,\epsilon M_F/(\sigma\eta^2).$$

Proof. **Step S7.44: Cauchy.** On $D(s-2\sigma,r/2)$, which has margin $\sigma/2$ in $|\cdot|_{p,r}$ inside $D(s-\sigma,r)$ (Lemma 3.33), $|DX_G|_{p\to p_+,r} \le \frac{2}{\sigma}|X_G|_{D(s-\sigma,r)}$ for $G \in \{R(t),F\}$, and the same for parameter differences (Lemma 3.25).

Step S7.45: Bound. $|X_{R(t)}|^\lambda \le \max(3\epsilon, 6\epsilon)$ on $D(s,r)$ (Lemmas 7.8, 7.10), so by Remark 7.4, $|[X_{R(t)}, X_F]|^\lambda_r \le 2 \cdot \frac{2}{\sigma} \cdot 6\epsilon \cdot M_F$ on $D(s-2\sigma, r/2) \supset D(s-3\sigma, 4\eta r)$ ($4\eta \le \frac{1}{2}$); then $|\cdot|_{p_+,\eta r} \le \eta^{-2}|\cdot|_{p_+,r}$. □

Source: P96-K p. 130–131 ("$|[X_R, X_F]|_r \le |DX_R|_{r,r}|X_F|_r + |DX_F|_{r,r}|X_R|_r$", "$|Y|^\lambda_{\eta r} \le \eta^{-2}|Y|^\lambda_r$").

7.7. The KAM step.

Theorem 7.16 (One KAM step)**.** *Assume step data (Definition 7.9) and the smallness condition (16); in the modified case assume also $2\beta \le p+1-p_+$ for the Lipschitz parts. Let $C_2 = 896+168c_F$. Then there are a real holomorphic family $\Phi(\cdot,\xi) : D_+ = D(s-5\sigma, \eta r) \to D(s-4\sigma, 2\eta r)$ and N_+, P_+ such that, for all $\xi \in \Pi$:*

- *(i) $H\circ\Phi = N_+ + P_+ + \hat{a}_0(\xi)$ on D_+, with the weak conjugacy of Lemma 7.13(iii); Φ is symplectic ($\varpi_{\mathcal{P}}(D\Phi W, D\Phi W') = \varpi_{\mathcal{P}}(W, W')$), $D\Phi(w)$ maps $\mathcal{P}^{p_+}$ onto itself, Φ is real on real points, and $\Phi(w+\tau_k) = \Phi(w) + \tau_k$ for $k \in \mathbb{Z}^2$ (Lemma 7.11(iv));*
- *(ii) $P_+ \in \mathcal{S}_{p_+}(D_+)$ is real and periodic with*

$$|X_{P_+}|^\lambda_{\eta r, D_+} \le C_2\Big(\frac{\epsilon^2}{\alpha\sigma^t\eta^2} + \eta\epsilon\Big); \tag{17}$$

- *(iii) $|\omega_+ - \omega| + \lambda|\omega_+ - \omega|^{\mathcal{L}} \le 6\pi\epsilon$ and $\sup_j (j+2)^{p_+-p}\big(|\Omega_{+,j} - \Omega_j| + \lambda|\Omega_{+,j} - \Omega_j|^{\mathcal{L}}\big) \le 3\epsilon$; hence (ω_+, Ω_+) satisfies (14) with $M_0 + 3\epsilon$ and $M_+ \le M + (6\pi+3)\epsilon/\lambda$ when $\lambda > 0$;*
- *(iv) (Divisors.) For $K \ge 1$ and $0 < \alpha_+ \le \alpha$ with $6\pi(1+K)(1+K^\tau)\epsilon \le \alpha - \alpha_+$: $|\langle k, \omega_+\rangle + \langle l, \Omega_+\rangle| \ge \alpha_+\langle l\rangle/A_k$ on Π for $(k,l) \in \mathcal{Z}$ with $|k|_1 \le K$, whenever the same holds with α for ω, Ω; this holds for the Dirichlet weight $\langle l\rangle^{\mathrm{w}}$ of Lemma 6.6(ii) as well as for $\langle l\rangle$;*
- *(v) (Transformation.) $|\Phi - \mathrm{id}|^\lambda_{p_+,r,D_+} \le 2M_F$ and $|D\Phi - I|^\lambda_{p\to p_+,r,D_+} \le 16M_F/\sigma$, with $M_F = c_F\epsilon/(\alpha\sigma^{t-1})$.*

Proof. **Step S7.46: Construction.** Lemmas 7.8, 7.10, 7.11, 7.12, 7.13 give $\Phi = \phi^1$, (i), the formula for X_{P_+}, and (v) (Lemma 7.11(i)–(ii)).

Step S7.47: First term. $Y = X_P - X_R$ on $D(s-3\sigma, 4\eta r) \subset D(s, 4\eta r)$ has $|Y|^\lambda_{\eta r} \le 128\eta\epsilon$ (Lemma 7.8(iv)); Lemma 7.14 gives $\le 896\eta\epsilon$.

Step S7.48: Second term. Lemmas 7.15 and 7.14, integrated over $t \in [0,1]$: $\le 7\cdot 24\,\epsilon M_F/(\sigma\eta^2) = 168c_F\epsilon^2/(\alpha\sigma^t\eta^2)$.

Step S7.49: Frequencies. (iii) is Lemma 7.10; the Lipschitz constants follow from $\lambda|\hat{\omega}|^{\mathcal{L}} \le 6\pi\epsilon$, and $(j+2)^{p_+-p} \ge j^{1/2}$.

Step S7.50: Divisors. $|\langle k, \hat{\omega}\rangle + \langle l, \hat{\Omega}\rangle| \le |k|_1\, 6\pi\epsilon + 2\cdot 3\epsilon \le 6\pi(1+|k|_1)\epsilon \le (\alpha-\alpha_+)/(1+K^\tau) \le (\alpha-\alpha_+)\langle l\rangle/A_k$ for $|k|_1 \le K$, as $A_k \le 1+K^\tau$, $\langle l\rangle \ge 1$; subtract from the old lower bound. □

Source: P96-K §3, pp. 128–132: (8)–(15). Deviations: explicit constants ($c_0 = 16c_F$, C_2); the field θ-component carries $(2\pi)^{-1}$ (period one), so $|\hat{\omega}| \le 6\pi\epsilon$ where P96-K has $|\hat{\omega}| \lessdot |X_P|$; the domains $D(s-3\sigma, 4\eta r)$ for Taylor and commutators.

Remark 7.17 (Notation clash in P96-K). P96-K defines the operator norm $|L|_{r,s}$ from $(\mathcal{P}^{a,p}, |\cdot|_{p,s})$ to $(\mathcal{P}^{a,\bar{p}}, |\cdot|_{\bar{p},r})$ (p. 129) but applies it to $D\Phi^\nu$ (p. 134, "$|D\Phi^\nu|_{r_0,r_\nu,D_\nu} \le 2$"), which is not smoothing; there the norm must be the one from $|\cdot|_{p,r_\nu}$ to $|\cdot|_{p,r_0}$. W6 uses the latter explicitly (Lemma 8.4). Imprecision, not an error.

7.8. **The action–angle substitution and the initial Hamiltonian.** Following review ruling W4-R3, W5 carries W3's normal form (Theorem 5.25, Remark 5.30) into the form $N+P$ on $D(s,r)$ and checks that $D(s,r)$ lands in W3's ball B_{ρ_1}. Write $\mathfrak{W}=\mathfrak{W}_{a,p}$, $\mathfrak{W}^q=\mathfrak{W}_{a,q}$, with the form Ω of Lemma 5.4, and $H'=\Lambda+\bar G+G^{\mathrm{nor}}+K$ on B_{ρ_1}. For $\zeta\in\mathbb{C}$ with $\operatorname{Re}\zeta>0$, $\sqrt{\zeta}$ is the principal branch.

Lemma 7.18 (Action–angle substitution)**.** *For $\xi\in\mathbb{C}^2$ and $w=(\theta,y,u,v)\in\mathcal{P}$ with $\operatorname{Re}(\xi_i+y_i)>0$ $(i=1,2)$ define $\mathcal{A}_\xi(w)\in\mathfrak{W}$ by*

$$\mathcal{A}_\xi(w)_{\pm i}=\sqrt{\xi_i+y_i}\,e^{\mp 2\pi i\theta_i}\quad(i=1,2),$$
$$\mathcal{A}_\xi(w)_J=z_{J-2}=\tfrac{u_{J-2}+iv_{J-2}}{\sqrt2},\quad \mathcal{A}_\xi(w)_{-J}=\bar z_{J-2}=\tfrac{u_{J-2}-iv_{J-2}}{\sqrt2}\quad(J\ge3).$$

(i) $(\xi,w)\mapsto\mathcal{A}_\xi(w)$ is holomorphic on its (open) domain, 1-periodic in each θ_i, and maps real points with real ξ to $\mathfrak{W}_{\mathbb{R}}=\{\varrho w=w\}$. At $y=0$, $u=v=0$, real θ and $\xi\in(0,\infty)^2$ it is the linear torus: $\mathcal{Z}^{-1}\mathcal{A}_\xi(\theta,0,0,0)=K^0_\xi(\theta)$ (Convention 1.13, Lemma 5.3).

(ii) (Domain.) Let $0<s\le\frac18$, $r>0$, $t>0$, $|\xi|_\infty\le t$, $\operatorname{Re}\xi_i\ge 2r^2$, and $c_A=2200$. Then $\mathcal{A}_\xi(D(s,r))\subset B_\rho$ with $\rho^2=c_A(t+r^2)$; in particular $\mathcal{A}_\xi(D(s,r))\subset B_{\rho_1}$ whenever $c_A(t+r^2)<\rho_1(m)^2$.

(iii) (Derivative.) For every $q\in\mathbb{R}$, $D\mathcal{A}_\xi(w)$ is a linear bijection $\mathcal{P}^q\to\mathfrak{W}^q$ with inverse $W\mapsto(\Theta,Y,U,V)$,

$$\Theta_i=\frac{W_{-i}e^{-2\pi i\theta_i}-W_ie^{2\pi i\theta_i}}{4\pi i\sqrt{\xi_i+y_i}},\quad Y_i=\sqrt{\xi_i+y_i}\,(W_ie^{2\pi i\theta_i}+W_{-i}e^{-2\pi i\theta_i}),$$
$$U=\tfrac{\hat W_++\hat W_-}{\sqrt2},\qquad V=\tfrac{\hat W_+-\hat W_-}{i\sqrt2}$$

$(\hat W_\pm=(W_{\pm J})_{J\ge3})$, and $\Omega(D\mathcal{A}_\xi W,D\mathcal{A}_\xi W')=\varpi_{\mathcal{P}}(W,W')$ for $W\in\mathcal{P}^q$, $W'\in\mathcal{P}^{q'}$, $q+q'\ge0$.

(iv) (Normal form.) On the domain of (ii) with $\mathcal{A}_\xi(D(s,r))\subset B_{\rho_1}$,

$$H'\circ\mathcal{A}_\xi=e(\xi)+N_\xi+P_\xi,\qquad e(\xi)=\langle\lambda_J,\xi\rangle+\tfrac12\langle A\xi,\xi\rangle,\qquad N_\xi=\langle\omega^0(\xi),y\rangle+\sum_{j\ge1}\Omega^0_{j+2}(\xi)z_j\bar z_j,$$

with ω^0,Ω^0 of Definition 9.3 ($\omega^0=\lambda_J+A\xi$, $\Omega^0_J=\lambda_J+\frac{3}{2\pi\lambda_J}\langle b,\xi\rangle$), and $P_\xi=\tilde Q+G^{\mathrm{nor}}\circ\mathcal{A}_\xi+K\circ\mathcal{A}_\xi$, $\tilde Q=\frac12\langle Ay,y\rangle+\sum_{j\ge1}\frac{3\langle b,y\rangle}{2\pi\lambda_{j+2}}z_j\bar z_j$.

(v) (Conjugacy.) $P_\xi\in\mathcal{S}_{p+1}(D(s,r))$, real for real ξ, with $X_{P_\xi}=X_{\tilde Q}+D\mathcal{A}_\xi^{-1}\big(X_{G^{\mathrm{nor}}}+X_K\big)\circ\mathcal{A}_\xi$; and for every $w\in D(s,r)$,

$$X_{H'}(\mathcal{A}_\xi w)=D\mathcal{A}_\xi(w)\big(X_{N_\xi}(w)+X_{P_\xi}(w)\big)\qquad\text{in }\mathfrak{W}^{p-1},$$

where $X_{H'}=X_\Lambda+X_{\bar G}+X_{G^{\mathrm{nor}}}+X_K$ with $X_\Lambda(w')_l=-i\lambda_l'w_l'$ on all of $\mathfrak{W}$.

(vi) (Weight factors.) No weight conversion is needed: the normal weights of $\mathcal{P}$ and of $\mathfrak{W}$ are both $\omega_{a,q}(J)$ at the Dirichlet label (Remark 6.22), so W3's factors 3^pe^{2a}, $3^{p+1}e^{2a}$ (which compare with P96-K's weights j^qe^{aj}) do not occur. The remaining factors are $\|\hat w\|\le\|u\|+\|v\|\le\sqrt2\|\hat w\|$ (sum norm of $D(s,r)$ versus $\mathfrak{W}$), used in (ii) and Proposition 7.19, and $\|x\|_E\le\|\mathcal{Z}x\|\le\sqrt2\|x\|_E$ (Lemma 5.3), used only by W8 to return to `WaveSpace`.

Proof. **Step S7.51: Holomorphy and reality.** $\sqrt{\cdot}$ is holomorphic on $\operatorname{Re}>0$ ((W5-E4)) and is composed with the holomorphic $(\xi,y)\mapsto\xi_i+y_i$; the exponentials are entire and the normal part is linear. For real data, $\mathcal{A}_{-i}=\overline{\mathcal{A}_i}$ and $\bar z=\overline{z}$. At the torus, $\mathcal{Z}^{-1}$ gives $q_i=\sqrt2\operatorname{Re}(\sqrt{\xi_i}e^{-2\pi i\theta_i})=\sqrt{2\xi_i}\cos2\pi\theta_i$, $p_i=\sqrt2\operatorname{Im}(\dots)=-\sqrt{2\xi_i}\sin2\pi\theta_i$.

Step S7.52: Domain. $|\xi_i + y_i| < t + r^2$ and $|e^{\mp 2\pi i\theta_i}| \le e^{2\pi s}$, so the tangential part has $\sum_{|l|\le 2}|l|^{2p}e^{2a|l|}|w_l|^2 \le 2(e^2+4e^4)e^{4\pi s}(t+r^2) \le 2173(t+r^2)$ ($a=p=1$, $e^{4\pi s}\le e^{\pi/2}$); the normal part has $\|\hat w\|^2 = \|u\|^2+\|v\|^2 < r^2$.

Step S7.53: Derivative. $\partial\mathcal{A}_{\pm i} = \big(\frac{Y_i}{2\sqrt{\xi_i+y_i}} \mp 2\pi i\sqrt{\xi_i+y_i}\Theta_i\big)e^{\mp 2\pi i\theta_i}$; solving the 2×2 system gives the inverse. The form: on each tangential pair $i(W_iW'_{-i} - W_{-i}W'_i) = 2\pi(\Theta_iY'_i - Y_i\Theta'_i)$ by expansion, and on normal pairs $i(Z_j\bar Z'_j - \bar Z_jZ'_j) = U_jV'_j - V_jU'_j$; the series converge for $q+q'\ge 0$ by (9).

Step S7.54: Normal form. Substitute $w_iw_{-i} = \xi_i + y_i$ ($i=1,2$), $w_Jw_{-J} = z_{J-2}\bar z_{J-2}$ into $\Lambda = \sum\lambda_jw_jw_{-j}$ and $\bar G = \frac12\langle AI, I\rangle + \langle BI, Z\rangle$ (Theorem 5.25(iii), $(BI)_J = \frac{3}{2\pi\lambda_J}\langle b, I\rangle$, Theorem 2.10) with $I = \xi + y$, and expand (A symmetric).

Step S7.55: Fields. For $G \in \{G^{\mathrm{nor}}, K\}$ (fields into $\mathfrak{W}^+$, Theorem 5.25(ii)–(iii)): $D(G\circ\mathcal{A})W = \Omega(X_G(\mathcal{A}w), D\mathcal{A}W) = \varpi_{\mathcal{P}}(D\mathcal{A}^{-1}X_G(\mathcal{A}w), W)$ by (iii), and $D\mathcal{A}^{-1}$ maps $\mathfrak{W}^{p+1}$ to $\mathcal{P}^{p+1}$. $\tilde Q$ is an explicit polynomial; its field $(\frac{1}{2\pi}(Ay + \beta'(z,\bar z)), 0, \beta v, -\beta u)$ with $\beta_j = \frac{3\langle b,y\rangle}{2\pi\lambda_{j+2}}$, $\beta'_i = \sum_j \frac{3b_i}{2\pi\lambda_{j+2}}z_j\bar z_j$, lies in $\mathcal{P}^{p+1}$ since $(j+2)|\beta_j| \le \frac{3}{2\pi}|b|_1|y|$. Holomorphy: compositions. Reality: H', $\mathcal{A}_\xi$ are real.

Step S7.56: Conjugacy. For $W \in \mathcal{P}^{p+1}$: $D(H'\circ\mathcal{A})(w)W = \Omega(X_{H'}(\mathcal{A}w), D\mathcal{A}W)$, using $D\Lambda(w')W'' = \Omega(X_\Lambda(w'), W'')$ for $w' \in \mathfrak{W}$, $W'' \in \mathfrak{W}^{p+1}$ (Lemma 5.8, the pairing of $\mathfrak{W}^{p-1}$ with $\mathfrak{W}^{p+1}$). By (iv) and Lemma 7.13(iii)'s computation for N_ξ, the left side is $\varpi_{\mathcal{P}}(X_{N_\xi}(w) + X_{P_\xi}(w), W) = \Omega(D\mathcal{A}(X_{N_\xi}+X_{P_\xi}), D\mathcal{A}W)$ by (iii) with $q=p-1$, $q'=p+1$. As $D\mathcal{A}$ maps $\mathcal{P}^{p+1}$ onto $\mathfrak{W}^{p+1}$ and Ω is nondegenerate against $\mathfrak{W}^{p+1}$ (test with unit vectors, Lemma 5.4), the identity follows. □

Source: KP96 §7, Step 1, p. 25 ("symplectic polar and real coordinates $q_j = \sqrt{2(\xi_j+y_j)}e^{-ix_j}$", $I = \xi + y$, $\omega(\xi) = \alpha + A\xi$, $\Omega(\xi) = \beta + B\xi$); P96-W p. 290. Conversions: KP96's $q_j = \sqrt{2}z_j$; period-one angles $x = 2\pi\theta$; W0's $z = (q+ip)/\sqrt{2}$. The constant c_A and the domain check are ours (assigned to W5 by review ruling W4-R3; W3 Remark 5.30 assigned it to W4).

Proposition 7.19 (The initial perturbation; KP96 §7, Step 3)**.** *Let $m>0$, $a=p=1$, and ρ_1, C_*, C'_K be W3's constants (§5.9). Let $0<s\le\frac18$, $0<r\le 1$, $t>0$, $\rho_\xi>0$, and $\Pi\subset\mathbb{R}^2$ such that every $\zeta\in\mathbb{C}^2$ with $|\zeta-\xi|_\infty<\rho_\xi$ for some $\xi\in\Pi$ satisfies $\operatorname{Re}\zeta_i \ge 2r^2$ and $|\zeta|_\infty \le t$, and $c_A(t+r^2) < \rho_1^2$. Put*

$$\epsilon_{\mathrm{in}} = (1+2C_*)\,r^2 + C'_K\big(c_A(t+r^2)\big)^{5/2}\Big(\frac{3}{r} + \frac{2\sqrt{t+r^2}}{r^2}\Big).$$

Then $P_\xi \in \mathcal{S}_{p+1}(D(s,r))$ is a real periodic family on Π, and for every $\lambda\ge 0$

$$|X_P|^\lambda_{r,D(s,r)} \le \epsilon_{\mathrm{in}}\Big(1 + \frac{4\lambda}{\rho_\xi}\Big).$$

The frequencies ω^0, Ω^0_{j+2} satisfy (13) (hence (14)) with the constants of Lemma 6.6, and $|\omega^0(\xi) - \omega^0(\xi')| \le N_A|\xi-\xi'|$.

Proof. **Step S7.57: The polynomial part.** On $D(s,r)$: $|Ay| \le N_Ar^2 \le \frac35 r^2$ (Lemma 9.4(ii)); $|\beta'_i| \le \frac{3}{2\pi\cdot 3}\sum_j|z_j\bar z_j| \le \frac{1}{2\pi}\|z\|\|\bar z\| \le \frac{r^2}{4\pi}$ (weights ≥ 1, $\|z\|, \|\bar z\| \le r/\sqrt{2}$ by (10)); so the Θ-component is $\le r^2/8$. The Y-component vanishes. The normal components: $\|\beta u\|_{p+1} \le \sup_J J|\beta_{J-2}|\|u\| \le \frac{3}{2\pi}\cdot\frac32 r^2\|u\|$, so they contribute $\le \frac34 r^2$. Total $\le r^2$.

Step S7.58: G^{nor}. Its field depends on $\hat w$ only and is normal: $\|X_{G^{\mathrm{nor}}}\|_+ \le C_*\|\hat w\|^3 < C_*r^3$ (Theorem 5.25(iii)); by the inverse formula of Lemma 7.18(iii), $\|U\|_{p+1} + \|V\|_{p+1} \le 2\|(\hat W_+, \hat W_-)\|_+$, contributing $2C_*r^2$.

Step S7.59: K. $\|X_K(\mathcal{A}w)\|_+ \le C'_K\rho^5$ with $\rho^2 = c_A(t+r^2)$ (Lemma 7.18(ii)). In the inverse formula, $|W_{\pm i}| \le \|W\|_+/e$ (weights $\ge e^2$), $|\sqrt{\xi_i+y_i}| \ge \sqrt{\mathrm{Re}(\xi_i+y_i)} \ge r$, $|\sqrt{\xi_i+y_i}| \le \sqrt{t+r^2}$, $e^{2\pi s} \le e^{\pi/4}$; so $|\Theta| \le \frac{2e^{\pi/4}}{4\pi er}\|W\|_+ \le \frac{1}{r}\|W\|_+$, $|Y| \le \frac{2e^{\pi/4}}{e}\sqrt{t+r^2}\|W\|_+ \le 2\sqrt{t+r^2}\|W\|_+$, and the normal part $\le 2\|W\|_+$. In $|\cdot|_{p+1,r}$: $\le \|W\|_+(\frac{1}{r} + \frac{2\sqrt{t+r^2}}{r^2} + \frac{2}{r})$.

Step S7.60: Parameters. The fields of $\tilde{Q}$, $G^{\mathrm{nor}} \circ \mathcal{A}_\zeta$ do not depend on ζ. For fixed w, $\zeta \mapsto X_{K\circ\mathcal{A}_\zeta}(w)$ is holomorphic on the ρ_ξ-neighbourhood of Π (explicit formula) and bounded there by the same constant. For $\xi, \xi' \in \Pi$ with $|\xi - \xi'| < \rho_\xi/2$ the segment has a $\rho_\xi/2$-polydisc around each point inside the ρ_ξ-polydisc around ξ, so Lemma 3.25 gives the Lipschitz constant $2\epsilon_{\mathrm{in}}/\rho_\xi$; if $|\xi - \xi'| \ge \rho_\xi/2$ the difference is $\le 2\epsilon_{\mathrm{in}} \le 4\epsilon_{\mathrm{in}}|\xi - \xi'|/\rho_\xi$.

Step S7.61: Frequencies. Lemma 6.6(i) and Lemma 9.4(ii). □

Source: KP96 §7, Step 3, p. 26 ("$|\tilde{Q}| = O(r^4)$, $|R| = O(r^{g\lambda} + r^4)$ … $|X_P|_{r,D(1,r)} + \alpha|X_P|^{\mathcal{L}}_{r,D(1,r)} = O(r^2)$", parameter domain $U_{-4r^2}\Xi_r$, "Cauchy with respect to ξ"), in the wave form of P96-W p. 290. With KP96's $t = r^{2\lambda}$, $\lambda = \frac{2}{3}$, and $\rho_\xi \ge \alpha/2$: $\epsilon_{\mathrm{in}} = O(r^2)$ and $|X_P|^{\alpha/M} = O(r^2)$ for $M \ge 1$. Constants ours. The choice of r, t, ρ_ξ, Π belongs to W8.

7.9. Constants and ledger.

Constants, in the order chosen. (1) m; $a = p = 1$; τ (W7 fixes $\tau = 24$). (2) From W3, depending on m: ρ_1, C_*, C'_K; $c_A = 2200$ (absolute). (3) From W4: K_S, $C_1 = 7 + 10K_S < 36$, $b_{\mathrm{W}}(\tau)$ (absolute for fixed τ). (4) $c_F = 9C_1 b_{\mathrm{W}}(\tau)$, $c_0 = 16c_F$, $C_2 = 896 + 168c_F$, $t = 2\tau + 4$ (= 52 for $\tau = 24$), and for the modified (preparatory) step $b^\natural(\tau)$, $c_F^\natural = 9C_1 b^\natural(\tau)$, $c_0^\natural = 16c_F^\natural$, $C_2^\natural = 896 + 168c_F^\natural$, $t^\natural = 2\tau + 5$ (= 53): absolute for fixed τ, independent of m. (5) Per step: $s, \sigma, \eta, r, \alpha, \lambda, \Pi$, frequencies with M, then ϵ subject to (16). The only m-dependence of the step itself is through the frequencies (M, M_0) and the divisor hypotheses; Proposition 7.19 depends on m through ρ_1, C_*, C'_K, N_A, λ_1, λ_2.
Results. Definitions 7.1, 7.3, 7.9; Lemmas 7.2, 7.5, 7.6, 7.8, 7.10, 7.11, 7.12, 7.13, 7.14, 7.15, 7.18; Theorem 7.16; Proposition 7.19; Remarks 7.4, 7.7, 7.17. External: (W5-E1), (W5-E2).
Source issues. (a) P96-K p. 140: the λ-norm estimate with modified divisors needs $2\beta \le \bar{p} - \check{p}$ (SOURCE_GAP after Lemma 7.6); suspected erratum, avoided by W7's $\beta = \frac{1}{4}$. (b) P96-K p. 130: new field asserted without derivation (gap, supplied). (c) P96-K p. 134: operator norm notation (Remark 7.17). (d) P96-K p. 141: W4 used with $\check{p}$ silently (Lemma 7.5). (e) W3 Remark 5.30, "Return trip": the KAM tori of the Theorem D route are a priori only in $\mathfrak{W}^{p+1/2}$; W3's Corollary 5.29 handles them (Remark 8.15).

8. Milestone W6: iteration, convergence, invariance

This part iterates the KAM step of W5 (P96-K §4, pp. 132–135, and the preparatory step of §6, pp. 140–142) and delivers, for the wave equation with $J = \{1, 2\}$: a parameter set Π_* whose complement is contained in W7's excluded set, frequencies ω_* with W7's Diophantine and closeness properties, and for each $\xi \in \Pi_*$ an invariant torus Ψ_ξ in W3's Birkhoff coordinates, holomorphic on a complex strip, real, continuous in (ξ, θ), close to the linear torus, and invariant in the weak sense that W8 turns into clause (4) of `KAM.Wave.nlw2`.
Route. (1) One modified KAM step with the divisors $\hat{\alpha}\hat{w}(l)/A_k$ on $\Pi\backslash\Xi$ (the preparatory step), after which fields gain only half a weight ($p_+ = p + \frac{1}{2}$). (2) The iteration of W5's step with geometric σ_ν, $\eta_\nu^3 = \epsilon_\nu/(\alpha_\nu\sigma_\nu^t)$, levels $\alpha_\nu \downarrow \check{\alpha}/2$, removing at stage ν the zones with $|k| > K_{\nu-1}$. (3) Convergence of the compositions on the torus $\{y = 0, z = \bar{z} = 0\}$ only (no normal derivatives are needed), and of the frequencies. (4) Invariance: the conjugacy of each step is stated weakly (W5 Lemma 7.13(iii)); at points of the torus it becomes a strong identity in $\mathcal{P}^{p-1}$, which passes to the limit because X_N is bounded from $\mathcal{P}$ to $\mathcal{P}^{p-1}$.

Notation. P96-K's exponent $\kappa = \frac{4}{3}$ of §4 (p. 132) is written $\varkappa$ here; $\kappa = 2$ is reserved for the spectral exponent of W7 (W7 Remark 9.32(b)). Fixed throughout: $m > 0$, $a = p = 1$, $\tau = 24$, $t = 2\tau + 4 = 52$, and the constants c_F, c_0, C_2 of W5 (§7.9); the preparatory (modified) step uses $t^\natural = 53$, $c_F^\natural, c_0^\natural, C_2^\natural$ instead (Definition 7.9). Melnikov conditions are stated with the Dirichlet weight $\langle l\rangle$ of W7 (Definition 9.1); by Lemma 6.6(ii) they imply W4's conditions in the P96-K index at the same level.

8.1. **Sequences and the iterative lemma.**

Definition 8.1 (Iteration sequences)**.** Given $0 < s_0 \le 1$, $r_0 > 0$, $0 < \alpha_0 \le 1$, $M_0 > 0$ and $\epsilon_0 > 0$, put $\sigma_0 = s_0/40$ and for $\nu \ge 0$

$$\sigma_\nu = \sigma_0 2^{-\nu}, \quad s_{\nu+1} = s_\nu - 5\sigma_\nu, \quad \alpha_\nu = \tfrac{\alpha_0}{2}(1 + 2^{-\nu}), \quad M_\nu = M_0(2 - 2^{-\nu}), \quad \lambda_\nu = \frac{\alpha_\nu}{M_\nu},$$

$$\eta_\nu = \Big(\frac{\epsilon_\nu}{\alpha_\nu \sigma_\nu^t}\Big)^{1/3}, \quad r_{\nu+1} = \eta_\nu r_\nu, \quad \epsilon_{\nu+1} = 2C_2\,\epsilon_\nu\eta_\nu = \frac{2C_2\,\epsilon_\nu^{\varkappa}}{(\alpha_\nu\sigma_\nu^t)^{\varkappa-1}}\ (\varkappa = \tfrac{4}{3}), \quad D_\nu = D(s_\nu, r_\nu),$$

$$\gamma_0 = \big(2^{t+4}C_2 + c_0\big)^{-3}, \qquad K_0 = \big(96\pi\gamma_0\sigma_0^t\big)^{-1/(\tau+1)}, \qquad K_\nu = K_0 2^\nu, \quad K_{-1} = 0.$$

Then $s_\nu \downarrow s_\infty = \frac{3}{4}s_0$, $\sigma_\nu \le \min(\frac{1}{4}, s_\nu/6)$, $\lambda_\nu \ge \lambda_0/4$, $K_0 \ge 1$, and $K_\nu \ge 2^\nu$.

Source: P96-K §4, p. 132 ("$\alpha_\nu = \frac{\alpha_0}{2}(1+2^{-\nu})$, $M_\nu = M_0(2-2^{-\nu})$, … $\epsilon_{\nu+1} = \frac{c_1\epsilon_\nu^\kappa}{(\alpha_\nu\sigma_\nu^t)^{\kappa-1}}$, $\sigma_{\nu+1} = \frac{\sigma_\nu}{2}$, $\eta_\nu^3 = \frac{\epsilon_\nu}{\alpha_\nu\sigma_\nu^t}$, where $\kappa = \frac{4}{3}$", $\sigma_0 = s_0/40$, (17), $K_\nu = K_0 2^\nu$ with $K_0^{\tau+1} = 1/(c_1\gamma_0)$); period-one units; $c_1 = 2C_2$; our K_0 includes σ_0^t because our smallness is $\epsilon_0 \le \gamma_0\alpha_0\sigma_0^t$.

Lemma 8.2 (Iterative lemma)**.** *Let the data of Definition 8.1 be given, $p_+ = p + \frac{1}{2}$, and $\epsilon_0 \le \gamma_0\alpha_0\sigma_0^t$. Suppose at stage $\nu \ge 0$: a set $\Pi_\nu \subset \mathbb{R}^2$; frequencies (ω_ν, Ω_ν) on a set containing Π_ν satisfying (14) with $M_\omega + M_\Omega \le M_\nu$; on Π_ν, for all $(k,l) \in Z$ (W7's index set),*

$$|\langle k, \omega_\nu\rangle + \langle l, \Omega_\nu\rangle| \ge \alpha_\nu\langle l\rangle/A_k; \tag{18}$$

and a real periodic family $P_\nu \in \mathcal{S}_{p_+}(D_\nu)$ on Π_ν with $|X_{P_\nu}|^{\lambda_\nu}_{r_\nu, D_\nu} \le \epsilon_\nu$. Put $q_\nu = \epsilon_\nu/(\alpha_\nu\sigma_\nu^t)$. Then $q_\nu \le \gamma_0 2^{-\nu}$ and $q_{\nu+1} \le q_\nu/4$, and W5's step (Theorem 7.16, unmodified case) applies with $s_\nu, \sigma_\nu, \eta_\nu, r_\nu, \alpha_\nu, \lambda_\nu$; it yields $\Phi_{\nu+1} : D_{\nu+1} \to D_\nu$ (for $\xi \in \Pi_\nu$), $N_{\nu+1}$ with $(\omega_{\nu+1}, \Omega_{\nu+1})$ on Π_ν, and $P_{\nu+1}$, such that with

$$\Pi_{\nu+1} = \Pi_\nu \setminus \bigcup_{(k,l)\in Z,\ |k|>K_\nu} \mathcal{R}_{kl}^{\nu+1}, \qquad \mathcal{R}_{kl}^{\nu+1} = \Big\{\xi \in \Pi_\nu : |\langle k, \omega_{\nu+1}\rangle + \langle l, \Omega_{\nu+1}\rangle| < \alpha_{\nu+1}\frac{\langle l\rangle}{A_k}\Big\},$$

all hypotheses hold at stage $\nu+1$, provided that (18) for $k = 0$ holds at stage $\nu+1$ (it is supplied by Lemma 8.3). Moreover, with $M_{F,\nu} = c_F\epsilon_\nu/(\alpha_\nu\sigma_\nu^{t-1}) = c_F q_\nu\sigma_\nu$,

$$|\Phi_{\nu+1} - \mathrm{id}|^{\lambda_\nu}_{p_+, r_\nu, D_{\nu+1}} \le 2M_{F,\nu}, \quad |D\Phi_{\nu+1} - I|^{\lambda_\nu}_{p\to p_+, r_\nu, D_{\nu+1}} \le \frac{16M_{F,\nu}}{\sigma_\nu} \le 2^{-\nu-2},$$

$$|\omega_{\nu+1} - \omega_\nu| + \lambda_\nu|\omega_{\nu+1} - \omega_\nu|^{\mathcal{L}} \le 6\pi\epsilon_\nu, \qquad \sup_j (j+2)^{1/2}\big(|\Omega_{\nu+1,j} - \Omega_{\nu,j}| + \lambda_\nu|\cdot|^{\mathcal{L}}\big) \le 3\epsilon_\nu.$$

Proof. **Step S8.1: Smallness.** By induction, $q_\nu \le \gamma_0 2^{-\nu} \le \gamma_0$. Then $\eta_\nu = q_\nu^{1/3} \le \gamma_0^{1/3} \le \frac{1}{8}$ and $q_\nu \le c_0^{-3}$, i.e. $\epsilon_\nu = \alpha_\nu\sigma_\nu^t q_\nu \le \alpha_\nu\sigma_\nu^t\eta_\nu^2/c_0$: this is (16). $\lambda_\nu \le \alpha_\nu/M_\nu \le \alpha_\nu/(M_\omega + M_\Omega)$. Theorem 7.16 applies.

Step S8.2: New error. By (17), $\frac{\epsilon_\nu^2}{\alpha_\nu\sigma_\nu^t\eta_\nu^2} = \epsilon_\nu\eta_\nu$, so $|X_{P_{\nu+1}}|^{\lambda_\nu}_{r_{\nu+1}, D_{\nu+1}} \le 2C_2\epsilon_\nu\eta_\nu = \epsilon_{\nu+1}$; as $\lambda_{\nu+1} \le \lambda_\nu$ the same bounds the $\lambda_{\nu+1}$-norm. $q_{\nu+1} = \frac{2C_2\epsilon_\nu\eta_\nu 2^t}{\alpha_{\nu+1}\sigma_\nu^t} \le 2^{t+2}C_2 q_\nu^{4/3} \le 2^{t+2}C_2\gamma_0^{1/3}q_\nu \le q_\nu/4$ (using $\alpha_\nu \le 2\alpha_{\nu+1}$ and $\gamma_0^{1/3} \le 2^{-t-4}C_2^{-1}$).

Step S8.3: Lipschitz constants. By Theorem 7.16(iii) the Lipschitz constants grow by at most $(6\pi+3)\epsilon_\nu/\lambda_\nu = (6\pi+3)q_\nu\sigma_\nu^t M_\nu \le 22\gamma_0 2^{-\nu}M_\nu \le 2^{-\nu-2}M_\nu$, and $M_\nu(1+2^{-\nu-2}) \le M_{\nu+1}$. Tails stay finite.

Step S8.4: Divisors. With $K = K_\nu \ge 1$, $(1+K)(1+K^\tau) \le 4K^{\tau+1}$ and $6\pi\cdot 4K_\nu^{\tau+1}\epsilon_\nu \le 24\pi K_0^{\tau+1}2^{\nu(\tau+1)}\gamma_0 2^{-\nu}\alpha_0\sigma_0^t 2^{-\nu t} = \frac{\alpha_0}{4}2^{-\nu(\tau+4)} \le \alpha_\nu - \alpha_{\nu+1} = \alpha_0 2^{-\nu-2}$. Theorem 7.16(iv) gives (18) at $\nu+1$ on Π_ν for $0<|k|\le K_\nu$ (with the Dirichlet weight); for $|k|>K_\nu$ it holds on $\Pi_{\nu+1}$ by construction.

Step S8.5: Transformations. Theorem 7.16(v); $16M_{F,\nu}/\sigma_\nu = 16c_F q_\nu \le 16c_F\gamma_0 2^{-\nu} \le 2^{-\nu-2}$. □

Source: P96-K Iterative Lemma and its proof, pp. 132–133, and Estimates (18), p. 133. Deviations: explicit constants; the $k=0$ conditions are tracked separately (Lemma 8.3); the fields gain $p_+ - p = \frac12$ (P96-K §6, p. 141).

Lemma 8.3 (The conditions with $k=0$)**.** *In the setting of Lemma 8.2, suppose $|\langle l,\Omega_0\rangle| \ge 2\beta_0\langle l\rangle$ for $l\in\Lambda$ on Π_0, with $\beta_0\ge\alpha_0$ and $8\epsilon_0\le\beta_0$. Then for every ν, (18) with $k=0$ holds on Π_ν, and $\sup_j (j+2)^{1/2}|\Omega_{\nu,j}-\Omega_{0,j}| \le \sum_{\mu<\nu}3\epsilon_\mu \le 4\epsilon_0$.*

Proof. $\epsilon_{\mu+1}\le\epsilon_\mu/4$ (from $q_{\mu+1}\le q_\mu/4$ and $\alpha_{\mu+1}\sigma_{\mu+1}^t \le \alpha_\mu\sigma_\mu^t$), so $\sum_\mu 3\epsilon_\mu \le 4\epsilon_0$. For $|l|\le 2$, $|\langle l,\Omega_\nu-\Omega_0\rangle| \le 8\epsilon_0 \le \beta_0 \le \beta_0\langle l\rangle$, so $|\langle l,\Omega_\nu\rangle| \ge \beta_0\langle l\rangle \ge \alpha_\nu\langle l\rangle/A_0$. □

Source: ours; P96-K p. 137 ("$\mathcal{R}'_k(\alpha)$ is empty for $k=0$") and W7 Lemma 9.12.

8.2. **Convergence.**

Proposition 8.4 (Iteration and convergence; P96-K Theorem A, proof)**.** *Let the data of Definition 8.1 be given with $\epsilon_0 \le \gamma_0\alpha_0\sigma_0^t$, $p_+ = p+\frac12$, a set $\Pi_{-1}\subset\mathbb{R}^2$, frequencies (ω_0,Ω_0) on Π_{-1} satisfying (14) with $M_\omega + M_\Omega \le M_0$, the $k=0$ bound of Lemma 8.3 on Π_{-1}, and a real periodic family $P_0\in\mathcal{S}_{p_+}(D_0)$ on Π_{-1} with $|X_{P_0}|^{\lambda_0}_{r_0,D_0}\le\epsilon_0$. Let $\Pi_0 = \Pi_{-1}\setminus\bigcup_{(k,l)\in Z,\,k\ne0}\mathcal{R}^0_{kl}$ with $\mathcal{R}^0_{kl}$ built from (ω_0,Ω_0) at level α_0 on Π_{-1}, define Π_ν by Lemma 8.2, $\Pi_* = \bigcap_\nu\Pi_\nu$, and $\Phi^\nu = \Phi_1\circ\cdots\circ\Phi_\nu : D_\nu\to D_0$ (for $\xi\in\Pi_{\nu-1}$; $\Phi^0=\mathrm{id}$). Then:*

- *(i) (Parameters.) $\Pi_{-1}\setminus\Pi_* = \bigcup_{\nu\ge0}\bigcup_{(k,l)\in Z,\,|k|>K_{\nu-1}}\mathcal{R}^\nu_{kl}$, with $\mathcal{R}^\nu_{kl}\subset\Pi_{\nu-1}$ the zone of (ω_ν,Ω_ν) at level α_ν. If Π_{-1} is Borel and the $\omega_\nu,\Omega_{\nu,j}$ are continuous, every Π_ν and Π_* are Borel.*
- *(ii) (Frequencies.) $\omega_\nu\to\omega_*$, $\Omega_\nu\to\Omega_*$ uniformly on Π_*, with $|\omega_*-\omega_0|\le 8\pi\epsilon_0$, $|\omega_*-\omega_0|^{\mathcal{L}} \le 64\pi\epsilon_0/\lambda_0$, $\sup_j(j+2)^{1/2}|\Omega_{*,j}-\Omega_{0,j}|\le 4\epsilon_0$ and the Lipschitz constant of the latter $\le 16\epsilon_0/\lambda_0$; the same bounds hold for every (ω_ν,Ω_ν) on $\Pi_{\nu-1}$. For $\xi\in\Pi_*$ and all $(k,l)\in Z$, $|\langle k,\omega_*\rangle + \langle l,\Omega_*\rangle| \ge \frac{\alpha_0}{2}\langle l\rangle/A_k$.*
- *(iii) (Transformations on the torus.) Let $\iota(\theta)=(\theta,0,0,0)$ and $T_{s'}=\{|\mathrm{Im}\,\theta|<s'\}$. For $\xi\in\Pi_*$ the maps $\theta\mapsto\Phi^\nu(\iota\theta)$ are holomorphic on T_{s_∞}, periodic and real in the sense below, and converge uniformly on $T_{s_\infty}\times\Pi_*$ to Φ_*, holomorphic on T_{s_∞}, periodic and real, jointly continuous in (θ,ξ), with values in $\overline{D(s_0-4\sigma_0,2\eta_0 r_0)}\subset D_0$ and*

$$|\Phi_*(\theta)-\iota\theta|_{p_+,r_0} \le e_\Phi := 5c_F\frac{\epsilon_0}{\alpha_0\sigma_0^{t-1}} \qquad (\theta\in T_{s_\infty}).$$

$\partial_\theta\Phi^\nu\circ\iota\to\partial_\theta\Phi_*$ uniformly on $T_{s_0/2}$. Here a map $\Phi: T_{s'}\to\mathcal{P}$ is called periodic *if its error $E=\Phi-\iota$ is 1-periodic in each θ_i, i.e. $\Phi(\theta+k)=\Phi(\theta)+\tau_k$ ($\tau_k=(k,0,0,0)$, $k\in\mathbb{Z}^2$); and* real *if $\Phi(\theta)\in\mathcal{P}_\mathbb{R}$ for $\theta\in\mathbb{R}^2$ (real angle, action and normal components). Writing $\Phi_*(\theta)=(\theta+X(\theta),y(\theta),u(\theta),v(\theta))$, the angle error X and y,u,v are 1-periodic and real on $\mathbb{R}^2$.*

- *(iv) (Operator bounds.) For $w\in D_\nu$: $|D\Phi^\nu(w)W|_{q,r_0}\le 2|W|_{q,r_\nu}$ for $q\in\{p,p_+\}$; $D\Phi^\nu(w)$ is symplectic and maps $\mathcal{P}^{p_+}$ onto $\mathcal{P}^{p_+}$.*

Proof. **Step S8.6: Induction.** At $\nu = 0$, (18) holds on Π_0: for $k \neq 0$ by the removal, for $k = 0$ by Lemma 8.3. Lemma 8.2 with Lemma 8.3 gives all stages; $\epsilon_\nu \leq 4^{-\nu}\epsilon_0$.

Step S8.7: Parameters. (i) is the recursive definition (with $K_{-1} = 0$, so at $\nu = 0$ all $k \neq 0$). Borel: each zone is relatively open in a Borel set (W7 (W7-E4)), countable unions.

Step S8.8: Frequencies. Sum the increments of Lemma 8.2: $\sum 6\pi\epsilon_\nu \leq 8\pi\epsilon_0$, $\sum 6\pi\epsilon_\nu/\lambda_\nu \leq 24\pi \cdot \frac{4}{3}\epsilon_0 \cdot \frac{2}{\lambda_0}$ (as $\lambda_\nu \geq \lambda_0/4$); the Ω sums likewise with 3 in place of 6π. The divisor bound passes to the limit since $\alpha_\nu \downarrow \alpha_0/2$ and the left side is a finite sum of uniformly convergent terms.

Step S8.9: Domains. $\iota(T_{s_\infty}) \subset D_\nu$ for all ν; Theorem 7.16 gives $\Phi_{\nu+1}(D_{\nu+1}) \subset D(s_\nu - 4\sigma_\nu, 2\eta_\nu r_\nu) \subset D_\nu$, so Φ^ν is defined and holomorphic on D_ν for $\xi \in \Pi_{\nu-1}$; it is real and satisfies $\Phi^\nu(w + \tau_k) = \Phi^\nu(w) + \tau_k$ (Theorem 7.16(i) for each factor), so $\theta \mapsto \Phi^\nu(\iota\theta)$ is periodic and real in the sense of (iii).

Step S8.10: Operator bounds. For $W \in \mathcal{P}^q$: $|D\Phi_\mu W|_{q,r_{\mu-1}} \leq (1 + 2^{-\mu-1})|W|_{q,r_{\mu-1}} \leq (1 + 2^{-\mu-1})|W|_{q,r_\mu}$ by Lemma 8.2 and Lemma 7.2; multiply along the chain (each factor evaluated at the appropriate point of D_μ): $\prod_{\mu\geq 1}(1 + 2^{-\mu-1}) \leq e^{1/2} < 2$. Symplecticity and surjectivity on $\mathcal{P}^{p+}$ compose (Theorem 7.16(i)).

Step S8.11: Cauchy sequence. For $w = \iota\theta$: $\Phi^{\nu+1}(w) - \Phi^\nu(w) = \int_0^1 D\Phi^\nu(w_\tau)(\Phi_{\nu+1}w - w)\,d\tau$, $w_\tau = w + \tau(\Phi_{\nu+1}w - w) \in D_\nu$ (convex), so $|\Phi^{\nu+1}(w) - \Phi^\nu(w)|_{p+,r_0} \leq 2 \cdot 2M_{F,\nu} = 4c_F q_\nu \sigma_\nu$, and $\sum_\nu 4c_F q_\nu \sigma_\nu \leq 4c_F q_0 \sigma_0 \sum_\nu 8^{-\nu} \leq 5c_F q_0 \sigma_0 = e_\Phi$.

Step S8.12: Limit. The errors $\Phi^\nu \circ \iota - \iota$ are holomorphic, 1-periodic and real on $\mathbb{R}^2$; their uniform limit $\Phi_* - \iota$ is holomorphic (Lemma 3.27), 1-periodic and real on $\mathbb{R}^2$ ($\mathcal{P}_\mathbb{R}$ is closed); derivatives converge uniformly on $T_{s_0/2}$ (balls of radius $s_0/8$ with doubles inside T_{s_∞}, max norm in θ). The image bound: $\Phi^\nu(\iota\theta) = \Phi_1(\cdot) \in D(s_0 - 4\sigma_0, 2\eta_0 r_0)$ for $\nu \geq 1$.

Step S8.13: Continuity in ξ. Each $\Phi_{\nu+1}(w, \xi)$ is Lipschitz in ξ uniformly in w (Lemma 7.11(i)) and continuous in w, hence jointly continuous; compositions and uniform limits of jointly continuous maps are jointly continuous. □

Source: P96-K proof of Theorem A, pp. 134–135 ("the Φ^ν converge uniformly on $\bigcap D_\nu \times \Pi_\nu = D(s/2) \times \Pi_\alpha$", "$|D\Phi^\nu|_{r_0,r_\nu,D_\nu} \leq \prod(1 + 2^{-\mu-2}) \leq 2$", "$|\langle k, \omega_*\rangle + \langle l, \Omega_*\rangle| \geq \frac{\alpha}{2}\frac{\langle l\rangle_d}{A_k}$", p. 135). Deviations: only continuity (not Lipschitz continuity) of Φ_* in ξ is proved, which is what W8 needs (the Lipschitz family is a declared weakening, Remark 11.7); P96-K's Lipschitz extension (Lemma A.2) is not used (W7 works on the domains $\Pi_{\nu-1}$); operator norms as in Remark 7.17.

8.3. **Invariance.**

Lemma 8.5 (Strong conjugacy at regular points)**.** *Let $\Phi : O \to \mathcal{P}$ be C^1 on an open $O \subset \mathcal{P}$ with $D\Phi(w)$ symplectic on $\mathcal{P}$ and onto $\mathcal{P}^{p+}$ from $\mathcal{P}^{p+}$, and let H, H_+ be functions with fields $X_H : \Phi(O) \to \mathcal{P}^{p-1}$, $X_{H_+} : O \to \mathcal{P}^{p-1}$ such that $\varpi_\mathcal{P}(X_H(\Phi w), D\Phi(w)W) = \varpi_\mathcal{P}(X_{H_+}(w), W)$ for all $W \in \mathcal{P}^{p+}$. If $X_{H_+}(w) \in \mathcal{P}$, then $X_H(\Phi w) = D\Phi(w)X_{H_+}(w)$.*

Proof. With $A = X_{H_+}(w) \in \mathcal{P}$: $\varpi_\mathcal{P}(A, W) = \varpi_\mathcal{P}(D\Phi A, D\Phi W)$, so $\varpi_\mathcal{P}(X_H(\Phi w) - D\Phi A, W') = 0$ for all $W' \in \mathcal{P}^{p+}$. Test with the unit vectors (they lie in $\mathcal{P}^{p+}$; Step S7.2). □

Source: ours.

Proposition 8.6 (Invariant tori in KAM coordinates)**.** *In the setting of Proposition 8.4, let $H_0 = N_0 + P_0$ and $\nu_* = \omega_*/(2\pi)$. For $\xi \in \Pi_*$ and $\theta \in T_{s_0/2}$,*

$$X_{H_0}\big(\Phi_*(\theta)\big) = \partial_\theta\Phi_*(\theta)\,\nu_* := \sum_{i=1,2} \nu_{*,i}\,\partial_{\theta_i}\Phi_*(\theta) \qquad \text{in } \mathcal{P}^{p-1},$$

and both sides lie in $\mathcal{P}$; here $X_{N_0}(w) = (\frac{\omega_0}{2\pi}, 0, \Omega_0 v, -\Omega_0 u)$.

Proof. **Step S8.14: Weak chain.** Compose the function identities $H_\mu \circ \Phi_{\mu+1} = H_{\mu+1} + c_\mu$ on $D_{\mu+1}$ (Theorem 7.16(i); $c_\mu = \hat{a}_0$ constant in w; $H_\mu = N_\mu + P_\mu$; $\Phi_{\mu+1}(D_{\mu+1}) \subset D_\mu$) to get $H_0 \circ \Phi^\nu = H_\nu + \sum_{\mu<\nu} c_\mu$ on D_ν. Differentiate this scalar identity at $w \in D_\nu$ in a direction $W \in \mathcal{P}^{p+}$. The left side is $DH_0(\Phi^\nu w)\, D\Phi^\nu(w)W$ with $D\Phi^\nu(w)W \in \mathcal{P}^{p+}$ (Proposition 8.4(iv)); by the derivative formulas of Step S7.40 ($DN_0(w')W'' = \varpi_{\mathcal{P}}(X_{N_0}(w'), W'')$ for $w' \in \mathcal{P}$, $W'' \in \mathcal{P}^{p+}$, and $DP_0 W'' = \varpi_{\mathcal{P}}(X_{P_0}, W'')$) it equals $\varpi_{\mathcal{P}}(X_{H_0}(\Phi^\nu w), D\Phi^\nu(w)W)$. The right side is $\varpi_{\mathcal{P}}(X_{H_\nu}(w), W)$ by the same formulas. No extension of $D\Phi^\nu$ to $\mathcal{P}^{p-1}$ is used.

Step S8.15: Torus points. At $w = \iota\theta$, $X_{H_\nu}(w) = \iota'(\omega_\nu/2\pi) + X_{P_\nu}(w) \in \mathcal{P}$ ($\iota'(c) = (c,0,0,0)$). Lemma 8.5 with Proposition 8.4(iv): $X_{H_0}(\Phi^\nu \iota\theta) = \partial_\theta(\Phi^\nu \circ \iota)(\theta)\frac{\omega_\nu}{2\pi} + D\Phi^\nu(\iota\theta)X_{P_\nu}(\iota\theta)$.

Step S8.16: Limit. $|D\Phi^\nu X_{P_\nu}|_{p,r_0} \le 2\epsilon_\nu \to 0$; $\partial_\theta(\Phi^\nu \circ \iota) \to \partial_\theta \Phi_*$ and $\omega_\nu \to \omega_*$; $\Phi^\nu \iota\theta \to \Phi_*(\theta)$ in $\mathcal{P}$, and X_{H_0} is continuous $D_0 \to \mathcal{P}^{p-1}$ (X_{N_0} is bounded linear $\mathcal{P} \to \mathcal{P}^{p-1}$ plus a constant by Lemma 6.16(i), X_{P_0} is continuous into $\mathcal{P}^{p+}$). Pass to the limit in $\mathcal{P}^{p-1}$. The right side lies in $\mathcal{P}$. □

Source: P96-K p. 134 ("$|X_H \circ \Phi^\nu - D\Phi^\nu \cdot X_{N_\nu}| \ll |X_{P_\nu}|$, whence in the limit $X_H \circ \Phi = D\Phi \cdot X_{\omega_*}$"). SOURCE_GAP (supplied): the source's estimate uses $X_H \circ \Phi^\nu$ in the norm of $\mathcal{P}$, which is not defined since X_N is unbounded; the identity holds in $\mathcal{P}^{p-1}$ and is obtained from the weak conjugacy at torus points.

8.4. The preparatory step and the interface to W7.

Definition 8.7 (W6 constants)**.** In this order, after W7's constants (Definition 9.6) and W5's: $s = \frac{1}{8}$, $\sigma_P = s/40$, $s_0 = s - 5\sigma_P = \frac{7}{8}s$, $\sigma_0 = s_0/40$; $M = M_\omega + M_\Omega$ of Lemma 6.6(i) and $M_0 = 2M$; γ_0 of Definition 8.1;

$$\gamma_1 = \min\Big\{\gamma_0\sigma_0^t,\ \frac{\theta_0}{425(1+M)},\ \frac{\beta_*}{8}\Big\}, \qquad \gamma = \min\Big\{\frac{1}{40},\ (c_0^\natural)^{-3}\sigma_P^{t^\natural},\ \frac{\theta_0}{425(1+M)},\ \Big(\frac{\gamma_1\sigma_P^{t^\natural/3}}{2C_2^\natural}\Big)^{3/4}\Big\}.$$

All depend on m only (through M, θ_0, β_*).

Lemma 8.8 (Preparatory step; P96-K §6)**.** *Let $\Pi \subset \{|\xi|_\infty \le \frac{1}{2}\}$ be Borel, $0 < \alpha \le \alpha_0$ (W7 (C4)), $\hat{\alpha} = \alpha^{44/47}$, $\check{\alpha} = \alpha^{48/47}$, and Ξ W7's preparatory set (Definition 9.8). Let $r > 0$ and a real periodic family $P \in \mathcal{S}_{p+1}(D(s,r))$ on $\Pi \setminus \Xi$, with the unperturbed frequencies (ω^0, Ω^0), and a number ϵ (a chosen positive majorant) satisfy*

$$0 < \epsilon \le \gamma\alpha, \qquad |X_P|^\lambda_{r,D(s,r)} \le \epsilon, \qquad \lambda = \alpha/M$$

(e.g. Proposition 7.19, whose $\epsilon_{\mathrm{in}}(1 + 4\lambda/\rho_\xi)$ is positive). Put $\eta_P = (\epsilon/(\hat{\alpha}\sigma_P^{t^\natural}))^{1/3}$, $r_0 = \eta_P r$. Then one step of Theorem 7.16 in the modified case ($\beta = \frac{1}{4}$, $p_+ = p + \frac{1}{2}$, level $\hat{\alpha}$) applies on $\Pi \setminus \Xi$ and gives a real holomorphic family $\Phi_P : D_0 = D(s_0, r_0) \to D(s, r)$, symplectic with $D\Phi_P$ onto $\mathcal{P}^{p+}$, with $(N+P) \circ \Phi_P = N_0 + P_0 + \mathrm{const}$, the weak conjugacy of Lemma 7.13(iii), and:

- *(i) the divisor hypotheses hold: for $\xi \in \Pi \setminus \Xi$ and $(k,l) \in Z$, $|\langle k, \omega^0\rangle + \langle l, \Omega^0\rangle| \ge \hat{\alpha}\hat{w}(l)/A_k$ (since $\xi \notin S_{kl}$, Definition 9.8(i)), which is (15) with $\beta = \frac{1}{4}$ for $l = \pm(e_I - e_J)$ and ($\mathrm{H}_{\hat{\alpha}}$) otherwise (Dirichlet weights dominate P96-K's, Lemma 6.6(ii));*
- *(ii) the shift: $\omega_0 = \omega^0 + \hat{\omega}_P$, $\Omega_{0,J} = \Omega^0_J + \hat{\Omega}_{P,J}$ with $|\hat{\omega}_P| + \lambda|\hat{\omega}_P|^{\mathcal{L}} \le 6\pi\epsilon$ and $J\big(|\hat{\Omega}_{P,J}| + \lambda|\hat{\Omega}_{P,J}|^{\mathcal{L}}\big) \le 3\epsilon$ (the $\varepsilon_1 j^{-1}$ part of W7's (H2), $\bar{p} - p = 1$); (ω_0, Ω_0) satisfy (14) with $M_\omega + M_\Omega \le M_0 = 2M$;*
- *(iii) $|\Phi_P - \mathrm{id}|_{p_+,r,D_0} \le M_{F,P} := c_F^\natural \epsilon/(\hat{\alpha}\sigma_P^{t^\natural - 1})$, and Φ_P is Lipschitz in ξ in the norm $|\cdot|_{p,r}$;*
- *(iv) the new perturbation satisfies $\sup|X_{P_0}|_{p_+,r_0,D_0} \le \epsilon_0^\flat := 2C_2^\natural \epsilon^{4/3}(\hat{\alpha}\sigma_P^{t^\natural})^{-1/3} \le \gamma_1\check{\alpha}$;*
- *(v) (sharp size, Lipschitz part included) with $\lambda_0 = \check{\alpha}/M_0$: $|X_{P_0}|^{\lambda_0}_{r_0,D_0} \le \epsilon_0^\flat = 2C_2^\natural \epsilon^{4/3}(\hat{\alpha}\sigma_P^{t^\natural})^{-1/3} \le \gamma_1\check{\alpha}$.*

Proof. **Step S8.17: Data.** $\sigma_P \le \min(\frac14, s/6)$; $q_P = \epsilon/(\hat\alpha\sigma_P^{t^\natural}) \le \gamma\alpha/(\hat\alpha\sigma_P^{t^\natural}) \le \gamma/\sigma_P^{t^\natural} \le (c_0^\natural)^{-3}$ ($\alpha \le \hat\alpha$), so (16) holds with $\eta_P \le \frac18$; $\lambda = \alpha/M \le \hat\alpha/M$. The frequencies satisfy (13) on Π (Lemma 6.6).

Step S8.18: Divisors. (i): $\Xi = \bigcup_{(k,l)\in Z} S_{kl}$, so for $\xi \in \Pi \setminus \Xi$ and every $(k,l) \in Z$ the defining inequality of S_{kl} fails, which is the displayed bound; this uses only the unperturbed frequencies and Definition 9.8(i), no admissible family. For W7's Dirichlet weights: for $l = \pm(e_I - e_J)$, $\hat w = |I-J|\min(I,J)^{-1/4}$, which is (15) with $\beta = \frac14$ (W5 uses $\min(i,j)+2 = \min(I,J)$); for other l, $\hat w = \langle l\rangle \ge$ P96-K's weight. $p+1-p_+ = \frac12 \ge 2\beta$.

Step S8.19: Step. Theorem 7.16 in the modified case, with both parts of Lemma 7.6 ($2\beta \le p+1-p_+$). The frequency shifts are $\hat\omega = [b]$, $\hat\Omega_j = [Q]_{jj}$ of R, which is the Taylor polynomial of P with $p+1$ smoothing, so their λ-bounds use only Lemma 7.8 and Theorem 6.19(iv) with $\bar p - p = 1$; no divisor enters. M_0: the Lipschitz constants grow by $(6\pi+3)\epsilon/\lambda \le 22\gamma M \le M$.

Step S8.20: Transformation. (iii): the sup bound is the sharper estimate of Step S7.25 of Lemma 7.11, $|\phi^1 - \mathrm{id}|_{p_+,r} \le M_F$ on $U \supset D_0$ (not the λ-bound $2M_F$ of Theorem 7.16(v)), with $M_F = M_{F,P}$ in the modified case (Definition 7.9); the Lipschitz property is the Lipschitz part of Theorem 7.16(v), and $|\cdot|_{p,r} \le |\cdot|_{p_+,r}$.

Step S8.21: Size. By (17) (modified case: $C_2^\natural$, $t^\natural$) with $\eta_P^3 = q_P$: $\sup|X_{P_0}| \le 2C_2^\natural\epsilon\eta_P = 2C_2^\natural\epsilon^{4/3}(\hat\alpha\sigma_P^{t^\natural})^{-1/3}$, and with $\epsilon \le \gamma\alpha$: $\le 2C_2^\natural\gamma^{4/3}\sigma_P^{-t^\natural/3}\alpha^{4/3}\hat\alpha^{-1/3} = 2C_2^\natural\gamma^{4/3}\sigma_P^{-t^\natural/3}\check\alpha \le \gamma_1\check\alpha$, since $\frac43 - \frac13\cdot\frac{44}{47} = \frac{48}{47}$.

Step S8.22: Lipschitz part. By Lemma 7.6(ii) ($2\beta = \frac12 \le \varpi = \frac12$), (17) holds for the full λ-norm with $\lambda = \alpha/M$; as in Step S8.21, $|X_{P_0}|^\lambda_{r_0,D_0} \le 2C_2^\natural\epsilon\eta_P = \epsilon_0^\flat \le \gamma_1\check\alpha$, and $\lambda_0 = \check\alpha/(2M) \le \lambda$ gives (v). □

Source: P96-K §6, pp. 140–141 ("we precede the KAM iteration by one modified KAM step … $\hat\alpha = \alpha^{1-3w}$", "$|X_{P_0}|^\lambda_{\tilde p,\eta r} \le \frac{1}{(\hat\alpha\sigma^t)^{\varkappa-1}}(|X_P|^\lambda_{\tilde p,r})^\varkappa$" — printed with κ, our $\varkappa = \frac43$ — "$|X_{P_0}|^\lambda_{\tilde p,\eta r} \le \gamma\check\alpha$, $\check\alpha = \alpha^{1+w}$"). With W7's $w = \frac{1}{47}$ (weight exponent $\beta = \frac14$ separated from the loss $\varpi = \frac12$, Remark 7.7): $1-3w = \frac{44}{47}$, $1+w = \frac{48}{47}$; P96-K's own choice $w = \varpi/(4\kappa+4-\varpi)$ with $\beta = \varpi$ is not used. P96-K p. 141: "the KAM scheme now starts from scratch, with parameters $\check\alpha$ and $\check p$".

Definition 8.9 (The wave iteration)**.** In the setting of Lemma 8.8 with (v), apply Proposition 8.4 with s_0, σ_0 of Definition 8.7, $r_0 = \eta_P r$, $\alpha_0^{\mathrm{it}} = \check\alpha$, $M_0 = 2M$, a starting size $\epsilon_0 \in [\epsilon_0^\flat, \gamma_1\check\alpha]$ (a parameter; $\epsilon_0^\flat$ of Lemma 8.8(iv), which is > 0 because $\epsilon > 0$; so $\epsilon_0 > 0$, and $\eta_P > 0$, $r_0 = \eta_P r > 0$), $\Pi_{-1} = \Pi\setminus\Xi$, (ω_0, Ω_0), P_0, and $\beta_0 = \beta_*$. Write Π_*, ω_*, Ω_*, Φ_*, (ω_ν, Ω_ν) on $\Pi_{\nu-1}$, and $\mathcal{R}^\nu_{kl}$ for the outputs.

Lemma 8.10 (Applicability)**.** *The hypotheses of Proposition 8.4 hold for Definition 8.9.*

Proof. $\epsilon_0 \le \gamma_1\check\alpha \le \gamma_0\sigma_0^t\alpha_0^{\mathrm{it}}$; $|X_{P_0}|^{\lambda_0} \le \epsilon_0^\flat \le \epsilon_0$ by (iv)–(v); $k=0$: by W7 Lemma 9.5(ii), $|\langle l, \Omega^0\rangle| \ge 3\beta_*\langle l\rangle$, and the shift gives $|\langle l, \hat\Omega_P\rangle| \le 2\cdot 3\epsilon/3 \le 2\gamma\alpha \le \beta_*$ (W7 (C4): $\alpha \le \alpha_0 \le \beta_*/4$), so $|\langle l, \Omega_0\rangle| \ge 2\beta_*\langle l\rangle$; $\beta_* \ge \check\alpha$ (W7 (C4)) and $8\epsilon_0 \le 8\gamma_1 \le \beta_*$. □

Source: ours.

Lemma 8.11 (Lipschitz frequencies; W7's (H1)–(H2))**.** *Work in Definition 8.9 and put $D_\nu = \Pi_{\nu-1}$ (Borel, $\subset \Pi$), with $\Omega_{\nu,J}$ at the Dirichlet label $J \ge 3$ and K_ν of Definition 8.1. Then the family $(D_\nu, \omega_\nu, \Omega_\nu, K_\nu)_{\nu\ge0}$ is admissible for (Π, α) in the sense of Definition 9.9, with*

$$\varepsilon_\omega = 6\pi\gamma\alpha + 8\pi\epsilon_0, \quad \varepsilon_1 = 3\gamma\alpha, \quad \varepsilon_2 = 4\epsilon_0,$$
$$\theta = 425(1+M)\max(\gamma,\gamma_1) \le \theta_0,$$

where the three constants $\varepsilon_\omega, \varepsilon_1, \varepsilon_2$ are $\le \alpha$, and $\theta \le \theta_0$; θ depends on m only and is fixed before α (Definition 8.7). Lipschitz constants are with respect to $|\cdot|_\infty$ on $\mathbb{R}^2$, as in W7.

Proof. **Step S8.23: Sizes.** $\omega_\nu - \omega^0 = \hat\omega_P + (\omega_\nu - \omega_0)$: $\le 6\pi\epsilon + 8\pi\epsilon_0$ (Lemma 8.8(ii), Proposition 8.4(ii)); $|\Omega_{\nu,J} - \Omega^0_J| \le 3\epsilon/J + 4\epsilon_0/J^{1/2}$. With $\epsilon \le \gamma\alpha$, $\epsilon_0 \le \gamma_1\check\alpha$, $\check\alpha \le \alpha$, $6\pi\gamma \le \frac12$ and $8\pi\gamma_1 \le \frac12$: all $\le \alpha$.

Step S8.24: Lipschitz. $|\hat\omega_P|^{\mathcal L} \le 6\pi\epsilon/\lambda \le 6\pi\gamma M$; $|\omega_\nu - \omega_0|^{\mathcal L} \le 64\pi\epsilon_0/\lambda_0 = 128\pi M\epsilon_0/\check\alpha \le 128\pi\gamma_1 M$; $J|\hat\Omega_{P,J}|^{\mathcal L} \le 3\gamma M$, so $J^{1/2}$ times it is $\le 3\gamma M$; $(J)^{1/2}|\Omega_{\nu,J} - \Omega_{0,J}|^{\mathcal L} \le 16\epsilon_0/\lambda_0 \le 32\gamma_1 M$. The totals are $\le (6\pi + 128\pi)\max(\gamma,\gamma_1)M < 425\max(\gamma,\gamma_1)M$ for ω and $\le 35\max(\gamma,\gamma_1)M$ for Ω, using $J^{-1} \le J^{-1/2}$; and $425(1+M)\max(\gamma,\gamma_1) \le \theta_0$ by Definition 8.7.

Step S8.25: Borel. $\Pi \setminus \Xi$ is Borel directly: each $S_{kl} = \{\xi \in \Pi : |\phi_{kl}(\xi)| < \hat\alpha\hat w(l)/A_k\}$ with $\phi_{kl} = \langle k, \omega^0\rangle + \langle l, \Omega^0\rangle$ affine in ξ (finitely many j with $l_j \ne 0$), hence $S_{kl} = \Pi \cap U_{kl}$ with U_{kl} open, and Z is countable ((W7-E4)); no admissible family is used. Then Proposition 8.4(i) gives the Borel sets $D_\nu = \Pi_{\nu-1}$. □

Source: P96-K Theorem A, p. 122 ("$|\omega_\nu - \omega| + \frac{\alpha}{M}|\omega_\nu - \omega|^{\mathcal L} \le c\epsilon$"), Corollary C, p. 123 (smaller γ); W7 Remark 9.10.

Lemma 8.12 (Exclusion)**.** *In Definition 8.9, $\Pi \setminus \Pi_* \subset \mathcal E$, W7's excluded set for the admissible family of Lemma 8.11; the thresholds are $K_\nu \ge 2^\nu$ and the levels $\alpha_\nu \le \check\alpha$.*

Proof. $\Pi \setminus \Pi_* = \Xi \cup \big((\Pi \setminus \Xi) \setminus \Pi_*\big)$, and by Proposition 8.4(i) the second set is the union of zones $\mathcal R^\nu_{kl}$ at level $\alpha_\nu \le \alpha_0^{\mathrm{it}} = \check\alpha$, $|k| > K_{\nu-1}$, built from (ω_ν, Ω_ν) on $D_\nu = \Pi_{\nu-1}$ with the Dirichlet weight; each is contained in W7's $R^\nu_{kl} = R_{kl}(\check\alpha)$. □

Source: P96-K Theorem A, p. 122 ("$\Pi \setminus \Pi_\alpha \subset \bigcup \mathcal R^\nu_{kl}(\alpha)$", $|k| > K_0 2^{\nu-1}$) and p. 142 ("the KAM iteration now starts with the parameter set $\Pi_0 = \Pi \setminus \Xi_{\hat\alpha}$ and parameter $\check\alpha$").

Lemma 8.13 (Final frequencies)**.** *For $\xi \in \Pi_*$: $|\langle k, \omega_*(\xi)\rangle| \ge \check\alpha/(2A_k)$ for $k \ne 0$, and $|\omega_*(\xi) - \omega^0(\xi)|_\infty \le \varepsilon_\omega$. Moreover ω_* is Lipschitz on Π_* (constant $\le N_A + \theta$).*

Proof. Proposition 8.4(ii) with $l = 0$ and $\alpha_0^{\mathrm{it}} = \check\alpha$; the size and Lipschitz bounds of Lemma 8.11 pass to the limit; ω^0 has Lipschitz constant N_A (Lemma 9.4(ii)). □

Source: P96-K p. 135 ("$|\langle k, \omega_*\rangle + \langle l, \Omega_*\rangle| \ge \frac{\alpha}{2}\frac{\langle l\rangle_d}{A_k}$").

8.5. **Output to W8.**

Theorem 8.14 (Invariant tori in Birkhoff coordinates)**.** *In Definition 8.9, with $H_{\mathrm{in}} = N_\xi + P_\xi$ of Lemma 7.18(iv) on $D(s,r)$ (so that Proposition 7.19 supplies P), let for $\xi \in \Pi_*$ and $\theta \in T_{s_0/2}$*

$$\tilde\Psi_\xi(\theta) = \mathcal A_\xi\big(\Phi_P(\Phi_*(\theta))\big) \in \mathfrak W, \qquad \nu_*(\xi) = \omega_*(\xi)/(2\pi).$$

(i) $\tilde\Psi_\xi$ is holomorphic on $T_{s_0/2}$ with values in B_{ρ_1} (if $c_A(t + r^2) < \rho_1^2$, Lemma 7.18(ii)), 1-periodic, and $\varrho\tilde\Psi_\xi(\theta) = \tilde\Psi_\xi(\theta)$ for real θ; its normal part lies in $\mathfrak W^{p+1/2}$.

(ii) $(\xi,\theta) \mapsto \tilde\Psi_\xi(\theta)$ is continuous on $\Pi_* \times \mathbb R^2$, and ν_* is Lipschitz on Π_*.

(iii) (Invariance.) For $\xi \in \Pi_*$, $\theta \in \mathbb R^2$: $\partial_\theta\tilde\Psi_\xi(\theta)\,\nu_* = X_{H'}(\tilde\Psi_\xi(\theta))$ in $\mathfrak W^{p-1}$, $X_{H'} = X_\Lambda + X_{\bar G} + X_{G^{\mathrm{nor}}} + X_K$ (Lemma 7.18(v)).

(iv) (Closeness.) With $e_{\mathrm{tor}} = M_{F,P} + e_\Phi = \frac{c_F^\natural\epsilon}{\hat\alpha\sigma_P^{t^\natural-1}} + \frac{5c_F\epsilon_0}{\check\alpha\sigma_0^{t-1}}$ (e_Φ of Proposition 8.4(iii), with the parameter ϵ_0 of Definition 8.9), $\Phi_P(\Phi_*(\theta)) = (\theta + X(\theta), y(\theta), u(\theta), v(\theta))$ with $|X| \le e_{\mathrm{tor}}$, $|y| \le r^2 e_{\mathrm{tor}}$, $\|u\|_{a,p+1/2} + \|v\|_{a,p+1/2} \le r e_{\mathrm{tor}}$ on $T_{s_0/2}$; hence $\tilde\Psi_\xi(\theta)_{\pm i} = \sqrt{\xi_i + y_i(\theta)}\, e^{\mp 2\pi i(\theta_i + X_i(\theta))}$ and $\|\hat{\tilde\Psi}\| \le r e_{\mathrm{tor}}$, to be compared with the linear torus $\mathcal A_\xi(\iota\theta)$ (Lemma 7.18(i)).

(v) (Real analyticity.) If Γ is W3's map, then $\theta \mapsto \mathcal Z^{-1}\Gamma(\tilde\Psi_\xi(\theta))$ restricted to $\mathbb R^2$ is `AnalyticOnNhd Real` into $E_{a,p}$ (Lean: `WaveSpace` for $a = p = 1$ after the transport $\tau_{1,1}$).

Proof. **Step S8.26: Holomorphy.** Φ_* (Proposition 8.4(iii)) maps $T_{s_0/2}$ into D_0, Φ_P is holomorphic $D_0 \to D(s,r)$, and $\mathcal{A}_\xi$ is holomorphic on $D(s,r)$; reality and periodicity compose. The normal part of $\Phi_P\Phi_*$ is in $\mathcal{P}^{p_+}$ (Proposition 8.4(iii) and $\Phi_P - \mathrm{id} \in \mathcal{P}^{p_+}$).

Step S8.27: Continuity. Proposition 8.4(iii), Lemma 8.8(iii), and joint continuity of $(\xi, w) \mapsto \mathcal{A}_\xi(w)$; ν_* by Lemma 8.13.

Step S8.28: Invariance, prep step. By Proposition 8.6, $A := X_{H_0}(\Phi_*\theta) = \partial_\theta \Phi_* \nu_* \in \mathcal{P}$. Lemma 7.13(iii) for the preparatory step and Lemma 8.5 give $X_{H_{\mathrm{in}}}(\Phi_P\Phi_*\theta) = D\Phi_P\, A = \partial_\theta(\Phi_P \circ \Phi_*)\nu_*$ (chain rule).

Step S8.29: Invariance, Birkhoff. Lemma 7.18(v) at $w = \Phi_P\Phi_*\theta$ and the chain rule.

Step S8.30: Closeness. Lemma 8.8(iii) and Proposition 8.4(iii), with $|\cdot|_{q,r} \le |\cdot|_{q,r_0}$ $(r_0 \le r)$, unpacking the norm $|\cdot|_{p_+,r}$.

Step S8.31: Analyticity. Γ is analytic, hence holomorphic, on B_{ρ_1} (Remark 5.26); so $\mathcal{Z}^{-1}\Gamma\tilde{\Psi}_\xi : T_{s_0/2} \to E^{\mathbb{C}}_{a,p}$ is holomorphic, periodic and real on reals (Lemma 5.3, Theorem 5.21(iv)); Corollary 3.29 (stated for $E^{\mathbb{C}}_{1,1}$, the case $a = p = 1$). □

Source: P96-K Theorem A ("each embedding is real analytic on $|\mathrm{Im}\, x| < s/2$", $|\Phi - \Phi_0| \le c\epsilon/\alpha$), KP96 §7, Steps 4–5 ("$\Phi_r : \mathbb{T}^n \times \Pi_{r,\alpha} \to D(1,r)$", translated embedding); P96-W p. 290. Deviation: the construction gives values in $\mathfrak{W}^{p+1/2}$, not $\mathfrak{W}^{p+1}$ (P96-K's $\mathcal{P}^{a,\bar{p}}$), since after the preparatory step only $\check{p} = \bar{p} - \varpi$ is available; on real θ the invariance (iii) puts the points in $\mathfrak{W}^+$ a posteriori (Corollary 5.29).

Remark 8.15 (What W8 consumes from W6). For each shell (W8 chooses r, t, ρ_ξ, Π and α subject to Proposition 7.19, $\epsilon_{\mathrm{in}}(1 + 4\lambda/\rho_\xi) \le \gamma\alpha$ with $\lambda = \alpha/M$, $c_A(t + r^2) < \rho_1^2$, $\Pi \subset \{|\xi|_\infty \le \frac{1}{2}\}$ Borel, $\alpha \le \alpha_0$):

(a) a Borel set $\Pi_* \subset \Pi$ with $\Pi \setminus \Pi_* \subset \mathcal{E}$ (Lemma 8.12), so W7 Theorem 9.22 and Corollary 9.27 apply with the admissible family of Lemma 8.11;
(b) a Lipschitz frequency map $\nu_* = \omega_*/2\pi$ on Π_* with $|\langle k, \omega_*\rangle| \ge \check{\alpha}/(2A_k)$ and $|\omega_* - \omega^0| \le \varepsilon_\omega \le \alpha$ (Lemma 8.13; W7 Lemmas 9.29, 9.31);
(c) tori $\tilde{\Psi}_\xi$ with (i)–(v) of Theorem 8.14: holomorphic on the strip $T_{s_0/2}$, $s_0/2 = \frac{7}{128}$; real; jointly continuous; invariant in $\mathfrak{W}^{p-1}$; closeness in terms of e_{tor}; real analytic after Γ. The parameter is the action: $\mathcal{Z}^{-1}\mathcal{A}_\xi(\iota\theta) = K^0_\xi(\theta)$.

Starting size. All of (a)–(c) hold for every value of the parameter $\epsilon_0 \in [\epsilon_0^\flat, \gamma_1\check{\alpha}]$ of Definition 8.9: Lemmas 8.10, 8.11, 8.12, 8.13 and Theorem 8.14 use ϵ_0 only through $|X_{P_0}|^{\lambda_0} \le \epsilon_0^\flat \le \epsilon_0$ (Lemma 8.8(v)) and $\epsilon_0 \le \gamma_1\check{\alpha}$, and θ is the same for all of them. With the sharp choice $\epsilon_0 = \epsilon_0^\flat$ the closeness constant tends to zero with ϵ/α: since $\alpha^{4/3}\hat{\alpha}^{-1/3} = \check{\alpha}$,

$$\frac{\epsilon_0^\flat}{\check{\alpha}} = 2C_2^\natural \sigma_P^{-t^\natural/3}\Big(\frac{\epsilon}{\alpha}\Big)^{4/3}, \qquad M_{F,P} = c_F^\natural \sigma_P^{1-t^\natural}\alpha^{3/47}\frac{\epsilon}{\alpha},$$

$$e_{\mathrm{tor}} \le c_F^\natural \sigma_P^{1-t^\natural}\frac{\epsilon}{\alpha} + 10 c_F C_2^\natural \sigma_0^{1-t}\sigma_P^{-t^\natural/3}\Big(\frac{\epsilon}{\alpha}\Big)^{4/3} \qquad (t = 52,\ t^\natural = 53).$$

Return to Cartesian coordinates. Theorem 8.14(i),(iii) (restricted to $\theta \in \mathbb{R}^2$) is exactly the hypothesis of W3's Corollary 5.29: $\tilde{\Psi}_\xi$ maps into B_{ρ_1}, is differentiable into $\mathfrak{W}$, and is invariant in $\mathfrak{W}_{a,p-1}$. That corollary puts the torus points in $\mathfrak{W}^+$ and gives the coefficient system for $\mathcal{Z}^{-1}\Gamma \circ \tilde{\Psi}_\xi$, which is clause (4) (Lemma 3.3). *What W8 must still supply.* (1) Injectivity of $DK_I(\theta)$ and injectivity mod $\mathbb{Z}^2$ from (iv) (Cauchy estimates on the strip turn e_{tor} into C^1 bounds on X, y). (2) The $o(r(I))$ closeness: from (iv), $|\tilde{\Psi}_{\pm i} - \sqrt{\xi_i}e^{\mp 2\pi i\theta_i}| \le |y|/\sqrt{\xi_i} + 2\pi\sqrt{\xi_i}|X|e^{2\pi|X|}$ on real θ, which is $o(\sqrt{|\xi|})$ when $r^2 e_{\mathrm{tor}}/\xi_i \to 0$ and $e_{\mathrm{tor}} \to 0$ along the shells. (3) Patching, family injectivity, and the choice of $\alpha(t)$ (W7 uses $\alpha(t) = t^{95/64}$).

8.6. **Ledger.**
Results. Definitions 8.1, 8.7, 8.9; Lemmas 8.2, 8.3, 8.5, 8.8, 8.10, 8.11, 8.12, 8.13; Propositions 8.4, 8.6; Theorem 8.14; Remark 8.15. W7's forward labels `prep-step`, `freq-lip`, `exclusion`, `final-freq` are provided, with W7's $\beta = \frac{1}{4}$, $\hat{\alpha} = \alpha^{44/47}$, $\check{\alpha} = \alpha^{48/47}$.
Order of constants. $m \to$ W3 (ρ_1, C_*, C'_K), W7 ((C1)–(C5): β_*, θ_0, α_0, …), W4/W5 (C_1, $b_{\mathrm{W}}(24)$, c_F, c_0, C_2, $b^\natural(24)$, $c_F^\natural$, $c_0^\natural$, $C_2^\natural$) $\to s = \frac{1}{8}$, σ_P, s_0, σ_0, M, γ_0, K_0, γ_1, γ (Definition 8.7) $\to$ per shell: Π, α, r, t, ρ_ξ (W8) $\to \epsilon$, η_P, r_0 and the sequences. Every constant before the shell data depends on m only.

9. Milestone W7: measure of the resonance set

This part proves the measure estimate of P96-K Theorem D (p. 124; proof §§5–6, pp. 135–142) in the only case the wave equation needs: two tangential frequencies ($n = 2$), linear growth of the normal frequencies ($d = 1$), spectral exponent $\kappa = 2$, smoothing gain $\bar{p} - p = 1$, and *affine* unperturbed frequencies, namely those of the corrected partial Birkhoff normal form of Part 0 (Theorem 2.10). It then turns the estimate into the two measure-theoretic conclusions of the frozen headline `KAM.Wave.nlw2`: density one of the parameter set in the actions (clause (1)) and Diophantine frequencies with the stated limit (clause (2)).
What is excluded and why the coarse estimate is not enough. The KAM iteration (W5–W6) runs with parameters $\xi \in \mathbb{R}^2$, the tangential actions, and removes at each step the parameters at which a divisor $\langle k, \omega\rangle + \langle l, \Omega\rangle$, $|l| \le 2$, is smaller than $\alpha\langle l\rangle/A_k$. On a box of side t in the actions, the excluded measure is at most $C\,t\,\alpha^\mu$ (P96-K's form $c\rho^{n-1}\alpha^\mu$ with $n = 2$), so the *fraction* excluded is $C\alpha^\mu/t$. The perturbation size forces $\alpha \gtrsim t^{3/2}$ (KP96 p. 27, (8), in the variable t of Remark 9.28). P96-K Theorem B gives $\mu = \kappa/(\kappa+1) = \frac{2}{3}$, so $\alpha^\mu/t \gtrsim 1$: no density. Theorem D precedes the iteration by one step with weaker divisors on the normal differences $\Omega_i - \Omega_j$ (weight $|i-j|\min(i,j)^{-\beta}$), paid for by a loss of ϖ in the smoothing exponent; this raises the exponent to $\tilde{\mu} = \kappa/(\kappa+1-\beta/4)$. P96-K takes $\beta = \varpi$. We take the loss $\varpi = \frac{1}{2}$ but the weight exponent $\beta = \frac{1}{4}$, because the parameter-Lipschitz bound of the preparatory step needs $2\beta \le \varpi$ (W5, Lemma 7.6(ii) and Remark 7.7). Then $\tilde{\mu} = \frac{32}{47} > \frac{2}{3}$, and the fraction tends to zero along suitable $\alpha(t)$.
Route. All estimates are made in the *frequency coordinates* $\zeta = \omega^0(\xi) = \lambda_J + A\xi$, where the tangential part of a divisor is the linear function $\langle k, \zeta\rangle$; the affine change $\xi \mapsto \zeta$ multiplies Lebesgue measure by $|\det A|$ (Lemma 9.4). A single slicing lemma (Lemma 9.11) bounds the measure of every resonance zone along a coordinate line. The zones are grouped as in P96-K: $l = 0$; l with no sign change (Λ^+, finitely many per k after an emptiness lemma); and the differences $l = \pm(e_i - e_j)$ (Λ^-, infinitely many per k), which are grouped by $i - j$ and absorbed for large j by the asymptotics $\Omega_i - \Omega_j = (i-j)(1 + O(j^{-2}))$. Finitely many small (k, l) escape the slicing lemma; for them the nondegeneracy condition (A_3) of Part 0 is used. We do not use P96-K's Lipschitz extension (Lemma A.2): the slicing lemma works on the domain of the frequency maps.
Standing data. Standing assumption 1.1 ($m > 0$) and the conventions of §1 are in force: $\lambda_j = \sqrt{j^2 + m}$, actions $I_j = |z_j|^2$, period-one angles with $\nu = \omega/(2\pi)$ (Convention 1.13), and the corrected A, B of Theorem 2.10 (the printed P96-W values are 16 times too large, E-PDE-W-1, and are not used). Vectors in $\mathbb{R}^2$ carry the max norm $|\cdot|_\infty$, linear maps of $\mathbb{R}^2$ the induced operator norm $\|\cdot\|_\infty$ (maximal absolute row sum), and Leb is Lebesgue measure on $\mathbb{R}^2$ (Lean `volume` on `Fin 2 -> Real`). Integer vectors $k \in \mathbb{Z}^2$ carry $|k| = |k_1| + |k_2|$, the norm of the frozen `KAM.IsDiophantine`. Diameters are diam_∞. The two tangential modes are $J = \{1, 2\}$ (Lean index $i \in$ `Fin 2` is the Dirichlet mode $i + 1$); the normal modes are labelled by their Dirichlet index $j \ge 3$.

9.1. **External facts.**

(W7-E1) *Slicing (Tonelli).* For a Borel set $S \subset \mathbb{R}^2$,
$\mathrm{Leb}(S) = \int_{\mathbb{R}} \mathrm{Leb}_1\{t : (t, y) \in S\}\,dy = \int_{\mathbb{R}} \mathrm{Leb}_1\{t : (y, t) \in S\}\,dy$. (Mathlib:

`MeasureTheory.volume_preserving_finTwoArrow` to pass from `Fin 2 -> Real` to $\mathbb{R}\times\mathbb{R}$, `MeasureTheory.Measure.prod_apply`, `MeasureTheory.Measure.prod_apply_symm`, `MeasureTheory.lintegral_mono`, `MeasureTheory.setLIntegral_const`.)

(W7-E2) *Sets of small diameter on the line.* A set $S\subset\mathbb{R}$ with $|t-t'|\le d$ for all $t,t'\in S$ satisfies $\mathrm{Leb}_1(S)\le d$ (it lies in $[\inf S,\inf S+d]$; `Real.volume_le_diam`, `Real.volume_Icc`).

(W7-E3) *Lebesgue measure under affine maps.* For an invertible linear T of $\mathbb{R}^2$ and $c\in\mathbb{R}^2$, $\mathrm{Leb}(T(S)+c)=|\det T|\,\mathrm{Leb}(S)$. (`MeasureTheory.Measure.addHaar_image_linearMap`, `MeasureTheory.measure_preimage_add`, `LinearMap.det`.)

(W7-E4) *Borel sets.* If $D\subset\mathbb{R}^2$ is Borel and $f:D\to\mathbb{R}$ is continuous on D, then $\{\xi\in D:|f(\xi)|<\eta\}$ is Borel: by `continuousOn_iff'` applied to the open target interval $(-\eta,\eta)$ there is an open $U\subset\mathbb{R}^2$ with $f^{-1}(-\eta,\eta)\cap D=U\cap D$, which is Borel by `IsOpen.measurableSet` and `MeasurableSet.inter` (for f defined on the subtype D, pass first to `ContinuousOn` via `continuousOn_iff_continuous_domRestrict`). No openness of D is needed. Countable unions, intersections and differences of Borel sets are Borel (`MeasurableSet.iUnion`, `MeasurableSet.diff`); $\mathbb{Z}^2$ and the finitely supported l with $|l|\le 2$ form countable index sets (`Encodable`). Measure is countably subadditive and monotone (`MeasureTheory.measure_iUnion_le`, `MeasureTheory.measure_mono`).

(W7-E5) *Lattice sums.* $\#\{k\in\mathbb{Z}^2:|k|=N\}=4N$ for $N\ge 1$, and $\sum_{N\ge1}N^{-2}=\pi^2/6$ (`hasSum_zeta_two`). Hence $Z_3:=\sum_{k\in\mathbb{Z}^2\setminus 0}|k|^{-3}=4\zeta(2)=2\pi^2/3<7$.

(W7-E6) *Max-norm balls.* For $h>0$, `Metric.ball (0 : Fin 2 -> Real) h` is $(-h,h)^2$ (`ball_pi`, `Real.ball_eq_Ioo`); $\mathrm{Leb}\big(\prod_i(a_i,b_i)\big)=\prod_i(b_i-a_i)$ (`Real.volume_pi_Ioo`).

(W7-E7) *Extended reals.* For $0<x\le y<\infty$ in $[0,\infty]$ and $s\le x$, $x/y\le 1$, $(y-s)/y=1-s/y$; a function squeezed between $1-e(h)$ and 1 with $e(h)\to 0$ tends to 1 (`ENNReal.div_le_of_le_mul`, `ENNReal.sub_div`, `tendsto_of_tendsto_of_tendsto_of_le_of_le`). Real powers: $x\mapsto x^\tau$ is monotone on $[0,\infty)$ for $\tau\ge0$ (`Real.rpow_le_rpow`), $x^ax^b=x^{a+b}$ for $x>0$ (`Real.rpow_add`); $\lceil x\rceil\le 2x$ for $x\ge1$ (`Nat.ceil_le_two_mul`).

9.2. **Indices, weights and the reindexing.**

Definition 9.1 (Integer data)**.** (i) $\mathbb{N}_3=\{3,4,\dots\}$; $l\in\mathbb{Z}^{(\mathbb{N}_3)}$ ranges over finitely supported integer vectors on $\mathbb{N}_3$, with $|l|=\sum_j|l_j|$, $\langle l,x\rangle=\sum_j l_jx_j$, and e_j the unit vectors. $\Lambda=\{l:1\le|l|\le2\}$.

(ii) $\Lambda^-=\{\pm(e_i-e_j):i>j\ge3\}$ and $\Lambda^+=\Lambda\setminus\Lambda^-$ (the vectors $\pm e_i$, $\pm2e_i$, $\pm(e_i+e_j)$, $i\ne j$).

(iii) $\langle l\rangle=\max\big(1,|\sum_j j\,l_j|\big)$ (Dirichlet labels). For $l\in\Lambda^+$, $\langle l\rangle=\sum_j j|l_j|\ge3$; for $l=\pm(e_i-e_j)$, $i>j$, $\langle l\rangle=i-j$.

(iv) $|l|_\delta=\sum_j|l_j|j^\delta$ for $\delta\in\mathbb{R}$; for $l\in\Lambda$, $|l|_{-1}\le|l|_{-1/2}\le2$.

(v) $Z=\{(k,l)\in\mathbb{Z}^2\times(\Lambda\cup\{0\}):(k,l)\ne(0,0)\}$.

(vi) The exponents are fixed: $\tau=24$, the smoothing loss $\varpi=\frac12$ of the preparatory step, and the weight exponent $\beta=\frac14$ (W5's β; not to be confused with the constant β_*); $A_k=1+|k|^\tau$ (so $1\le A_k\le 2|k|^\tau$ for $k\ne0$).

(vii) The *modified weight* is $\hat w(l)=(i-j)\,j^{-\beta}=(i-j)\,j^{-1/4}$ for $l=\pm(e_i-e_j)$, $i>j$, and $\hat w(l)=\langle l\rangle$ otherwise; $\hat w(l)\le\langle l\rangle$.

Source: P96-K pp. 121 ($\langle l\rangle_d$, A_k, Z), 135 (Λ), 136 ($|l|_\delta$), 138 ($\Lambda^\pm$), 140 ((23), modified estimate with $\min(i^\varpi,j^\varpi)$). Specialized to $n=2$, $d=1$; the choices $\tau=24$, $\varpi=\frac12$, and the weight exponent $\beta=\frac14\ne\varpi$ are ours (Remark 9.7).

Remark 9.2 (Reindexing $j\mapsto j+2$ and comparison with P96-K's weights). P96-K numbers the normal frequencies $\Omega^{\mathrm K}_{j'}$, $j'\ge1$, and its Assumption B asks $\Omega^{\mathrm K}_{j'}=j'+\dots$. For the wave equation with $J=\{1,2\}$ the normal coordinates are $u_{j'}=q_{j'+2}$, $v_{j'}=p_{j'+2}$ (W3, Remark 5.30; P96-W

p. 291 writes $\Omega_{j-n} = \lambda_j + \dots$ with $n = 2$), so

$$\Omega^{\mathrm{K}}_{j'} = \Omega_{j'+2}, \qquad j' = j - 2.$$

Assumption B holds in the form $\Omega^{\mathrm{K}}_{j'} = j' + 2 + \frac{m}{2(j'+2)} + \dots$; the constant 2 is one of P96-K's "fixed lower order terms" (p. 120). We keep the Dirichlet label j throughout. P96-K's weight is $\langle l\rangle^{\mathrm{K}} = \max(1, |\sum_j (j-2) l_j|)$; for $|l| \le 2$ one has $\langle l\rangle^{\mathrm{K}} = \langle l\rangle$ on Λ^- and $\langle l\rangle^{\mathrm{K}} \le \langle l\rangle \le \langle l\rangle^{\mathrm{K}} + 4 \le 5\langle l\rangle^{\mathrm{K}}$ on Λ^+. P96-K's modified weight is $(i-j)/(j-2)^{\varpi}$; with our exponent β its analogue is $(i-j)/(j-2)^{1/4}$, and $j^{-1/4} \le (j-2)^{-1/4} \le 3^{1/4} j^{-1/4}$ for $j \ge 3$. Consequently the Melnikov conditions of this part (with $\langle l\rangle$) imply P96-K's (with $\langle l\rangle^{\mathrm{K}}$) at the same α, and our modified conditions imply the condition (23) with exponent β in place of ϖ and $\hat\alpha$ replaced by $\hat\alpha/3^{1/4}$. W5 states the modified condition directly in Dirichlet labels, as $|i-j|/(\min(i,j)+2)^{\beta} = |I-J|/\min(I,J)^{\beta}$ (Lemma 7.6), which is exactly $\hat w$. W5–W6 must use the weights of Definition 9.1, or absorb these factors.

Source: P96-K pp. 120 (Assumption B), 121, 140; P96-W p. 291; W3 Remark 5.30.

9.3. **Unperturbed frequencies and frequency coordinates.** Write $\lambda_J = (\lambda_1, \lambda_2)$, $b = (\lambda_1^{-1}, \lambda_2^{-1})$, and let A be the matrix of Theorem 2.10: $A = \frac{3}{8\pi}(4bb^{\mathsf T} - \mathrm{diag}(b_1^2, b_2^2))$.

Definition 9.3 (Unperturbed frequencies)**.** For $\xi \in \mathbb{R}^2$ (the tangential actions $I = \xi$ of the normal form),

$$\omega^0(\xi) = \lambda_J + A\xi \in \mathbb{R}^2, \qquad \Omega^0_j(\xi) = \lambda_j + (B\xi)_j = \lambda_j + \frac{3}{2\pi\lambda_j}\langle b, \xi\rangle \qquad (j \ge 3).$$

Source: P96-W p. 287 ($\omega(I) = \alpha + AI$, $\Omega(I) = \beta + BI$) with the corrected A, B of Theorem 2.10 (E-PDE-W-1/2); KP96 p. 25, Step 1 ($\omega(\xi) = \alpha + A\xi$, $\Omega(\xi) = \beta + B\xi$).

Lemma 9.4 (Frequency coordinates)**.** *(i)* $A^{-1} = \frac{8\pi}{21}\begin{pmatrix} -3\lambda_1^2 & 4\lambda_1\lambda_2 \\ 4\lambda_1\lambda_2 & -3\lambda_2^2 \end{pmatrix}$, $|\det A| = D_A := \dfrac{63}{64\pi^2\lambda_1^2\lambda_2^2}$, *and* $A^{-1}b = \frac{8\pi}{21}\lambda_J$.

(ii) $\|A\|_\infty = N_A := \frac{3}{8\pi}\big(\frac{3}{\lambda_1^2} + \frac{4}{\lambda_1\lambda_2}\big) < \frac{3}{5}$ *and* $\|A^{-1}\|_\infty = L_A := \frac{8\pi}{21}\lambda_2(4\lambda_1 + 3\lambda_2) = N_A/D_A$.

(iii) ω^0 *is an affine bijection of* $\mathbb{R}^2$ *with inverse* $\zeta \mapsto A^{-1}(\zeta - \lambda_J)$. *In the frequency coordinate* $\zeta = \omega^0(\xi)$ *the normal frequencies are*

$$\Omega^0_j = \lambda_j + \frac{4}{7\lambda_j}\, s(\zeta), \qquad s(\zeta) = \langle \lambda_J, \zeta - \lambda_J\rangle = \frac{21}{8\pi}\langle b, \xi\rangle,$$

i.e. in P96-K's labels $\Omega^{\mathrm{K}}_{j'}(\zeta) = \lambda_{j'+2} + \frac{4}{7\lambda_{j'+2}}\langle \lambda_J, \zeta - \lambda_J\rangle$.

(iv) For Borel $F \subset \mathbb{R}^2$: $\mathrm{Leb}(\omega^0(F)) = D_A\,\mathrm{Leb}(F)$ *and* $\operatorname{diam}\omega^0(F) \le N_A \operatorname{diam} F$; *for* $\xi, \xi' \in \mathbb{R}^2$, $|\xi - \xi'|_\infty \le L_A |\omega^0(\xi) - \omega^0(\xi')|_\infty$.

Proof. **Step S9.1: Inverse.** Put $M = 4bb^{\mathsf T} - D^2$, $D^2 = \mathrm{diag}(b_1^2, b_2^2)$. Since $D^{-2}b = \lambda_J$ and $b^{\mathsf T}D^{-2}b = 2$, Sherman–Morrison gives $M^{-1} = -D^{-2} + \frac{4}{7}\lambda_J\lambda_J^{\mathsf T}$ (denominator $1 - 4\cdot 2 = -7$), so $A^{-1} = \frac{8\pi}{3}M^{-1}$ is the displayed matrix; multiply out $AA^{-1} = \mathrm{Id}$ as a check. Then $A^{-1}b = \frac{8\pi}{3}(-\lambda_J + \frac{8}{7}\lambda_J) = \frac{8\pi}{21}\lambda_J$. $\det A$ is Theorem 2.13(i).

Step S9.2: Norms. The row sums of $|A|$ are $\frac{3}{8\pi}b_i(3b_i + 4b_{i'})$ ($\{i, i'\} = \{1, 2\}$); the first is larger since $b_1 > b_2$, and $3/\lambda_1^2 + 4/(\lambda_1\lambda_2) < 3 + 2$ because $\lambda_1 > 1$, $\lambda_2 > 2$, so $N_A < \frac{15}{8\pi} < \frac{3}{5}$. The row sums of $|A^{-1}|$ are $\frac{8\pi}{21}\lambda_i(3\lambda_i + 4\lambda_{i'})$; the second exceeds the first by $\frac{8\pi}{21}\cdot 3(\lambda_2^2 - \lambda_1^2) > 0$. The identity $L_A = N_A/D_A$ is direct substitution.

Step S9.3: Normal frequencies in ζ. $\langle b, \xi\rangle = \langle b, A^{-1}(\zeta - \lambda_J)\rangle = \langle A^{-1}b, \zeta - \lambda_J\rangle = \frac{8\pi}{21}s(\zeta)$ (A symmetric); substitute in Definition 9.3: $\frac{3}{2\pi\lambda_j}\cdot\frac{8\pi}{21} = \frac{4}{7\lambda_j}$.

Step S9.4: Measure and distances. (W7-E3) with $T = A$, $c = \lambda_J$; the diameter and distance bounds are the operator norms. □

Source: ours (elementary linear algebra on Theorem 2.10). P96-K Lemma 5, p. 136, introduces $\zeta = \omega(\xi)$ as parameter; P96-W p. 291 gives Ω in I, not in ζ.

Lemma 9.5 (Unperturbed separation and asymptotics)**.** *Let* $|\xi|_\infty \le \frac{1}{2}$ *and* $\zeta = \omega^0(\xi)$. *Put* $\beta_* = \frac{1}{2\sqrt{9+m}}$ *and* $a_\kappa = \frac{m}{2} + 1$. *Then:*

(i) $|s(\zeta)| \le 1$ *and* $|\omega^0(\xi)|_\infty \le \lambda_2 + \frac{3}{10}$;
(ii) (normal separation) $|\langle l, \Omega^0(\xi)\rangle| \ge 3\beta_* \langle l\rangle$ *for every* $l \in \Lambda$;
(iii) ($\kappa = 2$) *for* $i > j \ge 3$, $\big|\Omega_i^0(\xi) - \Omega_j^0(\xi) - (i-j)\big| \le a_\kappa (i-j) j^{-2}$;
(iv) (Lipschitz in ζ) *for* $l \in \Lambda$, $\zeta \mapsto \langle l, \Omega^0\rangle$ *is affine with* $|\langle l, \Omega^0(\zeta)\rangle - \langle l, \Omega^0(\zeta')\rangle| \le \frac{4}{7}(\lambda_1 + \lambda_2)|l|_{-1}|\zeta - \zeta'|_\infty$.

Proof. **Step S9.5: Size of** s. $|\langle b, \xi\rangle| \le \frac{1}{2}(\lambda_1^{-1} + \lambda_2^{-1}) < \frac{3}{4}$ and $\frac{21}{8\pi} < 0.84$, so $|s| \le 1$ by Lemma 9.4(iii); $|A\xi|_\infty \le N_A/2 < \frac{3}{10}$.

Step S9.6: Plus type. By (i), $|\frac{4}{7\lambda_j} s| \le \frac{4}{21}$, so $\Omega_j^0 \ge \lambda_j - \frac{4}{21} \ge j - \frac{1}{3} \ge \frac{8}{9} j$ for $j \ge 3$. For $l \in \Lambda^+$ all nonzero l_j have one sign, so $|\langle l, \Omega^0\rangle| = \sum |l_j| \Omega_j^0 \ge \frac{8}{9}\langle l\rangle \ge 3\beta_* \langle l\rangle$ (as $3\beta_* < \frac{1}{2}$).

Step S9.7: Minus type. For $i > j$: $\Omega_i^0 - \Omega_j^0 = (\lambda_i - \lambda_j)\big(1 - \frac{4s}{7\lambda_i\lambda_j}\big)$, and $\frac{4|s|}{7\lambda_i\lambda_j} \le \frac{4}{84}$ since $\lambda_i\lambda_j > 12$. The function $f(x) = \sqrt{x^2 + m}$ has increasing derivative $f'(x) = x/f(x)$ (§1.2), so $\lambda_i - \lambda_j \ge (i-j) f'(j) \ge (i-j)\frac{3}{\sqrt{9+m}}$. Hence $\Omega_i^0 - \Omega_j^0 \ge \frac{1}{2} \cdot \frac{3(i-j)}{\sqrt{9+m}} = 3\beta_*(i-j)$.

Step S9.8: Asymptotics. $\lambda_n - n = m/(\lambda_n + n)$; since $\lambda_i - \lambda_j \le i - j$ ($f' < 1$) and $\lambda_n + n > 2n$,

$$\Big|\frac{m}{\lambda_j + j} - \frac{m}{\lambda_i + i}\Big| = \frac{m\,[(\lambda_i + i) - (\lambda_j + j)]}{(\lambda_i + i)(\lambda_j + j)} \le \frac{2m(i-j)}{4ij} \le \frac{m(i-j)}{2j^2}.$$

The s-term is $\frac{4|s|}{7}\big|\frac{1}{\lambda_i} - \frac{1}{\lambda_j}\big| = \frac{4|s|}{7}\frac{\lambda_i - \lambda_j}{\lambda_i\lambda_j} \le \frac{i-j}{j^2}$. Add.

Step S9.9: Lipschitz. $\langle l, \Omega^0(\zeta)\rangle = \langle l, \lambda\rangle + \frac{4}{7}\big(\sum_j l_j/\lambda_j\big) s(\zeta)$ and $|s(\zeta) - s(\zeta')| \le (\lambda_1 + \lambda_2)|\zeta - \zeta'|_\infty$; $|\sum l_j/\lambda_j| \le |l|_{-1}$. □

Source: P96-K Lemma 3, p. 136 ((i)–(ii) without explicit β), p. 138 ($|(\Omega_i - \Omega_j)/(i-j) - 1| \le a j^{-\kappa}$); P96-W p. 291 ($\kappa = 2$, with the printed $v = \frac{24}{\pi} b$; corrected value $\frac{3}{2\pi} b$, Remark 2.14). Constants ours.

9.4. Constants and the order in which they are chosen.

Definition 9.6 (W7 constants)**.** In this order:

(C1) $m > 0$ (given); $\tau = 24$, $\varpi = \frac{1}{2}$, $\beta = \frac{1}{4}$, $w = \beta/(4\kappa + 4 - \beta) = \frac{1}{47}$ (numbers).
(C2) Explicit in m: λ_1, λ_2, D_A, N_A, L_A (Lemma 9.4); $\beta_* = \frac{1}{2\sqrt{9+m}}$, $a_\kappa = \frac{m}{2} + 1$; $W = \lambda_2 + 1$; $\vartheta = \beta_*/W$; $c_\Lambda = \frac{4}{7}(\lambda_1 + \lambda_2) + 1$; $K_* = 16 c_\Lambda$; $j_* = (16 c_\Lambda)^2$.
(C3) The finite exceptional set X (Definition 9.15), $N_X = \#X$, and the positive numbers σ_X, c_X of Lemma 9.17; they depend on m only.
(C4) The Lipschitz smallness required of W6 and the smallness threshold of α:

$$\theta_0 = \frac{1}{L_A}\min\Big(\frac{1}{8}, \frac{\sigma_X}{2(K_* + 2)}\Big), \qquad \alpha_0 = \Big(\min\Big\{\frac{1}{27}, \frac{\beta_*}{4}, \frac{\vartheta c_X}{4K_*(K_* + 4)}\Big\}\Big)^{47/44}.$$

(C5) $C_P = Z_3\big(8 + 32\vartheta^{-3} + 16(3 + a_\kappa)\vartheta^{-2}\big) + 4N_X K_* \vartheta^{-1}\sigma_X^{-1}$, $C_I = Z_3\big(8 + 32\vartheta^{-3} + 16(2 + a_\kappa)\vartheta^{-2} + 64\vartheta^{-1}\big) + 4N_X K_*^2 \vartheta^{-1}\sigma_X^{-1}$, $C_7 = L_A(C_P + C_I)$, with $Z_3 = 2\pi^2/3$ (W7-E5).
(C6) For each application: a Borel parameter set Π, then $\alpha \in (0, \alpha_0]$, then $\hat\alpha = \alpha^{1-3w} = \alpha^{44/47}$ and $\check\alpha = \alpha^{1+w} = \alpha^{48/47}$ (so $\check\alpha \le \alpha \le \hat\alpha$).

Every constant in (C1)–(C5) depends on m only; none depends on Π, on α, or on the frequency families of Definition 9.9.

Source: ours; P96-K pp. 136–137 (β, ϑ, L_*, K_*), p. 140 ($\hat\alpha$), p. 142 ($\check\alpha$, w). P96-K's constants are monotone in Π (p. 123); ours are independent of Π.

Remark 9.7 (Why these choices). P96-K uses one exponent $\varpi < \min(\bar p - p, 1) = 1$ both for the smoothing loss of the preparatory step and for the weight of its modified divisors. The two roles separate. The loss $\varpi = \frac12$ fixes the decay $j^{-1/2}$ of the frequency drift after the preparatory step ((H2)) and the iteration exponent $\check p - p = \frac12$. The weight exponent β alone enters the measure of the preparatory zones (Proposition 9.19: exponent $\kappa/(\kappa+1-\beta)$). The parameter-Lipschitz bound of the preparatory step requires $2\beta \le \varpi$, since the Lipschitz constant of $1/\delta$ carries $\min(I,J)^{2\beta}$ (W5, Lemma 7.6(ii) and its SOURCE_GAP). We take $\beta = \frac14$. The optimizing w of P96-K p. 142 solves $(1-3w)\frac{8}{11} = (1+w)\frac23$, i.e. $w = \beta/(4\kappa+4-\beta) = \frac{1}{47}$; then $\hat\alpha^{8/11} = \check\alpha^{2/3} = \alpha^{32/47}$ and $\tilde\mu = \kappa/(\kappa+1-\beta/4) = \frac{32}{47}$ (Step S9.41). The relation $\check\alpha = \alpha^{4/3}\hat\alpha^{-1/3}$ of the preparatory step (W6, Lemma 8.8(iv)) holds: $\frac43 - \frac13\cdot\frac{44}{47} = \frac{48}{47}$. The value $\tau = 24$ makes every lattice sum below converge with the single bound $|k|^{-3}$ (the slowest term is $|k|\,A_k^{-1/6} \le |k|^{-3}$ in Proposition 9.20); P96-K's (22) would ask $\tau \ge (n+3)/\delta^* = 15$ for the post-preparation exponent $\delta = -\frac12$. The frozen headline only needs some $\tau > 1$. **W4–W6 must use** $\tau = 24$ (a larger τ is also fine for W7: all bounds use only $A_k \ge |k|^{24}$).

9.5. **Interface: the frequency families delivered by W6.**

Definition 9.8 (Resonance zones)**.** Let $\Pi \subset \{\xi : |\xi|_\infty \le \frac12\}$ be Borel and $0 < \alpha \le \alpha_0$.

(i) The *preparatory zones* are, for $(k,l) \in Z$,

$$S_{kl} = \Big\{\xi \in \Pi : \big|\langle k, \omega^0(\xi)\rangle + \langle l, \Omega^0(\xi)\rangle\big| < \frac{\hat\alpha\,\hat w(l)}{A_k}\Big\}, \qquad \Xi = \bigcup_{(k,l)\in Z} S_{kl}.$$

(ii) Given a Borel $D \subset \Pi$ and maps $\omega' : D \to \mathbb{R}^2$, $\Omega'_j : D \to \mathbb{R}$ ($j \ge 3$), the *zones* at level $a > 0$ are $R_{kl}(a) = \{\xi \in D : |\langle k, \omega'(\xi)\rangle + \langle l, \Omega'(\xi)\rangle| < a\langle l\rangle/A_k\}$.

Definition 9.9 (Admissible frequency family)**.** An *admissible family* (for Π, α) is a sequence $(D_\nu, \omega_\nu, \Omega_\nu, K_\nu)_{\nu\ge0}$ with $D_\nu \subset \Pi$ Borel, $\omega_\nu : D_\nu \to \mathbb{R}^2$, $\Omega_{\nu,j} : D_\nu \to \mathbb{R}$ ($j \ge 3$), reals $K_\nu \ge 2^\nu$, $K_{-1} := 0$, and numbers $\varepsilon_\omega, \varepsilon_1, \varepsilon_2 \in [0,\alpha]$, $\theta \in [0,\theta_0]$ such that for all ν, all $\xi, \xi' \in D_\nu$ and all $j \ge 3$:

(H1) $|\omega_\nu(\xi) - \omega^0(\xi)|_\infty \le \varepsilon_\omega$ and $|(\omega_\nu - \omega^0)(\xi) - (\omega_\nu - \omega^0)(\xi')|_\infty \le \theta|\xi - \xi'|_\infty$;

(H2) $|\Omega_{\nu,j}(\xi) - \Omega^0_j(\xi)| \le \varepsilon_1 j^{-1} + \varepsilon_2 j^{-1/2}$ and $|(\Omega_{\nu,j} - \Omega^0_j)(\xi) - (\Omega_{\nu,j} - \Omega^0_j)(\xi')| \le \theta j^{-1/2}|\xi - \xi'|_\infty$.

Its *iteration zones* are $R^\nu_{kl} = R_{kl}(\check\alpha)$ built from $(D_\nu, \omega_\nu, \Omega_\nu)$, and the *excluded set* is

$$\mathcal{E} = \Xi \cup \bigcup_{\nu\ge0} \bigcup_{(k,l)\in Z,\ |k|>K_{\nu-1}} R^\nu_{kl}.$$

Source: P96-K Theorem A, p. 122 (frequency maps ω_ν, Ω_ν, zones $\mathcal{R}^\nu_{kl}(\alpha)$, $|k| > K_0 2^{\nu-1}$), (19) p. 135, p. 132 ($K_\nu = K_0 2^\nu$, $K_0 \ge 1$), §6 pp. 140–142 (preparatory step, $\hat\alpha$, $\check\alpha$, weights $j^{\bar p - p} = j$ for the preparatory shift and $j^{\check p - p} = j^{1/2}$ after it, $\check p = \bar p - \varpi$). W6, Lemma 8.11, supplies (H1)–(H2) in exactly this form.

Remark 9.10 (What W6 must supply; forward references). W7 is a statement about arbitrary admissible families; it uses nothing about the KAM step. To apply it, W6 must prove, on each parameter domain Π used and for the master parameter α of that domain:

- `bpW:w6:prep-step` (forward): the preparatory step of P96-K §6 runs on $\Pi \setminus \Xi$, where the unperturbed frequencies satisfy the conditions of Corollary 9.23(i) (weights Remark 9.2; modified weight $|I-J|\min(I,J)^{-1/4}$, i.e. W5's $\beta = \frac14$, with loss $\frac12$, so that $2\beta \le \frac12$ and the parameter-Lipschitz part of the step holds), and its frequency shift obeys (H1)–(H2) with

the $\varepsilon_1 j^{-1}$ part ($\bar{p}-p=1$). Its new perturbation has size $\lesssim \epsilon^{4/3}\hat{\alpha}^{-1/3} \le \gamma_1\check{\alpha}$ with $\check{\alpha}=\alpha^{48/47}$ ($\frac{4}{3}-\frac{1}{3}\cdot\frac{44}{47}=\frac{48}{47}$).

- `bpW:w6:freq-lip` (forward): the frequencies ω_ν, Ω_ν of the iteration started after the preparatory step are defined on Borel D_ν and satisfy (H1)–(H2) with $\varepsilon_\omega, \varepsilon_1, \varepsilon_2 \le \alpha$ and Lipschitz constant $\theta \le \theta_0$ (P96-K Theorem A gives $|\omega_\nu-\omega| + \frac{\alpha}{M}|\omega_\nu-\omega|^{\mathcal{L}} \le c\epsilon \le c\gamma\alpha$; so this is a smallness condition on γ; this is our explicit alternative to P96-K Corollary C, pp. 123 and 140, which instead chooses K_0 large so that the exceptional zones use the unperturbed frequencies).
- `bpW:w6:exclusion` (forward): the final parameter set Π_* of the iteration satisfies $\Pi\setminus\Pi_* \subset \mathcal{E}$, with thresholds $K_\nu \ge 2^\nu$ and iteration levels $\alpha_\nu \le \check{\alpha}$ (so its zones lie in $R^\nu_{kl}(\check{\alpha})$).
- `bpW:w6:final-freq` (forward): on Π_* the final frequency ω_* satisfies $|\langle k,\omega_*\rangle| \ge \check{\alpha}/(2A_k)$ for $k\neq 0$ (P96-K p. 135) and $|\omega_*-\omega^0|_\infty \le \varepsilon_\omega$.

The Λ^- argument uses the drift bound (H2) only through $\sup_{i\ge j}|\Omega_{\nu,i}-\Omega^0_i| \le (\varepsilon_1+\varepsilon_2)j^{-1/2}$; the Lipschitz parts of (H1)–(H2) are the "Lipschitz dependence of the frequencies on the parameters" imported from W6.

9.6. **The slicing lemma and the elementary zone estimates.**

Lemma 9.11 (Slicing)**.** *Let $E\subset\mathbb{R}^2$ be Borel and nonempty with $\operatorname{diam}_\infty E \le \rho$, let $\phi: E\to\mathbb{R}$ be continuous on E, $i\in\{1,2\}$, $\sigma>0$, and suppose $|\phi(\zeta+te_i)-\phi(\zeta)| \ge \sigma|t|$ whenever ζ, $\zeta+te_i\in E$. Then for every $\eta>0$ the set $\{\zeta\in E: |\phi(\zeta)|<\eta\}$ is Borel and has measure at most $2\eta\rho/\sigma$.*

Proof. **Step S9.10: Borel.** (W7-E4).

Step S9.11: One line. Fix the other coordinate y and let $T_y=\{t:\zeta(t)\in E, |\phi(\zeta(t))|<\eta\}$, $\zeta(t)=te_i+ye_{i'}$. For $t,t'\in T_y$, $\sigma|t-t'| \le |\phi(\zeta(t))-\phi(\zeta(t'))| < 2\eta$, so $\operatorname{Leb}_1(T_y)\le 2\eta/\sigma$ (W7-E2).

Step S9.12: Integrate. $T_y=\emptyset$ unless y lies in the projection of E on the i'-th axis, which is contained in $[y_0,y_0+\rho]$ for $y_0=\inf$ of that projection. Tonelli (W7-E1) gives the measure $\le \int_{y_0}^{y_0+\rho} 2\eta/\sigma\,dy$. □

Source: P96-K Lemma 5, p. 137 (Fubini along a line). Deviation: coordinate lines and a lower Lipschitz bound on the set itself replace the source's lines $\zeta=rv+w$, $v\in\{\pm1\}^n$, and the Lipschitz extension of Lemma A.2; see Remark 9.32(a) for the constant.

From now on fix Π, $\alpha\in(0,\alpha_0]$ and an admissible family. Write $\rho=\operatorname{diam}_\infty\Pi$ and $\rho_\zeta=\operatorname{diam}_\infty\omega^0(\Pi)\le N_A\rho$. A *frequency pair* on a Borel $D\subset\Pi$ is either (ω^0,Ω^0) ("unperturbed", with ε's and θ equal to 0) or some (ω_ν,Ω_ν) on D_ν. For such a pair let $\phi_{kl}=\langle k,\omega'\rangle+\langle l,\Omega'\rangle$, viewed as a function of $\zeta\in\omega^0(D)$.

Lemma 9.12 (Normal separation; Melnikov conditions with $k=0$)**.** *For a frequency pair on D and $l\in\Lambda$: $|\langle l,\Omega'(\xi)\rangle| \ge 2\beta_*\langle l\rangle$ on D. In particular $|\langle l,\Omega'\rangle| \ge a\langle l\rangle/A_0$ for every $a\le 2\beta_*$, and $R_{0l}(a)=\emptyset$ and $S_{0l}=\emptyset$.*

Proof. **Step S9.13: Drift.** By (H2), $|\langle l,\Omega'-\Omega^0\rangle| \le 2(\varepsilon_1+\varepsilon_2) \le 4\alpha \le \beta_* \le \beta_*\langle l\rangle$ since $\alpha\le\hat{\alpha}\le\beta_*/4$ by (C4).

Step S9.14: Conclude. Lemma 9.5(ii). For S_{0l} and R_{0l}: $\hat{\alpha},\check{\alpha}\le\beta_*$ and $\hat{w}(l)\le\langle l\rangle$, $A_0=1$. □

Source: P96-K Lemma 3, p. 136, and "$\mathcal{R}'_k(\alpha)$ is empty for $k=0$", p. 137.

Lemma 9.13 (Nonempty zones have large $|k|$)**.** *For a frequency pair on D, $l\in\Lambda$ and $a\le\beta_*$: if $\{\xi\in D: |\phi_{kl}(\xi)|<a\langle l\rangle/A_k\}\neq\emptyset$ then $|k|\ge\vartheta\langle l\rangle$; in particular $k\neq 0$. The same holds for S_{kl} ($a=\hat{\alpha}$).*

Proof. **Step S9.15: Bound the tangential term.** $|\omega'|_\infty \le \lambda_2 + \frac{3}{10} + \alpha \le W$ (Lemma 9.5(i), (H1)), so $|\langle k,\omega'\rangle| \le W|k|$.

Step S9.16: Compare. At a point of the zone, $W|k| \ge |\langle l,\Omega'\rangle| - a\langle l\rangle \ge (2\beta_* - \beta_*)\langle l\rangle$ (Lemma 9.12). □

Source: P96-K Lemma 4, p. 136 ($\vartheta = \beta/(|\omega|_\Pi + 1)$).

Lemma 9.14 (Zone estimate)**.** *Let a frequency pair on D be given, $k \neq 0$, and $\psi : D \to \mathbb{R}$ with $|\psi(\xi) - \psi(\xi')| \le \frac{|k|}{8}|\omega^0(\xi) - \omega^0(\xi')|_\infty$. Then for $\eta > 0$,*

$$\mathrm{Leb}\{\xi \in D : |\langle k,\omega'(\xi)\rangle + \psi(\xi)| < \eta\} \le \frac{8\eta\rho_\zeta}{|k|\, D_A}.$$

This applies to $\psi = \langle l,\Omega'\rangle$ when $l = 0$, or when $l \in \Lambda$ and $|k| \ge 8c_\Lambda|l|_{-1/2}$, and to $\psi \equiv c$ constant.

Proof. **Step S9.17: Lower Lipschitz bound in ζ.** In ζ, $\langle k,\omega'\rangle = \langle k,\zeta\rangle + \langle k,h(\zeta)\rangle$ with $h = (\omega' - \omega^0)\circ(\omega^0)^{-1}$, and $|h(\zeta) - h(\zeta')|_\infty \le \theta L_A|\zeta - \zeta'|_\infty \le \frac{1}{8}|\zeta - \zeta'|_\infty$ by (H1), Lemma 9.4(iv) and $\theta \le \theta_0$. Choose i with $|k_i| \ge |k|/2$. Then $|\phi(\zeta + te_i) - \phi(\zeta)| \ge (\frac{1}{2} - \frac{1}{8} - \frac{1}{8})|k||t| = \frac{|k|}{4}|t|$.

Step S9.18: Slicing. Lemma 9.11 on $E = \omega^0(D)$ (diam $\le \rho_\zeta$) with $\sigma = |k|/4$, then Lemma 9.4(iv) to return to ξ.

Step S9.19: The admissible ψ. For $l \in \Lambda$, Lemma 9.5(iv) and (H2) give the Lipschitz constant in ζ

$$\tfrac{4}{7}(\lambda_1 + \lambda_2)|l|_{-1} + L_A\theta|l|_{-1/2} \le c_\Lambda|l|_{-1/2} \le |k|/8$$

($L_A\theta \le 1$, $|l|_{-1} \le |l|_{-1/2}$). □

Source: P96-K Lemma 5, pp. 136–137 (hypothesis $|k| \ge 8LM|l|_\delta$; normalization $|\omega' - \mathrm{id}|^{\mathcal{L}} \le \frac{1}{2}$ there, $\le \frac{1}{8}$ here because coordinate lines are used).

Definition 9.15 (Exceptional set)**.** X is the set of $(k,l) \in \mathbb{Z}^2 \times \Lambda$ with $1 \le |k| < K_*$, $\langle l\rangle < K_*/\vartheta$, and either $l \in \Lambda^+$ or $l = \pm(e_i - e_j)$ with $i > j$ and $j < j_*$.

Lemma 9.16 (Dichotomy)**.** *X is finite. If $(k,l) \in Z \setminus X$, $l \in \Lambda$, and a zone of Lemma 9.13 for (k,l) is nonempty, then $k \neq 0$, $\langle l\rangle \le |k|/\vartheta$ and $|k| \ge 8c_\Lambda|l|_{-1/2}$.*

Proof. **Step S9.20: Finite.** $|k| < K_*$ bounds k; on Λ^+, $\langle l\rangle \ge \max\{j : l_j \neq 0\}$, so the support lies in $[3, K_*/\vartheta)$; on Λ^-, $j < j_*$ and $i < j + K_*/\vartheta$.

Step S9.21: Large k. Lemma 9.13 gives $k \neq 0$ and $\langle l\rangle \le |k|/\vartheta$. If $|k| \ge K_* = 16c_\Lambda$ then $8c_\Lambda|l|_{-1/2} \le 16c_\Lambda \le |k|$.

Step S9.22: Small k. If $|k| < K_*$, then $\langle l\rangle < K_*/\vartheta$, so $(k,l) \notin X$ forces $l = \pm(e_i - e_j)$, $i > j \ge j_*$. Then $|l|_{-1/2} \le 2j^{-1/2} \le 2j_*^{-1/2} = 1/(8c_\Lambda)$, so $8c_\Lambda|l|_{-1/2} \le 1 \le |k|$. □

Source: P96-K Lemma 6 and the set $\mathcal{X}$, pp. 137, 139 (the source bounds $|l|_\sigma < L_*$; for $d = 1$, $\delta < 0$ the role of $|l|_\sigma$ is played here by $j < j_*$).

Lemma 9.17 (Exceptional zones)**.** *For $(k,l) \in X$ let $g_{kl}(\zeta) = \langle k,\zeta\rangle + \langle l,\Omega^0(\zeta)\rangle = c_{kl} + \langle p_{kl},\zeta\rangle$ (affine, Lemma 9.4(iii)), $p_{kl} = k + \frac{4}{7}(\sum_j l_j/\lambda_j)\lambda_J$. Then $g_{kl} \not\equiv 0$. Put*

$$\sigma_X = \min\{|p_{kl}|_\infty : (k,l) \in X,\ p_{kl} \neq 0\}, \qquad c_X = \min\{|\langle k,\lambda_J\rangle + \langle l,\lambda\rangle| : (k,l) \in X,\ p_{kl} = 0\}$$

(minimum over the empty set $:= 1$); both are positive. For a frequency pair on D, $(k,l) \in X$ and $0 < a \le \hat\alpha$, the zone $Y = \{\xi \in D : |\phi_{kl}| < a\langle l\rangle/A_k\}$ is empty if $p_{kl} = 0$, and $\mathrm{Leb}(Y) \le 4\rho_\zeta a K_*/(\vartheta\sigma_X D_A)$ if $p_{kl} \neq 0$.

Proof. **Step S9.23: Nondegeneracy.** $g_{kl}\circ\omega^0$ is the affine function $\xi\mapsto\langle k,\omega(\xi)\rangle+\langle l,\Omega(\xi)\rangle$ of Theorem 2.13(iii) (condition (A_3), $1\le|l|\le 2$); it is not identically zero, and neither is g_{kl} since ω^0 is bijective. If $p_{kl}=0$, g_{kl} is the constant $g_{kl}(\lambda_J)=\langle k,\lambda_J\rangle+\langle l,\lambda\rangle\neq 0$. X is finite, so $\sigma_X,c_X>0$.

Step S9.24: Constant case. $|\phi_{kl}-g_{kl}|\le|k|\varepsilon_\omega+2(\varepsilon_1+\varepsilon_2)\le(K_*+4)\alpha\le c_X/2$ and $a\langle l\rangle/A_k\le\hat\alpha K_*/\vartheta<c_X/2$, both by (C4) and $\alpha\le\hat\alpha$; so $Y=\emptyset$.

Step S9.25: Sloped case. Choose i with $|(p_{kl})_i|=|p_{kl}|_\infty\ge\sigma_X$. The difference $\phi_{kl}-g_{kl}$ has Lipschitz constant in ζ at most $L_A\theta(|k|+|l|_{-1/2})\le L_A\theta(K_*+2)\le\sigma_X/2$ by (H1)–(H2) and (C4). Lemma 9.11 with $\sigma=\sigma_X/2$ and $\eta=a\langle l\rangle/A_k\le aK_*/\vartheta$, and Lemma 9.4(iv). □

Source: P96-K Lemma 9 (last paragraph, p. 141: "a nontrivial affine function of ξ, so one has $|\mathcal{R}^s_{kl}(\hat\alpha)|\ll\hat\alpha$"), Lemma 10 p. 142 and Corollary C p. 140 (Assumption A for the finitely many $(k,l)\in\mathcal{X}$). See Remark 9.32(c) for the Lipschitz condition.

Lemma 9.18 (Counting)**.** *(i) For $N\ge1$, $\#\{l\in\Lambda^+:\langle l\rangle\le N\}\le 4N^2$. (ii) If $K_\nu\ge 2^\nu$ $(\nu\ge0)$ and $K_{-1}=0$, then $\#\{\nu\ge0:|k|>K_{\nu-1}\}\le|k|$ for $k\neq0$. (iii) $\sum_{k\neq0}|k|^{-3}=Z_3<7$.*

Proof. (i) $\pm e_i$: $2N$ vectors; $\pm2e_i$: at most N; $\pm(e_i+e_j)$, $i<j$, $i+j\le N$: at most N^2; and $N^2+3N\le4N^2$. (ii) $\nu=0$ always counts; $\nu\ge1$ counts only if $2^{\nu-1}<|k|$, and $2^e\ge e+1$ gives $e=\nu-1\le|k|-2$, at most $|k|-1$ values. (iii) (W7-E5). □

Source: ours (P96-K p. 138 counts Λ^+ by $\vartheta^{-2}|k|^2$; p. 139 sums over ν with $K_\nu=K_0 2^\nu$).

9.7. **The preparatory zones.**

Proposition 9.19 (Preparatory zones)**.** *Ξ is Borel and* $\mathrm{Leb}(\Xi)\le C_P\,\rho_\zeta\,D_A^{-1}\,\hat\alpha^{8/11}$.

Proof. Work with the unperturbed pair on Π. $S_{0l}=\emptyset$ (Lemma 9.12); fix $k\neq0$ and sum over l.

Step S9.26: $l=0$. Lemma 9.14 with $\psi=0$, $\eta=\hat\alpha/A_k$: $\mathrm{Leb}(S_{k0})\le8\rho_\zeta\hat\alpha/(|k|A_kD_A)$.

Step S9.27: Λ^+. For $l\in\Lambda^+$, $(k,l)\notin X$, $S_{kl}\neq\emptyset$: $\langle l\rangle\le|k|/\vartheta$ and Lemma 9.14 applies (Lemma 9.16), giving $\mathrm{Leb}(S_{kl})\le8\rho_\zeta\hat\alpha\langle l\rangle/(|k|A_kD_A)\le8\rho_\zeta\hat\alpha/(\vartheta A_kD_A)$. By Lemma 9.18(i) the sum is at most $32\rho_\zeta\hat\alpha|k|^2/(\vartheta^3A_kD_A)$.

Step S9.28: Λ^-, grouping. The zone of $(k,-(e_i-e_j))$ is that of $(-k,e_i-e_j)$; so it suffices to bound, for each k, the zones $S_{k,ij}$ of $l=e_i-e_j$, $i>j$, and to double. Put $n=i-j\ge1$ (the source's m). $S_{k,ij}\neq\emptyset$ forces $n\le|k|/\vartheta$. For $(k,e_i-e_j)\notin X$, Lemma 9.14 gives $\mathrm{Leb}(S_{k,ij})\le8\rho_\zeta\hat\alpha n/(j^{1/4}A_k|k|D_A)$. For all j, by Lemma 9.5(iii),

$$S_{k,ij}\subset Q_{knj}:=\Big\{\xi\in\Pi:|\langle k,\omega^0\rangle+n|<\frac{\hat\alpha n}{A_kj^{1/4}}+\frac{a_\kappa n}{j^2}\Big\},$$

and $Q_{knj}\subset Q_{knj_0}$ for $j\ge j_0$ (the threshold decreases in j).

Step S9.29: Λ^-, choice of j_0. Let $j_0=\lceil(A_k/\hat\alpha)^{4/11}\rceil$; $j_0\ge3$ since $\hat\alpha\le\frac{1}{27}<3^{-11/4}$. By Lemma 9.14 (with $\psi\equiv n$ for Q_{knj_0}), the union of the $S_{k,ij}$ with $i-j=n$ and $(k,e_i-e_j)\notin X$ is contained in $\bigcup_{3\le j<j_0,\,(k,e_i-e_j)\notin X}S_{k,ij}\cup Q_{knj_0}$, whose measure is

$$\le\sum_{3\le j<j_0}\frac{8\rho_\zeta\hat\alpha n}{j^{1/4}A_k|k|D_A}+\mathrm{Leb}(Q_{knj_0})\le\frac{8\rho_\zeta n}{|k|D_A}\Big(\frac{\hat\alpha}{A_k}\sum_{j=1}^{j_0}j^{-1/4}+\frac{a_\kappa}{j_0^2}\Big)\le\frac{8\rho_\zeta n}{|k|D_A}(3+a_\kappa)\Big(\frac{\hat\alpha}{A_k}\Big)^{8/11},$$

using $\sum_{j\le j_0}j^{-1/4}\le\frac43j_0^{3/4}$, $j_0\le2(A_k/\hat\alpha)^{4/11}$ (so $\frac43\hat\alpha j_0^{3/4}/A_k\le\frac43 2^{3/4}(\hat\alpha/A_k)^{8/11}\le3(\hat\alpha/A_k)^{8/11}$) and $j_0^{-2}\le(\hat\alpha/A_k)^{8/11}$ (W7-E7).

Step S9.30: Λ^-, sum over n. $\sum_{n\le|k|/\vartheta}n\le|k|^2/\vartheta^2$; doubling for the sign, the Λ^- zones of $\pm k$ not in X contribute at most $16\rho_\zeta(3+a_\kappa)|k|\vartheta^{-2}(\hat\alpha/A_k)^{8/11}D_A^{-1}$ per k.

Step S9.31: Exceptional zones. By Lemma 9.17 with $a = \hat\alpha$ (note $\hat w \le \langle l\rangle$), the zones with $(k,l) \in X$ have total measure at most $4N_X\rho_\zeta\hat\alpha K_*/(\vartheta\sigma_X D_A)$.

Step S9.32: Sum over k. With $A_k \ge |k|^{24}$: $|k|^{-1}A_k^{-1}$, $|k|^2A_k^{-1}$ and $|k|A_k^{-8/11}$ are all $\le |k|^{-3}$. Summing with Lemma 9.18(iii) and $\hat\alpha \le \hat\alpha^{8/11}$ gives the bound with C_P of (C5). Borel: Lemma 9.21. □

Source: P96-K Lemma 9, p. 141, with $\kappa = 2$ and the weight exponent $\beta = \frac14$ in place of ϖ: $\lambda = \kappa/(\kappa{+}1{-}\beta) = \frac{8}{11}$; our j_0 is the source's choice $j_0^{\kappa+1-\beta} = A_k/\hat\alpha$, rounded up. P96-K Lemma 8, p. 138, for the grouping.

9.8. **The iteration zones.**

Proposition 9.20 (Iteration zones). *The set $\mathcal{E}_{\rm it} = \bigcup_\nu \bigcup_{(k,l)\in Z,|k|>K_{\nu-1}} R_{kl}^\nu$ is Borel and*

$$\mathrm{Leb}(\mathcal{E}_{\rm it}) \le \rho_\zeta D_A^{-1}\Big(Z_3\big(8 + 32\vartheta^{-3} + 16(2+a_\kappa)\vartheta^{-2}\big)\check\alpha^{2/3} + 32Z_3\vartheta^{-1}(\varepsilon_1+\varepsilon_2) + 4N_XK_*^2\vartheta^{-1}\sigma_X^{-1}\check\alpha\Big).$$

Proof. Fix ν and k with $|k| > K_{\nu-1}$ (so $k \ne 0$); the pair is (ω_ν, Ω_ν) on D_ν, level $\check\alpha \le \beta_*$.

Step S9.33: $l = 0$ and Λ^+. As in Steps S9.26–S9.27 with $\check\alpha$ in place of $\hat\alpha$, these zones have measure at most $8\rho_\zeta\check\alpha/(|k|A_kD_A) + 32\rho_\zeta\check\alpha|k|^2/(\vartheta^3A_kD_A)$.

Step S9.34: Λ^-, grouping with drift. For $l = e_i - e_j$, $i > j$, $n = i - j \le |k|/\vartheta$ (else empty). By (H2) and Lemma 9.5(iii), $|\langle l,\Omega_\nu\rangle - n| \le 2(\varepsilon_1+\varepsilon_2)j^{-1/2} + a_\kappa n j^{-2}$, so

$$R_{k,ij}^\nu \subset Q_{knj} := \Big\{\xi \in D_\nu : |\langle k,\omega_\nu\rangle + n| < \frac{\check\alpha n}{A_k} + \frac{2(\varepsilon_1+\varepsilon_2)}{j^{1/2}} + \frac{a_\kappa n}{j^2}\Big\},$$

decreasing in j.

Step S9.35: Λ^-, choice of j_0. Let $j_0 = \lceil (A_k/\check\alpha)^{1/3}\rceil \ge 3$ ($\check\alpha \le \frac{1}{27}$). Lemma 9.14 (for $j < j_0$, $(k,l) \notin X$; and for Q_{knj_0} with $\psi \equiv n$):

$$\begin{aligned}\sum_{j<j_0}\mathrm{Leb}(R_{k,ij}^\nu) + \mathrm{Leb}(Q_{knj_0}) &\le \frac{8\rho_\zeta}{|k|D_A}\Big(\frac{j_0\check\alpha n}{A_k} + \frac{a_\kappa n}{j_0^2} + \frac{2(\varepsilon_1+\varepsilon_2)}{j_0^{1/2}}\Big)\\ &\le \frac{8\rho_\zeta}{|k|D_A}\Big((2+a_\kappa)n\Big(\frac{\check\alpha}{A_k}\Big)^{2/3} + 2(\varepsilon_1+\varepsilon_2)A_k^{-1/6}\Big).\end{aligned}$$

Step S9.36: Λ^-, sum over n and sign. Summing $n \le |k|/\vartheta$ and doubling: $16\rho_\zeta D_A^{-1}\big((2 + a_\kappa)\vartheta^{-2}|k|(\check\alpha/A_k)^{2/3} + 2\vartheta^{-1}(\varepsilon_1+\varepsilon_2)A_k^{-1/6}\big)$.

Step S9.37: Sum over ν and k. By Lemma 9.18(ii) each k occurs for at most $|k|$ values of ν. Multiplied by $|k|$, each per-k term is $\le |k|^{-3}$ times its coefficient ($|k|A_k^{-1/6} \le |k|^{-3}$ is the slowest, since $\tau = 24$); sum with Lemma 9.18(iii).

Step S9.38: Exceptional zones. Each $(k,l) \in X$ occurs for at most $|k| < K_*$ values of ν; Lemma 9.17 with $a = \check\alpha$. Borel: Lemma 9.21. □

Source: P96-K Lemma 10, p. 142 ($\mu = \kappa^* = \kappa/(\kappa+1) = \frac23$, contribution of order α from the preparatory shift of Ω); Lemma 8 and (21), pp. 138–139; proof of Theorem B, p. 139.

9.9. **The measure theorem.**

Lemma 9.21 (Measurability)**.** *Every zone S_{kl}, R^{ν}_{kl}, Q_{knj} is Borel, and so are Ξ, $\mathcal{E}_{\mathrm{it}}$, $\mathcal{E}$ and $\Pi \setminus \mathcal{E}$.*

Proof. **Step S9.39: Continuity.** ω^0, Ω^0_j are affine; $\omega_\nu - \omega^0$ and $\Omega_{\nu,j} - \Omega^0_j$ are Lipschitz on D_ν by (H1)–(H2); each ϕ_{kl} involves finitely many j, so it is continuous on its Borel domain.

Step S9.40: Unions. (W7-E4); the index sets $\mathbb{N} \times Z$ are countable. □

Source: ours (P96-K does not discuss measurability; its zones are open in Π after the extension of Lemma A.2).

Theorem 9.22 (Measure of the resonance set; P96-K Theorem D for the wave equation)**.** *Let $m > 0$ and the constants of Definition 9.6 be given. Let $\Pi \subset \{\xi \in \mathbb{R}^2 : |\xi|_\infty \le \frac{1}{2}\}$ be Borel with $\rho = \operatorname{diam}_\infty \Pi$, let $0 < \alpha \le \alpha_0$, and let an admissible family (Definition 9.9) be given. Then $\mathcal{E}$ is Borel and*

$$\mathrm{Leb}(\mathcal{E}) \le C_7\, \rho\, \alpha^{32/47}.$$

Equivalently, in the frequency coordinates, $\mathrm{Leb}(\omega^0(\mathcal{E})) \le (C_P + C_I) N_A \rho\, \alpha^{32/47}$. The constant C_7 depends on m only.

Proof. **Step S9.41: Assemble.** $\mathrm{Leb}(\mathcal{E}) \le \mathrm{Leb}(\Xi) + \mathrm{Leb}(\mathcal{E}_{\mathrm{it}})$ by (W7-E4). In Propositions 9.19–9.20 substitute $\hat\alpha^{8/11} = \alpha^{\frac{44}{47}\cdot\frac{8}{11}} = \alpha^{32/47}$, $\check\alpha^{2/3} = \alpha^{\frac{48}{47}\cdot\frac{2}{3}} = \alpha^{32/47}$, $\check\alpha \le \alpha \le \alpha^{32/47}$ and $\varepsilon_1 + \varepsilon_2 \le 2\alpha$ (all exponents of $\alpha \le 1$ compared).

Step S9.42: Return to ξ. $\rho_\zeta / D_A \le N_A \rho / D_A = L_A \rho$ (Lemma 9.4(ii)). □

Source: P96-K Theorem D, p. 124, and its proof, pp. 140–142, for $n = 2$, $d = 1$, $\kappa = 2$, $\bar p - p = 1$, loss $\varpi = \frac{1}{2}$ and weight exponent $\beta = \frac{1}{4}$ (the source has $\beta = \varpi$): $\tilde\mu = \kappa/(\kappa + 1 - \beta/4) = \frac{32}{47}$, $c\rho^{n-1} = c\rho$. The source's optimization of w gives $w = \beta/(4\kappa + 4 - \beta) = \frac{1}{47}$.

Corollary 9.23 (Melnikov conditions on the complement)**.** *In the setting of Theorem 9.22:*

- *(i) for $\xi \in \Pi \setminus \Xi$ and all $(k,l) \in Z$: $|\langle k, \omega^0(\xi)\rangle + \langle l, \Omega^0(\xi)\rangle| \ge \hat\alpha \hat w(l)/A_k$; in particular ($l = 0$) $|\langle k, \omega^0(\xi)\rangle| \ge \hat\alpha / A_k$ for $k \ne 0$;*
- *(ii) for $\xi \in D_\nu \setminus \bigcup_{|k| > K_{\nu-1}} R^\nu_{kl}$: $|\langle k, \omega_\nu\rangle + \langle l, \Omega_\nu\rangle| \ge \check\alpha \langle l\rangle / A_k$ for all $(k,l) \in Z$ with $|k| > K_{\nu-1}$;*
- *(iii) for every frequency pair, every ξ in its domain and $l \in \Lambda$ ($k = 0$): $|\langle l, \Omega'(\xi)\rangle| \ge 2\beta_* \langle l\rangle \ge \hat\alpha \langle l\rangle$.*

Proof. (i), (ii): negation of the zone definitions (Definition 9.8). (iii): Lemma 9.12. □

Source: P96-K (5) and (23), pp. 125, 140; Theorem A, p. 122.

9.10. **Action shells and density one.** The headline's clause (1) compares Γ with the positive quadrant $P = (0,\infty)^2$ in the max-norm ball $B_h = (-h,h)^2$. KP96 obtains density one by applying the KAM theorem on a sequence of shrinking action shells and taking the union (KP96 p. 28, Step 5). Here is the measure part of that patching.

Definition 9.24 (Shells)**.** Fix $q \in (0,1)$ and $t_0 \in (0, \frac{1}{2}]$; $t_k = q^k t_0$. The k-th shell is $\mathcal{R}_k = (0,t_k)^2 \setminus (0,t_{k+1})^2 = \{I \in (0,t_k)^2 : \max_i I_i \ge t_{k+1}\}$. For $a > 0$, $U_{-a}\mathcal{R}_k = \{I \in \mathcal{R}_k : \operatorname{dist}_\infty(I, \mathbb{R}^2 \setminus \mathcal{R}_k) > a\}$.

Source: KP96 p. 26, Step 3 ($\Xi_r = \{0 < \xi < r^{2\lambda}\}$, $U_{-\rho}$), p. 28, Step 5 ($R_r = \Xi_r \setminus \Xi_{\eta r}$, $R_r^- = U_{-\alpha} R_r$, $r_k = \eta^k r_0$). Our t_k is KP96's $r_k^{2\lambda}$ and $q = \eta^{2\lambda}$.

Lemma 9.25 (Shell geometry)**.** *(i) The shells $\mathcal{R}_l$, $l \ge k$, are pairwise disjoint Borel sets with union $(0,t_k)^2$, and $\mathrm{Leb}(\mathcal{R}_l) = (1-q^2)t_l^2$. (ii) For $0 < a \le t_k$, $\mathrm{Leb}(\mathcal{R}_k \setminus U_{-a}\mathcal{R}_k) \le 7 a t_k$. (iii) $\mathcal{R}_k \subset \{I : |I|_\infty \ge t_{k+1}\}$ and $\operatorname{diam}_\infty \mathcal{R}_k \le t_k$.*

Proof. **Step S9.43: Partition.** For $I \in (0,t_k)^2$, $M = \max_i I_i \in (0,t_k)$ and $t_l \downarrow 0$, so there is exactly one $l \geq k$ with $t_{l+1} \leq M < t_l$, i.e. $I \in \mathcal{R}_l$. $\mathrm{Leb}(\mathcal{R}_l) = t_l^2 - t_{l+1}^2$ (W7-E6).

Step S9.44: Boundary layer. If $I \in \mathcal{R}_k$ satisfies $a < I_i < t_k - a$ $(i = 1,2)$ and $\max_i I_i > t_{k+1} + a$, then a max-norm ball of radius slightly larger than a around I stays in $\mathcal{R}_k$ (all inequalities are strict), so $I \in U_{-a}\mathcal{R}_k$. The complement in $\mathcal{R}_k$ lies in four strips of measure $\leq at_k$ each and in $\{t_{k+1} \leq \max_i I_i \leq t_{k+1} + a\} \cap (0,t_k)^2$, of measure $\leq (t_{k+1}+a)^2 - t_{k+1}^2 \leq 3at_k$.

Step S9.45: Size. Immediate. □

Source: ours (KP96 p. 28 uses $|R_r \setminus C_r| \leq c\alpha^\mu r^{-2\lambda}|R_r|$ without the boundary layer).

Theorem 9.26 (Density one from shell estimates)**.** *Let q, t_k, $\mathcal{R}_k$ be as in Definition 9.24, and let $\Gamma \subset P$ be Borel with*

$$\epsilon_k := \frac{\mathrm{Leb}(\mathcal{R}_k \setminus \Gamma)}{\mathrm{Leb}(\mathcal{R}_k)} \longrightarrow 0 \qquad (k \to \infty).$$

Then, with `volume` *on* `Fin 2 -> Real` *and* $B_h =$ `Metric.ball 0 h`*,*

$$\lim_{h \downarrow 0} \frac{\mathrm{Leb}(\Gamma \cap B_h)}{\mathrm{Leb}(P \cap B_h)} = 1 \quad \text{in } [0,\infty],$$

which is clause (1) of `KAM.Wave.nlw2`*. Quantitatively, for $t_{k+1} < h \leq t_k$ the ratio lies in $[1 - q^{-2}\sup_{l \geq k} \epsilon_l,\ 1]$.*

Proof. **Step S9.46: Denominator.** For $h > 0$, $P \cap B_h = (0,h)^2$ and $\mathrm{Leb}(P \cap B_h) = h^2 \in (0,\infty)$ (W7-E6).

Step S9.47: Upper bound. $\Gamma \cap B_h \subset (0,h)^2$ since $\Gamma \subset P$; the ratio is ≤ 1.

Step S9.48: Lower bound. For $0 < h \leq t_0$ let k be the index with $t_{k+1} < h \leq t_k$. Then $(0,h)^2 \subset (0,t_k)^2 = \bigsqcup_{l \geq k} \mathcal{R}_l$ (Lemma 9.25(i)), so

$$\mathrm{Leb}\big((0,h)^2 \setminus \Gamma\big) \leq \sum_{l \geq k} \epsilon_l (1-q^2) t_l^2 \leq \bar{\epsilon}_k\, t_k^2, \qquad \bar{\epsilon}_k = \sup_{l \geq k} \epsilon_l,$$

and $t_k^2 < h^2/q^2$. Hence $\mathrm{Leb}(\Gamma \cap B_h) = h^2 - \mathrm{Leb}((0,h)^2 \setminus \Gamma) \geq h^2(1 - \bar{\epsilon}_k/q^2)$.

Step S9.49: Limit. As $h \downarrow 0$, $k \to \infty$ and $\bar{\epsilon}_k \to 0$; squeeze (W7-E7). In Lean the ratio is an `ENNReal` quotient with nonzero finite denominator, and the filter is `nhdsWithin 0 (Set.Ioi 0)`. □

Source: KP96 p. 28, Step 5 and p. 29 ("$|E \cap \Xi_{r_k}| \leq \sum_{l \geq k} |R_{r_l} \setminus C_{r_l}|$"), in the max-norm-ball form of the frozen clause (1); ours for the ratio with $P \cap B_h$.

Corollary 9.27 (Shell estimate from Theorem 9.22)**.** *Suppose that for every k there are $\alpha_k \in (0,\alpha_0]$ with $\alpha_k \leq t_k$ and a Borel set Π_k with $U_{-\alpha_k}\mathcal{R}_k \subset \Pi_k \subset (0,t_k)^2$, and an admissible family for (Π_k, α_k) with excluded set $\mathcal{E}_k$ such that $\Gamma \supset \mathcal{R}_k \cap (\Pi_k \setminus \mathcal{E}_k)$. Then*

$$\epsilon_k \leq \frac{7\alpha_k + C_7 \alpha_k^{32/47}}{(1-q^2)\, t_k} \leq \frac{(7 + C_7)\, \alpha_k^{32/47}}{(1-q^2)\, t_k}.$$

Hence $\epsilon_k \to 0$ whenever $\alpha_k^{32/47}/t_k \to 0$, and then Theorem 9.26 applies.

Proof. $\mathcal{R}_k \setminus \Gamma \subset (\mathcal{R}_k \setminus U_{-\alpha_k}\mathcal{R}_k) \cup \mathcal{E}_k$; apply Lemma 9.25(ii) and Theorem 9.22 with $\rho \leq t_k$ (Lemma 9.25(iii); $\Pi_k \subset \{|\xi|_\infty \leq \tfrac{1}{2}\}$ since $t_k \leq \tfrac{1}{2}$), and divide by $\mathrm{Leb}(\mathcal{R}_k) = (1-q^2)t_k^2$. □

Source: KP96 p. 27, (10), and p. 28, Step 5.

Remark 9.28 (Scaling with the action radius; admissible choices of α). On a box of side t in the actions (tangential amplitude $\sim\sqrt{t}$, i.e. $r(I)^2\sim t$ in the headline's $r(I)=\sqrt{I_1+I_2}$), the frequency box has side $\le N_A t$ and area $D_A t^2$, and the excluded fraction is $\le C\alpha^{32/47}/t$: the excluded set scales like $t\,\alpha^{32/47}$, not like the area t^2. In KP96's variables (p. 26, Step 3) $t=r^{2\lambda}$ with $\lambda=4/g=\frac{2}{3}$ and r the normal-domain radius, and the KAM smallness needs $\alpha\ge c_1 r^2=c_1 t^{3/2}$ (p. 27, (8)). Density needs $\alpha^{32/47}=o(t)$, i.e. $\alpha=o(t^{47/32})$. Since $\frac{3}{2}=\frac{48}{32}>\frac{47}{32}$ these are compatible, so the admissible window is $t^{3/2}\lesssim\alpha=o(t^{47/32})$; equivalently $\frac{32}{47}\cdot 2=\frac{64}{47}>2\lambda=\frac{4}{3}$, which is KP96's $\lambda<\bar\mu$ (p. 30, Step 8) and P96-W's $g=6>4+(4-\Delta)/\kappa=\frac{11}{2}$ (p. 291). An explicit choice is $\alpha(t)=t^{95/64}$, the midpoint exponent of the window (KP96's $\alpha=r^{95/48}$): then $\alpha^{32/47}/t=t^{95/94-1}=t^{1/94}\to 0$, $t^{3/2}/\alpha=t^{1/64}\to 0$, and $\alpha(t)\le t$ for $t\le 1$. The window is narrow (exponents $\frac{47}{32}$ to $\frac{48}{32}$); the earlier choice $t^{47/32}$, made for the weight exponent $\frac{1}{2}$, now sits on its upper edge and gives only $\alpha^{32/47}/t=1$, so it is no longer admissible. The choice belongs to W8 (it also controls the closeness $\Psi-\Psi_0$); W7 needs only $\alpha_k\le\min(\alpha_0,t_k)$ and $\alpha_k^{32/47}/t_k\to 0$. With P96-K Theorem B's $\mu=\frac{2}{3}$ one would need $\alpha=o(t^{3/2})$, which contradicts the KAM smallness: this is why Theorem D is used.

Source: KP96 pp. 26–30, Steps 3–8; P96-W p. 289 (Cantor Manifold Theorem) and p. 291; P96-K p. 123 (the coarse estimate is "not sufficient" for the wave equation).

9.11. **Diophantine frequencies and the frequency limit.**

Lemma 9.29 (From the $l=0$ condition to `KAM.IsDiophantine`). *Let $c>0$ and $\omega\in\mathbb{R}^2$ with $|\langle k,\omega\rangle|\ge c/A_k$ for all $k\in\mathbb{Z}^2\setminus 0$ ($A_k=1+|k|^{24}$). Then $\nu=\omega/(2\pi)$ satisfies `KAM.IsDiophantine (c/(4*pi)) 24 nu`, i.e.*

$$\frac{c/(4\pi)}{\left(\sum_i|k_i|\right)^{24}}\le\left|\sum_i k_i\nu_i\right|\qquad(k\in\mathbb{Z}^2,\ k\neq 0).$$

Proof. **Step S9.50: Base.** For $k\neq 0$, $S=\sum_i|k_i|$ is a positive integer, so $S\ge 1$, $S^{24}\ge 1$ and $A_k=1+S^{24}\le 2S^{24}$ (Lean: the real power `Real.rpow` with base $S\ge 1>0$; 24 is the real number τ).

Step S9.51: Rescale. $|\sum_i k_i\nu_i|=|\langle k,\omega\rangle|/(2\pi)\ge c/(2\pi A_k)\ge c/(4\pi S^{24})$. □

Source: ours. Frozen `KAM.IsDiophantine` (`KAM/Frozen/Defs/IsDiophantine.lean`): $\gamma/|k|_1^\tau\le|k\cdot\omega|$, $|k|_1=\sum|k_i|$, no 2π; the 2π enters through $\nu=\omega/(2\pi)$ (Convention 1.13). P96-K Remark 1, p. 122 (frequencies are Diophantine).

Remark 9.30 (Use in clause (2)). On the final parameter set of a shell, `bpW:w6:final-freq` gives $|\langle k,\omega_*\rangle|\ge\check\alpha/(2A_k)$, so Lemma 9.29 with $c=\check\alpha/2$ gives $\gamma_I=\check\alpha_k/(8\pi)>0$, $\check\alpha_k=\alpha_k^{48/47}$, for every I in that shell's part of Γ, with the single exponent $\tau=24>1$ of the headline. γ_I depends on the shell and tends to 0 as $I\to 0$, as the headline allows.

Lemma 9.31 (Frequency limit). *Let $\Gamma\subset\bigcup_k\mathcal{R}_k$ and $\omega_*:\Gamma\to\mathbb{R}^2$ with $\sup_{\Gamma\cap\mathcal{R}_k}|\omega_*-\omega^0|_\infty\le e_k$, $e_k\to 0$. Then $\nu=\omega_*/(2\pi)$ satisfies $\nu(I)\to\lambda_J/(2\pi)$ as $I\to 0$ in Γ, i.e. `Tendsto nu (nhdsWithin 0 Gamma) (nhds fun i => lam m hm (i : Nat).succPNat / (2 * pi))`.*

Proof. **Step S9.52: Shell index.** If $I\in\Gamma\cap\mathcal{R}_k$ and $|I|_\infty<t_{k_0+1}$, then $t_{k+1}\le|I|_\infty<t_{k_0+1}$ (Lemma 9.25(iii)), so $k>k_0$.

Step S9.53: Estimate. $|\nu(I)-\lambda_J/(2\pi)|_\infty\le(e_k+N_A|I|_\infty)/(2\pi)$; given $\epsilon>0$ choose k_0 with $\sup_{k>k_0}e_k<\pi\epsilon$ and then $|I|_\infty<\min(t_{k_0+1},\pi\epsilon/N_A)$. The Lean index i of `Fin 2` is the mode $i+1$ (`(i : Nat).succPNat`), so the limit vector is $(\lambda_1,\lambda_2)/(2\pi)$. □

Source: ours; KP96 p. 27, (9) ($|\tilde\omega_r-\omega|\le cr^2$) gives e_k for W8.

9.12. **Source check, deviations and suspected errata.**

Remark 9.32 (Pages read and findings). All P96-K pages cited (pp. 120–125, 131–142) and P96-W pp. 287–291 were read on the page images; KP96 pp. 22–31 on the author PDF. Findings in §§5–6 of P96-K:

(a) *Constant in Lemma 5, p. 137.* Printed: "$\{r : rv+w \in \Delta, |\phi(rv+w)| \le \delta\} \subseteq \{r : |r-r_0(w)| \le 4\delta|k|^{-1}\}$ … and hence $|\mathcal{R}^{\Delta}_{kl}(\alpha)| \le 4(\operatorname{diam}\Delta)^{n-1}\alpha\cdot\frac{\langle l\rangle_d}{A_k|k|}$ by Fubini's theorem." The r-set has length $8\delta/|k|$, and the coordinates $(r,w)\mapsto rv+w$, $v\in\{\pm1\}^n$, have Jacobian $|v|_2=\sqrt{n}$, so the factor 4 does not follow (already for $n=1$ the display gives 8). Harmless: only the form $c_3\alpha/A_k$ is used later. Our Lemma 9.14 has the factor $8\rho_\zeta/D_A$ with coordinate lines.

(b) *Symbol κ in §6, p. 140.* The display "$\ll\frac{1}{(\hat\alpha\sigma^t)^{\kappa-1}}(|X_P|^{\lambda}_{\check p,r})^{\kappa}$" uses the $\kappa=\frac43$ of §4 (p. 132, "where $\kappa=\frac43$"), not the spectral κ of Theorem D (p. 123, the unnumbered spectral-asymptotic display preceding Theorem B); only with $\frac43$ does $\hat\alpha=\alpha^{1-3w}$ yield $\check\alpha=\alpha^{1+w}$. Notational clash, not a mathematical error; it concerns W5–W6, not W7.

(c) *Lemma 10, p. 142: "the functions $\langle k,\omega'\rangle+\langle l,\Omega'\rangle$ are Lipschitz close of order α to nontrivial affine functions".* Theorem A gives sup-closeness $O(\alpha)$ but Lipschitz closeness only $|\omega_\nu-\omega|^{\mathcal{L}}\le c\epsilon M/\alpha\le c\gamma M$, a small constant, not $O(\alpha)$. What the argument needs is Lipschitz closeness small compared with the gradients of the finitely many affine functions of $\mathcal{X}$, which holds after shrinking γ depending on $\mathcal{X}$ (an explicit alternative to P96-K's Corollary C, pp. 123 and 140, which chooses K_0 large so that the exceptional zones use the unperturbed frequencies). The printed order-α assertion for the unweighted Lipschitz seminorm is not justified by these estimates. The proof needs only sufficiently small Lipschitz seminorm, made explicit here by $\theta\le\theta_0$ (Lemma 9.17, (C4)); the measure conclusion is retained.

(d) No other defect found in Lemmas 3–10, Theorem B, Corollary C, Theorem D. Checked in particular: (22) for $d=1$ reads $\tau\ge(n+3)\frac{\delta-1}{\delta}$, which for $\delta<0$ equals $(n+3)/\delta^*$ as the proof needs; the "whichever gives the better estimate" choice of j_0 in Lemma 8 does give $\alpha^{\kappa^*}A_k^{-\delta^*}$ when the two choices are used in complementary ranges of A_k; the exponent algebra of p. 142 ($w=\varpi/(4\kappa+4-\varpi)$, $\tilde\mu=\kappa/(\kappa+1-\varpi/4)$); and $\alpha\le\check\alpha^{\mu}$, which lets p. 142 drop the term α of Lemma 10.

Deviations introduced here: Dirichlet labels and $\langle l\rangle$ instead of P96-K's reindexed weights (Remark 9.2); coordinate-line slicing on the domain of the frequency maps instead of oblique lines after a Lipschitz extension; explicit constants and the fixed choices $\varpi=\frac12$, $\tau=24$; the weight exponent $\beta=\frac14$ of the modified divisors, distinct from the loss $\varpi=\frac12$ (P96-K takes them equal; with $\beta=\varpi$ the parameter-Lipschitz bound of the preparatory step fails, W5 SOURCE_GAP after Lemma 7.6), hence $w=\frac{1}{47}$ and $\tilde\mu=\frac{32}{47}$; a single drift hypothesis (H2) combining the preparatory shift (j^{-1}) and the iteration drift ($j^{-1/2}$); our own choices of j_0; the boundary layer of Lemma 9.25(ii); measurability (not discussed in the sources).

9.13. **Summary of the interface.**

Remark 9.33 (Consumed and supplied). *Consumed:* Theorem 2.10 (A, B), Theorem 2.13 ((i) $\det A$, (iii) (A_3)), Remark 2.14 ($\kappa=2$, re-proved quantitatively in Lemma 9.5); forward from W6: `bpW:w6:prep-step`, `bpW:w6:freq-lip`, `bpW:w6:exclusion`, `bpW:w6:final-freq` (Remark 9.10). *Supplied:* to W5–W6, the constants $\tau=24$, $\varpi=\frac12$, $\beta=\frac14$, $w=\frac{1}{47}$, $\hat\alpha=\alpha^{44/47}$, $\check\alpha=\alpha^{48/47}$, θ_0, α_0, the zone definitions and Corollary 9.23 (in particular the $k=0$ conditions); to W8, Theorem 9.22, Theorem 9.26 with Corollary 9.27 (clause (1)), Lemma 9.29 and Lemma 9.31 (clause (2)), Lemma 9.21 (Borel Γ), and Remark 9.28 (admissible $\alpha(t)$).

10. Milestone W8: shells, patching, family injectivity, and the proof of W-NLW2

This part assembles the frozen headline `KAM.Wave.nlw2` (Theorem 11.5) from milestones W0–W7. The KAM theorem of W5–W6 runs on one parameter domain at a time and yields, on a Borel set of actions ξ of almost full measure in that domain, invariant tori close to the linear tori. A single domain cannot give density one at the origin (W7, Remark 9.28), so, following KP96 §7, Step 5, the theorem is applied on a sequence of dyadic action shells, each trimmed by a margin equal to its own divisor level α_n, and the outputs are pieced together. Because the trimmed shells are open, pairwise disjoint and at positive distance from one another, the patched objects are continuous wherever each piece is: no compatibility across shell boundaries is needed.

What W8 adds to W6–W7. Remark 8.15 lists three items.

(1) *Embedding.* Cauchy estimates on the strip turn the angle error X into a $\frac{1}{2}$-Lipschitz periodic function; with the injectivity of W3's map $\Gamma^{\mathbb{R}}$ this gives injectivity modulo $\mathbb{Z}^2$ and injective derivatives (Lemma 10.11).

(2) $o(r)$ *closeness.* W6 states the closeness through $e_{\mathrm{tor}} = M_{F,P} + e_\Phi$, with $e_\Phi = 5c_F\epsilon_0/(\check\alpha\sigma_0^{51})$ for a starting size $\epsilon_0 \in [\epsilon_0^\flat, \gamma_1\check\alpha]$ that is a parameter of Definition 8.9 (Remark 8.15, "Starting size"), $\epsilon_0^\flat$ being the sharp post-preparatory size of Lemma 8.8(v). With the upper value $\epsilon_0 = \gamma_1\check\alpha$ the term e_Φ does not depend on the shell, while closeness of order $o(r)$ needs $e_{\mathrm{tor}} \to 0$ along the shells. We give the parameter a shell-dependent value $\epsilon_{0,n}$ in that interval, small relative to $\check\alpha_n$ (Lemma 10.4, Lemma 10.6); then $e_{\mathrm{tor}} \lesssim t_n^{1/64}$ (Lemma 10.7).

(3) *Patching and family injectivity.* Two tori that meet carry the same orbit, by uniqueness for the coefficient system (Lemma 10.18). In Birkhoff coordinates the phase of each tangential mode along that orbit winds at the rate ν_i, so the two frequencies coincide (Lemma 10.19). The frequency map is injective on the patched parameter set (Lemma 10.20): within a shell because the drift $\omega_* - \omega^0$ is θ_0-Lipschitz with $\theta_0 L_A \le \frac{1}{8}$, and across shells because of the margins.

The source obtains injectivity differently, from the Lipschitz estimates of the family in ξ (KP96 §7, Step 6, (11)); W6 proves only continuity in ξ (the Lipschitz family is a declared weakening, Remark 11.7), so W8 uses the frequency route above.

Standing data and notation. Standing assumption 1.1 ($m > 0$) and the conventions of §1 hold; $a = p = 1$. Vectors in $\mathbb{R}^2$ and $\mathbb{C}^2$ carry the max norm $|\cdot|$; Leb is `volume` on `Fin 2 -> Real`; the Lean index $i \in$ `Fin 2` is the Dirichlet mode $i+1$. Shells are indexed by $n \ge 0$ (W7's k); W6's iteration index is written μ; ν always denotes a frequency vector. The KAM exponent $2\tau + 4 = 52$ of W5–W6 is written as a number. From earlier parts we use: W3's ρ_1, C_*, C_K', C_F (§5.9), the maps $\mathcal{Z}$ (Lemma 5.3), Γ and $\Gamma^{\mathbb{R}} = \mathcal{Z}^{-1}\Gamma\mathcal{Z}$ (Theorem 5.25, Corollary 5.27) and $r_1 = \rho_1/\sqrt{2}$; W5's $c_A = 2200$ and $\mathcal{A}_\xi$ (Lemma 7.18); W6's $s = \frac{1}{8}$, $\sigma_P = s/40$, $s_0 = \frac{7}{8}s = \frac{7}{64}$, $\sigma_0 = s_0/40$, M, γ_0, γ_1, γ (Definition 8.7), c_F (iteration steps) and $c_F^\natural$, $C_2^\natural$, $t^\natural = 53$ (preparatory step) (§7.9); W7's $\tau = 24$, θ_0, α_0, C_7, L_A, N_A, ω^0, Ω^0 (Definitions 9.6, 9.3, Lemma 9.4); W2's M_m (Theorem 4.8); the transport $\tau_E = \tau_{1,1} \times \tau_{1,1} : E_{1,1} \to$ `WaveSpace`, a linear isometric bijection (Lemma 3.2(iv)) whose inverse $\kappa_{1,1} \times \kappa_{1,1}$ is `coeff` in each factor. For $I \in P = (0,\infty)^2$, $r(I) = \sqrt{I_1 + I_2}$ (Lean `Real.sqrt (I 0 + I 1)`). The frozen `linearTorus I hI` $= \tau_E \circ K_I^0$ (Definition 11.1, Convention 1.13) does not depend on the proof `hI`.

10.1. External facts.

(W8-E1) *Inner products along curves.* If $c_1, c_2 : \mathbb{R} \to \ell^2_{\mathbb{R}}$ are differentiable at t, then $s \mapsto \langle c_1(s), c_2(s)\rangle$ has derivative $\langle c_1'(t), c_2(t)\rangle + \langle c_1(t), c_2'(t)\rangle$ (`HasDerivAt.inner`); $\langle f, g\rangle = \sum_j f_j g_j$ for $f, g \in \ell^2_{\mathbb{R}}$ with absolutely summable terms (`lp.inner_eq_tsum`, `lp.summable_inner`); termwise sums and differences of summable series (`Summable.tsum_add`, `Summable.tsum_sub`); Cauchy–Schwarz (`real_inner_le_norm`).

(W8-E2) *Sign of the derivative.* A real function differentiable on an interval with nonpositive (nonnegative) derivative is antitone (monotone) there (`antitoneOn_of_deriv_nonpos`, `monotoneOn_of_deriv_nonneg`); derivatives of products and of $t \mapsto e^{ct}$ (`HasDerivAt.mul`, `Real.hasDerivAt_exp`).

(W8-E3) *Compactness.* A continuous map on $[-T, T]$ is bounded there (`IsCompact.exists_bound_of_continuousOn`, `isCompact_Icc`); analytic maps are continuous (`AnalyticAt.continuousAt`).

(W8-E4) *Complex exponential and integers.* $e^z = 1$ iff $z \in 2\pi i\mathbb{Z}$ (`Complex.exp_eq_one_iff`); $|e^{ix}| = 1$ and $|e^{ix} - 1| \le |x|$ for real x (`Complex.norm_exp_ofReal_mul_I`, `Real.norm_exp_I_mul_ofReal_sub_one_le`); a continuous function from a connected space ($\mathbb{R}$ or $\mathbb{R}^2$) with values in $\mathbb{Z} \subset \mathbb{R}$ is constant (`IsPreconnected.constant`, with `isPreconnected_univ`); if $t \mapsto tc$ is bounded on $\mathbb{R}$ then $c = 0$.

(W8-E5) *Square roots.* For $a > 0$ and $a + b \ge 0$: $|\sqrt{a+b} - \sqrt{a}| \le |b|/\sqrt{a}$ (from $(\sqrt{a+b} - \sqrt{a})(\sqrt{a+b} + \sqrt{a}) = b$; `Real.sq_sqrt`, `Real.sqrt_pos`).

(W8-E6) *Distance to a set.* $x \mapsto \operatorname{dist}(x, S)$ is 1-Lipschitz (`Metric.continuous_infDist_pt`), so $\{x : \operatorname{dist}(x, S) > a\}$ is open; it is 0 on S (`Metric.infDist_zero_of_mem`) and $\le |x - y|$ for $y \in S$ (`Metric.infDist_le_dist_of_mem`).

(W8-E7) *Real powers.* For $x > 0$: $x^a x^b = x^{a+b}$ (`Real.rpow_add`); $x \mapsto x^a$ is monotone on $[0, \infty)$ for $a \ge 0$ and $x^a \le x^b$ for $0 < x \le 1$, $a \ge b$ (`Real.rpow_le_rpow`, `Real.rpow_le_rpow_of_exponent_ge`); $x^a \ge 0$ for $x \ge 0$ (`Real.rpow_nonneg`); $x \mapsto x^a$ is continuous at 0 with value 0 for $a > 0$ (`Real.continuousAt_rpow_const`, `Real.zero_rpow`), hence tends to 0 along `nhdsWithin 0 (Set.Ioi 0)` (`nhdsWithin_le_nhds`).

(W8-E8) *Local definitions.* A map is continuous on S if each point of S has an open neighbourhood O such that it is continuous on $S \cap O$ (`continuousOn_of_locally_continuousOn`), and continuity on a set transfers to any map agreeing with it there (`ContinuousOn.congr`); continuous linear maps preserve `AnalyticOnNhd` (`ContinuousLinearMap.comp_analyticOnNhd`) and compose with `fderiv` by the chain rule (`fderiv_comp`, `ContinuousLinearMap.fderiv`); an inequality between limits of convergent sequences passes to the limit (`le_of_tendsto_of_tendsto'`).

10.2. **Shell data.** All numbers below are fixed after m, in the order written, and depend on m only.

Definition 10.1 (W8 constants)**.** With the constants listed in the standing data, put

$$E_1 = 1 + 2C_* + 6(2c_A)^{5/2}C'_K, \qquad E_3 = \Big(1 + \frac{8}{M}\Big)E_1, \qquad E_4 = c_F^{\natural}\gamma\,\sigma_P^{-52} + 5c_F\gamma_1\sigma_0^{-51},$$

$$E_5 = \frac{E_3E_4}{\gamma}, \qquad E_6 = 6\sqrt{2}\,C_F(2c_A)^{3/2}, \qquad E_7 = 162E_5 + E_6,$$

$$t_0 = \min\Big\{4^{-64},\ \alpha_0^{64/95},\ \frac{\rho_1^2}{8c_A},\ \Big(\frac{\gamma}{E_3}\Big)^{64},\ \Big(\frac{7}{256E_5}\Big)^{64}\Big\},$$

and the headline data

$$\tau = 24, \qquad r_0 = \sqrt{2t_0}, \qquad C_3 = 4\sqrt{c_A}, \qquad C = \max(1, M_m)\,C_3, \qquad \eta(r) = 2^{33/64}E_7\,r^{1/32}$$

(Lean: `Real.rpow`; only $r > 0$ matters).

Source: ours. The shell exponents follow KP96 §7: $t = r^{2\lambda}$ with $\lambda = \frac{2}{3}$ (p. 26, Step 3), i.e. the normal radius $r^2 = t^{3/2}$, and $\alpha = t^{95/64}$ (W7 Remark 9.28; KP96 p. 30, Step 8).

Definition 10.2 (Shells)**.** For $n \geq 0$ let $q = \frac{1}{2}$, $t_n = 2^{-n}t_0$, and $\mathcal{R}_n = (0,t_n)^2 \setminus (0,t_{n+1})^2$ (W7 Definition 9.24 with $q = \frac{1}{2}$). Put

$$\alpha_n = t_n^{95/64}, \quad r_n = t_n^{3/4} \text{ (so } r_n^2 = t_n^{3/2}), \quad \rho_{\xi,n} = \frac{\alpha_n}{2}, \quad \rho_n = \big(c_A(t_n + r_n^2)\big)^{1/2},$$

$$\epsilon_n^\sharp = E_3 t_n^{3/2}, \quad \kappa_n = \frac{\epsilon_n^\sharp}{\gamma\alpha_n} = \frac{E_3}{\gamma} t_n^{1/64}, \quad \epsilon_{0,n} = \kappa_n^{4/3}\gamma_1\check{\alpha}_n \ (\check{\alpha}_n = \alpha_n^{48/47}), \quad \hat{\alpha}_n = \alpha_n^{44/47},$$

and the trimmed shell $\Pi_n = U_{-\alpha_n}\mathcal{R}_n = \{\xi \in \mathcal{R}_n : \operatorname{dist}(\xi, \mathbb{R}^2 \setminus \mathcal{R}_n) > \alpha_n\}$.

Source: KP96 p. 28, Step 5 ($R_r = \Xi_r \setminus \Xi_{\eta r}$, $R_r^- = U_{-\alpha}R_r$, $r_k = \eta^k r_0$; "the C_{r_k} are disjoint and even have a pairwise positive distance to each other"); W7 Definition 9.24.

Lemma 10.3 (Shell data are admissible)**.** *For every $n \geq 0$:*

- *(i)* $0 < t_n \leq t_0 \leq 4^{-64}$; $\alpha_n \leq \min(\alpha_0, t_n)$; $r_n \leq 1$; $r_n^2 \leq \alpha_n/4$; $\kappa_n \leq 1$, *so* $\epsilon_n^\sharp \leq \gamma\alpha_n$; *and* $\rho_n^2 \leq 2c_A t_n < \rho_1^2/2 = r_1^2$.
- *(ii)* Π_n *is open (hence Borel),* $\Pi_n \subset \mathcal{R}_n \subset (0,t_n)^2 \subset \{|\xi| \leq \frac{1}{2}\}$, *and every* $\xi \in \Pi_n$ *has* $\alpha_n < \xi_i < t_n - \alpha_n$ $(i = 1,2)$. *If* $\xi \in \Pi_n$ *and* $\xi' \notin \mathcal{R}_n$ *(in particular* $\xi' \in \mathcal{R}_{n'}$, $n' \neq n$*), then* $|\xi - \xi'| > \alpha_n$.
- *(iii) The hypotheses of Proposition 7.19 hold with* $s = \frac{1}{8}$, $r = r_n$, $t = t_n$, $\rho_\xi = \rho_{\xi,n}$, $\Pi = \Pi_n$, *and its* ϵ_{in} *satisfies* $\epsilon_{\text{in}}(1 + 8/M) \leq \epsilon_n^\sharp$; *so the initial perturbation* P *satisfies* $|X_P|^\lambda_{r_n, D(s,r_n)} \leq \epsilon_n^\sharp$ *for* $\lambda = \alpha_n/M$.
- *(iv) The hypotheses of Lemma 8.8 hold with* $\Pi = \Pi_n$, $\alpha = \alpha_n$, $r = r_n$, $\epsilon = \epsilon_n^\sharp$ *(in particular* $0 < \epsilon_n^\sharp \leq \gamma\alpha_n$*), and those of Remark 8.15* ($c_A(t + r^2) < \rho_1^2$).

Proof. **Step S10.1: Sizes.** $t_n \leq t_0 \leq 4^{-64} \leq 1$. $\alpha_n = t_n^{95/64} \leq t_0^{95/64} \leq \alpha_0$ and $\alpha_n \leq t_n$ since $t_n \leq 1$ (W8-E7). $r_n^2 = t_n^{96/64} = t_n^{1/64}\alpha_n \leq 4^{-1}\alpha_n$ since $t_n^{1/64} \leq t_0^{1/64} \leq \frac{1}{4}$. $\kappa_n \leq (E_3/\gamma)t_0^{1/64} \leq 1$ by the fourth entry of t_0. $\rho_n^2 = c_A(t_n + r_n^2) \leq 2c_A t_n \leq 2c_A t_0 \leq \rho_1^2/4 < \rho_1^2/2$ by the third entry. This is (i).

Step S10.2: Trimmed shell. Π_n is open by (W8-E6), and contained in $\mathcal{R}_n$ because the distance vanishes on $\mathbb{R}^2 \setminus \mathcal{R}_n$. For $\xi \in \Pi_n$ the points $\xi - \xi_i e_i$ and $\xi + (t_n - \xi_i)e_i$ lie outside $\mathcal{R}_n \subset (0,t_n)^2$, at distances ξ_i and $t_n - \xi_i$ (if positive), so both exceed α_n (W8-E6). For $\xi' \notin \mathcal{R}_n$, $|\xi - \xi'| \geq \operatorname{dist}(\xi, \mathbb{R}^2 \setminus \mathcal{R}_n) > \alpha_n$. Shells are disjoint by Lemma 9.25(i). $(0,t_n)^2 \subset \{|\xi| \leq \frac{1}{2}\}$ as $t_n \leq \frac{1}{2}$. This is (ii).

Step S10.3: Proposition 7.19. Let $\xi \in \Pi_n$ and $\zeta \in \mathbb{C}^2$ with $|\zeta - \xi| < \alpha_n/2$. Then $\operatorname{Re}\zeta_i > \xi_i - \alpha_n/2 > \alpha_n/2 \geq 2r_n^2$ and $|\zeta_i| < t_n - \alpha_n + \alpha_n/2 < t_n$ by (ii); $c_A(t_n + r_n^2) < \rho_1^2$ by (i); $0 < r_n \leq 1$. With $t_n + r_n^2 \leq 2t_n$, $r_n = t_n^{3/4}$ and $t_n^{-3/4} \leq t_n^{-1}$,

$$\epsilon_{\text{in}} \leq (1 + 2C_*)t_n^{3/2} + C_K'(2c_A t_n)^{5/2}\big(3t_n^{-3/4} + 2\sqrt{2}\,t_n^{-1}\big) \leq \big(1 + 2C_* + 6(2c_A)^{5/2}C_K'\big)t_n^{3/2} = E_1 t_n^{3/2}.$$

With $\lambda = \alpha_n/M$, $4\lambda/\rho_{\xi,n} = 8/M$, so Proposition 7.19 gives $|X_P|^\lambda \leq (1 + 8/M)E_1 t_n^{3/2} = \epsilon_n^\sharp$. This is (iii).

Step S10.4: Lemma 8.8. Π_n is Borel in $\{|\xi| \leq \frac{1}{2}\}$, $0 < \alpha_n \leq \alpha_0$; the majorant $\epsilon = \epsilon_n^\sharp = E_3 t_n^{3/2}$ is positive, since $t_n > 0$ and $E_3 \geq E_1 \geq 1$, which is the positivity hypothesis $0 < \epsilon$ of Lemma 8.8; P is real periodic in $\mathcal{S}_{p+1}(D(s,r_n))$ (Proposition 7.19) with $|X_P|^{\alpha_n/M} \leq \epsilon_n^\sharp \leq \gamma\alpha_n$ by (i), (iii). This is (iv). □

Source: ours; checks the conditions listed in Remark 8.15 and Proposition 7.19 (KP96 p. 26, Step 3: parameter domain $U_{-4r^2}\Xi_r$, here the margin $\alpha_n \geq 4r_n^2$).

10.3. **The KAM iteration on one shell.**

Lemma 10.4 (Sharp size after the preparatory step)**.** *Fix n and run Lemma 8.8 with the data of Lemma 10.3(iv), in particular with the positive majorant $\epsilon = \epsilon_n^\sharp$. Then $\eta_P = (\epsilon_n^\sharp/(\hat\alpha_n\sigma_P^{53}))^{1/3}$ and the new perturbation satisfies, with $\lambda_0 = \check\alpha_n/(2M)$,*

$$|X_{P_0}|^{\lambda_0}_{r_0,D_0} \le \epsilon_{0,n}^\flat := 2C_2^\natural(\epsilon_n^\sharp)^{4/3}(\hat\alpha_n\sigma_P^{53})^{-1/3} \le \epsilon_{0,n} \le \gamma_1\check\alpha_n,$$

and $\epsilon_{0,n}^\flat > 0$.

Proof. **Step S10.5: The sharp bound.** Lemma 8.8(v) (the sharp size, Lipschitz part included) with $\epsilon = \epsilon_n^\sharp$ and $\alpha = \alpha_n$; its $\epsilon_0^\flat$ is $\epsilon_{0,n}^\flat$, positive because $\epsilon_n^\sharp > 0$.

Step S10.6: Comparison. With $\epsilon_n^\sharp = \kappa_n\gamma\alpha_n$ and $\frac{4}{3}-\frac{1}{3}\cdot\frac{44}{47} = \frac{48}{47}$, $\epsilon_{0,n}^\flat = \kappa_n^{4/3}\cdot 2C_2^\natural\gamma^{4/3}\sigma_P^{-53/3}\check\alpha_n$, and $2C_2^\natural\gamma^{4/3}\sigma_P^{-53/3} \le \gamma_1$ because $\gamma \le (\gamma_1\sigma_P^{53/3}/(2C_2^\natural))^{3/4}$ (Definition 8.7). Hence $\epsilon_{0,n}^\flat \le \kappa_n^{4/3}\gamma_1\check\alpha_n = \epsilon_{0,n}$, and $\epsilon_{0,n} \le \gamma_1\check\alpha_n$ as $\kappa_n \le 1$ (Lemma 10.3(i)). □

Source: P96-K §6, p. 141 ("$|X_{P_0}|^\lambda_{\check p,\eta r} \preccurlyeq \frac{1}{(\hat\alpha\sigma^t)^{\varkappa-1}}(|X_P|^\lambda_{\check p,r})^\varkappa$"), in the form of Lemma 8.8(v).

Definition 10.5 (The shell iteration)**.** For $n \ge 0$, the *shell iteration* is Definition 8.9 run with the shell data of Lemma 10.3(iv) and with the starting-size parameter $\epsilon_0 = \epsilon_{0,n}$, which lies in the admissible interval $[\epsilon_{0,n}^\flat, \gamma_1\check\alpha_n]$ by Lemma 10.4 (the other entries are W6's: $r_0 = \eta_P r_n$, $\alpha_0^{\mathrm{it}} = \check\alpha_n$, $M_0 = 2M$, $\Pi_{-1} = \Pi_n \setminus \Xi$, $\beta_0 = \beta_*$). Its outputs are written with a superscript (n): $\Pi_*^{(n)}$, $\omega_*^{(n)}$, $\nu_*^{(n)} = \omega_*^{(n)}/(2\pi)$, $\Phi_*^{(n)}$, $\Phi_P^{(n)}$, $\tilde\Psi_\xi^{(n)}$ (Theorem 8.14), the admissible family $(D_\mu^{(n)}, \omega_\mu^{(n)}, \Omega_\mu^{(n)}, K_\mu)_\mu$ and its excluded set $\mathcal{E}^{(n)}$ (Definition 9.9), and

$$e_{\mathrm{tor}}^{(n)} = \frac{c_F^\natural\epsilon_n^\sharp}{\hat\alpha_n\sigma_P^{52}} + \frac{5c_F\epsilon_{0,n}}{\check\alpha_n\sigma_0^{51}}, \qquad \varepsilon_\omega^{(n)} = 6\pi\gamma\alpha_n + 8\pi\epsilon_{0,n}.$$

Put $\Gamma_n = \Pi_*^{(n)}$.

Source: ours (W6 Definition 8.9 with a shell-dependent value of its parameter ϵ_0).

Lemma 10.6 (W6 with the sharp starting size)**.** *For every n, the conclusions of Lemma 8.10, Lemma 8.11 (with $\varepsilon_\omega = \varepsilon_\omega^{(n)}$, $\varepsilon_1 = 3\gamma\alpha_n$, $\varepsilon_2 = 4\epsilon_{0,n}$, the same $\theta = 425(1+M)\max(\gamma,\gamma_1) \le \theta_0$), Lemma 8.12, Lemma 8.13 and Theorem 8.14 hold for the shell iteration, with $e_{\mathrm{tor}} = e_{\mathrm{tor}}^{(n)}$ in Theorem 8.14(iv).*

Proof. **Step S10.7: Applicability.** Lemma 8.10 holds for every value of the parameter $\epsilon_0 \in [\epsilon_0^\flat, \gamma_1\check\alpha]$ of Definition 8.9; $\epsilon_{0,n}$ is such a value (Lemma 10.4).

Step S10.8: Frequencies. Lemma 8.11 for this value of the parameter, with $\varepsilon_\omega = 6\pi\gamma\alpha_n + 8\pi\epsilon_{0,n}$, $\varepsilon_2 = 4\epsilon_{0,n}$, and θ independent of ϵ_0.

Step S10.9: Exclusion and final frequencies. Lemmas 8.12 and 8.13 for this value.

Step S10.10: Tori. Theorem 8.14 for this value; in (iv) $e_{\mathrm{tor}} = c_F^\natural\epsilon_n^\sharp/(\hat\alpha_n\sigma_P^{52}) + 5c_F\epsilon_{0,n}/(\check\alpha_n\sigma_0^{51}) = e_{\mathrm{tor}}^{(n)}$. □

Source: W6 Definition 8.9 and Remark 8.15 ("Starting size": the results hold for every value of the parameter $\epsilon_0 \in [\epsilon_0^\flat, \gamma_1\check\alpha]$).

Lemma 10.7 (Shell errors tend to zero)**.** *For every n: $e_{\mathrm{tor}}^{(n)} \le E_4\kappa_n = E_5 t_n^{1/64} \le \frac{7}{256}$, and $\varepsilon_\omega^{(n)} \le \alpha_n/(70L_A) \le \alpha_n$.*

Proof. **Step S10.11:** e_{tor}. $c_F^\natural \epsilon_n^\sharp/(\hat\alpha_n \sigma_P^{52}) = c_F^\natural \sigma_P^{-52}\gamma\kappa_n \alpha_n^{3/47} \le c_F^\natural \sigma_P^{-52}\gamma\kappa_n$ and $5c_F\epsilon_{0,n}/(\check\alpha_n\sigma_0^{51}) = 5c_F\sigma_0^{-51}\gamma_1\kappa_n^{4/3} \le 5c_F\sigma_0^{-51}\gamma_1\kappa_n$ ($\alpha_n, \kappa_n \le 1$). Sum: $E_4\kappa_n = E_4(E_3/\gamma)t_n^{1/64} = E_5 t_n^{1/64} \le E_5 t_0^{1/64} \le \frac{7}{256}$ by the last entry of t_0.

Step S10.12: ε_ω. $\epsilon_{0,n} \le \gamma_1\check\alpha_n \le \gamma_1\alpha_n$, so $\varepsilon_\omega^{(n)} \le (6\pi\gamma + 8\pi\gamma_1)\alpha_n \le \frac{14\pi\theta_0}{425}\alpha_n$, since $\gamma, \gamma_1 \le \theta_0/(425(1+M))$ (Definition 8.7); and $\theta_0 \le 1/(8L_A)$ (W7 (C4)) gives $\frac{14\pi}{3400 L_A} < \frac{1}{70 L_A}$. Finally $1/(70L_A) \le 1$ because $L_A = N_A/D_A \ge \frac{8\pi}{21}\lambda_2(4\lambda_1 + 3\lambda_2) > 1$ (Lemma 9.4(ii)). □

Source: ours. Compare KP96 p. 27, (9): $|\Phi_r - \Phi_0|_r \le cr^2/\alpha$, which tends to 0 along the shells; our $\kappa_n \approx \epsilon_n^\sharp/\alpha_n$ plays the role of r^2/α.

Lemma 10.8 (Frequencies on one shell)**.** *For every n and $\xi, \xi' \in \Gamma_n$:*

(i) $|(\omega_*^{(n)} - \omega^0)(\xi) - (\omega_*^{(n)} - \omega^0)(\xi')| \le \theta_0|\xi - \xi'|$;
(ii) $|\omega_*^{(n)}(\xi) - \omega_*^{(n)}(\xi')| \ge \frac{7}{8L_A}|\xi - \xi'|$;
(iii) $|\omega_*^{(n)}(\xi) - \omega^0(\xi)| \le \varepsilon_\omega^{(n)} \le \alpha_n/(70L_A)$, *and* $|\langle k, \omega_*^{(n)}(\xi)\rangle| \ge \check\alpha_n/(2A_k)$ *for* $k \in \mathbb{Z}^2 \setminus 0$;
(iv) $\nu_*^{(n)}$ *is Lipschitz (in particular continuous) on* Γ_n, *and* Γ_n *is Borel.*

Proof. **Step S10.13: Limit of (H1).** $\Gamma_n = \bigcap_\mu \Pi_\mu \subset D_\mu^{(n)} = \Pi_{\mu-1}$ for all μ, so (H1) of Lemma 10.6 applies to ξ, ξ' at every stage with $\theta \le \theta_0$; $\omega_\mu^{(n)} \to \omega_*^{(n)}$ on Γ_n (Proposition 8.4(ii)); pass to the limit (W8-E8). This is (i).

Step S10.14: Lower bound. By Lemma 9.4(iv), $|\omega^0(\xi) - \omega^0(\xi')| \ge |\xi - \xi'|/L_A$ (here Theorem 2.13(i), $\det A \ne 0$, enters); subtract (i) and use $\theta_0 \le 1/(8L_A)$. This is (ii).

Step S10.15: Size and divisors. Lemma 8.13 via Lemma 10.6, and Lemma 10.7. This is (iii).

Step S10.16: Regularity. Theorem 8.14(ii) via Lemma 10.6; Borel by Proposition 8.4(i), with $\Pi_n \setminus \Xi$ Borel (Lemma 10.3(ii), Lemma 9.21) and the frequencies continuous (Step S8.25). This is (iv). □

Source: P96-K Theorem A, p. 122 (Lipschitz closeness of the frequencies), W7 (H1); KP96 p. 27, (9). The injectivity (ii) is ours.

10.4. **The tori of one shell.**

Lemma 10.9 (Shape of a shell torus)**.** *Let $n \ge 0$, $\xi \in \Gamma_n$, $\Psi = \tilde\Psi_\xi^{(n)}$, and write $\Phi_P^{(n)}\Phi_*^{(n)}(\theta) = (\theta + X(\theta), y(\theta), u(\theta), v(\theta))$ on $T_{s_0/2}$ (Theorem 8.14(iv)), $e = e_{\mathrm{tor}}^{(n)}$. Then:*

(i) X, y are holomorphic on $T_{s_0/2}$ with $|X| \le e$, $|y| \le r_n^2 e$, and the normal part of Ψ has $\|\hat\Psi(\theta)\| \le r_n e$ there;
(ii) for real θ, $y(\theta)$ and $X(\theta)$ are real, $\xi_i + y_i(\theta) \ge \frac34\alpha_n > 0$, and

$$\Psi_{\pm i}(\theta) = \sqrt{\xi_i + y_i(\theta)}\, e^{\mp 2\pi i(\theta_i + X_i(\theta))} \qquad (i = 1, 2)$$

with the positive square root;
(iii) $X(\theta + k) = X(\theta)$ for real θ and $k \in \mathbb{Z}^2$;
(iv) for real θ the real derivative satisfies $\|DX(\theta)\| \le \frac{128e}{7} \le \frac12$ (operator norm for the max norm), and $|X(\theta) - X(\theta')| \le \frac12|\theta - \theta'|$ for real θ, θ';
(v) $\|\Psi(\theta)\| < \rho_n \le r_1$ for real θ; hence $x(\theta) := \mathcal{Z}^{-1}\Psi(\theta) \in B_{r_1}^E$ is real.

Proof. **Step S10.17: Bounds.** (i) is Theorem 8.14(iv) (Lemma 10.6); X and y are components of the holomorphic map $\Phi_P^{(n)}\Phi_*^{(n)} - \iota$, and the normal part of $\mathcal{A}_\xi$ is linear in (u, v) with $\|\hat w\| \le \|u\| + \|v\|$ (Lemma 7.18(vi)), the $(a, p + \frac12)$-norms dominating the (a, p)-norms.

Step S10.18: Reality. Let θ be real. By Theorem 8.14(i), $\varrho\Psi = \Psi$, i.e. $\Psi_{-i} = \overline{\Psi_{+i}}$ (Lemma 5.3). By the formula of Theorem 8.14(iv), $\Psi_{+i}\Psi_{-i} = \xi_i + y_i$, so $\xi_i + y_i = |\Psi_{+i}|^2$ is real and y_i is real. Then $\xi_i + y_i \geq \alpha_n - r_n^2 e \geq \frac{3}{4}\alpha_n$ (Lemma 10.3(i),(ii), $e \leq 1$), the principal root is the positive root, and $\overline{\Psi_{+i}} = \Psi_{-i}$ gives $e^{2\pi i(\theta_i + \overline{X_i})} = e^{2\pi i(\theta_i + X_i)}$, i.e. $e^{-4\pi \operatorname{Im} X_i} = 1$ in modulus, so $\operatorname{Im} X_i = 0$ (W8-E4). This is (ii).

Step S10.19: Periodicity. Ψ is 1-periodic (Theorem 8.14(i)). By (ii) at θ and $\theta + k$, the phases agree: $e^{-2\pi i(k_i + X_i(\theta+k) - X_i(\theta))} = 1$, so $X_i(\theta + k) - X_i(\theta) \in \mathbb{Z}$ (W8-E4). As a function of real θ this is continuous and integer valued, hence constant (W8-E4), and of modulus $\leq 2e < 1$; so it is 0. This is (iii).

Step S10.20: Cauchy. For real θ, the ball $B(\theta, s_0/2) \subset \mathbb{C}^2$ lies in $T_{s_0/2}$, X is holomorphic there and bounded by e; Lemma 3.25 gives $\|DX(\theta)\| \leq e/(s_0/2) = 128e/7 \leq \frac{1}{2}$ (Lemma 10.7), and the Lipschitz bound on the convex set $\mathbb{R}^2$, with $\mathbb{R}^2 + B(0, s_0/2) = T_{s_0/2}$. The real derivative of $X|_{\mathbb{R}^2}$ is the restriction of the complex one. This is (iv).

Step S10.21: Ball. $\Phi_P^{(n)}\Phi_*^{(n)}(\theta) \in D(s, r_n)$ and $\mathcal{A}_\xi(D(s, r_n)) \subset B_{\rho_n}$ (Lemma 7.18(ii), $|\xi| \leq t_n$, $\xi_i \geq 2r_n^2$). $\rho_n \leq r_1$ by Lemma 10.3(i); $\|\mathcal{Z}^{-1}w\|_E \leq \|w\|$ and $\mathcal{Z}^{-1}$ maps real points to E (Lemma 5.3). This is (v). □

Source: ours; W6 Remark 8.15, item (1) ("Cauchy estimates on the strip turn e_{tor} into C^1 bounds on X"); KP96 §7, Step 7 ("X is close to the identity on $|\operatorname{Im} x| < \frac{1}{2}$").

Definition 10.10 (Shell embeddings)**.** For $n \geq 0$, $\xi \in \Gamma_n$ and $\theta \in \mathbb{R}^2$ put

$$K_\xi^{(n)}(\theta) = \tau_E\Big(\Gamma^{\mathbb{R}}\big(\mathcal{Z}^{-1}\tilde{\Psi}_\xi^{(n)}(\theta)\big)\Big) = \tau_E\big(\mathcal{Z}^{-1}\Gamma(\tilde{\Psi}_\xi^{(n)}(\theta))\big) \in \texttt{WaveSpace},$$

well defined by Lemma 10.9(v) and Corollary 5.27.

Source: P96-W p. 290 ("These tori carry the quasi-periodic motions $\gamma = \Gamma \circ \Psi$"); W6 Remark 8.15, return to Cartesian coordinates.

Lemma 10.11 (Each shell torus is an analytic torus embedding)**.** *For $n \geq 0$ and $\xi \in \Gamma_n$, `IsAnalyticTorusEmbedding` $K_\xi^{(n)}$ holds.*

Proof. Write $\Psi = \tilde{\Psi}_\xi^{(n)}$, $x = \mathcal{Z}^{-1}\Psi$, $K = K_\xi^{(n)}$.

Step S10.22: Periodicity. $\Psi(\theta + k) = \Psi(\theta)$ (Theorem 8.14(i)), so $K(\theta + k) = K(\theta)$.

Step S10.23: Analyticity. By Theorem 8.14(v), $\theta \mapsto \mathcal{Z}^{-1}\Gamma(\Psi(\theta))$ is `AnalyticOnNhd Real` on $\mathbb{R}^2$ into $E_{1,1}$; compose with the continuous linear τ_E (W8-E8).

Step S10.24: Injectivity modulo $\mathbb{Z}^2$. If $K(\theta) = K(\theta')$, then $\Gamma^{\mathbb{R}}(x(\theta)) = \Gamma^{\mathbb{R}}(x(\theta'))$ (τ_E injective), so $x(\theta) = x(\theta')$ by Corollary 5.27(iii) and Lemma 10.9(v), and $\Psi(\theta) = \Psi(\theta')$ ($\mathcal{Z}$ bijective). Comparing the $+i$ components with Lemma 10.9(ii): the moduli agree and $e^{-2\pi i(\theta_i + X_i(\theta) - \theta_i' - X_i(\theta'))} = 1$, so $k := \theta + X(\theta) - \theta' - X(\theta') \in \mathbb{Z}^2$ (W8-E4). Put $\theta'' = \theta' + k$. By Lemma 10.9(iii), $X(\theta'') = X(\theta')$, hence $\theta - \theta'' = X(\theta'') - X(\theta)$ and, by (iv) there, $|\theta - \theta''| \leq \frac{1}{2}|\theta - \theta''|$. So $\theta'' = \theta$ and $\theta - \theta' = k$.

Step S10.25: Derivative. Let $\theta, h \in \mathbb{R}^2$ with `fderiv Real K θ h` $= 0$. By Corollary 3.29 applied to $\tilde{K} = \mathcal{Z}^{-1}\Gamma \circ \Psi$ on $T_{s_0/2}$ (holomorphic: Step S8.31) and the chain rule (W8-E8), τ_E^{-1}`fderiv Real K θ h` $= \operatorname{re}\big(\mathcal{Z}^{-1}D\Gamma(\Psi(\theta))\, D\Psi(\theta)h\big)$. For real θ and h, $D\Psi(\theta)h = \lim_{s\to 0}(\Psi(\theta + sh) - \Psi(\theta))/s$ is a limit of real points, hence ϱ-fixed; $D\Gamma(w)$ commutes with ϱ at real w (differentiate $\Gamma \circ \varrho = \varrho \circ \Gamma$, Theorem 5.21(iv)); so the vector inside re is real and equals 0. $D\Gamma(\Psi(\theta))$ is invertible on $\mathfrak{W}$

(Theorem 5.21(iii), $\|\Psi(\theta)\| < \rho_1 \le \rho_0$) and $\mathcal{Z}^{-1}$ is injective, so $D\Psi(\theta)h = 0$. Its $+i$ component is, by Lemma 10.9(ii) (differentiating along the real line $s \mapsto \theta + sh$),

$$e^{-2\pi i\phi_i}\Big(\frac{Dy_i(\theta)h}{2\sqrt{\xi_i + y_i(\theta)}} - 2\pi i\sqrt{\xi_i + y_i(\theta)}\,\big(h_i + DX_i(\theta)h\big)\Big), \qquad \phi_i = \theta_i + X_i(\theta),$$

with $Dy_i(\theta)h$, $DX_i(\theta)h$ real. Its vanishing forces the imaginary part of the bracket to vanish, so $h_i = -DX_i(\theta)h$ for $i = 1, 2$; then $|h| \le \|DX(\theta)\||h| \le \frac{1}{2}|h|$ (Lemma 10.9(iv)), so $h = 0$. □

Source: P96-K Theorem A, p. 122 ("each embedding is real analytic"); KP96 §7, Steps 4 and 7 ("real analytic embedding"; inverse X^{-1} of the angular component). The injectivity argument through the $\frac{1}{2}$-Lipschitz angle error and the injectivity of $\Gamma^{\mathbb{R}}$ is ours (W3 Remark 5.30, return trip).

Lemma 10.12 (Closeness and size on one shell)**.** *For $n \ge 0$, $\xi \in \Gamma_n$ and $\theta \in \mathbb{R}^2$, with K^0_ξ the linear torus,*

$$\|K^{(n)}_\xi(\theta) - \tau_E K^0_\xi(\theta)\| \le E_7\, t_n^{33/64}, \qquad \|K^{(n)}_\xi(\theta)\| \le 2\rho_n.$$

Proof. Write $\Psi = \tilde\Psi^{(n)}_\xi$, $w = \Psi(\theta)$, $w^0 = \mathcal{A}_\xi(\iota\theta)$, $x = \mathcal{Z}^{-1}w$, $x^0 = \mathcal{Z}^{-1}w^0 = K^0_\xi(\theta)$ (Lemma 7.18(i)), $e = e^{(n)}_{\mathrm{tor}}$.

Step S10.26: Tangential modes. By Lemma 10.9(ii) and $w^0_{\pm i} = \sqrt{\xi_i}e^{\mp 2\pi i\theta_i}$,

$$\begin{aligned}|w_{\pm i} - w^0_{\pm i}| \le |\sqrt{\xi_i + y_i} - \sqrt{\xi_i}| + \sqrt{\xi_i}\,|e^{\mp 2\pi i X_i} - 1| &\le \frac{|y_i|}{\sqrt{\xi_i}} + 2\pi\sqrt{\xi_i}|X_i| \\ &\le \frac{r_n^2 e}{\sqrt{\alpha_n}} + 2\pi\sqrt{t_n}\,e \le (1+2\pi)\sqrt{t_n}\,e\end{aligned}$$

by (W8-E5), (W8-E4), Lemma 10.3(ii), and $r_n^2/\sqrt{\alpha_n} = t_n^{97/128} \le t_n^{1/2}$.

Step S10.27: Phase-space distance. The tangential weights of $\mathfrak{W}_{1,1}$ are $|l|^2e^{2|l|}$, so the tangential part of $w - w^0$ has norm $\le (2(e^2+4e^4))^{1/2}(1+2\pi)\sqrt{t_n}\,e \le 22(1+2\pi)\sqrt{t_n}\,e$; the normal part of w^0 is 0 and that of w has norm $\le r_n e \le \sqrt{t_n}\,e$ (Lemma 10.9(i)). Hence $\|x - x^0\|_E \le \|w - w^0\| \le 162\sqrt{t_n}\,e$ (Lemma 5.3).

Step S10.28: Birkhoff map. By Corollary 5.27(ii) and Lemma 3.5, $\|\Gamma^{\mathbb{R}}(x) - x\|_E \le \|\Gamma^{\mathbb{R}}(x) - x\|_{E_{1,2}} \le 6\sqrt{2}C_F\|x\|_E^3 \le 6\sqrt{2}C_F\rho_n^3 \le E_6 t_n^{3/2}$ (Lemma 10.9(v), $\rho_n^2 \le 2c_A t_n$).

Step S10.29: Closeness. τ_E is isometric, so $\|K^{(n)}_\xi(\theta) - \tau_E K^0_\xi(\theta)\| \le 162\sqrt{t_n}\,e + E_6 t_n^{3/2} \le (162E_5 + E_6)t_n^{1/2+1/64}$ by Lemma 10.7 and $t_n^{3/2} \le t_n^{33/64}$.

Step S10.30: Size. $6\sqrt{2}C_F\|x\|_E^2 \le 6\sqrt{2}C_F\rho_1^2/2 = 6\sqrt{2}/320 < 1$ ($\rho_1^2 = 1/(160C_F)$), so $\|K^{(n)}_\xi(\theta)\| = \|\Gamma^{\mathbb{R}}(x)\|_E \le 2\|x\|_E < 2\rho_n$. □

Source: KP96 §7, Step 7, p. 30 ("$\frac{1}{\sqrt{2}}|\tilde q' - \tilde q| = |\sqrt{I+y} - \sqrt{I}| \le |y| \min \sqrt{I_j}^{-1}$", "$\|\hat q\| \le \|u\| + \|v\|$"), measured here at the same angle (no radial reparametrization); W3 Remark 5.30 ("the closeness $\|\Gamma^{\mathbb{R}}(x) - x\| \le 6\sqrt{2}C_F\|x\|^3$ is $o(r)$").

Lemma 10.13 (Invariance and continuity on one shell)**.** *Let $n \ge 0$.*

(i) For $\xi \in \Gamma_n$, $\theta \in \mathbb{R}^2$, $j \in \mathbb{N}_+$, with $K = K^{(n)}_\xi$ and $\nu = \nu^{(n)}_*(\xi)$, the two `HasDerivAt` *statements of clause (4) of* `nlw2` *hold: $t \mapsto$* `coeff (K (θ + t • ν)).1 j` *has derivative λ_j* `coeff (K θ).2 j` *at 0, and $t \mapsto$* `coeff (K (θ + t • ν)).2 j` *has derivative $-\lambda_j$* `coeff (K θ).1 j − cubicGrad m hm (K θ).1 j` *at 0.*

(ii) $(\xi, \theta) \mapsto K^{(n)}_\xi(\theta)$ is continuous on $\Gamma_n \times \mathbb{R}^2$.

Proof. **Step S10.31: Weak invariance.** By Theorem 8.14(i),(iii) (Lemma 10.6), restricted to $\theta \in \mathbb{R}^2$, $\Psi = \tilde{\Psi}_\xi^{(n)}$ maps into B_{ρ_1}, is differentiable into $\mathfrak{W}$ (holomorphic), real, and satisfies $D\Psi(\theta)\nu = X_{H'}(\Psi(\theta))$ in $\mathfrak{W}_{a,p-1}$ with $X_{H'} = X_\Lambda + X_{\bar{G}} + X_{G^{\mathrm{nor}}} + X_K$ (Theorem 5.25).

Step S10.32: Transport. Corollary 5.29 gives the coefficient system (2) along ν for $(Q,P) = \mathcal{Z}^{-1}\Gamma \circ \Psi$: $D_\nu Q_j = \lambda_j P_j$, $D_\nu P_j = -\lambda_j Q_j - G_q(Q)_j$.

Step S10.33: Frozen form. $\kappa_{1,1}$ of the components of $K(\theta+t\nu)$ is $(Q,P)(\theta+t\nu)$; the derivative at $t=0$ of a coordinate of the differentiable curve $t \mapsto (Q,P)(\theta+t\nu)$ is the coordinate of $D_\nu(Q,P)(\theta)$ (Lemma 3.3); and `cubicGrad m hm (K θ).1 j` $= G_q(Q(\theta))_j$ (Corollary 3.16(iv)), $\lambda_j =$ `lam m hm j`. This is (i).

Step S10.34: Continuity. Theorem 8.14(ii) (Lemma 10.6): $(\xi,\theta) \mapsto \tilde{\Psi}_\xi^{(n)}(\theta)$ is continuous on $\Gamma_n \times \mathbb{R}^2$ into $\mathfrak{W}$, with values in $\mathcal{Z}(B^E_{r_1})$ (Lemma 10.9(v)); $\mathcal{Z}^{-1}$ and τ_E are continuous linear and $\Gamma^{\mathbb{R}}$ is analytic, hence continuous, on $B^E_{r_1}$ (Corollary 5.27(i)). This is (ii). □

Source: P96-W p. 290; W6 Remark 8.15 (return to Cartesian coordinates: "which is clause (4)"); W3 Remark 5.30, return trip.

10.5. **Patching.**

Definition 10.14 (The patched family)**.** Let $\Gamma = \bigcup_{n\ge 0}\Gamma_n$ (Definition 10.5). For $I \in \mathbb{R}^2$ let $n(I)$ be the unique n with $I \in \mathcal{R}_n$ when $I \in (0,t_0)^2$ (Lemma 9.25(i)). Define

$$\nu(I) = \begin{cases} \nu_*^{(n(I))}(I), & I \in \Gamma, \\ 0, & \text{otherwise}, \end{cases} \qquad K(I)(\theta) = K_I(\theta) = \begin{cases} K_I^{(n(I))}(\theta), & I \in \Gamma, \\ 0, & \text{otherwise}. \end{cases}$$

Together with r_0, C, τ, η of Definition 10.1, these are the witnesses of `nlw2`. (Lean: $n(I)$ by `Nat.find` on $t_{n+1} \le \max_i I_i$; the case split by `if h : I ∈ Γ`, decidable classically.)

Source: KP96 p. 28, Step 5 ("Define the embedding ... by piecing together the corresponding definitions on each component. ... Similarly for $\tilde{\omega}$").

Lemma 10.15 (Locality of the patching)**.** *Let $O_n = \Pi_n$ (open). Then the O_n are pairwise disjoint, $\Gamma_n \subset O_n$, and $\Gamma \cap O_n = \Gamma_n$. For $I \in \Gamma_n$: $n(I) = n$, $\nu(I) = \nu_*^{(n)}(I)$ and $K_I = K_I^{(n)}$. Consequently $\nu = \nu_*^{(n)}$ on $\Gamma \cap O_n$ and $K = K^{(\cdot)}$ on $(\Gamma \cap O_n) \times \mathbb{R}^2$.*

Proof. **Step S10.35: Disjointness.** $O_n \subset \mathcal{R}_n$ (Lemma 10.3(ii)) and the $\mathcal{R}_n$ are disjoint (Lemma 9.25(i)); $\Gamma_n \subset \Pi_n$ by construction ($\Pi_* \subset \Pi_{-1} = \Pi_n \setminus \Xi$).

Step S10.36: Identification. If $I \in \Gamma \cap O_n$, then $I \in \Gamma_{n'}$ for some n', and $\Gamma_{n'} \subset \mathcal{R}_{n'}$ meets $O_n \subset \mathcal{R}_n$ only if $n' = n$. For $I \in \Gamma_n \subset \mathcal{R}_n$, $n(I) = n$, and the definitions unfold. □

Source: KP96 p. 28 ("These definitions are correct, since the C_{r_k} are disjoint and even have a pairwise positive distance to each other. So Φ is certainly continuous in I").

Remark 10.16 (Shell boundaries). No consistency of the shell constructions across shell boundaries is needed or claimed: the parameter sets Γ_n lie in the open trimmed shells $O_n = U_{-\alpha_n}\mathcal{R}_n$, which are pairwise disjoint and at distance $> \alpha_n$ from $\mathbb{R}^2 \setminus \mathcal{R}_n$ (Lemma 10.3(ii)). Each point of Γ therefore has an open neighbourhood O_n on which Γ, ν and K coincide with the single-shell objects, and continuity on Γ is local (Lemma 10.17). The price is the boundary layer $\mathcal{R}_n \setminus O_n$, of measure $\le 7\alpha_n t_n = o(\mathrm{Leb}\,\mathcal{R}_n)$ (Lemma 9.25(ii)), already accounted for in Corollary 9.27.

Lemma 10.17 (Regularity of the patched family)**.** *Γ is Borel, $\Gamma \subset \{I : (\forall i,\ 0 < I_i) \wedge r(I) < r_0\}$, ν is continuous on Γ, and $(I,\theta) \mapsto K_I(\theta)$ is continuous on $\Gamma \times \mathbb{R}^2$.*

Proof. **Step S10.37: Borel.** Countable union of the Borel Γ_n (Lemma 10.8(iv); (W7-E4)).

Step S10.38: Inclusion. $\Gamma_n \subset (0,t_n)^2 \subset (0,t_0)^2$, and $I_1 + I_2 < 2t_0 = r_0^2$ there, so $r(I) < r_0$ (`Real.sqrt_lt_sqrt`).

Step S10.39: Continuity of ν. For $I \in \Gamma_n$, O_n is an open neighbourhood with $\nu = \nu_*^{(n)}$ on $\Gamma \cap O_n = \Gamma_n$, continuous by Lemma 10.8(iv); apply (W8-E8).

Step S10.40: Continuity of K. For $(I,\theta) \in \Gamma_n \times \mathbb{R}^2$, $O_n \times \mathbb{R}^2$ is an open neighbourhood on which $(I',\theta') \mapsto K_{I'}(\theta')$ agrees on $(\Gamma \times \mathbb{R}^2) \cap (O_n \times \mathbb{R}^2) = \Gamma_n \times \mathbb{R}^2$ with the map of Lemma 10.13(ii); apply (W8-E8). □

Source: KP96 p. 28, Step 5; W7 Lemma 9.21.

10.6. **Family injectivity.**

Lemma 10.18 (Uniqueness for the coefficient system)**.** *Let $Y, Y' : \mathbb{R} \to$ `WaveSpace` be analytic coefficient solutions (Definition 4.4) with $Y(0) = Y'(0)$. Then $Y = Y'$.*

Proof. Write $Y_i(t) = (Y\,t).i$, $\omega_j = \omega_{1,1}(j)$, and $g(y)_j = \omega_j$ `cubicGrad m hm y j` for $y \in \ell^2_{\mathbb{R}}$.

Step S10.41: Raw equations. By Corollary 3.16(iv), $g(y)_j = j^{-1}\mathcal{G}^w(y)_j$, so $g : \ell^2_{\mathbb{R}} \to \ell^2_{\mathbb{R}}$ with $\|g(y) - g(y')\| \le 3C_m\|y - y'\| \max(\|y\|, \|y'\|)^2$ (part (iii) there). By Lemma 4.6, Y_1, Y_2 are differentiable into $\ell^2_{\mathbb{R}}$ with, after multiplying the coefficient identities by ω_j, $(Y_1')_j = \lambda_j (Y_2)_j$ and $(Y_2')_j = -\lambda_j (Y_1)_j - g(Y_1)_j$; likewise for Y'.

Step S10.42: Regularity. The sequences $(\lambda_j (Y_2)_j) = Y_1'$ and $(\lambda_j (Y_1)_j) = -Y_2' - g(Y_1)$ lie in $\ell^2_{\mathbb{R}}$, and the same for Y'.

Step S10.43: Energy identity. Put $D_i = Y_i - Y_i'$ and $f(t) = \|D_1(t)\|^2 + \|D_2(t)\|^2$. By (W8-E1), f is differentiable with $f' = 2\langle D_1, D_1'\rangle + 2\langle D_2, D_2'\rangle$, and

$$\langle D_1, D_1'\rangle = \sum_j (D_1)_j \lambda_j (D_2)_j, \qquad \langle D_2, D_2'\rangle = -\sum_j (D_2)_j \lambda_j (D_1)_j - \langle D_2, g(Y_1) - g(Y_1')\rangle,$$

all series absolutely convergent by Step S10.42 (products of two ℓ^2 sequences). Hence $f' = -2\langle D_2, g(Y_1) - g(Y_1')\rangle$.

Step S10.44: Differential inequality. Fix $T > 0$ and $R \ge \|Y(t)\|, \|Y'(t)\|$ on $[-T,T]$ ((W8-E3); Y, Y' are analytic, hence continuous). With $L = 3C_m R^2$, Cauchy–Schwarz and $\|Y_1\| \le \|Y\|$ give $|f'| \le 2L\|D_2\|\|D_1\| \le Lf$ on $[-T,T]$.

Step S10.45: Gronwall. $h_+(t) = e^{-Lt} f(t)$ has $h_+' = e^{-Lt}(f' - Lf) \le 0$ on $[0,T]$ and $h_-(t) = e^{Lt} f(t)$ has $h_-' \ge 0$ on $[-T,0]$ (W8-E2); with $h_\pm(0) = f(0) = 0$ and $h_\pm \ge 0$, both vanish. Hence $f = 0$ on $[-T,T]$ for every T, so $D_1 = D_2 = 0$ and $Y = Y'$. □

Source: ours. P96-W uses no uniqueness statement; the energy argument is the standard one for the semilinear wave equation, in coefficient form.

Lemma 10.19 (Winding)**.** *Let $a, a' : \mathbb{R} \to (0,\infty)$ and $\phi, \phi' : \mathbb{R} \to \mathbb{R}$ be continuous, $\nu_0, \nu_0' \in \mathbb{R}$, with $\phi(t) - t\nu_0$ and $\phi'(t) - t\nu_0'$ bounded, and $a(t)e^{-2\pi i\phi(t)} = a'(t)e^{-2\pi i\phi'(t)}$ for all t. Then $\nu_0 = \nu_0'$.*

Proof. **Step S10.46: Moduli and phases.** Taking moduli, $a = a'$; dividing, $e^{-2\pi i(\phi - \phi')} = 1$, so $\phi(t) - \phi'(t) \in \mathbb{Z}$ for all t (W8-E4).

Step S10.47: Constant. $\phi - \phi'$ is continuous and integer valued on $\mathbb{R}$, hence a constant c (W8-E4).

Step S10.48: Rates. $t(\nu_0 - \nu_0') = c - (\phi(t) - t\nu_0) + (\phi'(t) - t\nu_0')$ is bounded in t, so $\nu_0 = \nu_0'$ (W8-E4). □

Source: ours.

Lemma 10.20 (The frequency map is injective on Γ)**.** *If $I, I' \in \Gamma$ and $\nu(I) = \nu(I')$, then $I = I'$.*

Proof. Let $I \in \Gamma_n$, $I' \in \Gamma_{n'}$; by Lemma 10.15, $2\pi\nu(I) = \omega_*^{(n)}(I)$ and $2\pi\nu(I') = \omega_*^{(n')}(I')$.

Step S10.49: Same shell. If $n = n'$, Lemma 10.8(ii) gives $0 \geq \frac{7}{8L_A}|I - I'|$.

Step S10.50: Different shells. If $n \neq n'$, then $I' \notin \mathcal{R}_n$ and $I \notin \mathcal{R}_{n'}$, so $|I - I'| > \max(\alpha_n, \alpha_{n'})$ (Lemma 10.3(ii)). By Lemma 9.4(iv) and Lemma 10.8(iii),

$$|\omega_*^{(n)}(I) - \omega_*^{(n')}(I')| \geq \frac{|I - I'|}{L_A} - \frac{\alpha_n + \alpha_{n'}}{70L_A} > \frac{\max(\alpha_n, \alpha_{n'})}{L_A}\Big(1 - \frac{2}{70}\Big) > 0,$$

contradicting $\nu(I) = \nu(I')$. □

Source: ours; KP96 p. 28, Step 6 ("Otherwise, $J \in C_{r_l}$ with $l > k$. Then $|I - J| \geq \alpha(r_k)$") for the separation of patches.

Proposition 10.21 (Family injectivity)**.** *For all $I, I' \in \Gamma$ and $\theta, \theta' \in \mathbb{R}^2$: if $K_I(\theta) = K_{I'}(\theta')$, then $I = I'$ and $\theta - \theta' \in \mathbb{Z}^2$. This is clause (6) of* `nlw2`*.*

Proof. Let $I \in \Gamma_n$, $I' \in \Gamma_{n'}$, $\nu = \nu(I)$, $\nu' = \nu(I')$, $\Psi = \tilde{\Psi}_I^{(n)}$, $\Psi' = \tilde{\Psi}_{I'}^{(n')}$, with the functions X, y of Lemma 10.9 for Ψ and X', y' for Ψ'.

Step S10.51: Orbits. K_I and $K_{I'}$ are `AnalyticOnNhd Real` on $\mathbb{R}^2$ (Lemma 10.11) and satisfy the invariance (7) with ν, ν' (Lemma 10.13(i), Lemma 10.15). By Lemma 4.5, $Y(t) = K_I(\theta + t\nu)$ and $Y'(t) = K_{I'}(\theta' + t\nu')$ are analytic coefficient solutions, and $Y(0) = Y'(0)$. By Lemma 10.18, $Y = Y'$.

Step S10.52: Birkhoff coordinates. For every t, $\Gamma^{\mathbb{R}}(\mathcal{Z}^{-1}\Psi(\theta + t\nu)) = \Gamma^{\mathbb{R}}(\mathcal{Z}^{-1}\Psi'(\theta' + t\nu'))$ (τ_E injective), both arguments in $B_{r_1}^E$ (Lemma 10.9(v)). By Corollary 5.27(iii) and bijectivity of $\mathcal{Z}$, $\Psi(\theta + t\nu) = \Psi'(\theta' + t\nu')$ for all $t \in \mathbb{R}$.

Step S10.53: Frequencies. Fix $i \in \{1, 2\}$ and compare the $+i$ components by Lemma 10.9(ii): with $a(t) = \sqrt{I_i + y_i(\theta + t\nu)}$, $\phi(t) = \theta_i + t\nu_i + X_i(\theta + t\nu)$ and the primed analogues, $a, a' > 0$ and ϕ, ϕ' are continuous, $|\phi(t) - t\nu_i| \leq |\theta_i| + e_{\mathrm{tor}}^{(n)}$ and likewise for ϕ' (Lemma 10.9(i)). Lemma 10.19 gives $\nu_i = \nu_i'$. So $\nu(I) = \nu(I')$.

Step S10.54: Actions. $I = I'$ by Lemma 10.20.

Step S10.55: Angles. Now $K_I(\theta) = K_I(\theta')$, and the third conjunct of `IsAnalyticTorusEmbedding` K_I (Lemma 10.11) gives $k \in \mathbb{Z}^2$ with $\theta - \theta' = k$. □

Source: P96-W Main Theorem, pp. 272–273 ("a Lipschitz continuous embedding Φ" of the family, whose restriction to each torus is an embedding), Cantor Manifold Theorem and Remark 1, pp. 288–289 ("Lipschitz continuous embedding $\Psi : \mathcal{T}[\mathcal{C}] \to \mathcal{E}$"; "a parametrization of each torus in which the flow is linear"); KP96 §7, Steps 5–6. Deviation: the source obtains injectivity from the Lipschitz estimates (11) of KP96 p. 28; here it comes from uniqueness of orbits, the winding of the tangential phases, and the injectivity of the frequency map, since W6 provides only continuity in ξ.

10.7. **The proof of W-NLW2.**

Theorem 10.22 (Proof of W-NLW2)**.** *Theorem 11.5 (`KAM.Wave.nlw2`) holds exactly as frozen: for every $m > 0$ the witnesses r_0, C, τ, η of Definition 10.1 and Γ, ν, K of Definition 10.14, chosen in this order and depending on m only, satisfy every conjunct of the frozen statement.*

Proof. Fix $m > 0$ (`hm : 0 < m`). The existential witnesses are introduced in the frozen order $r_0, C, \tau, \eta, \Gamma, \nu, K$; each is defined from m and the constants of W3–W7 alone (Definitions 10.1, 10.2, 10.5, 10.14), before any I, θ or θ_0.

Step S10.56: Positivity of r_0, C; $\tau > 1$. $t_0 > 0$ as a minimum of five positive numbers, so $r_0 = \sqrt{2t_0} > 0$; $C_3 = 4\sqrt{c_A} > 0$ and $C = \max(1, M_m)C_3 \geq C_3 > 0$ (Corollary 4.11); $\tau = 24 > 1$.

Step S10.57: $\eta \geq 0$ on $(0, r_0)$. For $r > 0$, $r^{1/32} \geq 0$ and $E_7 \geq 0$ (W8-E7).

Step S10.58: $\eta \to 0$. $r \mapsto 2^{33/64} E_7 r^{1/32}$ tends to 0 along `nhdsWithin 0 (Set.Ioi 0)` (W8-E7).

Step S10.59: `MeasurableSet` Γ. Lemma 10.17, Step S10.37.

Step S10.60: $\Gamma \subseteq \{I \mid (\forall i, 0 < I_i) \wedge r(I) < r_0\}$. Lemma 10.17, Step S10.38.

Step S10.61: `ContinuousOn` ν Γ. Lemma 10.17, Step S10.39.

Step S10.62: `ContinuousOn (Function.uncurry K) (`$\Gamma \times^s$ `Set.univ).` Lemma 10.17, Step S10.40.

Step S10.63: Clause (1): density one. Apply Corollary 9.27 with $q = \frac{1}{2}$, the $t_n \leq \frac{1}{2}$ of Definition 10.2, $\alpha_n \in (0, \alpha_0]$, $\alpha_n \leq t_n$ (Lemma 10.3(i)), $\Pi_n = U_{-\alpha_n}\mathcal{R}_n$ (Borel, $\subset (0, t_n)^2$, Lemma 10.3(ii)), the admissible family of the shell iteration with excluded set $\mathcal{E}^{(n)}$ (Lemma 10.6, from Lemma 8.11), and $\mathcal{R}_n \cap (\Pi_n \setminus \mathcal{E}^{(n)}) \subset \Pi_n \setminus \mathcal{E}^{(n)} \subset \Gamma_n \subset \Gamma$ (Lemma 8.12 via Lemma 10.6). It bounds the relative defect of shell n by $\frac{(7+C_7)\alpha_n^{32/47}}{(3/4)t_n} = \frac{4}{3}(7 + C_7)t_n^{1/94} \to 0$ ($\frac{95}{64} \cdot \frac{32}{47} = \frac{95}{94}$; W7 Remark 9.28). Theorem 9.26 (with $\Gamma \subset P$ Borel, Lemma 10.17) gives the `Tendsto` statement of clause (1) in $[0, \infty]$, along `nhdsWithin 0 (Set.Ioi 0)`.

Step S10.64: Clause (2), Diophantine. Let $I \in \Gamma$, $n = n(I)$. By Lemma 10.8(iii), $|\langle k, \omega_*^{(n)}(I)\rangle| \geq (\check{\alpha}_n/2)/A_k$ for $k \neq 0$; Lemma 9.29 with $c = \check{\alpha}_n/2$ and $\nu(I) = \omega_*^{(n)}(I)/(2\pi)$ (Lemma 10.15) gives `KAM.IsDiophantine` γ_I `24 (`ν `I)` with $\gamma_I = \check{\alpha}_n/(8\pi) > 0$, chosen after I as the frozen order allows (W7 Remark 9.30).

Step S10.65: Clause (2), frequency limit. Apply Lemma 9.31 to $\Gamma \subset \bigcup_n \mathcal{R}_n$ and $\omega_* = 2\pi\nu$ on Γ, which equals $\omega_*^{(n)}$ on $\Gamma \cap \mathcal{R}_n = \Gamma_n$, with $e_n = \varepsilon_\omega^{(n)} \leq \alpha_n \to 0$ (Lemma 10.8(iii)): `Tendsto` ν `(nhdsWithin 0` Γ`) (nhds fun i => lam m hm (i : Nat).succPNat / (2 * pi)).`

Step S10.66: Clause (3), embedding. Let I, `hI`, $I \in \Gamma$, $n = n(I)$. $K_I = K_I^{(n)}$ (Lemma 10.15) and Lemma 10.11 give `IsAnalyticTorusEmbedding (K I).`

Step S10.67: Clause (3), closeness. For every θ, Lemma 10.12 with $\tau_E K_I^0(\theta) =$ `linearTorus I hI` θ gives $\|K_I(\theta) -$ `linearTorus I hI` $\theta\| \leq E_7 t_n^{33/64}$. Since $I \in \mathcal{R}_n$, $t_n = 2t_{n+1} \leq 2\max_i I_i \leq 2(I_1 + I_2) = 2r(I)^2$ (Lemma 9.25(iii)), so $E_7 t_n^{33/64} \leq E_7 2^{33/64} r(I)^{33/32} = \eta(r(I))\, r(I)$ (W8-E7).

Step S10.68: Clause (3), size. By Lemma 10.12, $\|K_I(\theta)\| \leq 2\rho_n \leq 2\sqrt{2c_A t_n} \leq 2\sqrt{2c_A}\sqrt{2}\, r(I) = C_3 r(I) \leq C\, r(I)$.

Step S10.69: Clause (4), invariance. For $I \in \Gamma$, θ, j: Lemma 10.13(i) with $K_I = K_I^{(n)}$, $\nu(I) = \nu_*^{(n)}(I)$ (Lemma 10.15).

Step S10.70: Clause (5), PDE solutions. Corollary 4.11 with $C_3 = 4\sqrt{c_A}$, the set Γ, and the maps ν, K: its hypotheses are Steps S10.66, S10.68 (with C_3) and S10.69. It yields clause (5) verbatim with $C = \max(1, M_m)C_3$, the same C as in Step S10.68.

Step S10.71: Clause (6), family injectivity. Proposition 10.21.

Step S10.72: Assembly. The conjunction of Steps S10.56–S10.71, in the frozen order of conjuncts, is the body of `nlw2` for the chosen witnesses. □

Source: P96-W Main Theorem pp. 272–273 and its proof pp. 289–291 ("We can now prove the Main Theorem"), for $J = \{1, 2\}$, in the declared weakened form of Remark 11.7; KP96 §7, Steps 4–8, pp. 27–31.

Frozen conjunct	W8 step	Consumed results
$0 < r_0$, $0 < C$, $1 < \tau$	S10.56	Def. 10.1; Cor. 4.11
$\eta \geq 0$ on $(0, r_0)$; $\eta \to 0^+$	S10.57, S10.58	(W8-E7)
`MeasurableSet` Γ; $\Gamma \subseteq \ldots$	S10.59, S10.60	Lem. 10.17, 10.8(iv); Prop. 8.4(i); Lem. 9.21
`ContinuousOn` ν Γ	S10.61	Lem. 10.15, 10.8(iv); Thm. 8.14(ii)
`ContinuousOn (uncurry K)`	S10.62	Lem. 10.15, 10.13(ii); Thm. 8.14(ii); Cor. 5.27(i)
(1) density one	S10.63	Thm. 9.26, Cor. 9.27, Thm. 9.22, Rem. 9.28; Lem. 8.11, 8.12 via Lem. 10.6; Lem. 10.3
(2) Diophantine, γ_I per I	S10.64	Lem. 8.13 via 10.6; Lem. 10.8(iii), 9.29
(2) $\nu(I) \to \lambda_J/2\pi$ as $I \to 0$ in Γ	S10.65	Lem. 9.31, 10.8(iii), 10.7
(3) analytic torus embedding	S10.66	Lem. 10.11, 10.9; Thm. 8.14(i),(iv),(v); Cor. 5.27(iii), Thm. 5.21(iii); Lem. 3.25, Cor. 3.29
(3) $o(r)$ closeness to `linearTorus`	S10.67	Lem. 10.12, 10.7, 10.4; Lem. 7.18(i),(vi); Cor. 5.27(ii); Lem. 5.3, 9.25(iii)
(3) $\|K_I\| \leq C\, r(I)$	S10.68	Lem. 10.12, 10.3(i); Cor. 4.11
(4) coordinatewise invariance	S10.69	Lem. 10.13(i); Thm. 8.14(iii); Cor. 5.29; Lem. 3.3, Cor. 3.16(iv)
(5) PDE solutions, same C	S10.70	Cor. 4.11, Thm. 4.9
(6) family injectivity	S10.71	Prop. 10.21; Lem. 10.18, 10.19, 10.20; Lem. 4.5, 4.6; Cor. 5.27(iii); Lem. 9.4(iv)

10.8. **Source check, deviations and ledger.**

Remark 10.23 (Pages read and deviations). P96-W pp. 271–273 (PDF 4–6) and pp. 288–291 (PDF 21–24) and KP96 pp. 27–30 (author version) were read for this part. No defect was found on these pages beyond those already recorded (E-PDE-W-1/2). Deviations introduced here:

(a) *Starting size.* Each shell gives W6's parameter ϵ_0 (Definition 8.9, Remark 8.15) the value $\epsilon_{0,n} = \kappa_n^{4/3}\gamma_1\check{\alpha}_n \in [\epsilon_{0,n}^{\flat}, \gamma_1\check{\alpha}_n]$, with $\epsilon_{0,n}^{\flat}$ the sharp size of Lemma 8.8(v) (Definition 10.5, Lemma 10.4); with the upper value $\gamma_1\check{\alpha}$ the term e_Φ would be a constant and clause (3) would not be $o(r)$.

(b) *Injectivity.* Obtained from orbit uniqueness, winding and injectivity of ν rather than from Lipschitz estimates in ξ (KP96 p. 28, (11)), which W6 does not provide.

(c) *Closeness at the same angle.* KP96 §7, Step 7 compares after the radial reparametrization X^{-1}; the frozen clause (3) compares $K_I(\theta)$ with $K_I^0(\theta)$ at the same θ, and the bound $O(r^{33/32})$ is obtained directly.

(d) *Explicit shells.* $q = \frac{1}{2}$, $r_n^2 = t_n^{3/2}$, $\alpha_n = t_n^{95/64}$, margins α_n (KP96: η, $r^{2\lambda} = t$, the geometric mean choice of Step 8).

Results. Definitions 10.1, 10.2, 10.5, 10.10, 10.14; Lemmas 10.3, 10.4, 10.6, 10.7, 10.8, 10.9, 10.11, 10.12, 10.13, 10.15, 10.17, 10.18, 10.19, 10.20; Proposition 10.21; Theorem 10.22 (the root); Remark 10.16. External: (W8-E1)–(W8-E8). *Labels.* No forward references to unwritten parts; the headline label `bpW:wh:nlw2` is defined in the next section.

11. The frozen headline W-NLW2

Status: FROZEN (owner decision D41; Lean text in `KAM/Frozen/Wave/`; `nlw2` is `DRAFT_SORRY`).

This section states the headline proposed in the scoping report (`docs/explorations/pde-kam-and-leapfrogging.md`, §2.3), adopted for the source/design phase by owner decision D30, with every definition written out in the conventions of §1. After independent statement audit, the owner approved it (decision D41): the theorem W-NLW2 (`KAM.Wave.nlw2`) and its eleven definitions (`lam`, `sineMode`, `ellWeight`, `WaveSpace`, `coeff`, `reconU`, `reconV`, `cubicGrad`, `linearTorus`, `IsAnalyticTorusEmbedding`, `IsClassicalSolution`) are frozen in `KAM/Frozen/Wave/` (`Defs/` and

`Theorems/`); the definitions are frozen whole, and `nlw2` is frozen with the manifest-bound body `by sorry` (`DRAFT_SORRY`) that the proof campaign replaces. W-NLW2 is *the D30 first headline (a declared weakening of P96-W's Main Theorem for $J = \{1, 2\}$)*; it is not the full specialized Main Theorem. The weakenings are listed in Remark 11.7.

11.1. **Definitions.** Fix $m > 0$, the weights $a = p = 1$ (Convention 1.5), the phase space $E = E_{1,1}$ with norm $\|(q,p)\|_E = \max(\|q\|_{1,1}, \|p\|_{1,1})$, the reconstruction U_m, V_m (Definition 1.7) and the cubic gradient G_q (Definition 1.9). Let $P = (0,\infty)^2$, $r(I) = \sqrt{I_1 + I_2}$ for $I \in P$, and $B_h = \{I \in \mathbb{R}^2 : \max_i |I_i| < h\}$.

Definition 11.1 (Linear torus)**.** For $I \in P$, $K_I^0 : \mathbb{R}^2 \to E$ is the map of Convention 1.13.

Definition 11.2 (Analytic torus embedding)**.** A map $K : \mathbb{R}^2 \to E$ is an *analytic torus embedding* if it is $\mathbb{Z}^2$-periodic, real analytic (each point has a neighbourhood on which K is the restriction of a holomorphic map from a complex neighbourhood in $\mathbb{C}^2$ to the complexification $E_{\mathbb{C}}$), injective modulo $\mathbb{Z}^2$ ($K(\theta) = K(\theta') \Rightarrow \theta - \theta' \in \mathbb{Z}^2$), and has injective derivative $DK(\theta) : \mathbb{R}^2 \to E$ at every θ.

Definition 11.3 (Derivative along a frequency)**.** For $\nu \in \mathbb{R}^2$ and differentiable K, $D_\nu K(\theta) = DK(\theta)\nu$.

Definition 11.4 (Diophantine)**.** $\nu \in \mathbb{R}^2$ is (γ, τ)-Diophantine if $|k \cdot \nu| \geq \gamma |k|_1^{-\tau}$ for all $k \in \mathbb{Z}^2 \setminus \{0\}$, $|k|_1 = |k_1| + |k_2|$. (Same as the frozen track-K definition `KAM.IsDiophantine`.)

11.2. **Statement.**

Theorem 11.5 (W-NLW2, frozen)**.** *For every $m > 0$ there exist $r_0 > 0$, $C > 0$, $\tau > 1$, a function $\eta : (0, r_0) \to [0, \infty)$ with $\eta(r) \to 0$ as $r \downarrow 0$, a Borel set $\Gamma \subset \{I \in P : r(I) < r_0\}$, a continuous map $\nu : \Gamma \to \mathbb{R}^2$, and maps $K_I = (Q_I, P_I) : \mathbb{R}^2 \to E$ ($I \in \Gamma$) such that $(I, \theta) \mapsto K_I(\theta)$ is continuous on $\Gamma \times \mathbb{R}^2$ and:*

(1) (Density one in actions.) $\displaystyle\lim_{h \downarrow 0} \frac{\mathrm{Leb}_2(\Gamma \cap B_h)}{\mathrm{Leb}_2(P \cap B_h)} = 1.$

(2) (Two independent frequencies.) *For every $I \in \Gamma$ there is $\gamma_I > 0$ such that $\nu(I)$ is (γ_I, τ)-Diophantine, and $\nu(I) \to (\lambda_1, \lambda_2)/(2\pi)$ as $I \to 0$ in Γ.*

(3) (Small embedded tori.) *Each K_I is an analytic torus embedding, and for all $I \in \Gamma$*

$$\sup_\theta \|K_I(\theta) - K_I^0(\theta)\|_E \leq \eta(r(I))\, r(I), \qquad \sup_\theta \|K_I(\theta)\|_E \leq C\, r(I).$$

(4) (Exact invariance.) *For every $I \in \Gamma$, $\theta \in \mathbb{R}^2$ and $j \geq 1$,*

$$D_{\nu(I)} Q_{I,j}(\theta) = \lambda_j P_{I,j}(\theta), \qquad D_{\nu(I)} P_{I,j}(\theta) = -\lambda_j Q_{I,j}(\theta) - G_q(Q_I(\theta))_j.$$

(5) (PDE solutions.) *For every $I \in \Gamma$ and $\theta_0 \in \mathbb{R}^2$, the function $u(t,x) = U_m\big(Q_I(\theta_0 + t\nu(I))\big)(x)$ is real analytic in a neighbourhood of every point of $\mathbb{R} \times [0, \pi]$, is a classical solution of (1) on $\mathbb{R}$, satisfies $u_t(t, \cdot) = V_m\big(P_I(\theta_0 + t\nu(I))\big)$, and*

$$\sup_{t \in \mathbb{R},\, x \in [0,\pi]} \big(|u| + |u_x| + |u_t|\big) \leq C\, r(I).$$

(6) (Family injectivity.) *For all $I, I' \in \Gamma$ and $\theta, \theta' \in \mathbb{R}^2$, if $K_I(\theta) = K_{I'}(\theta')$ then $I = I'$ and $\theta - \theta' \in \mathbb{Z}^2$. Thus distinct parameters give disjoint tori, and each torus is embedded (with (3)).*

Remark 11.6 (Reading of the statement).

- *Quantifier order.* $r_0, C, \tau, \eta, \Gamma, \nu, K$ are chosen after m and before I and θ_0; γ_I may degenerate as $I \to 0$. No uniformity in m is claimed.
- *Invariance* in (4) is the coefficient system (2) along the linear flow on the torus (period-one angles, Convention 1.13); by Proposition 2.6(i) it yields (5) once the required time regularity is checked (W2). The cubic term is the physical force by (4).

- *Normalization independence.* The statement uses only the fixed real Hamiltonian, the standard actions $I_j = (q_j^2 + p_j^2)/2$ (through K_I^0) and period-one angles; it contains none of the printed P96-W constants. The frequency asymptotics in (2) are only the leading term; the twist A of Theorem 2.10 enters the proof, not the statement.
- *Family injectivity* (6) is proved in W8 from the source's family construction: the tori are images of the parameter tori under a single map that is injective on the union of the parameter tori (P96-W Main Theorem pp. 272–273, "embedding" of the family; Cantor Manifold Theorem and Remark 1 pp. 288–289).
- *Deviations from the source* (to be audited): pure cubic nonlinearity; $J = \{1, 2\}$; $a = p = 1$; period-one angles; Borel density-one set instead of a Cantor set (isolated points are allowed, cf. KP96 p. 24, Remark 2); linear stability, Whitney regularity, and the Lipschitz extension of P96-W Remark 2 p. 289 are omitted; the closeness to K_I^0 is a little-$o(r)$ bound rather than the $O(r^\sigma)$, $\sigma > 1$, of the radial map of the Cantor Manifold Theorem (P96-W p. 289), because the linearizing parametrization of Remark 1 there has a worse estimate (KP96 §7, Step 7).

Remark 11.7 (Declared weakenings). Relative to P96-W's Main Theorem specialized to $J = \{1, 2\}$, W-NLW2 declares the following weakenings (orchestrator ruling W-A2, recorded for owner review). *Deferred together* (D30 Q3): linear stability of the tori; Whitney regularity of the family; and the Lipschitz regularity of the family in Cartesian coordinates, i.e. the source's Lipschitz Cantor manifold (P96-W pp. 272–273, 288–289); W-NLW2 requires only continuity of $(I, \theta) \mapsto K_I(\theta)$ on $\Gamma \times \mathbb{R}^2$. *Retained*: each torus is an analytic embedding (3), and the family is injective (6), so the union of the tori is a disjoint union of embedded tori; what is not asserted is that the induced map on $\Gamma \times \mathbb{T}^2$ is a Lipschitz (bi-Lipschitz-parametrized) embedding. *Also declared*: Borel density-one parameter set instead of a Cantor set; $o(r)$ closeness instead of $O(r^\sigma)$; pure cubic nonlinearity, $a = p = 1$, period-one angles (Remark 11.6).

Source: P96-W Main Theorem pp. 272–273 (condition (4) holds for $J = \{1, 2\}$), Lemma 1 p. 278, Cantor Manifold Theorem and Remark 1 p. 289, assembly pp. 290–291; KP96 Theorem 2 p. 19 and §7 pp. 25–31; P96-K Theorems A, D.